

Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency

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Abstract

Observer-Patch Holography describes finite patches that record shared boundaries and repair disagreement. On a declared twelve-port branch, normalized repair response gives a rank-three metric quotient. Signed records and conservative seam currents are dense in its continuous Euclidean completion. Conservative Fibonacci record populations, with complete reads inside a shrinking metric radius and a declared model clock, have causal orders and normalized counts converging to flat $1 + 3$ spacetime. Their ordering fraction in interior causal diamonds tends to $1/10$. Segment-covering estimates control the finite causal and volume errors. For nonzero closed convex pointed displacement cones, operational boost covariance selects Lorentz shape; accessible interval counts reconstruct proper-time ratios. A separate positive scalar action on source-metric populations has controlled continuum detector readings. Modular covariance and a round celestial limit give Lorentz-frame kinematics. Positive null translations and finite tomography reconstruct stress tensors under locality, covariance and Ward hypotheses. With same-family physical stress identification, fixed-cap entropy stationarity, controlled small-ball and curvature limits, universal coupling, a vacuum reference and an independent scale, the Einstein equation follows up to a cosmological constant. Finite MaxEnt identities also determine the fixed-screen capacity-transfer law and de Sitter shock sign.

Derivation map

Normalized repair response gives one positive rank-three source Gram quotient E . Signed cumulative records and conservative seam words embed densely in its Euclidean completion. Its unit directions form S^2 ; adjoining a scalar line gives $\mathbb{R} \oplus E$ with quadratic form $t^2 - \|x\|^2$. This connects the source metric to a four-dimensional Lorentz carrier without selecting a physical event clock or interpreting every algebraic direction as a measured signal.

The conditional implications on this common carrier are

$$\begin{aligned}
& \begin{array}{l} \text{nonzero closed convex pointed cone} \\ \text{finite irreducible operational rotations} \\ \text{one continuous boost direction} \end{array} \implies \begin{array}{l} \text{Lorentz displacement cone} \\ \text{up to time orientation} \end{array}, \\
& \begin{array}{l} \text{conservative Fibonacci populations} \\ \text{complete local reads and } \Delta=a/c \end{array} \implies \begin{array}{l} \text{flat causal-order limit} \\ \text{raw count-volume convergence} \end{array}, \\
& \begin{array}{l} \text{controlled order/count limit} \\ \text{accessible interval records} \\ \text{one reference interval} \end{array} \implies \text{proper-time ratios from counts}, \\
& \begin{array}{l} \text{positive source-metric scalar action} \\ \text{controlled sampling and smooth data} \end{array} \implies \begin{array}{l} \text{continuum detector bounds} \\ \text{and finite oscillator quantization} \end{array}, \\
& \text{positive translations and finite null tomography} \implies \text{conditional conserved } \langle T_{ab} \rangle, \\
& \begin{array}{l} \text{physical stress and universal coupling} \\ \text{entropy, small-ball and tensor-curvature limits} \end{array} \implies G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle.
\end{aligned}$$

These implications have distinct hypotheses. In particular, operational boost covariance acts on admissible histories and interventions; an algebraic action on the celestial sphere does not establish it. The finite proper carrier rotations and a continuous boost along one axis suffice to select the closed convex pointed causal cone by Theorem A.25. This theorem selects cone shape under its covariance premise, not a one-step read stencil or field equation.

The constructive event family uses actual conservative source records. Put $\varphi = (1 + \sqrt{5})/2$ and $L = 2/\sqrt{\varphi + 2}$. For consecutive Fibonacci numbers $q = F_n$ and $p = F_{n+1}$, with $n \geq 4$, its q^3 spatial records have a source-metric assignment to equal-volume cells with error $H_q = 2\sqrt{3}L/q$. Each event reads every preceding-layer record inside radius $a_q = L/\sqrt{q}$, including its own preceding record, and the model layer duration is $\Delta_q = a_q/c$, with $c > 0$. Segment subdivision bounds the reachable radius between $c(1 - 2h_q/a_q)T$ and cT . Since $h_q \leq H_q$ and $H_q/a_q \rightarrow 0$, these cones agree in the limit away from their null boundary. Proposition A.20 proves normalized raw counts converging to $dt d^3x$ volume. On fixed interior timelike intervals the ordering fraction tends to $1/10$. These are analytic limits; the finite path construction, the layered order, the flat reference constants and the count-clock enclosure are formalized in Lean, and the finite readings of the family and of its carrier realization are recorded in Section A.9.

The population, complete-neighbour read rule, model tick and choice of one counted event per site and layer are supplied. Independent exact replay checks conservative currents, Gram-metric neighbours, actual writer versions, schedule changes and transmitted interventions at finite cutoffs. It does not turn those four controls into an asymptotic experiment or identify the tick with a matter clock. Under the same count limit, the observer readout $(|I|/|J|)^{1/4}$ converges to the proper-time ratio of two interior timelike diamonds. It reads accessible ancestry and counts, with a reference interval fixing units; its decoder needs no coordinate labels or oscillator. Theorem A.26 and Proposition A.27 state its limit and finite-error conditions.

Exact order embeddings form a separate finite interface. For a source spatial readback and positive height scale, a parent-edge speed bound maps authenticated ancestry into the future cone. Equal-height spatial separation and strict spacelike separation of increasing-height nonancestors give the converse, hence an injective order embedding. The source height is the attained longest authenticated-parent-chain length. These finite statements do not identify height with proper time, or element number with volume. A causal-set faithful embedding also requires the appropriate population density[1, 2]. The controlled source-record construction instead proves vanishing cone and count errors; it need not reflect null comparisons exactly at each finite resolution. Neither route selects curved topology or the physical interpretation of its readouts. Under the relevant continuum causality hypotheses, causal order and volume determine conformal geometry and scale[12, 13].

The minimal source-direction Einstein-shape theorem places supplied symmetric geometry and stress fields on the same finite event type. It asks for balance on nine fixed algebraic source direc-

tions, four discrete step maps, Ward/Bianchi identities, and step connectedness. Exact tomography derives all-null balance, the pointwise metric ambiguity, its constancy, and

$$\text{geometry}_{ab} = \kappa \text{stress}_{ab} + \Lambda \eta_{ab}.$$

The all-null relation is therefore a conclusion rather than a premise. The minimal package contains no normalization, universal-source, vacuum-calibration, or Newton-constant field. The nine-set is algebraic, not selected by provenance or invariant as a set under Lorentz transformations. No provenance edge selects a direction, tensor field, step, or balance. The ambient carrier dimension and signature are constructed independently by the source-causal carrier theorem; the poset dimension, fields, and their physical interpretation remain separate.

Identifying **geometry** with smooth curvature and **stress** with physical stress-energy requires the source-causal continuum, count-volume, stress, small-ball, coupling, scale, and remainder receipts on one family, together with either a tensor-curvature reconstruction converging to the Einstein tensor or the independent continuum small-ball/null-balance identification. Scalar-curvature convergence alone is diagnostic. The full **composedEinsteinBranch** remains available for that conditional physical composition; it does not construct its source-derived tower.

Stress reconstruction consumes same-source null-stress matching, locality/covariance, and the weak Ward balance; finite null tomography does not supply them. The celestial S^2 and the frame hyperboloid do not by themselves populate a spacetime. The source-causal continuum and Einstein conclusions are conditional theorems on their named receipts. Of the local-stress inputs, one finite precursor is in hand at one energy component: a face-local discrete Poynting law on the committed twelve-port carrier, exact at every step of the scaled Maxwell evolution, whose face sum is the committed global energy balance and whose density-flux pair is one conserved current in the source-free case (**Lean/Screen/LocalEnergyBalance.lean**). It supplies no momentum or stress components, covariance, or continuum limit, and the identification of the screen field energy with the modular energy the composition consumes is read against one declared clause: the clock-and-energy calibration row of the four-law surface, whose reference is the Gibbs state of a declared energy at one declared inverse temperature β . That row is inhabited by the screen field energy, and against it the modular energy ledger of any state is exactly β times its field-energy expectation plus $\log Z$, the cap first law takes the Clausius form $S(p) - S(\tau) = \beta(\langle H \rangle_p - \langle H \rangle_\tau) - D(p||\tau)$ with the offset cancelled, the slope and the reference are determined by the ledger, the energy is free exactly up to an additive constant, and β is determined by the reference once the family and the step are declared; a two-point reference outside the Gibbs form has equal energy expectations with unequal ledgers, so the affine reading is specific to that clause. Nothing derives that the record reference is Gibbs and nothing selects β ; the identification stays conditional on the declared row (**Lean/Thermodynamics/GibbsReferenceEnergyIdentification.lean**). The collar modular surface does not supply that inverse temperature. Its one thermal object, the reference on the realized branch of the declared maximum-entropy constraint, with modular Hamiltonian $K = \lambda \cdot S$ on the four joint sectors, has spectral weights equal to the Gibbs state of K at $\beta = 1$ in modular units, and for every nonzero β they equal the Gibbs state of K/β at β , so every temperature declaration inhabits the calibration row for the collar reference and two declarations with different β both do; a declared energy with two distinct levels pins β , a declared β pins the energy up to a constant, and the collar reference is Gibbs for an energy H at β exactly when $K = \beta H + \text{const.}$ The collar functional is KMS at parameter one for the inner flow of K and at parameter β for the flow of K/β , so the KMS parameter and the inverse temperature trade against the energy scale. The identification of the collar weights with the record reference of the cap first law stays declared (**Lean/Thermodynamics/CollarTemperatureReading.lean**).

The sourced field equations on the committed complex are consistent in the following exact sense. The hopping load and hopping current of a declared port path satisfy the committed continuity equation at every port and every step, and a finite family of hopping charges satisfies it with total load equal to the sum of the charges. Along the scaled Ampère evolution with any current, the Gauss constraint propagates from one step to the next exactly when the sources satisfy the continuity equation, so a field history that solves the Ampère evolution with a family of hopping charges and starts from Gauss-consistent data stays Gauss-consistent at every step. The complete criterion is an equivalence: for declared sources, a field history solving the Ampère evolution and the Gauss constraint at every step exists if and only if the initial load is neutral and the sources are conserved; the existence direction is an explicit second-order recursion in the temporal gauge started from the canonical Coulomb field of the initial load. Because the port boundary conserves total load on the closed carrier, one hopping charge alone has a Gauss-consistent history only when its charge is zero; a join of a worldline to the field sector therefore carries a neutral partner or a neutralising background. A neutral pair of crossing charges, with integer Green potential checked against the committed Green and incidence tables by a kernel check, inhabits the scaled Maxwell bundle with a nonzero current, and the unit crossing of every seam satisfies the continuity equation by a kernel check that fixes the sign convention between the crossing current and the incidence operator (`Lean/Screen/SeamChargeContinuity.lean`). The neutral-partner route is also realized in one finite action rather than by appending an independently prescribed background after variation. For $h \neq 0$, the source-free Maxwell window plus the two opposite minimal couplings and two clock actions is invariant under endpoint-vanishing gauge functions. With the seam-potential endpoints fixed, its field variations are equivalent to the interior sourced Ampère equations at $m < N$ and Gauss equations at $n < N + 1$. Replacing either path over a closed two-step interval while fixing the fields and other path gives an exact off-shell action difference, not by itself a path equation of motion. At $q = 1$, $h = 1/2$, the explicit Coulomb-started packet is stationary in the field slots, but for every $N \geq 1$ delaying either its positive seam-0 crossing or its negative seam-29 crossing, while fixing the fields and the other path, changes the same action by $5/12$ for every pair of clock-unit values within the fixed quadratic clock-action form. The charge-conjugate results therefore fail the declared closed-two-step exchange-stationarity condition in both path slots; applying both canonical exchanges simultaneously changes the action by $5/6$. These are only the displayed discrete exchanges, not continuum first-variation claims or a no-go for other paths, fields, or action shapes. The counterexample is specific to that Coulomb start. On the window $N = 1$, a second exact rational temporal-gauge history with the same crossings solves the sourced equations and leaves the same action unchanged under every in-window closed two-step replacement of either path. The exhaustive route check has two zero-difference endpoint routes and two triangular detours whose clock difference -4 is cancelled by interaction difference $+4$. The sources are nonzero and opposite. At the separately declared carrier unit $\tau = 3$, both crossings are timelike; identifying $\tau = h$ would instead make a crossing spacelike (`Lean/Screen/NeutralPairCoupledAction.lean`, `Lean/Screen/NeutralPairJointStationaryWitness.lean`). The step, charges, paths, rational history, clock unit, and action shapes are declared; the sign of the committed load relative to a physical charge, source selection, boundedness of the sourced history, a minimum or stability theorem, laboratory calibration, and any continuum limit are outside this statement.

The three-dimensional source metric has an exact local construction. For the declared twelve-port repair mean, the centered equal-port response kernel has an exact spectral decomposition. Dividing by its slow response before taking the infinite-step limit selects a rank-three projector. Its range in the twelve-port counting space, equipped with the induced Gram form, is an intrinsic Euclidean carrier. On the conditional signed cumulative port-record/load module, the limiting form has a three-dimensional radical on the real six-control extension. The integer module has no

nonzero metric kernel, embeds densely in the resulting quotient, and completes to the same abstract continuous carrier, up to the unique isometry that preserves the port labels. The source load map is exactly onto $\mathbb{Z}^6$, with the antipodally equal loads as its kernel. Conservative seam boundaries have image

$$D_6 = \{z \in \mathbb{Z}^6 : \sum_i z_i \equiv 0 \pmod{2}\},$$

with Smith invariants $(1, 1, 1, 1, 1, 2)$. The pullback response metric makes this submodule dense in the same completion; the ordinary six-dimensional lattice metric on D_6 is a different metric and is not used. Cumulative record addition acts simply transitively on the exact record carrier and by isometries on its completion. The sixty directed seam labels form one proper carrier-rotation orbit and give the corresponding homogeneous internal convolution under the stated naturality and unique-minimizer premises. The thirteen finite covariance/isometry table identities in the proper-action subpacket are checked with Lean’s `native_decide`; they consequently trust the native compiler and runtime as well as the kernel and are not kernel-only receipts.

Every finite-step kernel has rank eleven on the centered space and rank six on the signed sector. The limiting response form supplies the rank-three metric and the completion topology. The rotation-covariant seam average selects a spatial operator under the stated feasibility and minimization hypotheses. A physical propagation law additionally requires the population, accepted dynamical moves and clock attachment. The source metric alone does not choose a preferred Cartesian frame or identify record addition with physical translation.

The Lorentz arrow is a composition with two independent inputs on one refinement tower: the finite cap-normal Bisognano–Wichmann certificate, whose clauses are cap-normal density and non-degeneracy, BW framing, support-order faithfulness, geometric support-flow group and continuity control, held-out oriented cross-ratio convergence, independently normalized geometric 2π -KMS comparison, and wrong-normalization separation, and an independently produced complete multiresolution algebra-state comparison package on the same tower. Geometry does not manufacture the modular state, and bare finite consensus implies neither input. The Einstein arrow consumes one common scaling family, universal coupling, a vacuum reference, and independent scale readouts.

The finite boundary-circle diagnostic uses half-filled ground states of anti-periodic nearest-neighbour hopping with Fermi velocity one, on rings of 16, 32, 64, 128 sites. Summing every fitting odd hopping range of the half-ring modular Hamiltonian gives median relative profile errors 0.004808, 0.001152, 0.0002850, 0.00007106 against the conformal profile. A fixed range-nine extraction instead increases from 0.0001124 at 64 sites to 0.0003381 at 128. Both extractions use the same Hamiltonian and state. The omitted contribution at each bond is bounded algebraically by $2 \sum_{r>9} r |h_{i,i+r}|$ over fitting odd ranges, with $i = j + (1 - r)/2$ at bond j ; its evaluation here is numerical. These profile and transport cross-ratio comparisons establish finite diagnostics, not a KMS identity on a common cap algebra or a Cauchy-limit certificate (`code/geometry/modular_clock_instrumentation.py`).

The formal soldering layer separates event readback from frame choice. Once a coincidence quotient and invariant raw readback are supplied, the readback descends uniquely. A time-oriented affine Lorentz overlap cocycle and a compatible chart-coordinate family make Lorentz intervals chart independent, future-null directions covariant, and observer rest readbacks isometric under transport. The coordinate family must be identified with the quotient-descended readback; the source Gram quotient supplies its candidate spatial metric rather than a physical localization map.

Clock comparisons have a corresponding conditional interface. On a unit-timelike affine history, positive clock increments equal dimensionless proper time. Two ordered common events fix the positive-affine relation between two declared clock scales; a third gives a cross-multiplication check with no new fit parameter. These identities do not construct the clock histories or attach their

rates to the source dynamics. The count clock provides a distinct, retrospective duration readout on the declared source-record family, with explicit resolution requirements when durations are accumulated along a moving worldline.

A kinematic mass-shell packet is machine-checked on the declared Hermitian Lorentz module (`Lean/Geometry/MassShellKinematics.lean`). For one declared mass parameter m and any future unit-timelike frame u , the scaled vector $p = m u$ has Lorentz square exactly m^2 , and its scalar coordinate obeys $p_0^2 = m^2 + |\mathbf{p}|^2$. For positive m the scalar coordinate dominates m , with equality exactly when the spatial coordinates vanish, and at the standard frame it equals m . Every oriented Lorentz map carries the construction to the transported frame with the same Lorentz square, so the shell identity holds in every oriented chart, and a future null vector satisfies the zero-mass identity $v_0^2 = |\mathbf{v}|^2$. The mass parameter is a declared hypothesis on the declared module, stated in its dimensionless convention with unit invariant speed. The packet identifies no coordinate with a physical energy readout and makes no physical rest-energy claim: the clock and energy calibration anchors and the observer-to-physical-spacetime attachment are not supplied.

A separate finite-graph precursor makes the discrete inverse-square route premise-complete. For a supplied integer layering, steady source law, shell equidistribution, and shell count cn^2 through a finite depth, exact divergence bookkeeping gives outward flux per shell edge $Q/(cn^2)$, with n^2 times that flux independent of the shell. An explicit three-vertex chain inhabits the premises. The same formalization also proves why the depth cutoff cannot be hidden: on a closed finite carrier, a globally steady drained source has zero total load, while an n^2 shell count imposed at every positive layer forces $c = 0$. These are finite conditional identities, not an identification of layer with physical radius, flux with gravity, or source with mass density.

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1 Recovered Core: Foundations and Structural Branches

Observer-Patch Holography is a reconstruction program for fundamental physics. On the declared screen-first global-support branch, physical data are charted by a horizon screen S^2 , and the effective world is encoded by the mutual consistency of overlapping patch descriptions. This support screen is distinct from a local carrier boundary and a federation overlap nerve. On the producer branch, S^2 enters only after the carrier-to-support map satisfies the spherical incidence, mesh, cross-ratio, normalization, and refinement receipts. The legacy finite reference fixture does not supply that map because it has no composable seam triangle and its higher-overlap condition is vacuous. The source-derived producer instead supplies a twelve-chart, thirty-seam, twenty-face incidence nerve, nonvacuous seam-triangle cocycles, confluent phase repair, and a controlled oriented- S^2 support limit on one declared tower. Event, modular, scale, and laboratory attachments require separate premises. This recovered-core section develops fixed-cutoff overlap repair, the separated cofinal refinement-limit consensus bridge, collar recovery, and the geometry-readout route to Lorentz frame kinematics and conditional Einstein dynamics.

1.1 Overview

Overlap consistency first turns local patch states into a constrained gluing problem. The fixed-cutoff branch gives collar decompositions, edge centers, and finite repair dynamics on the declared physical quotient. Certified read-from provenance generates a finite informational poset and its exact longest authenticated-parent-chain height. Independently, the exact rank-three source Gram quotient and a real axis define a four-dimensional ambient target carrier with Lorentz inertia $(1, 3)$; source height enters only through the temporal coordinate of a scaled event placement. This is not an intrinsic dimension estimate for the finite poset, and the source unit directions are exactly the celestial S^2 and future-null-ray boundary. This removes a freely declared rank-four chart and fitted signature from the finite precursor. Modular flow on caps independently supplies Lorentz-frame kinematics and the three-dimensional observer-frame hyperboloid. Conservative Fibonacci-record populations, complete reads inside a shrinking metric radius, and a declared model clock give a separate flat $1 + 3$ causal and count-volume limit. This supplied law does not identify a physical source. The exact-embedding route to physical spacetime requires source-selected placements with two-way cone/order agreement; the controlled route permits uniformly vanishing causal discrepancy. Both require compatible physical clocks, count calibration and continuum geometry. Smooth Einstein promotion further requires same-family tensor-curvature convergence or the independent continuum small-ball/null-balance identification; scalar-curvature convergence alone is diagnostic. Positive null translations supply directional charges, and finite null tomography reconstructs a symmetric tensor when its dependent-family linearity relations hold. Its interpretation as a local, covariant, conserved stress tensor additionally requires the declared same-source null-stress, locality/covariance, and weak-Ward supplies. Fixed-cap generalized-entropy stationarity, a uniform small-diamond limit, universal coupling, a vacuum reference, and independent scale readouts then imply the Einstein equation. Bare finite consensus stops at quotient normal forms, and the cap-normal H^3 chart stops at observer frames. Neither statement populates a spacetime.

The local pixel ratio $P \equiv a_{\text{cell}}/\ell_\star^2$, the cosmic record capacity $N_\star = \log D_\star$, and the selected no- G scale certificate enter only on their declared quantitative branches. The source-causal continuum and Einstein theorems do not depend on the Standard Model matter or sector-completion branches.

1.2 Model and Axioms

1.2.1 Observers and access model

The support-local algebra-state-record reduct of an observer patch is $O_{\text{red}} = (P_O, \mathcal{A}(P_O), \rho_O, R_O)$, where:

- $P_O \subset S^2$ is a connected screen patch (the observer's access region).
- $\mathcal{A}(P_O)$ is the von Neumann algebra associated to P_O .
- ρ_O is the local state, obtained by restricting the global state to $\mathcal{A}(P_O)$.
- R_O is a finite record algebra generated by stable internal record projectors within P_O .

An operational observer additionally carries overlap interface algebras and restriction maps, allowed update and repair instruments, and checkpoint data used for continuation. Observers are internal self-reading patterns in the global state. A patch and its compatible marginal supply a geometric or algebra-state chart. An operational observer requires the interface, record, repair, and checkpoint conditions.

1.2.2 Screen, patches, and algebra net

We work in a single static patch with a horizon screen S^2 . Each connected subregion $P \subset S^2$ is assigned a von Neumann algebra $\mathcal{A}(P)$. The net satisfies isotony:

$$P \subset Q \implies \mathcal{A}(P) \subset \mathcal{A}(Q).$$

A global state ω is a positive linear functional on the inductive-limit algebra. Overlap consistency is imposed algebraically: for overlaps $P_1 \cap P_2$, ω restricted to $\mathcal{A}(P_1 \cap P_2)$ is the same from either side.

1.2.3 The three OPH axioms

This paper states the three OPH axioms because both specialist derivations share their observer-overlap foundation. Branch-local hypotheses are written at the theorem that consumes them. The support-visible BW scaling theorem handles the scaling step. The source-causal continuum and Einstein theorems use the additional collar-recovery, edge-normalization, generalized-entropy, common-domain, stress, scaling, universal-coupling, vacuum-reference, and scale-readout interfaces stated in their own hypotheses; none of those interfaces is an axiom.

The screen axiom below defines the screen-first support branch. The producer branch reaches the same axiom only after its carrier-to-support receipts establish the global support screen.

Axiom 1 (oriented observer-patch federation and spherical support). There exists an observer patch net on an oriented spherical screen: at every finite resolution each local carrier has twelve primitive boundary ports forming the vertices of an oriented triangular boundary with 30 edges and 20 faces, combinatorially the boundary of an icosahedron, and carriers join through typed seams and coherent triple overlaps, refine to an oriented spherical support, and expose local state, readback, records, repair moves, and checkpoints. Their complete infinitesimal port response is represented faithfully on a finite-dimensional unitary response space and is closed under the commutators generated by ordered response composition. Formally, for every regulator r in a directed system there is a typed object $\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$: a finite patch/overlap

category with an isotone local algebra net and central record algebras; a federation of finite carriers, each with twelve primitive pairwise-orthogonal central port projections summing to one and a boundary packet $K = (P, E, F, o)$ with $|P| = 12$, $|E| = 30$, $|F| = 20$, degree-five ports, five-cycle links, and coherently oriented two-face edges, isomorphic to the icosahedral boundary complex with no preferred labels; seam algebras with unital restrictions and coherent triple-overlap cocycles forming the nerve N_r ; a finite edge-midpoint refinement support S_r realized orientation-preservingly in S^2 with mesh tending to zero; and a source-bound degree-one bridge $b_r : N_r \rightarrow S_r$, all commuting with refinement. Each carrier also has $V_{r,i} = \mathbb{R}[P_{r,i}]$, a finite-dimensional complex Hilbert space $H_{r,i}$, and an injective real-linear map

$$D_{r,i} : V_{r,i} \longrightarrow \mathfrak{u}(H_{r,i}).$$

The image $\mathfrak{g}_{r,i} = D_{r,i}(V_{r,i})$ is closed under commutators and complete for the declared quotient-visible infinitesimal port response. The primitive port probes span $V_{r,i}$, no additional public response direction is omitted, and $-\text{Tr}(D_{r,i}(v)D_{r,i}(w))$ is positive definite. The response construction is natural under admissible presentation equivalence and refinement. The axiom does not select a particular response table, inverse-port law, Lie type, global group, particle interpretation, or metric content.

Axiom 2 (observer agreement). Observers operating on the screen agree on the meaning of the data they jointly interpret. Formally, the interpretation map $\mathcal{J}_r : \text{Data}_r \rightarrow \text{Meaning}_r$ from observer-accessible data to operational meanings is natural with respect to every visible overlap restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map: every declared data-access diagram for accepted public data commutes after interpretation, and $\mathcal{J}_r \circ c_{s \rightarrow r} = C_{s \rightarrow r} \circ \mathcal{J}_s$ across resolutions. The axiom constrains accepted shared data only, and the domain it quantifies over is supplied by the Axiom 1 interfaces rather than by the axiom itself.

Accepted reversible overlap transports form a groupoid $\mathcal{O}_r$. For a complete carrier chart o , its closed paths $\text{Hol}_r(o)$ map onto the orientation-preserving incidence automorphisms $\text{Aut}^+(K_{r,i})$. For every proper carrier automorphism a , Axiom 2 supplies a closed path γ_a and one projective implementer $[U_a] \in \text{PU}(H_{r,i})$ satisfying

$$\text{Ad}_{[U_a]} D_{r,i}(v) = D_{r,i}(a \cdot v).$$

Writing

$$G_{D,r,i}^0 = \langle \exp(tD_{r,i}(v)) : t \in \mathbb{R}, v \in V_{r,i} \rangle^0,$$

the implementer is endogenous:

$$[U_a] = [g_a c_a], \quad g_a \in G_{D,r,i}^0, \quad c_a \in C_{U(H_{r,i})}(\mathfrak{g}_{r,i}).$$

Closed paths compose projectively and the construction commutes with refinement. A unitary lift need not be unique. Scalar phases and centralizer actions are permitted because they act trivially on the response algebra. An independent spectator carrying a matching projective action cannot replace the response-generated factor or contribute a public response direction. The closed path, port action, and implementer must arise from the same accepted source history in an executable realization.

The two response clauses support an abstract target-free compact Lie-type theorem. They do not certify that an executable producer emitted a particular current fixture. Such a certification must reconstruct the port generators, commutator, and projective implementers from ordered source histories without a named current model upstream. Axiom 2 does not imply global state

extension, repair termination, confluence, unique normal forms, record durability, Byzantine safety, dissemination, or commutation of state optimization with refinement; raw mismatch and repair precede acceptance.

Axiom 3 (conditional maximum randomness). Everything that observer agreement leaves unconstrained is maximally random. Formally, at finite regulator r a state is a compatible family of local normalized states on the accessible algebra net; $\mathcal{K}_r$ is the nonempty convex set of such families satisfying the finite observer-visible constraints supplied by Axioms 1 and 2 through an A1-generated constraint grammar with a factorization theorem; and, relative to an exact reference family τ_r , an A1-generated observer cover $\mathcal{G}_r$, and strictly positive exact weights $w_{r,P}$ constructed from quotient-visible A1 data by a rule natural under admissible presentation equivalence, the realized state is the information projection. The cover restriction map is injective on $\mathcal{K}_r$; equivalently, at smooth points no nonzero feasible tangent is invisible to the whole cover. The realized state is

$$\rho_r = \arg \min_{\rho \in \mathcal{K}_r} \sum_{P \in \mathcal{G}_r} w_{r,P} D(\rho_{r,P} \| \tau_{r,P}).$$

The same projection rule applies separately to each declared optimizer object type (ontic state, inference state, or transition distribution), and statements about one type do not transfer to another. This is equivalent to weighted local entropy maximization when every local reference density is identity-proportional in its declared trace. "Random" means least informative relative to the declared reference and the complete agreement constraint set. The axiom selects one state inside one fixed feasible space. It does not compare unrelated state spaces or field contents, does not imply optimizer compatibility across refinement, and does not imply recovery, alignment, or mixing; each such statement carries its own theorem, interface, or countermodel.

The complete formal basis, including its constraint map and non-implications, is given in the [canonical three-axiom reference](#).

The closure principle. The axioms describe the observer screen. The global closure principle states that the simulating and the simulated description are one system. Every quantity that has both a construction-side reading and a readback-side reading must take the same value once a typed bridge proves that the two readings denote one invariant. The equality is forced by self-identity. Constructing that bridge and the return map is part of the physics. Existence, uniqueness, and stability are separate determinacy tests on the resulting equation. The name given to such an equation carries no mathematical weight: after the typed identification, unequal readings would describe two systems rather than the single self-referential universe.

The local screen-grain proposal

$$P = \varphi + \sqrt{\pi}/A_T(P),$$

with $A_T(P)$ the inverse coupling returned by the declared source map, has one interval-certified fixed point on its declared analytic domain. Interpreting that root as the laboratory coupling additionally requires a target-independent choice of the physical map, a typed bridge identifying the two sides as readings of one quantity, and same-scheme hadronic spectral transport to the Thomson endpoint. The displayed map's certified fixed point sits at $P = 1.6309720959$ and returns $A_T = 136.9948352$; the certified gauge-width variant's fixed point sits at $P = 1.6309682414$ and returns 137.0356601. Through the same outer equation the measured 2022 CODATA coupling $\alpha^{-1} = 137.035999177(21)$ reads back the grain value $P = 1.6309682094$. The comparisons stand at 3.0×10^{-4} relative for the displayed map and at 2.5×10^{-6} for the gauge-width variant, as

diagnostics of the declared source maps. The latter absolute residual is 3.3904×10^{-4} , compared with experimental standard uncertainty 2.1×10^{-8} ; this is not agreement at measured precision.

The direct global proposal is

$$N = \log M_0(\mathfrak{U}_N).$$

The bounded completion class defined by base agreement, positivity, and the carrier bound does not select a unique cosmic value. Universal all-rung membership in a complete A1–A3 capacity-source contract and the corresponding executable-to-Lean bridge are absent. The incomplete antecedent determines no direct N , and no theorem extends the bounded-class conclusion to the complete source class. A positive result must complete the source antecedent and prove one physical zero, either within the three-axiom source contract or through a separately named stronger source law. A separate common-load branch starts from

$$N_0 = \pi \exp\left(\frac{6\pi}{P \alpha_U(P)}\right).$$

On the finite collar branch, the declared total reserve expectation $P/4$ and six-class equidistribution give presence probability $P/24$ for each declared class. If one class is physically selected as the blocked event, its scalar-weighted presence receipt is discharged, and that normalized collar-survival factor is proved to act on the global capacity, the corresponding candidate is

$$N_{\text{pres}} = N_0 \left(1 - \frac{P}{24}\right), \quad \ln \frac{N_{\text{pres}}}{\pi} = \frac{6\pi}{P \alpha_U(P)} + \ln \left(1 - \frac{P}{24}\right).$$

The exponential alternative $N_{\text{Pois}} = N_0 e^{-P/24}$ requires a separate mean-count or continuum carrier. Exact neutral, one-class, and six-class-total actions share the declared local datum, obey positive composition and finite cut-count regrouping, and disagree globally. The finite source therefore selects no action or blocked-event semantics. The source class determines no named-law value, and its horizon branch has no capacity object. A stronger source-derived action would require the screen/electroweak bridge, physical common-load bridge, physical $\mathbb{Z}_6$ seam action, scalar-weighted receipt, and horizon-record identification. On the source-forward numerical branch,

$$N_0 = 3.5321 \times 10^{122},$$

so the finite-presence candidate is 3.2921×10^{122} and the exponential candidate is 3.3001×10^{122} . The full weighted Planck base- Λ CDM chain gives the comparison coordinate 3.3129×10^{122} . The respective residuals are -0.63 percent and -0.39 percent. Both comparisons were exposed before the branch choice and carry no predictive weight.

Collar recoverability, generalized-entropy structure, and quantum focusing are not axioms: they are classified gravity interfaces with their own theorems, countermodels, and required inputs, consumed field by field where the Einstein composition needs them. Sector content is not selected by any axiom: the conditional rank-45 tensor candidate does not establish physical families or a local Spin carrier, while scalar multiplicity and extra-sector completeness lack physical attachments. Each theorem below lists the declared completions it consumes.

1.2.4 Closure values and theorem-local technical data

The quantitative branches discussed here use one incomplete declared-map coordinate, one proposed capacity coordinate, and one selected no- G scale certificate:

$$P \equiv a_{\text{cell}}/\ell_{\star}^2, \quad N_{\text{CRC}} \equiv \log \dim \mathcal{H}_{\text{tot}}, \quad \gamma_{\star} \equiv \frac{\ell_{\star} \nu_{\text{Cs}}}{c}, \quad B_{\star} \equiv \frac{3\pi}{\ell_{\star}^2}.$$

Here $P = P_\star$ denotes the unique mathematical root of the selected declared outer/inner map. No measured coupling enters that solve. Its physical Thomson-limit interpretation requires the same-scheme hadronic spectral transport, target-independent selection of that map, and a typed bridge proving that the two coordinates read the same physical quantity. A positive global coordinate would be defined by a source-only correctable public-record packet and the stable whole-fiber saturation equation $\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}$, with $N_\star = \log D_\star$. The independent weak/Higgs common-load bridge gives $N_{\text{bridge}} = \pi \exp[6\pi/(P\alpha_U(P))]$. On the source-forward branch $P = P_{\text{fwd}}$, and that bridge is $N_0 = 3.532131543418936 \dots \times 10^{122}$, while the full weighted Planck base- Λ CDM comparison coordinate is $N_\Lambda = 3.31292709806038 \dots \times 10^{122}$, a mismatch of about 6.6 percent. On the finite collar branch, declared total reserve expectation $P_{\text{fwd}}/4$ and six-class equidistribution give presence probability $P_{\text{fwd}}/24$ for each class. If one class is physically selected, its scalar-weighted receipt is discharged, and its normalized survival factor acts on global capacity, it gives $N_{\text{pres}} = N_0(1 - P_{\text{fwd}}/24) = 3.292097877326465 \dots \times 10^{122}$. The alternative $N_0 e^{-P_{\text{fwd}}/24} = 3.300072225377652 \dots \times 10^{122}$ requires a separate mean-count or continuum carrier. Neither global action is selected: exact positive compositional completions share the local survival datum and disagree globally. The named-law branch is not evaluable on this source class, and its horizon branch has no capacity object. Both comparisons are retrospective. Capacity-level neutrino side estimates tied to that branch value are comparison-only bookkeeping terms separate from the rejected weighted-cycle construction. The selected scale certificate supplies $\gamma_\star$, with $B_\star$ as the SI curvature display of the same branch. Once supplied, $\ell_\star^2 = 3\pi/B_\star$ becomes the usual Planck-area display after $G_{\text{SI}} = c^3 \ell_\star^2 / \hbar$. These are quantitative-branch quantities, not extra axioms.

Observation enters as an external test of the source-defined branches. In the OPH reverse-engineering reading, the universe is a closed mathematical fixed structure, and each readback object is specified independently of its measured target. For N , this object is set-valued before whole-fiber scalarization and its uniqueness theorem is the exact finite-size slack law, not a Banach contraction. Measurements may define comparison coordinates; they construct neither closure. The local map has a certified unique root. Global uniqueness requires the regulator-stable condition $s(D_\star) = 0 < s(D)$ elsewhere. A bounded target-clean counterfamily does not entail that condition under shared base agreement, positivity, the carrier bound, and executable finite controls. The exact verdict applies to this bounded completion class. The complete A1–A3 source contract has not been shown to contain the countermodels at every rung, and no executable-to-Lean bridge establishes that membership. Direct N is not evaluable on the incomplete source antecedent, and the stronger source-class verdict does not follow. A positive result needs a complete source antecedent, one physical zero, and a universe-level carrier attachment.

The spacetime and Einstein derivation uses the same convention as a compression estimate. A quantitative row enters that estimate only when the declared source map has no dependency path from the measured target or a calibrated proxy. If p_i upper-bounds the conditional accidental hit probability of row i conditional on the other accepted rows, then

$$P_{\text{acc}} \leq \prod_i p_i.$$

No discriminating target-independent prediction enters this compression estimate. The electroweak checks use known targets and are comparison-only. Other positive results are theorem-level, conditional, comparison-only, or non-discriminating.

If both missing physical identifications are supplied, the two dimensionless coordinates determine dimensionless geometry, not an SI scale by themselves. On the de Sitter branch,

$$N_\star = \frac{3\pi}{\Lambda_\star \ell_\star^2}, \quad P_\star = \frac{a_{\text{cell}}}{\ell_\star^2},$$

so

$$\Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}, \quad \Lambda_\star a_{\text{cell}} = \frac{3\pi P_\star}{N_\star}.$$

Equivalently, for the scale product $\mathcal{B}_\ell := \Lambda_\star N_\star$,

$$\mathcal{B}_\ell \ell_\star^2 = 3\pi, \quad \mathcal{B}_\ell a_{\text{cell}} = 3\pi P_\star.$$

These invariants survive the rescaling $\ell_\star^2 \mapsto \lambda^2 \ell_\star^2$, $a_{\text{cell}} \mapsto \lambda^2 a_{\text{cell}}$, and $\Lambda_\star \mapsto \lambda^{-2} \Lambda_\star$, while $\mathcal{B}_\ell \mapsto \lambda^{-2} \mathcal{B}_\ell$. Thus the SI value of $\mathcal{B}_\ell$, the SI area $\ell_\star^2$, and G_{SI} require the selected scale certificate; they are not derived from $P_\star$ and $N_\star$ alone. If the selected no- G scale certificate supplies $\gamma_\star$, equivalently $B_\star$, then

$$\gamma_\star = \frac{\ell_\star \nu_{\text{Cs}}}{c}, \quad B_\star = \mathcal{B}_\ell = \Lambda_\star N_\star, \quad \ell_\star^2 = \frac{3\pi}{B_\star}.$$

The branch certificate used for the SI gravity row is

$$B_\star = 3.60787391468 \dots \times 10^{70} \text{ m}^{-2},$$

hence

$$\ell_\star^2 = 2.61228030237 \dots \times 10^{-70} \text{ m}^2.$$

Rounded capacity displays such as $N_\Lambda \simeq 3.313 \times 10^{122}$ are useful cosmological labels, but they are not high-precision inputs for the Newton row; with the Planck-2018 curvature benchmark, using only the rounded capacity would shift the Newton row at the 8.4×10^{-4} level.

The capacity normalization uses the de Sitter static-patch entropy [48]. If

$$N_{\text{patch}} = \left(\frac{r_{\text{dS}}}{\ell_P} \right)^2$$

denotes the bare horizon area ratio, then

$$N_{\text{scr}} = S_{\text{dS}} = \frac{A_{\text{dS}}}{4\ell_P^2} = \pi N_{\text{patch}} = \frac{3\pi}{\Lambda \ell_P^2}.$$

Using the Planck-2018 late-time de Sitter scale [50] gives $N_{\text{patch}} \simeq 1.05 \times 10^{122}$, $N_{\text{scr}} \simeq 3.313 \times 10^{122}$, and $\Lambda \ell_P^2 \simeq 2.845 \times 10^{-122}$.

The controlled scaling step for the Lorentz, null-modular, and Einstein statements is the support-visible BW scaling theorem. Fixed-cutoff regulator bookkeeping, quasi-local propagation, endpoint-Lipschitz control, and collar decomposition are cited from their preceding theorems.

Local information projection at the regulator scale. At the regulator scale ℓ_{UV} , consider the Axiom-3 subbranch carrying a global-extension interface and identity-proportional local references for which the state-determining weighted local objective is represented by one finite density matrix ω . The selected state maximizes the corresponding weighted local entropy subject to:

1. A finite set $\{O_a\}$ of gauge-invariant local operator densities, each supported on a ball of radius $\leq r_0 = O(\ell_{\text{UV}})$.
2. Constraint equations $\langle \sum_x O_a(x) \rangle = C_a$, one homogeneous global sum per density label a , where x runs over the cells of the UV lattice. The constraints are the N_{con} global sums, not one independent constraint per cell, so the number of independent constraints and of Lagrange multipliers is N_{con} (plus the optional global constraints below) on every lattice, independent of the cell count.
3. Optionally, a finite number of global constraints (total energy, charge).

This is the finite-matrix subbranch needed to derive the local Gibbs form of Lemma 2.6 from Axiom 3 with a cutoff-independent multiplier count. The global extension and the objective representation are explicit branch premises.

Clarification (conditional maximum randomness and thermal equilibrium). Axiom 3 is local state selection relative to an exact reference and a complete visible constraint set. It does not state that the universe is in thermal equilibrium. Non-equilibrium physics is modeled in the enlarged *local-multiplier* family obtained by replacing the homogeneous multipliers λ_a with slowly varying fields $\lambda_a(x)$, equivalently by imposing per-cell constraints $\langle O_a(x) \rangle = c_a(x)$. That enlarged family has dimension proportional to the number of UV cells, so it is a function-space approximation regime with gradient error terms. The refinement-stable branch and the induced maps $R_{\ell \rightarrow L}$ below concern the homogeneous N_{con} -parameter branch under the separate optimizer-pushforward interface. Controlled violations of exact Markov additivity appear under the explicit mixing hypotheses stated later in Section 2.3. Equilibrium is an approximation regime with explicit error terms.

Rotationally invariant constraint branch. Constraint sets are $\text{SO}(3)$ -invariant on S^2 .

Gauge as overlap redundancy at fixed cutoff. Overlap identifications are not unique; the freedom that leaves overlap observables invariant forms a local groupoid.

Central-defect subbranch. Assume the overlap centers are identified with one fixed abelian unitary coefficient group Z_Σ on which overlap transport acts trivially. Choose unitary overlap implementers with $U_{ji} = U_{ij}^{-1}$. On triple overlaps, when the failure of strict coherence is central, define

$$\Omega_{ijk} := U_{ij}U_{jk}U_{ki} = z_{ijk}, \quad z_{ijk} \in Z_\Sigma,$$

equivalently $U_{ij}U_{jk} = z_{ijk}U_{ik}$. The defect is data of the implementer lift; writing only $\text{Ad}(z_{ijk})$ would erase it because conjugation by a central element is the identity.

Collar double-scaling hypothesis. There is one refinement family with $\delta \rightarrow 0$, $\ell_{\text{UV}} \rightarrow 0$, and uniform finite-range and strong conditional Gibbs mixing constants such that

$$I(A_\delta : D_\delta \mid B_\delta)_\omega \leq c|\partial C|_{\text{UV}}e^{-\delta/\xi}, \quad \frac{\delta}{\xi} - \log |\partial C|_{\text{UV}} \longrightarrow +\infty.$$

The second condition controls the complete boundary-prefactored envelope. The weaker ratio $\delta/\ell_{\text{UV}} \rightarrow \infty$ does not imply it. Section 2.3 gives the finite-range theorem, explicit constants, a sufficient logarithmic schedule, and the finite-receipt fields. On the exact central-interface branch the CMI is zero without this mixing premise.

Geometric-subnet BW lift. The continuum Lorentz statement is asked only on the support-visible extracted geometric subnet for caps, together with the controlled collar/refinement limit that carries the Markov and recovery remainders explicitly. The finite cap regulators are type-I algebras, but the scaling-limit observer algebra may leave that class, so the theorem is automorphism-level on the emitted geometric cap pair and the special type-I generator form is $K_C = 2\pi B_C + Z_C$ with Z_C central. The spacetime and Einstein paper proves the support-visible BW cap automorphism only when one cofinal branch carries the finite cap-normal certificate and the complete multiresolution algebra-state comparison package. It also proves the cap-normal H^3 chart: $q(\Omega) = (1, \Omega)$, $n_C = (\cot \alpha, \csc \alpha \mathbf{c})$, $n_{gC} = \Lambda_g n_C$, and $H^3 \simeq \text{SO}^+(3, 1)/\text{SO}(3)$, with cap half-spaces but no event-position consequence. Bare confluence underdetermines topology, dimension, and modular normalization by explicit countermodels. Separately, authenticated provenance generates the finite informational order and canonical source height. The exact rank-three source Gram quotient and an independent real axis define the ambient 1+3 Lorentz target carrier with an S^2 null boundary. After

multiplication by a supplied positive `timeScale`, source height supplies only the temporal coordinate of an event placement. It is not an execution rank, intrinsic poset dimension, physical clock, or physical event time. The null-net-standardness packet supplies derived positive translations, while the source-causal continuum certificate keeps physical order, spatial placement, history-family custody, count-volume calibration, independent dimension, manifoldlikeness, topology, uniqueness, and curvature convergence explicit. The closure packets prove finite null tomography and finite entropy/MaxEnt identities; local stress, Ward conservation, generalized-entropy stationarity, smooth curvature, and the composed Einstein reading remain conditional on their named analytic and physical supplies. No source-derived family carrying the complete continuum packet is constructed (Theorems 4.3c–4.3h).

Derived quasi-local propagation and modular-locality control. At scale ℓ_{UV} , the automorphism group generated by the MaxEnt generator of Lemma 2.6, or more generally by any branch generator lying in the same bounded-support algebraic closure, has a finite Lieb–Robinson velocity v_{LR} : for local operators A, B supported on regions separated by distance d ,

$$\|[A(t), B]\| \leq c\|A\|\|B\| \min(|R_A|, |R_B|) e^{-(d-v_{LR}|t|)/\xi}$$

for $d > v_{LR}|t|$, where $\xi = O(\ell_{UV})$.

This is the branch-internal control statement that turns the quasi-local structure of the local-Gibbs branch into explicit support control for time-evolved operators. It is part of the third-axiom branch stated at regulator scale.

Approximate modular covariance on the fixed-cutoff realized branch. Under the fixed-cutoff realized presentation, Lemma 2.6 (local Gibbs), the derived quasi-local propagation bound above, and the collar double-scaling plus mixing hypotheses of Section 2.3, the modular flow $\sigma_t^{\omega, C}$ maps $\mathcal{A}(R)$ into a slightly thickened region algebra:

$$\sigma_t^{\omega, C}(\mathcal{A}(R)) \subseteq \mathcal{A}(R^{+v_{\text{mod}}|t|}) \quad \text{up to error } \eta(d - v_{\text{mod}}|t|),$$

where R^{+s} denotes the s -neighborhood thickening, v_{mod} is a "modular propagation velocity" controlled by v_{LR} and local norm bounds, and d is the distance from R to ∂C .

Heuristic motivation. On the reduced-modular-locality branch of Lemma 4.1a-b, modular additivity localizes the nonadditive defect of $K_C = -\log \rho_C$ to the collar. The associated propagation receipt then bounds support spreading under $e^{iK_C t}$. In the rate-controlled collar limit of Section 2.3, the thickening vanishes in macroscopic units. This remains heuristic support for the tangent-limit package; a Lieb–Robinson bound for the global Gibbs generator alone does not prove locality of the reduced modular Hamiltonian.

Continuum-limit heuristic. Define the induced region flow

$$f_t^C(R) := \lim_{\ell_{UV} \rightarrow 0} R^{+v_{\text{mod}}|t|}.$$

Then $\sigma_t^{\omega, C}(\mathcal{A}(R)) = \mathcal{A}(f_t^C(R))$ becomes exact in the continuum limit, with error controlled by

$$\eta(\delta) \lesssim 2\sqrt{c|\partial C|_{UV}} e^{-\delta/(2\xi)}$$

when CMI is measured in nats, with the full rate condition imposed on the prefactor.

The discussion above is heuristic support for the tangent-limit package used in Theorem 4.2.

Refinement-stable information-projection branch. Choose a family of coarse-graining channels $\Phi_{\ell \rightarrow L}$ between UV scale ℓ and IR scale L satisfying the separately declared optimizer-pushforward and refinement-tail interface. Then on the realized branch one may write

$$\Phi_{\ell \rightarrow L}(\omega_\ell(\lambda)) = \omega_L(R_{\ell \rightarrow L}(\lambda)),$$

for an induced map $R_{\ell \rightarrow L}$ on the common finite-dimensional homogeneous multiplier family. The existence of $R_{\ell \rightarrow L}$ as a self-map of that same family is the load-bearing cross-scale premise. Reusing the density labels a does not establish it. Coarse-graining a finite-range Gibbs family generically generates interactions outside the retained density list. The closure defect, the moment-matching I-projection realizing $R_{\ell \rightarrow L}$, and its trace-norm residual bound are constructed in Definition 2.6a and Lemma 2.6b. Exact vanishing of the defect along the realized branch remains part of the declared optimizer-pushforward interface. Under that interface, the regulator states lie in a shared multiplier family, and the realized branch is a persistent trajectory or invariant subset under the induced refinement maps. This state-side persistence statement does not upgrade fixed-cutoff edge labels to a transportable refinement-persistent sector colimit.

Fixed-cutoff realized presentation. At a UV scale ℓ_{UV} , local patch algebras are type-I with finite-dimensional Hilbert spaces, and gauge-as-gluings is realized as a boundary action on a lifted finite-dimensional presentation. Section 2.3 proves the resulting fixed-cutoff collar package directly from overlap consistency on this realized presentation, while distinguishing the lifted fixed-point algebra from the sector-preserving collar algebra used in the edge-center (EC) theorem.

External mathematical inputs include SSA and recovery theorems (Petz 1986, 1988; Fawzi and Renner 2015). The Einstein route also uses the fixed-volume area-variation identity for small geodesic balls and the local quadratic-polarization argument in the tensor upgrade. The null modular bridge and small-ball kernel are carried internally on the compact surface rather than imported from a separate EFT small-ball law.

Recovered-core theorem boundary. The Lorentz, null-modular, and Einstein recovered core runs through the support-visible automorphism theorem. Fixed-cutoff statements remain regulator statements. Full citations appear in the References.

1.2.5 Notation

- ρ_C : reduced state on cap C .
- $K_C := -\log \rho_C^\omega$: modular Hamiltonian of the reference state.
- B_C : geometric generator of the cap-preserving conformal dilation.
- $S_{\text{gen}}(C)$: generalized entropy on a cap.
- ℓ_{UV} : UV length scale of the refined screen net.
- δ : collar width around a cap boundary.

1.3 Information-Theoretic Tools

1.3.1 Strong subadditivity and Markov states

For any tripartite state ρ_{ABC} ,

$$I(A : C \mid B) := S(AB) + S(BC) - S(B) - S(ABC) \geq 0.$$

Exact Markov states satisfy $I(A : C \mid B) = 0$ and admit a recovery map:

$$\rho_{ABC} = (\text{id}_A \otimes \mathcal{R}_{B \rightarrow BC})(\rho_{AB}).$$

1.3.2 Approximate recovery

If $I(A : C \mid B) \leq \varepsilon$ in nats, there exists a CPTP recovery map $\mathcal{R}$ with

$$\|\rho_{ABC} - (\text{id}_A \otimes \mathcal{R})(\rho_{AB})\|_1 \leq 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}.$$

1.3.3 Collar refinement and sufficient mechanisms

Fix a cap $C \subset S^2$ with boundary circle. For collar width δ define

$$B_\delta := \{x \in S^2 : \text{dist}(x, \partial C) \leq \delta\}, \quad A_\delta := C \setminus B_\delta, \quad D_\delta := (S^2 \setminus C) \setminus B_\delta.$$

Then $S^2 = A_\delta \cup B_\delta \cup D_\delta$ with A_δ and D_δ interacting only through B_δ . The collar double-scaling hypothesis is the requirement that $I(A_\delta : D_\delta \mid B_\delta) \rightarrow 0$ in the refinement limit. At fixed regulator scale, the same finite-dimensional realized presentation supplies the patch-net and overlap-gluing data used below. We therefore isolate that realized presentation first, then separate the exact-Markov and quantitative routes.

Finite quotient ensemble theorem. Fix a finite regulator r . The physical presentation space is Σ_r , the presentation redundancy groupoid is Γ_r , and the finite physical quotient is

$$Q_r = \Sigma_r / \Gamma_r, \quad \pi_r : \Sigma_r \rightarrow Q_r.$$

The quotient removes nonphysical presentation data: gauge representatives, port relabelings, mesh labels, shard or worker identifiers, queue order, repair schedule identifiers, retry counters, timestamps unless declared semantic, hidden carrier coordinates, and inert ancillary labels. If only settled configurations carry probability, the probability space is the normal-form subset

$$N_r = n_r(Q_r).$$

The map n_r is a normal-form map, not a probability law. Any physical branch inherits this firewall: it must declare the quotient-intrinsic source law or action before a normal form can be read as a selection or prediction claim.

Observable algebras and reference states. In the finite classical case the quotient observable algebra is

$$\mathcal{O}_r = \ell^\infty(Q_r).$$

In the finite quantum case the physical algebra is a declared quotient algebra $\mathcal{A}_r^{\text{phys}}$ with state

$$\omega_r(A) = \text{Tr}(\rho_r A).$$

When the reference object is obtained from a finite lifted carrier, the load-bearing data are not an abstract groupoid cardinality alone. They are a tracially pointed quotient

$$(\mathcal{A}_{r,b}^{\text{phys}}, \tau_{r,b}^0), \quad \mathcal{A}_{r,b}^{\text{phys}} = z_{r,b} B(\tilde{\mathcal{H}}_r)^{G_r} z_{r,b},$$

where $U_r : G_r \rightarrow U(\tilde{\mathcal{H}}_r)$ is the compact gauge action and $z_{r,b}$ is the central projection for the declared boundary or superselection sector. The reference trace is

$$\tau_{r,b}^0(A) = \frac{\text{Tr}_{\tilde{\mathcal{H}}_r}(A)}{\text{Tr}_{\tilde{\mathcal{H}}_r}(z_{r,b})}.$$

If

$$z_{r,b}\tilde{\mathcal{H}}_r \cong \bigoplus_{\alpha} V_{\alpha} \otimes M_{\alpha},$$

with $d_{\alpha} = \dim V_{\alpha}$ and $m_{\alpha} = \dim M_{\alpha}$, then

$$\mathcal{A}_{r,b}^{\text{phys}} \cong \bigoplus_{\alpha} I_{V_{\alpha}} \otimes B(M_{\alpha}), \quad p_{r,\alpha} = \frac{d_{\alpha}m_{\alpha}}{\sum_{\beta} d_{\beta}m_{\beta}}.$$

These are the induced central-sector weights only after the carrier representation and boundary sector have been fixed.

OPH quotient ensemble. An OPH quotient ensemble is specified by a quotient-intrinsic base weight and action

$$m_r : Q_r \rightarrow \mathbb{R}_{>0}, \quad S_r : Q_r \rightarrow \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q)e^{-S_r(q)}, \quad Z_r = \sum_{q \in Q_r} w_r(q), \quad \mu_r(q) = Z_r^{-1}w_r(q).$$

Equivalently, one may state an intrinsic projective prior ν_r on Q_r and set $\mu_r = (n_r)_{\#}\nu_r$. Uniform quotient counting, uniform representative counting pushed to the quotient, groupoid weights, and tracial central-sector weights are different physical claims. The paper must declare which one is being used.

Normal-form projector non-selection. For any retraction $N : Q \rightarrow Q_{\text{nf}}$ onto a subset $Q_{\text{nf}} \subseteq Q$, that is, any map whose restriction to Q_{nf} is the identity, the induced map on laws

$$\mathcal{C}_Q(\mu) = N_{\#}\mu$$

is idempotent:

$$\mathcal{C}_Q^2 = \mathcal{C}_Q.$$

Every law supported on Q_{nf} is fixed; both statements use the retraction property. Therefore settlement or canonicalization never selects a unique physical probability law by itself.

Selection-gap corollary. Let $X \subseteq Q_{\text{nf}}$ be a finite set of quotient-normal candidates distinguished by visible invariants. Normal-form data determine X and its quotient-visible invariants, but they do not choose a member of X . If two laws μ, ν are supported on X and concentrate on different candidates, both are fixed by $\mathcal{C}_Q$. Unique sector selection therefore requires source data: an intrinsic action with a unique minimizer, a declared physical ensemble, or a refinement-stable gap certificate. A defect or holonomy classification can classify possible sectors without choosing the physical sector, and a contraction or repair generator can certify convergence toward a declared target without creating the target law.

Finite MaxEnt quotient ensemble. For finite Q , positive m , and quotient observables $F_1, \dots, F_k$, maximizing

$$\mathcal{H}_m(\nu) = - \sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

subject to

$$\sum_q \nu(q) = 1, \quad \sum_q \nu(q) F_a(q) = c_a$$

has the full-support solution, when the feasible full-support surface is nonempty,

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

Boundary optima obey the same formula after restricting to their support. On a finite noncommutative quotient algebra with faithful reference state σ_r ,

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\text{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

The finite constraint ledger must name every $F_{r,a}$, its units and support, the target expectation and source, sector or zero-mode treatment, refinement transformation, and proof that no run output or observational output entered the source definition.

Refinement compatibility and RG closure. For $s \succeq r$, let $c_{sr} : Q_s \rightarrow Q_r$ be the physical coarse map. Exact compatibility of weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q': c_{sr}(q')=q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant $\alpha_{sr} > 0$ independent of q . Then

$$(c_{sr})_{\#} \mu_s = \mu_r.$$

If the one-step defects are

$$\delta_{k+1,k} = \|(c_{k+1,k})_{\#} \mu_{k+1} - \mu_k\|_{\text{TV}},$$

then

$$\|(c_{nr})_{\#} \mu_n - \mu_r\|_{\text{TV}} \leq \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

For exponential-family refinement, exact closure requires the fine conditional free energy

$$G_{sr,\theta}(q_r) = -\log \mathbb{E}_{m_s^0} [\exp[-\theta \cdot F_s(Q_s)] \mid c_{sr}(Q_s) = q_r]$$

to equal $\kappa_{sr}(\theta) + R_{sr}(\theta) \cdot F_r(q_r)$. If the residual is uniformly bounded by ε , the induced total-variation defect is bounded by $\tanh \varepsilon$.

Implementation invariance and representative lifting. If implementations A, B have quotient bijections $h_r : Q_r^A \rightarrow Q_r^B$ satisfying

$$m_r^B(h_r q) = m_r^A(q), \quad S_r^B(h_r q) = S_r^A(q), \quad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then

$$(h_r)_\# \mu_r^A = \mu_r^B.$$

For tracially pointed quantum quotients the corresponding equivalence is a trace-preserving quotient equivalence. It is invariant under unitary intertwiners preserving the gauge action and sector, and under inert trivial ancillas $A \mapsto A \otimes I_{\text{anc}}$. It is not invariant under arbitrary changes of gauge-representation multiplicities.

If an implementation stores representatives, a representative-level law must be a conditional lift

$$\tilde{\mu}_r(x) = \mu_r(\pi_r x) \kappa_r(x \mid \pi_r x), \quad \sum_{x: \pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then $(\pi_r)_\# \tilde{\mu}_r = \mu_r$. Uniform representative sampling yields orbit-size weights and is physical only if representative counting is the declared base measure.

Quotient-lumpable kernels and sampler correctness. A representative kernel $\tilde{P}(x, y)$ descends to Q_r only when

$$P_Q(q, q') = \sum_{y: \pi(y)=q'} \tilde{P}(x, y)$$

is independent of the chosen representative $x \in \pi^{-1}(q)$. For $w(q) = m(q)e^{-S(q)}$ and proposal $R(q, q')$ with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min \left\{ 1, \frac{w(q')R(q', q)}{w(q)R(q, q')} \right\}$$

gives detailed balance

$$\mu(q)R(q, q')a(q, q') = \mu(q')R(q', q)a(q', q).$$

Repair-informed proposals must include the Hastings asymmetry term; otherwise the stationary law is generically changed.

Repair generators are not selectors. A repair generator of the form

$$L_{\text{rep}} = \sum_C c_C (I - E_C)$$

is a relaxation or sampling object after a law has been selected. Conditional expectations E_C are defined on $L^2(X_r, \pi_r)$, so the reference law π_r is input. On overlapping collars the expectations need not commute. The correct finite gap certificate is the Poincare constant

$$\kappa_r = \inf_{f \perp 1} \frac{\sum_C \|(I - E_C)f\|^2}{\|f\|^2}.$$

If local fiber rates have a positive lower bound γ_* , then

$$L_{\text{rep}} \geq \gamma_* \kappa_r (I - P_0).$$

Finite repair completeness gives $\kappa_r > 0$ at fixed regulator. A uniform refinement lower bound $\inf_r \kappa_r > 0$ is a separate theorem or receipt.

Finite evidence accuracy. For bounded coarse observables O , if

$$\|\hat{\mu}_s - \mu_s\|_{\text{TV}} \leq \epsilon_{\text{samp}}$$

and the refinement defects sum to ϵ_{ref} , then

$$\left| \mathbb{E}_{\hat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O] \right| \leq 2\|O\|_{\infty}(\epsilon_{\text{samp}} + \epsilon_{\text{ref}}).$$

Continuum-facing observables require a realization map and correlation Cauchy bound in addition to a finite histogram.

Physical-vacuum requirement. A stationary sampler is not a physical vacuum. For any faithful target law one can build a positive transfer operator with that law as ground state, so positivity alone is not a selector. A physical-vacuum interpretation requires source Euclidean slab data

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

whose conductance $J_r(q, q') = J_r(q', q) \geq 0$, local potential V_r , and slab thickness $a_{t,r}$ are derived without using the target law or sampler output. With connected event graph,

$$(H_r^E f)(q) = \frac{1}{m_r^0(q)} \sum_{q'} J_r(q, q') (f(q) - f(q')) + V_r(q) f(q)$$

is self-adjoint and bounded below on $L^2(Q_r, m_r^0)$; its finite Feynman–Kac semigroup is positivity improving. Perron–Frobenius gives a unique positive normalized ground state Ω_r , and the finite vacuum law is

$$\mu_r^{\text{vac}}(q) = |\Omega_r(q)|^2 m_r^0(q).$$

For $T_r = e^{-a_{t,r}(H_r^E - E_{0,r})}$, the Doob kernel is stochastic and detailed-balanced with μ_r^{vac} . A continuum interpretation additionally requires reflection positivity or equivalent reconstruction plus refinement compatibility of the transfer family.

Primordial and cosmological prediction firewall. A screen covariance contains incomplete radial information. The complete one-shell map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

has an infinite-dimensional kernel that persists under positivity. OPH physical primordial interpretation requires the source-only stress, single-clock, entropy-repair, curvature-evolution, adiabatic-mode, phase-coherence, physical mode, radial-null-space, and forward-projection receipts together with a scale-natural physical dilation intertwiner or complete radial cross-covariance tomography. A finite radial prior produces a conditional continuation. Observable CMB comparison also requires declared source, solver, dataset, covariance, nuisance, data-use, and pooled-reducer provenance.

Interpretation classes and required receipts. Every ensemble-facing run records its ensemble id, interpretation class, regulator, representative schema, gauge action, canonicalizer, base measure, action coefficients, coarse maps, zero-mode projector, amplitude convention, sampler, smoothing

policy, source provenance, and explicit nonclaims. The seed belongs to the run receipt rather than the ensemble definition. The interpretation classes are

seed, proposal, or repair noise,
 conventional reference ensemble,
 OPH-native quotient ensemble,
 OPH physical vacuum,
 OPH primordial field,
 observable cosmological prediction.

The evidence bundle must keep separate receipts for stationary-law schedule invariance, detailed balance of the aggregate kernel, and pathwise partition invariance. Deterministic replay of semantic random streams or a canonical serial chain is useful, but it is not pathwise partition invariance. Smoothing must preserve raw coefficients, raw spectra, smoothing kernels, smoothed coefficients, smoothed spectra, and hashes of each stage; it is not part of S_r unless explicitly declared.

Derived regulator realization. Choose a finite UV cellulation at scale ℓ_{UV} and let R be a finite union of cells. Hilbertize each cell's finite local data to a finite-dimensional space $\tilde{\mathcal{H}}_i \cong \mathbb{C}^{n_i}$. The extended algebra before quotienting by overlap redundancy is

$$\tilde{\mathcal{A}}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R), \quad \tilde{\mathcal{H}}_R = \bigotimes_{i \in R} \tilde{\mathcal{H}}_i.$$

Overlap-preserving changes of local trivialization act only on cut data. On each finite-dimensional chart, their unitary image has compact closure, so one may represent the boundary gluing redundancy by a compact group $G_{\partial R} \subset U(\tilde{\mathcal{H}}_R)$. The lifted boundary-invariant endomorphism algebra is

$$A_{\text{inv}}(R) = \tilde{\mathcal{A}}(R)^{G_{\partial R}} = \mathcal{B}(\tilde{\mathcal{H}}_R)^{G_{\partial R}}.$$

This is the fixed-cutoff realized presentation used throughout the collar analysis. The EC theorem works on the invariant-state realization and uses the sector-preserving induced collar algebra there; it does not apply to the entire fixed-point algebra on the unreduced tensor product. When the consensus paper speaks of quotient repair, the physical repair law is defined on this fixed-point / overlap-invariant data and any representative-level map is only a lift of that quotient update. Descent to the gauge quotient is therefore built into the physical-algebra formulation as quotient-level repair: the local map is $\text{locRep}_\lambda : Q \rightarrow Q$, and the global map is the finite quotient normal-form operator

$$\text{Rep}_\lambda = \overline{\text{nf}}_\lambda : Q \rightarrow Q.$$

On the declared fixed-cutoff collar branch, the local repair step itself is read from exact Markov splice or a declared Petz/Fawzi–Renner recovery channel. The declared fixed-cutoff branch adds repair completeness, together with control on the declared Petz domain where that branch is used. The consensus paper proves those clauses on a nontrivial rooted-tree packet-net export: finite packet labels live on a rooted tree, hidden labels are quotient-only, weighted parent-copy repair strictly lowers Φ , normal forms are exactly consistent packet assignments, and the classical full-support Petz channel is CPTP and trace-norm contractive on its positive-support-gap domain. Gauge-invariant observables factor through the quotient normal form on that verified branch. The declared touched-overlap acceptance contract makes Φ the finite-patch Lyapunov functional for primitive repair proposals on the scalar branch, but a proposal becomes a physical step only through the transactional

acceptance layer: snapshot-current read sets, unchanged-outside-write validation, preserved boundary/sector/holonomy data, and exact well-founded descent. The read set is validation-complete: it contains every support of every measure term that can change under the write. Overlapping primitive proposals are aggregated by connected conflict component; the restriction-compatible union-collar glue supplies the canonical aggregate payload, while primitive members of that component do not commit independently. On the same fixed-cutoff quantum lift, the terminal expectation functional on each declared physical observable algebra is therefore unique even when microscopic representative lifts differ by gauge labels globally or by sector labels inside one quotient-local glued state. The unqualified coding statement at this stage is only a finite constraint-code statement: the overlap-net codewords are the globally consistent states $C = \Phi^{-1}(0)$. A bare graph does not determine code distance: the same graph can carry constant readouts with distance 1 or repetition constraints with distance $|V|$. Topological-code distance/min-cut, Knill–Laflamme resilience, spectral-gap convergence, BFT wall-clock liveness, and hardware search-work reduction therefore require their own certificates and are not imported by the core overlap graph. Exact descent gives termination. Confluence follows from an independently checked conflict-component diamond on the physical quotient, protected-support and protected-conflict completeness, and repair completeness. The theorem therefore gives one schedule-independent terminal quotient normal form from a fixed initial quotient state. A stronger same-boundary conclusion requires a preserved boundary/sector map and at most one consistent quotient extension in that boundary fiber. The layered functional carrier in the consensus paper proves the finite multi-edge, multi-step witness for $H_B \wedge H_{\text{fib}}$; the functional selected-fiber theorem gives the rooted obstruction-check form by constructing the single candidate from the boundary/root data and then checking all non-tree, sector, and holonomy obstructions. On that branch, multiple same-boundary candidate interiors can be present, inconsistent candidates are eliminated, and all surviving candidates share one quotient normal form. Normal-form hashes are public equality receipts after quotienting, not selectors among physically distinct minimizing endpoints. If accepted schedules terminate at different observer-facing quotient normal forms without a declared holonomy or higher-gauge obstruction, that repair law is outside the OPH consensus theorem. The neutral result of Ref. [14] packages this quantifier split exactly: under observation preservation and completeness, agreement modulo a silent equivalence of normal endpoints reached from all same-boundary sources is equivalent to injectivity of the induced boundary map on the consistent quotient. This cross-source criterion, same-source confluence, weak normalization, and all-schedule liveness are separate obligations. The named layered and functional carriers discharge the cross-source premise only on their stated domains. For separated cofinal refinement systems, the consensus paper adds an inverse-limit bridge: if finite-stage restriction maps commute with the quotient normal-form maps and holonomy maps, and if compatible families are visibly separated on cofinal finite stages, then finite normal forms and holonomy obstructions assemble into unique refinement-limit classes with finite-stage witnesses for nonzero holonomy. It also adds the RG-facing comparison: when a chosen coarse-graining channel shadows the finite-stage normal forms and obstruction maps with declared defects ε^n and ε^h , reconciling first and then coarse-graining gives the same macroscopic law readout as coarse-graining first and then reconciling, up to $\max\{\varepsilon^n, \varepsilon^h\}$; exact naturality is the zero-defect case. Exact normal-form naturality is discharged by a repair-morphism witness, and holonomy naturality by the corresponding cochain map. For distributed implementations, the same paper proves that a worker run presents one finite universe only when it starts from one global carrier and each event projects to a legal monolithic repair path, a physical stutter, or a certified rollback to an earlier committed projection. The required certificate names the global graph, initial state, partition, cut interfaces, observer registry, run/config/code hashes, linearized commits, restart roots, final monolithic normal form, and recomputed readout. Executor order is separate from observer history. Observer-clock naturality

additionally requires a history-augmented quotient with semantic event keys independent of worker ID, repair iteration, queue position, timestamp, and retry metadata; a global observer registry descended from local registry groupoids with disjoint patch/cap/future namespaces and explicit lineage arrows; a state-preserving observer-algebra extraction functor from terminal augmented normal forms; a support-visible cap chart map natural with that extraction; and an operational clock instrument whose readout is calibrated up to the declared affine reparameterization and residual bound. The listed objects or their proof-producing finite certificates are prerequisites. In their absence, modular-order/readback naturality is a named premise rather than a consequence of implementation bookkeeping. The same quotient boundary applies to neutral-bulk readout. For each shard s , raw record rows Σ_s must first be quotient by the declared presentation groupoid Γ_s and then sent through the terminal normal-form map n_s , producing

$$X_s = n_s(\Sigma_s/\Gamma_s).$$

Interface maps $\tau_{ts} : U_{st} \rightarrow U_{ts}$ between these terminal charts must satisfy the identity, inverse, cocycle, and zero-cycle-holonomy checks before rows from different shards are compared. A global neutral readout is the quotient

$$Q_{\text{vis}} = \left(\bigsqcup_s X_s \right) / \sim_\tau,$$

not the concatenation of shard-local tables. Channel features $F_{s,c}$ descend to Q_{vis} only when their domains and values are transported by the same atlas. For a complete declared channel set $\mathcal{C}_\star$, the neutral distance is

$$d_{\text{neu}}(x, y) = \left(\sum_{c \in \mathcal{C}_\star} w_c d_c(F_c(x), F_c(y))^p \right)^{1/p}.$$

Here $p \geq 1$, and $\mathcal{C}_\star$ is finite. An infinite declared channel set is allowed only when the displayed weighted p -sum is finite for every compared pair. Under that finite-distance premise this is stated first as a quotient-visible pseudometric. It becomes a metric only after quotienting zero-distance feature-collision classes or proving that the channels jointly separate points. Pairwise available-channel comparison is explicitly excluded: missingness is handled only by complete cases, a fixed quotient-visible missing symbol, or a train-only imputation map labelled as an imputed-representation metric. The certificate records a finite channel list or a pairwise weighted- ℓ^p summability witness. Presentation changes (gauge representatives, port relabelings, observer order, schedules, and shard partitions) preserve the metric only when they induce a bijection of Q_{vis} together with channel isometries; refinement claims additionally require a cofinally vanishing distance tail modulus. Finite Euclidean claims require the double-centered Gram certificate

$$B = -\frac{1}{2}H(D^{\circ 2})H \succeq 0$$

and noisy runs must report negative spectral mass, rank/effective-rank data, held-out stress, and planted/shuffled controls. Statistical certificates split independent generative shard batches before preprocessing; chart alignment, scaling, imputation, weights, graph construction, dimension choice, and thresholds are train/validation objects, and a test batch, seed, boundary condition, trajectory family, duplicate, or descendant appearing in more than one split blocks the claim. These theorems establish at most a common finite quotient metric or pseudometric. Identifying it with a physical Riemannian bulk or with Lorentzian/Einstein spacetime (via the source-causal continuum certificate of Theorem 4.3e, under its physical-order, refinement, count-volume, independent-dimension, manifoldlikeness, topology, and curvature receipts) requires the separate modular/geometric branch

below. The same consensus surface is classified explicitly as a finite-state exact theorem package plus this controlled inverse-limit bridge: decidable normal-form computation with the Lyapunov step bound, automatic approximate stability only through the collar-local splice and record controls, a conditional fair-block contraction branch for long-run noisy approximate consensus, and a verified rooted-tree packet-net subdomain. Computational universality for growing patch-net families belongs to the consensus paper's expressive-power boundary and is not a dependency for the recovered-core branches.

Fixed-cutoff packet closure map and invariant simplex. On any declared fixed-cutoff packet quotient Q whose repair relation is terminating and confluent, the normal-form map

$$N : Q \rightarrow Q_{\text{nf}}$$

is well-defined and schedule-independent. It induces an affine OPH closure map on the probability simplex over packet states,

$$\mathcal{C}_Q : \Delta(Q) \rightarrow \Delta(Q), \quad \mathcal{C}_Q \left(\sum_{q \in Q} p_q \delta_q \right) = \sum_{q \in Q} p_q \delta_{N(q)}.$$

Since Q is finite, $\Delta(Q)$ is a nonempty compact convex set and $\mathcal{C}_Q$ is a continuous affine self-map. Its image $\Delta(Q_{\text{nf}})$ is invariant, and $\mathcal{C}_Q^2 = \mathcal{C}_Q$. This is an actual closure map on the fixed-cutoff packet-closed quotient branch. It is not the full Appendix/habitat closure-map theorem for arbitrary OPH state-and-law data; the first operation not internalized at that level is the general habitat lift from this finite packet quotient to the full compact-convex observer-supporting sector.

Finite quotient ensemble theorem. The closure map above is an idempotent normal-form projector. It sends any input law to terminal quotient normal forms, and every law supported on those normal forms is a fixed point. Hence the physical probability law at regulator r must be supplied as an additional quotient-intrinsic object. Let

$$Q_r = \Sigma_r / \Gamma_r$$

be the finite physical quotient, or let $N_r = n_r(Q_r)$ be the quotient-normal-form subset when only settled configurations are intended to carry probability. The declared OPH quotient ensemble is specified by an intrinsic base weight and action

$$m_r : Q_r \rightarrow \mathbb{R}_{>0}, \quad S_r : Q_r \rightarrow \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q) e^{-S_r(q)}, \quad Z_r = \sum_{q \in Q_r} w_r(q), \quad \mu_r(q) = Z_r^{-1} w_r(q).$$

Equivalently, one may state an intrinsic projective prior ν_r on Q_r and set $\mu_r = (n_r)_\# \nu_r$. The choice of m_r , S_r , or ν_r is load-bearing: uniform quotient counting, uniform representative counting pushed to the quotient, finite-groupoid weights, and trace weights of a declared quotient algebra are generally different.

Finite MaxEnt form. For a finite Q , positive m , and quotient observables $F_1, \dots, F_k$, maximizing

$$\mathcal{H}_m(\nu) = - \sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

over the affine constraint surface

$$\sum_q \nu(q) = 1, \quad \sum_q \nu(q) F_a(q) = c_a$$

has a unique full-support maximizer of the form below when the feasible set contains a full-support point (equivalently, the prescribed moment vector is in the relative interior of the attainable moment set):

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

This is the finite-dimensional Lagrange-multiplier argument with strict concavity of $\mathcal{H}_m$. If no full-support feasible point exists, the unique optimizer lies on a proper face; the same finite-multiplier formula holds only after restricting the state space and observables to the optimizer's support (or globally as an extended exponential family with infinite penalties). On a finite noncommutative quotient algebra, when the feasible set contains a faithful state, the corresponding density operator with faithful reference state σ_r is

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\text{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

Refinement and projective compatibility. For $s \succeq r$, let $c_{sr} : Q_s \rightarrow Q_r$ be the physical coarse-graining map. Exact compatibility of the weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q': c_{sr}(q')=q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant $\alpha_{sr} > 0$ independent of q ; necessarily $\alpha_{sr} = Z_s/Z_r$. The proof is just the equality

$$(c_{sr})_{\#} \mu_s(q) = Z_s^{-1} \sum_{q': c_{sr}(q')=q} m_s(q') e^{-S_s(q')}$$

compared with $Z_r^{-1} m_r(q) e^{-S_r(q)}$. For the selected-prior route, if

$$c_{sr} \circ n_s = n_r \circ c_{sr}, \quad (c_{sr})_{\#} \nu_s = \nu_r,$$

then functoriality gives $(c_{sr})_{\#} (n_s)_{\#} \nu_s = (n_r)_{\#} \nu_r$. Along a chain, finite-stage compatibility defines a unique projective-limit probability measure on the cylinder σ -algebra. If the one-step defects are

$$\delta_{k+1,k} = \|(c_{k+1,k})_{\#} \mu_{k+1} - \mu_k\|_{\text{TV}},$$

then total-variation contraction gives

$$\|(c_{nr})_{\#} \mu_n - \mu_r\|_{\text{TV}} \leq \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

Implementation invariance and representative lifting. If two implementations A, B carry quotient bijections $h_r : Q_r^A \rightarrow Q_r^B$ with

$$m_r^B(h_r q) = m_r^A(q), \quad S_r^B(h_r q) = S_r^A(q), \quad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then $(h_r)_{\#} \mu_r^A = \mu_r^B$. If an implementation stores representatives $\pi_r : \Sigma_r \rightarrow Q_r$, a representative-level law must be a conditional lift

$$\tilde{\mu}_r(x) = \mu_r(\pi_r x) \kappa_r(x \mid \pi_r x), \quad \sum_{x: \pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then $(\pi_r)_{\#} \tilde{\mu}_r = \mu_r$. Uniform representative sampling weights quotient states by orbit size and is physical only if representative counting is the declared base measure. A representative Markov

kernel must also be quotient-lumpable: the transition probability to an orbit cannot depend on which representative of the source orbit is used.

Sampler correctness. A production sampler is a separate interface from a deterministic settler:

$$\text{settle}(q) \rightarrow n_r(q), \quad \text{sample}(\mu_r) \rightarrow q.$$

For $w(q) = m(q)e^{-S(q)}$ and a proposal kernel $R(q, q')$ with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min \left\{ 1, \frac{w(q')R(q', q)}{w(q)R(q, q')} \right\}$$

gives detailed balance

$$\mu(q)R(q, q')a(q, q') = \mu(q')R(q', q)a(q', q),$$

and hence $\mu P = \mu$. Irreducibility and aperiodicity give uniqueness and convergence on the finite state space. If a repair-informed proposal is $q' = R_{\text{repair}}(q) + \eta$ with density g , the Hastings ratio must include

$$\frac{g(q - R_{\text{repair}}(q'))}{g(q' - R_{\text{repair}}(q))},$$

dropping this term generally changes the stationary law. For a lazy reversible finite chain with spectral gap γ ,

$$\|P^n(x, \cdot) - \mu\|_{\text{TV}} \leq \frac{1}{2} \sqrt{\mu(x)^{-1} - 1} (1 - \gamma)^n,$$

and the usual autocorrelation bound follows from the nontrivial spectral radius. Exact small regulators should therefore enumerate Q_r , w_r , P_r , detailed-balance error, stationarity error, refinement residuals, lumpability/orbit-size checks, and the exact spectral gap.

Gaussian screen continuation. For a real reduced screen field space V_r after background and dipole modes are removed, let K_r be positive definite and impose fixed expected quadratic release energy,

$$\mathbb{E}[q] = 0, \quad \mathbb{E}[q^\top K_r q] = d_r A_r, \quad d_r = \dim V_r.$$

The unique maximum-entropy density relative to the declared Euclidean volume is

$$p_r(q) = \frac{\sqrt{\det K_r}}{(2\pi A_r)^{d_r/2}} \exp \left[-\frac{1}{2A_r} q^\top K_r q \right], \quad \text{Cov}(q) = A_r K_r^{-1}.$$

This is the relative-entropy proof against the displayed Gaussian. If instead every sample obeys $q^\top K_r q = d_r A_r$, the MaxEnt law is uniform on the ellipsoid, not Gaussian. Under a linear coarse field map $C_{sr} : V_s \rightarrow V_r$, exact Gaussian refinement is the covariance criterion

$$A_r K_r^{-1} = C_{sr} (A_s K_s^{-1}) C_{sr}^\top.$$

For harmonic truncation, choose real modes $Y_{\ell m}$, $2 \leq \ell \leq L$, set

$$K_L Y_{\ell m} = \kappa_\ell Y_{\ell m}, \quad \kappa_\ell = [\ell(\ell + 1)]^{1+\theta/2},$$

and sample independent coefficients

$$a_{\ell m} \sim \mathcal{N}(0, A/\kappa_\ell).$$

Projection from L' to L then discards high modes and is exactly refinement-compatible. For irregular meshes, covariance should be primary under coarse-graining:

$$\Sigma_r = C_{sr} \Sigma_s C_{sr}^\top, \quad K_r = A_r \Sigma_r^{-1}.$$

Equivalently, when eliminating fine variables from a precision matrix

$$K_s = \begin{pmatrix} K_{cc} & K_{cf} \\ K_{fc} & K_{ff} \end{pmatrix},$$

the coarse precision is the Schur complement

$$K_r = K_{cc} - K_{cf}K_{ff}^{-1}K_{fc},$$

not the principal block K_{cc} .

Physical vacuum and primordial premises. A stationary sampler is not a physical vacuum: the identity transition kernel leaves every measure stationary. A finite vacuum theorem needs a physical transfer operator. If T_r is real symmetric, positive definite, and strictly positive, Perron–Frobenius gives a simple largest eigenvalue $\lambda_{0,r}$ with positive eigenvector Ω_r . Then

$$H_r = -a_t^{-1} \log(T_r / \lambda_{0,r})$$

is positive semidefinite, Ω_r is the unique zero-energy state, and the Doob kernel

$$P_r(x, y) = \frac{T_r(x, y)\Omega_r(y)}{\lambda_{0,r}\Omega_r(x)}$$

is stochastic and detailed-balanced with

$$\mu_r(x) = \frac{\Omega_r(x)^2}{\sum_z \Omega_r(z)^2}.$$

Continuum identification further requires reflection positivity or an equivalent reconstruction certificate plus refinement compatibility of the transfer family. A screen covariance alone does not determine an unrestricted primordial curvature spectrum. Even the complete noiseless one-shell sequence for the thin-shell map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

determines the three-dimensional correlation only on $0 \leq s \leq 2\chi_\star$ and has an infinite-dimensional kernel that persists under positivity. Source-derived uniqueness requires a physical dilation-intertwiner theorem or complete radial cross-covariance tomography. A finite prior gives a conditional continuation. Conventional free-scalar and lattice-gauge baselines may therefore be recorded. They instantiate neither an OPH-native physical vacuum nor a primordial source without these premises.

2 Finite Screen Spectrum Theorem Package

The screen-level spectrum theorem supplies the finite source object used by the cosmology branch. It does not supply temperature-temperature, temperature-polarization, polarization-polarization, lensing, likelihoods, or a physical cosmic microwave background interpretation. Those require the Boltzmann and data-contract theorems.

Definition 2.1 (finite screen regulator). *For regulator r , a finite screen regulator is*

$$\mathfrak{S}_r = (\mathcal{T}_r, M_r, \Gamma_r, \mathcal{Q}_r, C_{sr}, J_{rs}),$$

where $\mathcal{T}_r$ is a finite cellulation of the screen, $M_r \succ 0$ is the area/quadrature mass matrix, Γ_r is the hidden-presentation groupoid, $\mathcal{Q}_r = \Sigma_r/\Gamma_r$ is the physical quotient, and C_{sr} , J_{rs} are the coarse-graining and interpolation maps. Shape regularity and quadrature convergence require

$$f^\top M_r g \rightarrow \int_{S^2} f g d\Omega$$

with an explicit band-limited error bound

$$\left| f^\top M_r g - \int f g d\Omega \right| \leq \varepsilon_{M,r}(L) \|f\|_{H^s} \|g\|_{H^s}.$$

Patch count alone is therefore not an angular-resolution certificate.

Definition 2.2 (geometric screen scalar). *Let the quotient-visible collar-volume readout be*

$$J_{X,r}(x) = \lambda_r(x) \sqrt{\det \sigma_{AB,r}(x)} > 0,$$

and let $\bar{J}_{X,r}$ be emitted by an independently defined homogeneous or frozen-background branch, not by a CMB fit. Define

$$q_{0,r} = \frac{1}{3} \log \frac{J_{X,r}}{\bar{J}_{X,r}}.$$

On a certified spherical chart, with $B_r = [1, n_x, n_y, n_z]$,

$$\Pi_{<2,r} = B_r (B_r^\top M_r B_r)^{-1} B_r^\top M_r, \quad q_r = (I - \Pi_{<2,r}) q_{0,r}.$$

On an irregular screen, B_r is replaced by the certified generalized eigenprojector onto the finite $\ell = 0, 1$ subspace. Hard-coded feature z -scores or observer-summary weights are diagnostic proxies, not q_r .

Proposition 2.3 (quotient invariance). *If a recharting U preserves the screen inner product and geometric readout,*

$$U^\top M'_r U = M_r, \quad q'_{0,r} = U q_{0,r}, \quad B'_r = U B_r,$$

then $\Pi'_{\geq 2,r} U = U \Pi_{\geq 2,r}$, and $q'_r = U q_r$.

Proof. Substitution gives

$$\begin{aligned} \Pi'_{<2,r} U &= U B_r (B_r^\top U^\top M'_r U B_r)^{-1} B_r^\top U^\top M'_r U \\ &= U B_r (B_r^\top M_r B_r)^{-1} B_r^\top M_r = U \Pi_{<2,r}. \end{aligned}$$

Subtracting from U gives the high-mode identity. □

Definition 2.4 (physical scalar precision and action). *The scalar precision K_r must have a physical origin on $V_r = \text{im } \Pi_{\geq 2,r}$. Allowed branches are:*

$$K_r^{\text{Hess}} = \nabla^2 \Phi_r \Big|_{q=0},$$

for a quotient-visible mismatch or release free energy; a repair Dirichlet form

$$K_r^{\text{diss}} = H_r - T_r^\top H_r T_r, \quad T_r^\top H_r T_r \preceq H_r;$$

or the reversible generator form

$$K_r^{\text{Dir}} = -\frac{1}{2}(L_r + L_r^{\dagger H}).$$

A raw nonsymmetric repair matrix or caller-supplied κ is not a precision operator. The absolute normalization must be fixed independently, for example by

$$K_0 Y_{\ell m} = \ell(\ell + 1) Y_{\ell m}$$

on the $\theta = 0$ branch. Once normalized,

$$S_{\text{scr},r}[q] = \frac{1}{2A_{q,r}} \langle q, K_r q \rangle_r = \frac{1}{2A_{q,r}} q^\top M_r K_r q.$$

Theorem 2.5 (finite MaxEnt screen covariance). *Let $K_r e_i = \kappa_i e_i$ with an M_r -orthonormal basis of V_r , $\kappa_i > 0$, and $d_r = \dim V_r$. Among continuous densities in coefficients $q = \sum_i a_i e_i$ with $\mathbb{E}[a_i] = 0$ and*

$$\frac{1}{2} \mathbb{E} \langle q, K_r q \rangle_r = E_{q,r}^{\text{src}},$$

the unique entropy maximizer is

$$d\mu_r(q) = Z_r^{-1} \exp \left[-\frac{1}{2A_{q,r}} \langle q, K_r q \rangle_r \right] dq, \quad A_{q,r} = \frac{2E_{q,r}^{\text{src}}}{d_r},$$

and

$$\mathbb{E}[a_i a_j] = \delta_{ij} \frac{A_{q,r}}{\kappa_i}.$$

Proof. The Euler–Lagrange equation for entropy with normalization and quadratic-energy constraints gives a Gaussian density with inverse temperature β_r . The expected energy is

$$E_{q,r}^{\text{src}} = \frac{1}{2} \sum_i \kappa_i \frac{1}{\beta_r \kappa_i} = \frac{d_r}{2\beta_r}.$$

Thus $A_{q,r} = \beta_r^{-1} = 2E_{q,r}^{\text{src}}/d_r$. Strict concavity of entropy gives uniqueness. $\square$

Remark 2.6 (discrete quotient branch). *For a discrete finite quotient with intrinsic base weight $m_r(q)$, the exact Gibbs law is*

$$\mu_r(q) = Z_r^{-1} m_r(q) e^{-\beta_r E_r(q)}.$$

A Gaussian statement on that branch requires a separate Laplace or central-limit theorem with controlled higher-order terms.

Definition 2.7 (source release energy). *Let m_r^0 be the tracially pointed base weight on the finite physical collar quotient. Let $F_{r,a}$ be a finite ledger of primitive collar observables: scalar occupancy, protected-center presence, released volume, conserved source charges, and the release clock. The ledger excludes screen energy, sky spectra, CMB residuals, likelihoods, fitted amplitudes, and measurement-calibrated proxies. For source-emitted constraint values $c_{r,a}^{\text{col}}$, the selected release law is*

$$\nu_r^{\text{rel}}(x) = \frac{m_r^0(x) \exp[-\sum_a \lambda_{r,a} F_{r,a}(x)]}{Z_r}, \quad \mathbb{E}_{\nu_r^{\text{rel}}} F_{r,a} = c_{r,a}^{\text{col}}.$$

Strict concavity of relative entropy gives uniqueness on full support. A boundary optimum uses the same form on its certified support. The constraint ledger, base weight, source DAG, multipliers, support, and refinement maps belong to the receipt.

For this independently selected quotient ensemble on released collar normal forms, define

$$E_{q,r}^{\text{src}} = \frac{1}{2} \int \langle q_r(x), K_r q_r(x) \rangle_r d\nu_r^{\text{rel}}(x).$$

The receipt must expose ν_r^{rel} , the base measure, K_r , d_r , $E_{q,r}^{\text{src}}$, $A_{q,r}$, no-observation ancestry, and the same operator normalization used in the action. MaxEnt alone does not determine amplitude.

Proposition 2.8 (microscopic-law necessity). *For fixed positive K_r , the Gaussian family $Z(A)^{-1} \exp[-\langle q, K_r q \rangle / (2A)]$ exists for every $A > 0$. The quadratic MaxEnt form therefore leaves one positive scale free. The source release law in Definition 2.7, or an equivalent source-only vacuum law, supplies the energy that fixes $A_{q,r} = 2E_{q,r}^{\text{src}}/d_r$.*

Theorem 2.9 (rotational screen spectrum). *If the continuum precision K_θ commutes with the $SO(3)$ action on scalar functions, then by Schur's lemma each $\mathcal{H}_\ell$ is an eigenspace. For the exact conformal-shell family,*

$$\Lambda_\ell(\theta) = \frac{\Gamma(\ell + 2 + \theta/2)}{\Gamma(\ell - \theta/2)}, \quad \ell \geq 2.$$

Then $\Lambda_\ell(0) = \ell(\ell + 1)$, and the MaxEnt covariance gives

$$C_\ell^q = A_q \frac{\Gamma(\ell - \theta/2)}{\Gamma(\ell + 2 + \theta/2)}.$$

Consequently $\mathcal{D}_\ell^q = \ell(\ell + 1)C_\ell^q / (2\pi) \sim (A_q/2\pi)\ell^{-\theta}$.

Definition 2.10 (source-derived conformal precision). *Let $L_r \succeq 0$ be the normalized detailed-balanced scalar collar Dirichlet operator on V_r , with continuum limit $-\Delta_{S^2}$. Set*

$$B_r = (L_r + \tfrac{1}{4}I)^{1/2}, \quad K_{\theta,r} = \frac{\Gamma(B_r + \frac{3}{2} + \frac{\theta}{2})}{\Gamma(B_r - \frac{1}{2} - \frac{\theta}{2})}.$$

For $-2 < \theta < 4$ on the retained $\ell \geq 2$ branch, this operator is positive. The gamma recurrence gives $K_{0,r} = L_r$ exactly. The finite receipt carries detailed balance, positivity, the $\ell(\ell + 1)$ normalization, anisotropy splitting, and refinement residuals.

Theorem 2.11 (finite refinement error). *Suppose that for $2 \leq \ell \leq L$*

$$\left| \frac{\lambda_{\ell m,r}}{\Lambda_\ell(\theta)} - 1 \right| \leq \varepsilon_{K,r}(L), \quad \left| \frac{A_{q,r}}{A_q} - 1 \right| \leq \varepsilon_{A,r}, \quad \varepsilon_{K,r} < 1.$$

Then $C_{\ell m,r}^q = A_{q,r}/\lambda_{\ell m,r}$ satisfies

$$\left| \frac{C_{\ell m,r}^q}{A_q/\Lambda_\ell(\theta)} - 1 \right| \leq \frac{\varepsilon_{A,r} + \varepsilon_{K,r}}{1 - \varepsilon_{K,r}}.$$

Proof. Write $A_{q,r} = A_q(1 + a_r)$ and $\lambda_{\ell m,r} = \Lambda_\ell(1 + b_{\ell m,r})$. Then

$$\frac{C_{\ell m,r}^q}{A_q/\Lambda_\ell} = \frac{1 + a_r}{1 + b_{\ell m,r}},$$

and the displayed bound follows from $|a_r| \leq \varepsilon_A$, $|b_{\ell m,r}| \leq \varepsilon_K$. □

Theorem 2.12 (refinement-semigroup tilt). *Let R_b be the scalar refinement map for scale ratio $b > 1$. If one isolated scalar covariance mode has positive eigenvalue $\lambda(b)$, $\lambda(b_1 b_2) = \lambda(b_1)\lambda(b_2)$, and λ is continuous, then there is a unique real θ with*

$$\lambda(b) = b^{-\theta}, \quad \theta = -\frac{\log \lambda(b)}{\log b}.$$

Proof. Set $g(t) = -\log \lambda(e^t)$. The semigroup law gives $g(t+s) = g(t) + g(s)$. Continuity makes $g(t) = \theta t$, hence the result. $\square$

Definition 2.13 (edge-center reserve generator receipt). *Write $s = \log b$ for logarithmic refinement thickness. Let $u_{\text{full}}(s)$ be the scalar-conditioned covariance-survival cocycle across a full oriented collar, and let $u_q(s)$ be its source-facing half-collar restriction. The receipt requires*

$$u(s+t) = u(s)u(t), \quad u(0) = 1,$$

strong continuity at zero, the source-derived full-collar density

$$-u'_{\text{full}}(0) = \frac{P_\star}{24},$$

and the orientation-reversal coarea identity

$$-u'_q(0) = \frac{1}{2}[-u'_{\text{full}}(0)] = \frac{P_\star}{48}.$$

These are infinitesimal generator statements. A one-step survival probability has exponent $-\log u_q(\log b)/\log b$.

Theorem 2.14 (edge-center tilt and repair-clock reconciliation). *Under Definition 2.13,*

$$u_q(s) = e^{-\theta s}, \quad \theta = \frac{P_\star}{48}, \quad n_s = 1 - \frac{P_\star}{48}.$$

In the coordinate $\theta = \kappa_{\text{rep}}(P_\star - \varphi)$, the same branch has

$$\kappa_{\text{rep}}^{\text{edge}} = \frac{P_\star}{48(P_\star - \varphi)}.$$

The value e is a separate diagnostic hypothesis unless an additional identity equates it with this source-derived coordinate.

Proof. For $g(s) = -\log u_q(s)$, the cocycle law gives the continuous Cauchy equation. Hence $g(s) = \theta s$. Differentiation at zero and the half-collar identity give $\theta = P_\star/48$. The formula for $\kappa_{\text{rep}}^{\text{edge}}$ is algebraic. $\square$

Theorem 2.15 (homogeneous radial source family). *Assume source-stress closure, a single clock, freezeout, multicenter consistency, and translation/rotation covariance. Let $C_\zeta = M_{\Delta_\zeta^2}$ on the common physical $d \ln k$ mode basis. If a scale-natural source embedding transports refinement to physical dilation and its finite operator residual converges to*

$$D_s^{-1} C_\zeta D_s = e^{-\theta s} C_\zeta, \quad (D_s f)(k, \hat{k}) = f(e^{-s} k, \hat{k}),$$

then every positive measurable representative satisfies

$$\Delta_\zeta^2(k) = A_\zeta(k/k_\star)^{-\theta}$$

almost everywhere. Continuity gives the pointwise identity. A single shell has an infinite-dimensional radial kernel, including positive ambiguities; complete radial cross-covariances give the independent spherical-Hankel tomography route.

Proof. Conjugation of the multiplication operator gives $\Delta_\zeta^2(e^s k) = e^{-\theta s} \Delta_\zeta^2(k)$ almost everywhere. In logarithmic coordinates, rational-translation invariance on a common full-measure set forces $e^{\theta t} \Delta_\zeta^2(e^t)$ to be constant almost everywhere. The one-shell and tomography statements follow from the correlation-restriction and spherical-Hankel theorems in the primordial bridge packet. $\square$

Theorem 2.16 (thin-shell power-law lift). *Assume $q(\hat{n}) = Z_\star \Pi_{\ell \geq 2} \zeta_\star(R_\star \hat{n})$ and*

$$\Delta_\zeta^2(k) = A_\zeta(k/k_\star)^{-\theta}.$$

Then

$$C_\ell^q = 4\pi Z_\star^2 \int d \ln k \Delta_\zeta^2(k) j_\ell^2(k R_\star),$$

and, for $-2 < \theta < 4$,

$$A_q = \pi^{3/2} Z_\star^2 A_\zeta(k_\star R_\star)^\theta \frac{\Gamma(1 + \theta/2)}{\Gamma(3/2 + \theta/2)}.$$

Thus

$$A_\zeta = \frac{A_q}{\pi^{3/2} Z_\star^2 (k_\star R_\star)^\theta} \frac{\Gamma(3/2 + \theta/2)}{\Gamma(1 + \theta/2)}.$$

Proposition 2.17 (finite-window bound). *Let $\Psi_\ell(k) = \int dr W(r) j_\ell(kr)$, $W \geq 0$, $\int W dr = 1$, with mean $R_\star$ and variance σ_R^2 . If*

$$\delta_\ell(k) = \frac{k^2 \sigma_R^2}{2} \sup_{r \in \text{supp } W} |j_\ell''(kr)|,$$

then

$$|C_{\ell,W}^q - C_{\ell,\text{shell}}^q| \leq 4\pi Z_\star^2 \int d \ln k \Delta_\zeta^2(k) \left[2\delta_\ell(k) + \delta_\ell(k)^2 \right].$$

Proposition 2.18 (radial non-identifiability). *For a finite radial basis, $C = Ap$. If $A \in \mathbb{R}^{N_\ell \times N_k}$ has rank r , then $\dim \ker A = N_k - r$, and every $p + v$ with $v \in \ker A$ gives the same screen spectrum. A physical radial reconstruction therefore requires either a source theorem restricting p , or a declared prior with a published null basis, resolution kernels, positivity checks, and prior-sensitivity report.*

Theorem 2.19 (promotion contract for the thin-shell power-law pipeline). *Under the declared thin-shell lift and power-law source family above, one hash-locked source DAG may emit*

$$(q_r, S_{\text{scr},r}, \theta, A_{q,r}, C_{\ell,r}^q, A_{\zeta,r}, \Delta_{\zeta,r}^2)$$

as a conditional primordial packet when the geometric-scalar, collar-law, release-energy, reserve-generator, conformal-precision, refinement, source-stress, single-clock, freezeout, adiabaticity, isocurvature, phase-coherence, thin-shell, radial-null, finite-window, and forward- residual receipts pass on that DAG. Any measurement, likelihood, fitted parameter, or measurement-calibrated ancestor fails the source packet. Physical temperature and polarization spectra require the independent Boltzmann, recombination, nuisance, covariance, and likelihood gates. This contract is specific to the displayed pipeline. A different source-only radial reconstruction or forward projection may use another fully typed contract, provided it certifies its own injectivity or null space, window and convergence control, forward residual, and no-target-leak provenance.

Target 2.20 (requirements for a source-derived screen spectrum). *A concrete source-derived spectrum requires the following receipt families before primordial interpretation:*

1. *geometry, scalar quotient, low-mode projector, scalar precision, and operator normalization;*

2. *quotient ensemble selection, scalar release energy, and MaxEnt screen covariance;*
3. *the infinitesimal reserve generator, scalar refinement tilt, and angular screen spectrum with a finite error budget;*
4. *source-stress, clock, freezeout, radial-window, radial-null, forward-residual, and transfer fire-wall receipts.*

The theorem and algorithm packet is complete for the declared thin-shell power-law pipeline. No finite source DAG that passes this receipt set is supplied, and the list is not a no-go for a differently typed radial pipeline.

2.0.1 Source-only primordial bridge theorem

The finite screen covariance theorem is a screen theorem. To read it as a source-only primordial curvature spectrum one must add the following bridge objects and receipts. The dependency chain is

$$\boxed{\text{nf}(\mathcal{Q}_r) \rightarrow (\chi_r, u_r, h_r) \rightarrow \text{BackgroundCurvatureStatus}_r \rightarrow (g, T_I) \rightarrow q_\zeta = \zeta_\rho \rightarrow \text{single clock} \rightarrow \text{freeze-out} \rightarrow \text{coherent growing mode} \rightarrow \Delta_\zeta^2(k).$$

No successful fit of a screen C_ℓ or a TT spectrum may replace one of these hypotheses.

Definition (background curvature branch label). The primordial bridge carries a finite record

$$\text{BackgroundCurvatureStatus}_r = (\text{BranchType}, \kappa, I_K, I_{\Omega_K}, \text{Basis}, \text{Top})$$

where

$$\text{BranchType} \in \{\text{FLATExact}, \text{FLATAssumed}, \text{OpenCurved}, \text{ClosedCurved}, \text{Unresolved}\}.$$

FLATExact is allowed only after a direct theorem or conditional CMH receipt. FLATAssumed may run flat kernels but its outputs remain assumption-conditioned. UNRESOLVED blocks physical interpretation. The H^3 observer-facing atlas is not this record; this record comes from the clocked FLRW slice and its spatial Levi-Civita holonomy or declared branch input.

Definition (source stress readout). On a reconstructed branch with relational metric g_{ab} , let

$$S_{\text{src}} = \sum_I S_I[g, \Psi_I], \quad T_{ab}^I = -\frac{2}{\sqrt{-g}} \frac{\delta S_I}{\delta g^{ab}}, \quad T_{ab}^{\text{tot}} = \sum_I T_{ab}^I,$$

where I ranges over every active sector, including ordinary matter, radiation, neutrinos, OPH anomaly or repair sectors, and any auxiliary source kept in the run. At finite regulator this is a quotient-visible map

$$\text{StressRead}_r : \text{nf}(\mathcal{Q}_r) \longrightarrow \{T_{r,ab}^I\}_I.$$

All source stress components must be read from the same global normal form as the collar scalar.

Source-stress closure theorem. Assume each S_I is relationally diffeomorphism invariant, the source equations hold, sector exchanges satisfy

$$\nabla_a T_I^{ab} = Q_I^b, \quad \sum_I Q_I^b = 0,$$

and StressRead_r descends to the physical quotient and converges. Then

$$\nabla_a T_{\text{tot}}^{ab} = 0.$$

On the recovered Einstein branch,

$$G_{ab} + \Lambda g_{ab} = 8\pi G T_{ab}^{\text{tot}}.$$

The evidence bundle must also report the non-circular comparison

$$E_T^{ab} = T_{\text{src}}^{ab} - \frac{G^{ab} + \Lambda g^{ab}}{8\pi G};$$

using the geometric right-hand side alone is a geometry diagnostic, not a source-only primordial receipt.

Total-energy frame and curvature covector. When T^a_b has a unique future timelike unit eigenvector u^a , define

$$T^a_b u^b = -\rho u^a, \quad h_{ab} = g_{ab} + u_a u_b,$$

and decompose

$$T^{ab} = \rho u^a u^b + p h^{ab} + \pi^{ab}, \quad u_a \pi^{ab} = 0, \quad \pi^a_a = 0.$$

Let $\Theta = \nabla_a u^a$ and let σ_{ab} be the shear. Total energy conservation gives

$$\dot{\rho} + \Theta(\rho + p) + \sigma_{ab} \pi^{ab} = 0, \quad \dot{f} := u^a \nabla_a f.$$

On an expanding branch, define

$$p_{\text{eff}} = p + \frac{\sigma_{ab} \pi^{ab}}{\Theta};$$

then $\dot{\rho} + \Theta(\rho + p_{\text{eff}}) = 0$.

Theorem (exact total-curvature evolution). Work on an open flow domain on which $\Theta \neq 0$, $\dot{\rho} \neq 0$, and $\rho + p_{\text{eff}} \neq 0$, so every quotient below is defined. Let

$$\dot{\alpha} = \Theta/3, \quad \zeta_a = \nabla_a \alpha - \frac{\dot{\alpha}}{\dot{\rho}} \nabla_a \rho,$$

and

$$\Gamma_a^{\text{eff}} = \nabla_a p_{\text{eff}} - \frac{\dot{p}_{\text{eff}}}{\dot{\rho}} \nabla_a \rho.$$

Then, exactly and without a long-wavelength approximation,

$$\mathcal{L}_u \zeta_a = -\frac{\Theta}{3(\rho + p_{\text{eff}})} \Gamma_a^{\text{eff}}.$$

Hence a barotropic effective source, $p_{\text{eff}} = p_{\text{eff}}(\rho)$, conserves ζ_a .

Collar scalar corollary. Assume that the quotient record emits a connected uniform-total-density cut and a collar-adapted spatial metric

$$h = \lambda^2 dr^2 + \sigma_{AB} (dx^A + \beta^A dr)(dx^B + \beta^B dr), \quad \lambda > 0, \quad \sigma_{AB} \succ 0.$$

The block-determinant identity gives

$$\sqrt{\det h} = \lambda \sqrt{\det \sigma_{AB}}.$$

The native readout premise is that J_X and its frozen reference are these spatial volume densities, with the same reference-density convention. The formula $J_X = \lambda \sqrt{\det \sigma_{AB}}$ without this metric

attachment does not establish that premise. For relational rods Lie-dragged by the total-energy flow, the attached volume Jacobian satisfies

$$\frac{d}{d\tau} \frac{1}{3} \log J_X = \frac{\Theta}{3}.$$

Relative to a frozen background,

$$q_\zeta = \frac{1}{3} \log \frac{J_X}{\bar{J}_X}.$$

On the uniform-total-density cut,

$$q_\zeta = \zeta_\rho + \text{constant}.$$

After removing the monopole and dipole,

$$q = \Pi_{\ell \geq 2} q_\zeta = \Pi_{\ell \geq 2} \zeta_\rho, \quad Z_q = 1.$$

This is the source-only quotient scalar on the OPH screen; by itself it has type SCREENCURVATURE, not PRIMORDIALCURVATURE.

Definition (rank-one single-clock branch). Let

$$\mathcal{F}(x) = (\rho, p_{\text{eff}}, \{\rho_I, p_I, Q_I\}_{I=1}^N)$$

be the scalar source map on the source moduli space. A branch is rank-one single-clock when $\text{rank } d\mathcal{F} = 1$, $d\rho \neq 0$, every level set of ρ is connected, all sector velocities agree with u^a , and orthogonal momentum transfer vanishes. An approximate single-clock branch may instead carry an explicit interval bound on that transfer, but the bound must then be propagated into the transfer and curvature-evolution residuals. At finite cutoff this requires a quotient-visible clock χ_r and a factorization

$$\mathcal{F}_r = \hat{\mathcal{F}}_r \circ \chi_r.$$

Theorem (single-clock adiabaticity). On an exact rank-one single-clock branch, and for the dynamically nondegenerate sectors I, J under discussion with $\dot{\rho}_I \neq 0$ and $\dot{\rho}_J \neq 0$,

$$p_{\text{eff}} = p_{\text{eff}}(\rho), \quad S_a^{IJ} = 3(\zeta_a^I - \zeta_a^J) = 0,$$

and the intrinsic nonadiabatic pressure and transfer covectors obey

$$\Gamma_a^I = 0, \quad \Xi_a^I = 0.$$

On the nonvanishing domain of the total-curvature theorem, therefore $\mathcal{L}_u \zeta_a = 0$. The proof is the chain rule through the single clock: every source scalar is a function of the same χ , and connected ρ -fibers remove residual branch labels.

Theorem (entropy repair gap). Let the linearized scalar repair/evolution block be

$$\delta Y_{n+1} = L_r \delta Y_n + \eta_n,$$

and let $P_\perp$ project away gauge directions, constrained zero modes, and the adiabatic growing direction. If there is a positive metric H_r such that

$$\|P_\perp L_r P_\perp\|_{H_r} \leq e^{-\gamma_r} < 1,$$

and the projected sector is not sourced by the complementary block,

$$P_\perp L_r = P_\perp L_r P_\perp,$$

then

$$\|P_{\perp}\delta Y_n\|_{H_r} \leq e^{-n\gamma_r} \|P_{\perp}\delta Y_0\|_{H_r} + \sum_{j=0}^{n-1} e^{-(n-1-j)\gamma_r} \|P_{\perp}\eta_j\|_{H_r}.$$

In the unforced case the orthogonal entropy, decaying, and repair scalar modes decay exponentially and converge to zero; they need not vanish after a finite number of steps. If the block-invariance identity is absent, the term $P_{\perp}L_r(1 - P_{\perp})\delta Y_n$ must instead be included in the forcing bound. Eigenvalues alone do not certify this theorem for nonnormal repair matrices; a norm bound or a discrete Lyapunov inequality

$$L_{\perp}^{\top} H L_{\perp} \preceq e^{-2\gamma} H$$

is required.

Theorem (exact and approximate growing-mode coherence). Let $X(k)$ collect the scalar initial-condition variables for curvature, radiation, baryons, CDM/anomaly, neutrinos, and repair-sector scalars, and let $v_+(k)$ be the normalized adiabatic growing vector. Assume the single-clock and repair-gap theorems hold through readout and the growing/orthogonal splitting is invariant. If the initial orthogonal component is zero and no forcing or independent stochastic source has an orthogonal component, then

$$X(k) = v_+(k)\zeta(k), \quad \mathcal{P}_X(k) = \mathcal{P}_{\zeta}(k)v_+(k)v_+(k)^{\dagger}.$$

Thus $\text{rank } \mathcal{P}_X(k) = 1$, source-side isocurvature fractions vanish, and all species share a deterministic growing-mode phase. Without those exact zero-component hypotheses, the repair-gap estimate gives only

$$X_n(k) = v_+(k)\zeta_n(k) + \varepsilon_{\perp,n}(k),$$

with $\varepsilon_{\perp,n}$ bounded by the preceding geometric-series inequality; strict contraction alone does not make it zero at finite readout. Finite runs must report λ_2/λ_1 , $\beta_{\text{iso}}(k)$, decaying-mode fraction, and phase-convention defects as source diagnostics.

Theorem (screen-to-primordial bridge). Assume the geometric record and causal-area metric receipts, the collar identity $q_{\zeta} = \zeta_{\rho}$, source-stress closure, rank-one normal form, positive repair gap, freeze-out, adiabatic growing-mode, isocurvature, and phase-coherence receipts all pass. Also assume the physical scale-bridge theorem: a source embedding with `source_embedding_hash`, a physical mode basis with `physical_mode_basis_id`, a calibrated comoving coordinate χ , mode-normalization and density-of-states receipts, and a no-post-hoc-calibration receipt, and a `BackgroundCurvatureStatusr` that is not `UNRESOLVED`. Then the low-mode-removed OPH screen scalar is the source-only primordial curvature field restricted to the declared shell or radial window:

$$q(\hat{n}) = \Pi_{\ell_{\text{src}} \geq 2} \int d\chi W(\chi) \zeta_{\star}(\chi \hat{n}).$$

For a thin shell, $q(\hat{n}) = \Pi_{\ell_{\text{src}} \geq 2} \zeta_{\star}(R_{\star} \hat{n})$. The forward spectrum is exactly

$$C_{\ell}^q = 4\pi \int_0^{\infty} d \ln k \Delta_{\zeta}^2(k) |\Psi_{\ell}^{\text{BranchType}}(k)|^2,$$

$$\Psi_{\ell}^{\text{BranchType}}(k) = \begin{cases} \int dr W(r) j_{\ell}(kr), & \text{BranchType} = \text{FLATEXACT or FLATASSUMED}, \\ \int dr W(r) \Phi_{\ell k}^{(-)}(r), & \text{BranchType} = \text{OPENCURVED}, \\ \int dr W(r) \Phi_{\ell k}^{(+)}(r), & \text{BranchType} = \text{CLOSEDCURVED}, \end{cases}$$

and for a thin shell

$$C_{\ell}^q = 4\pi \int_0^{\infty} d \ln k \Delta_{\zeta}^2(k) \left| \Psi_{\ell}^{\text{BranchType}}(k; R_{\star}) \right|^2.$$

At finite regulator the continuum integral is represented by the finite spectral measure

$$d\nu_r(k) = \sum_j w_{rj} \delta(k - k_{rj}),$$

so the actual finite evidence receipt is

$$C_{\ell_{\text{src}}, r}^q = 4\pi \sum_j w_{rj}^{\log k} \Delta_{\zeta, r}^2(k_{rj}) |\Psi_{\ell_{\text{src}}, r}^{\text{BranchType}}(k_{rj})|^2.$$

Every radial node, window, and mode bin is bound to the geometry hash, source-embedding hash, scale-certificate hash, boundary-condition hash, and mode-basis identifier, and the evidence bundle must emit radial kernel geometry compatibility receipt. The observed multipole L_{CMB} is not ℓ_{src} ; it is the output index of the line-of-sight transfer solver. Cap eigenmodes, inverse cap-opening angles, and relabeled unit strings remain diagnostic by the dimensional, angular, cap-mode, and unit-string no-go propositions.

Physical scale dependency. The k_{rj} grid in the preceding equation is not a free producer field. It is consumed only after

physical spatial k receipt

recomputes the comoving k -intervals from the physical geometry, scale certificate, scale factor, finite eigenvalues, normalization, and refinement residuals. The radial kernel and source-screen projection additionally require

screen to physical k association receipt.

The source covariance is a finite positive operator on the common physical mode projectors:

$$C_{\zeta, r} \succeq 0.$$

Homogeneity and isotropy on the safe band require

$$P_I C_{\zeta, r} P_I = \mathcal{P}_\zeta(k_I) P_I + E_I$$

with a declared residual bound, together with an off-diagonal leakage report

$$\sum_{I \neq J} |P_I C_{\zeta, r} P_J|^2.$$

The same physical projector family is used by primordial covariance and anomaly response:

common primordial anomaly mode basis receipt.

Mode freezeout and common initial surface are separate gates:

physical mode freezeout map receipt, *physical freezeout surface receipt.*

The latter contains initial data and normal derivatives on one common spacelike surface.

Radial prior, null space, and residual receipt. The map above is a forward projection, not a free inversion. With finite bins p_j for Δ_ζ^2 ,

$$A_{\ell j} = 4\pi \int_{\text{bin } j} d \ln k b_j(k) |\Psi_\ell(k)|^2, \quad C^q = A p.$$

The evidence bundle must declare the basis, support, prior p_0, Q , positivity conditions, singular spectrum of $AQ^{-1/2}$, effective rank, null basis, resolution kernels, prior sensitivity, and every forward residual

$$r_\ell = C_\ell^q - \hat{C}_\ell.$$

A finite prior, regularizer, positivity constraint, or square discretization selects one representative from a nonunique shell equivalence class. Its typed output is `CONDITIONALRADIALCONTINUATION`. A source-derived primordial packet using either radial route certified below requires the common source, scale, mode, null-space, positivity, ancestry, and non-fitting forward-residual receipts together with that route's exact uniqueness certificate. Another source-restricted injective construction may be used if it supplies its own orbit-separation theorem and the same physical and no-target-leak interfaces. If the geometry packet is imported, the result is *conditional physical*; a *source-native physical* result requires the native `CosmoGeomReadr` theorem.

Complete radial-lift theorem packet. The screen-to-radial map is a forward restriction with a nonunique unrestricted inverse. For a flat source window W and geometric normalization Z_q ,

$$C_{\ell,W}^q = 4\pi Z_q^2 \int_0^\infty d \ln k \Delta_\zeta^2(k) |\Psi_{\ell,W}(k)|^2, \quad \Psi_{\ell,W}(k) = \int W(dr) j_\ell(kr).$$

The packet below certifies two sufficient uniqueness routes: radial dilation and radial tomography. They are not asserted to be exhaustive or mutually exclusive. A source-derived parametric or dynamical law, an injective source-restricted basis, analytic or PDE constraints, or another fully typed orbit-separating construction remains admissible. A finite prior continuation has the type `CONDITIONALRADIALCONTINUATION`.

Theorem (one-shell correlation restriction and exact no-go). For a thin shell of radius R , the complete angular covariance satisfies

$$K_R(\mu) = Z_q^2 \xi_\zeta \left(R\sqrt{2-2\mu} \right) = \sum_{\ell \geq 0} \frac{2\ell+1}{4\pi} C_{\ell,R}^q P_\ell(\mu).$$

Hence even the noiseless sequence $\{C_{\ell,R}^q\}_{\ell \geq 0}$ determines the three-dimensional correlation only on $0 \leq s \leq 2R$. The full one-shell operator has an infinite-dimensional kernel, including positivity-preserving ambiguities. Explicitly, for any nonzero $g \in C_c^\infty((2R, \infty))$,

$$h_g(k) = \frac{2k^3}{\pi} \int_0^\infty r^2 g(r) j_0(kr) dr$$

has correlation perturbation $\xi_{h_g} = g$ and therefore changes no shell multipole. Choosing $B = |h_g| + w$ with any strictly positive integrable w makes B and $B + h_g$ two distinct positive spectra with the same complete shell covariance.

Proof. The spherical-Bessel addition theorem gives the displayed restriction identity. The map $\mu \mapsto R\sqrt{2-2\mu}$ covers exactly $[0, 2R]$. Radial Fourier inversion gives $\xi_{h_g} = g$ and is injective, so the kernel contains an infinite-dimensional copy of $C_c^\infty((2R, \infty))$. The positive pair differs by the same null element. $\square$

Corollary (exact low-mode-removed kernel). For a continuous signed correlation perturbation ξ_h , invisibility to every shell multipole $\ell \geq 0$ is equivalent to $\xi_h(s) = 0$ on $0 \leq s \leq 2R$. Invisibility to every retained OPH multipole $\ell \geq 2$ is equivalent to

$$\xi_h(s) = a + b \left(1 - \frac{s^2}{2R^2} \right), \quad 0 \leq s \leq 2R,$$

for constants a, b . Legendre completeness gives the first statement. Removing $\ell = 0, 1$ leaves the span of P_0 and P_1 invisible, which gives the second.

Producer theorem (source-refinement orbit to physical dilation). Let $\mathcal{H}_r^{\text{src}}$ be the scale-labelled scalar mode space generated by the cofinal refinement orbit of the released screen, including its scale label. A single angular cut is insufficient. Let U_r map the orbit unitarily onto the physical rank-one adiabatic source subspace. If

$$D_{s,r}U_r = U_r R_{s,r}, \quad C_{\zeta,r} = U_r C_{\text{src},r} U_r^*,$$

then

$$D_{s,r}^{-1} C_{\zeta,r} D_{s,r} - u_r(s) C_{\zeta,r} = U_r \left(R_{s,r}^{-1} C_{\text{src},r} R_{s,r} - u_r(s) C_{\text{src},r} \right) U_r^*.$$

Thus the source and physical operator residuals have equal norm. If the finite covariances converge strongly with a uniform norm bound, the finite dilation maps and their inverses converge strongly, $u_r(s) \rightarrow u(s)$, and this residual tends to zero, then $D_s^{-1} C_{\zeta} D_s = u(s) C_{\zeta}$ in the continuum. The commuting square states that source refinement followed by the physical embedding agrees with physical embedding followed by wavelength dilation, and it transfers the source covariance error exactly to the physical covariance.

Proof. Conjugate the source covariance by U_r and use the commuting square; unitary invariance gives the residual identity. For the limit, add and subtract the finite conjugated expression. Strong convergence, uniform boundedness, scalar convergence, and the norm-residual hypothesis send the three resulting terms to zero. $\square$

Theorem (source-dilation uniqueness). Let $C_{\zeta} = M_{\Delta_{\zeta}^2}$ on the common physical $L^2(d \ln k \, d\Omega_k)$ mode basis. Define

$$(D_s f)(k, \hat{k}) = f(e^{-s} k, \hat{k}).$$

If the source refinement packet proves

$$D_s^{-1} C_{\zeta} D_s = u_q(s) C_{\zeta}, \quad u_q(s+t) = u_q(s) u_q(t), \quad -\frac{d}{ds} \ln u_q(s) \Big|_{s=0} = \theta,$$

then $u_q(s) = e^{-\theta s}$ and there is $A > 0$ such that

$$\Delta_{\zeta}^2(k) = A k^{-\theta} \quad \text{almost everywhere.}$$

For a continuous source representative and pivot k_* ,

$$\boxed{\Delta_{\zeta}^2(k) = A_{\zeta} (k/k_*)^{-\theta}}, \quad A_{\zeta} = \Delta_{\zeta}^2(k_*).$$

Proof. Strong continuity solves the multiplicative Cauchy equation for u_q . Conjugation of the multiplication operator gives $\Delta_{\zeta}^2(e^s k) = e^{-\theta s} \Delta_{\zeta}^2(k)$ almost everywhere for each s . In logarithmic coordinates, $e^{\theta t} \Delta_{\zeta}^2(e^t)$ is invariant under all rational translations on one common full-measure set; bounded-transform convolution makes it constant almost everywhere. Continuity upgrades the identity to every k . $\square$

Theorem (exact flat thin-shell lift). On the source-dilation branch, for $-2 < \theta < 4$ and $\ell \geq 2$,

$$\int_0^\infty \frac{dx}{x} x^{-\theta} j_\ell^2(x) = \frac{\sqrt{\pi}}{4} \frac{\Gamma(1 + \theta/2)}{\Gamma(3/2 + \theta/2)} \frac{\Gamma(\ell - \theta/2)}{\Gamma(\ell + 2 + \theta/2)},$$

so

$$C_{\ell, R_\star}^q = A_q \frac{\Gamma(\ell - \theta/2)}{\Gamma(\ell + 2 + \theta/2)},$$

with the complete amplitude conversion

$$A_\zeta = \frac{A_q}{\pi^{3/2} Z_q^2 (k_\star R_\star)^\theta} \frac{\Gamma(3/2 + \theta/2)}{\Gamma(1 + \theta/2)}.$$

Proof. Insert the source power law into the exact forward map, set $x = kR_\star$, and apply the Weber–Schafheitlin square integral. The convergence strip follows from $j_\ell(x) \sim x^\ell/(2\ell+1)!!$ at zero and $j_\ell(x) = O(x^{-1})$ at infinity. $\square$

At the source-native pivot $k_\star R_\star = 1$ with $Z_q = 1$ and the source-law tilt θ , the conversion factor evaluates to approximately 0.16080676. It is an arithmetic consequence of the symbolic relation and supplies no source amplitude by itself.

Corollary (source-family injectivity and multipole consistency). Fix $\theta, Z_q, k_\star$, and a source window for which the forward coefficient is positive and finite. The forward map is injective in A_ζ on the family $\Delta_\zeta^2 = A_\zeta(k/k_\star)^{-\theta}$. For each retained multipole,

$$A_\zeta^{(\ell, W)} = \frac{C_{\ell, W}^q}{4\pi Z_q^2 k_\star^\theta \int d \ln k k^{-\theta} |\Psi_{\ell, W}(k)|^2}.$$

Every $A_\zeta^{(\ell, W)}$ must agree. A disagreement falsifies the declared window/dilation packet and cannot be repaired with an ℓ -dependent amplitude. Once the source law fixes θ , the screen has one amplitude left to determine, and the unused multipoles become consistency checks.

Theorem (finite-window bound). For $0 < \theta < 2\ell$, define

$$I_\ell(\theta) = \int d \ln x x^{-\theta} j_\ell^2(x), \quad J_\ell(\theta) = I_\ell(\theta - 2) - \left[\ell(\ell + 1) - \frac{\theta(\theta + 1)}{2} \right] I_\ell(\theta),$$

$$\eta_{\ell, W} = \frac{2\sqrt{J_\ell(\theta)}}{\theta} \int W(dr) |r^{\theta/2} - R_\star^{\theta/2}|.$$

Then

$$|C_{\ell, W}^q - C_{\ell, R_\star}^q| \leq 4\pi Z_q^2 A_\zeta k_\star^\theta \eta_{\ell, W} \left(2R_\star^{\theta/2} \sqrt{I_\ell(\theta)} + \eta_{\ell, W} \right).$$

Proof. In the Hilbert norm $\|f\|_\theta^2 = \int d \ln k k^{-\theta} |f(k)|^2$, $\|j_\ell(kr)\|_\theta = r^{\theta/2} \sqrt{I_\ell}$ and $\|\partial_r j_\ell(kr)\|_\theta = r^{\theta/2-1} \sqrt{J_\ell}$. The fundamental theorem of calculus, Minkowski, and $|||f||^2 - ||g||^2| \leq \|f - g\|(\|f\| + \|g\|)$ give the bound. $\square$

Theorem (finite singular-value and prior continuation). For a declared finite radial basis, write $C = Ap$. If $A \in \mathbb{R}^{N_\ell \times N_k}$ has rank r , then $\dim \ker A = N_k - r$, and every exact solution is $p = p_{\text{part}} + V_0 a$ for a right-null basis V_0 . Given a prior center p_0 and $Q \succ 0$, the unique minimizer of $\frac{1}{2}(p - p_0)^\top Q(p - p_0)$ subject to $Ap = C$ is

$$p_* = p_0 + Q^{-1} A^\top (A Q^{-1} A^\top)^+ (C - A p_0).$$

The run publishes the singular values, rank threshold, right-null basis, resolution operator, unresolved projector, positivity active set, and prior sensitivity. The output type is `CONDITIONALRADIALCONTINUATION`.

Theorem (unrestricted tomographic uniqueness). For

$$C_\ell(r, r') = \frac{2}{\pi} \int_0^\infty k^2 P_\zeta(k) j_\ell(kr) j_\ell(kr') dk,$$

the covariance operator on $L^2(r^2 dr)$ obeys

$$Q_\ell = \mathcal{H}_\ell^{-1} M_{P_\zeta} \mathcal{H}_\ell.$$

Thus a complete radial cross-covariance operator determines P_ζ uniquely almost everywhere. Fixing one radius r_0 , the family $C_\ell(r, r_0)$ known for all r suffices: its spherical-Hankel transform is $\sqrt{2/\pi} P_\zeta(k) j_\ell(kr_0)$, and the Bessel zeros are discrete. A finite collection of auto-spectra does not supply this tomography theorem.

Proof. Apply spherical-Hankel inversion in the first radial variable. Unitary conjugation exposes the multiplication operator; equal multiplication operators have equal multipliers almost everywhere. $\square$

Theorem (branch-typed radial boundary). The ordinary spherical-Bessel and gamma-function formulas apply to FLATEXACT and FLATASSUMED branches. Open and closed spatial branches use their declared hyperspherical eigenfunctions and spectral measures. An unresolved curvature branch blocks physical radial identification, and the flat thin-shell amplitude formula cannot certify a curved branch.

Physical source requirements. A source-derived primordial packet using either certified route above requires all source-stress, single-clock, freezeout, growing-mode, phase, physical-scale, source-amplitude, null-space, positivity, and non-fitting forward-residual receipts, together with its radial-dilation intertwiner or radial-tomography certificate. An alternative source construction must supply its own injectivity or orbit-separation theorem and the same physical and no-target-leak interfaces. A finite inverse selected before evaluation is a conditional radial continuation. It supplies no physical temperature or polarization spectrum.

Corollary (transfer firewall). Radial identification changes no temperature, polarization, lensing, recombination, foreground, nuisance, covariance, or likelihood conclusion. Those outputs require the independent transfer and likelihood system.

total stress closure receipt	$T_I^{ab}, Q_I^a, E_T, \nabla_a T^{ab}$
single clock normal form receipt	$\chi_r, \text{rank } d\mathcal{F}, \text{ connected fibers}$
entropy repair gap receipt	$L_\perp, H, \gamma, \text{ forcing/refinement bounds}$
curvature evolution receipt	$\zeta_a, \Gamma_a^{\text{eff}}, \text{ evolution residual, freeze bound}$
adiabatic mode receipt	$S_{IJ}, \Gamma_I, \Xi_I, \text{ growing-mode projection}$
isocurvature bound report	$\beta_{\text{iso}}(k) \text{ and source covariance}$
primordial phase coherence receipt	$\lambda_2/\lambda_1, \text{ phase and decaying-mode defects}$
screen to radial lift receipt	$W(r), R_\star, \Psi_\ell(k), \text{ source hashes}$
radial null space report receipt	singular values, null basis, resolution kernels
forward projection residual receipt	$r_\ell, \text{ aggregate norms, quadrature and solver errors.}$

Source-provenance and pooled-reducer firewall. Let the source artifact DAG have parent-to-child edges and let M be the set of measurement, residual, likelihood, posterior, fitted-parameter, or measurement-calibrated nodes. Define

$$\tau(v) = \mathbf{1}_{v \in M} \vee \bigvee_{p \in \text{Parents}(v)} \tau(p).$$

Then $\tau(v) = 1$ exactly when v has a forbidden measurement ancestor. The proof is induction over a topological ordering of the DAG. Every node claimed as source-derived for $\eta_R, \Gamma_{\text{rec}}, A_\zeta, q_{\text{IR}}, \ell_{\text{IR}}, B_A(k, a), \rho_A(a)$, and N_{CRC} must have $\tau = 0$. Unknown source metadata fails closed.

For nonlinear estimates, shards may not average nonlinear quantities. Each row must emit additive sufficient statistics in a commutative monoid $(\mathcal{M}, \oplus)$, for example

$$\sum w_i, \quad \sum w_i x_i, \quad \sum w_i x_i x_i^\top$$

or, for weighted response regression,

$$X^\top W X, \quad X^\top W y, \quad y^\top W y.$$

Associativity, commutativity, complete coverage, and duplicate removal make the pooled statistic partition-invariant; the deterministic estimator is then evaluated only after pooling.

Three-dimensional homogeneous lift. A rotationally invariant screen covariance about one observer is not a statistically homogeneous three-dimensional primordial field by itself. A frozen lift

$$\mathcal{L}_r : V_r^{\text{screen}} \rightarrow \mathcal{H}_r^{(3)}$$

must define

$$\Sigma_r^{(3)} = \mathcal{L}_r \Sigma_r^{\text{screen}} \mathcal{L}_r^*$$

and certify positivity, refinement convergence, rotation covariance, translation covariance or an equivalent multicenter consistency theorem, and explicit null-space treatment. If the limiting covariance commutes with translations and rotations, Fourier diagonalization gives

$$\langle \zeta_a(\mathbf{k}) \zeta_b^*(\mathbf{k}') \rangle = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{k}') P_{ab}(|\mathbf{k}|),$$

with $P_{ab}(k)$ positive semidefinite.

Transfer-ready initial data. The source-to-transfer map must emit a complete initial state

$$Y(k, \eta_i) = \mathcal{M}_{\text{init}}(k; \mathfrak{B}_{\text{src}}) \xi(k),$$

including metric, photon, polarization, baryon, neutrino, anomaly, recipient, and auxiliary response variables. Near a singular early-time point, regular Frobenius modes solve

$$Y' = \left(\frac{M_{-1}}{\eta} + M_0 + O(\eta) \right) Y, \quad Y = \eta^s (v_0 + v_1 \eta + \cdots), \quad M_{-1} v_0 = s v_0.$$

The branch must classify regular adiabatic growing modes, allowed isocurvature modes, decaying modes, gauge modes, and singular interaction modes. On the adiabatic branch,

$$S_{IJ} = 3(\zeta_I - \zeta_J) = 0$$

for all active sectors. The finite evidence receipt must bound the initial Hamiltonian and momentum constraints and the propagated residual

$$\epsilon_{\text{constraint}} = \sup_{k, \eta} \|\mathcal{C}(k, \eta) Y(k, \eta)\|.$$

Receipt contract. Every ensemble run must record its ensemble id, interpretation class, regulator, representative schema, gauge action, quotient canonicalizer, base measure definition, action

coefficients, coarse maps, zero-mode projector, amplitude convention, sampler kernel, partition-invariant random-event schema, smoothing policy, source provenance, and explicit nonclaims. The seed belongs to the run receipt rather than the ensemble definition. The interpretation classes are:

seed, proposal, or repair noise,
 conventional reference ensemble,
 OPH-native quotient ensemble,
 OPH physical vacuum,
 OPH primordial field,
 observable cosmological prediction.

Smoothing must preserve raw coefficients, raw spectra, smoothing kernels, smoothed coefficients, smoothed spectra, and hashes of each stage; it is not part of S_r unless explicitly declared. For bounded coarse observables O , if the implemented law satisfies $\|\hat{\mu}_s - \mu_s\|_{\text{TV}} \leq \epsilon_{\text{samp}}$ and the refinement defects sum to ϵ_{ref} , then

$$\left| \mathbb{E}_{\hat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O] \right| \leq 2\|O\|_{\infty}(\epsilon_{\text{samp}} + \epsilon_{\text{ref}}),$$

which is the finite evidence accuracy statement for this theorem.

Theorem 2.3 (EC from regulated overlap gluing). For a collar B_{δ} around a cap boundary Σ , there is a canonical decomposition

$$H_{B_{\delta}} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{G_{\Sigma}} = \bigoplus_{\alpha} (H_{b_L^{\alpha}} \otimes H_{b_R^{\alpha}}),$$

and the sector-preserving collar algebra

$$A_{\text{EC}}(B_{\delta}) := \bigoplus_{\alpha} (B(H_{b_L^{\alpha}}) \otimes B(H_{b_R^{\alpha}}))$$

with

$$Z(A_{\text{EC}}(B_{\delta})) = \bigoplus_{\alpha} \mathbb{C} \mathbf{1}_{\alpha},$$

such that $\mathcal{A}(A_{\delta}B_{\delta})$ acts only on $H_{b_L^{\alpha}}$ and $\mathcal{A}(B_{\delta}D_{\delta})$ acts only on $H_{b_R^{\alpha}}$ within each block.

Proof. Split the collar into half-collars B_L and B_R meeting on $\Sigma = \partial C$. By the realized regulator presentation above, the physical collar Hilbert space is the diagonal invariant subspace $(\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{G_{\Sigma}}$. Decompose each side into irreps:

$$\tilde{\mathcal{H}}_{B_L} = \bigoplus_{\alpha} (V_{\alpha} \otimes H_{b_L^{\alpha}}), \quad \tilde{\mathcal{H}}_{B_R} = \bigoplus_{\beta} (V_{\beta}^* \otimes H_{b_R^{\beta}}).$$

Then

$$\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R} = \bigoplus_{\alpha, \beta} (V_{\alpha} \otimes V_{\beta}^*) \otimes (H_{b_L^{\alpha}} \otimes H_{b_R^{\beta}}).$$

By Schur's lemma,

$$(V_{\alpha} \otimes V_{\beta}^*)^{G_{\Sigma}} \cong \begin{cases} \mathbb{C}, & \alpha = \beta, \\ 0, & \alpha \neq \beta. \end{cases}$$

Therefore the invariant subspace is

$$H_{B_\delta} = \bigoplus_{\alpha} (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

as claimed. The induced sector-preserving collar algebra is

$$A_{\text{EC}}(B_\delta) = \bigoplus_{\alpha} (B(H_{b_L^\alpha}) \otimes B(H_{b_R^\alpha})),$$

so the center is generated by the block projectors. Adjacent region algebras act on the left or right factor only because the gluing action is supported on Σ . QED.

Remark. On the central-defect subbranch, replace G_Σ by its central extension. The sector label α then ranges over irreps of the extension; the decomposition is unchanged.

We refer to the decomposition in Theorem 2.3 as **edge-center completion (EC)**.

Exact Markov route. If $I(A_\delta : D_\delta \mid B_\delta) = 0$, the Hayden–Jozsa–Petz–Winter (HJPW) structure theorem yields a blockwise factorization of the state over *some* direct-sum/tensor decomposition of $\mathcal{H}_{B_\delta}$; that decomposition is state-dependent, and HJPW does not by itself identify it with the preselected EC factors b_L^α, b_R^α of Theorem 2.3. The exact identity used for literal Markov-modular equalities below is the **EC-aligned** normal form

$$\rho_{A_\delta B_\delta D_\delta} = \bigoplus_{\alpha} p_{\alpha} (\rho_{A_\delta b_L^\alpha} \otimes \rho_{b_R^\alpha D_\delta}), \quad I_{\omega}(A_\delta : D_\delta \mid B_\delta) = 0,$$

and the identification of the HJPW factors with the EC factors is a genuine additional state condition, the **Markov-split alignment hypothesis (MSA)**. It is not implied by exact Markovity: with four qubits A, B_L, B_R, D and $\rho = \Phi_{AB_R} \otimes \Phi_{B_L D}$ (Bell pairs across the cut, one center block), one has $I(A : D \mid B) = 0$ but $I(A : B_R) = 2 \ln 2$, so the state is exactly Markov yet not of the displayed EC-aligned form, which would force $I(A : B_R) = 0$. All literal Markov-modular equalities below therefore assume exact Markovity *plus* MSA, or an explicitly stated idealized limit that supplies both; The spacetime and Einstein paper states MSA formally and records this counterexample. MSA is checkable state by state: The spacetime and Einstein paper proves it equivalent to a splitting of $\log \rho$ into commuting one-sided terms supported on Ab_L^α and $b_R^\alpha D$, to the existence of a ρ -preserving conditional expectation onto either one-sided edge algebra (a Takesaki commuting square over the collar center [57]), and to blockwise vanishing one-sided mutual information $I(Ab_L^\alpha : b_R^\alpha D) = 0$. On the local information-projection branch this reduces the derivation of MSA to a sharp condition on the effective Hamiltonian: every cross-cut coupling must act through the central sector label, as it does in lattice-gauge-type regulators where the interface energy is a function of the conserved flux. On that declared **central-interface branch** The spacetime and Einstein paper derives both MSA and exact collar Markovianity as theorems (descent of one-sided invariants and boundary charges through the edge-center quotient), and it adopts the central-interface normal form as an explicit named branch premise, so the literal Markov-modular identities hold on the declared branch without a further state hypothesis. This premise is independent of A1–A3: The spacetime and Einstein paper exhibits a finite regulator package that satisfies observer agreement, schedule-independent transactional repair, and an exactly closed refinement-channel family while its retained constraint family carries a boundary-invariant noncentral cross-cut coupling. Observer agreement and repair do not force the premise. Off the declared branch MSA is carried as an explicit hypothesis.

On the separately declared rate-controlled family obtained by pushing the optimizer forward, Theorem 2.3 supplies the kinematic edge-center decomposition. Exact Markovity is one idealized

route. The following lemma and theorem supply the quantitative decay route when the approximation is kept explicit.

Lemma 2.6 (interior local information projection implies local Gibbs form).

Under Axiom 3 on the subbranch with identity-proportional local references, state-determining aggregation equivalent to the declared entropy objective, homogeneous global-sum constraints $\langle \sum_x O_a(x) \rangle = C_a$, and the finite-dimensional regulator realization above, assume also that the constraint vector lies in the relative interior of the attainable moment set, so the selected state is faithful on the retained sector. Then the selected state has the finite-multiplier Gibbs form

$$\omega = \frac{e^{-H_{\text{eff}}}}{\text{Tr } e^{-H_{\text{eff}}}}, \quad H_{\text{eff}} = \sum_x \sum_a \lambda_a O_a(x) + H_{\text{sec}},$$

where the sum runs over UV cells x and density labels a , with one cell-independent multiplier λ_a per label. For the finite-range collar theorem, H_{sec} is either a sum of uniformly bounded finite-range charge densities or is central and fixed on the chosen superselection sector. A genuinely nonlocal noncentral term falls outside the theorem. Under this condition the effective Hamiltonian is finite-range in UV graph units, and the multiplier count matches the constraint count N_{con} plus the number of retained sector terms on every lattice.

Proof. On a finite-dimensional algebra, an interior entropy maximizer subject to linear constraints $\text{Tr}(\rho O_i) = c_i$ is given by finite Lagrange multipliers:

$$\rho = \frac{e^{-\sum_i \lambda_i O_i}}{\text{Tr } e^{-\sum_i \lambda_i O_i}}.$$

Strict concavity of von Neumann entropy ensures uniqueness. Here the constrained operators are the N_{con} global sums $O_a := \sum_x O_a(x)$ plus the retained sector terms, so the exponent has one shared multiplier per density label. MaxEnt gives the Gibbs form on this faithful/interior branch; the displayed locality restriction on H_{sec} is separate and load-bearing. A boundary constraint can instead force a rank-deficient optimizer (for example $\text{Tr}(\rho |1\rangle\langle 1|) = 0$ on $\mathbb{C}^2$); it has no finite full-algebra multiplier and must be treated on its support or by infinite penalties. Such a boundary branch does not supply the stagewise faithfulness used below. Had the constraints instead been imposed per cell, $\langle O_a(x) \rangle = c_a(x)$, the multipliers would be cell-dependent fields $\lambda_a(x)$ and the family would grow with the lattice; that enlarged family belongs to the non-equilibrium approximation regime of the clarification above and is not the branch used in the refinement argument. QED.

Definition 2.6a (closure defect and induced refinement map). Fix successive regulator scales ℓ (finer) and L (coarser), let $\{\omega_L(\lambda')\}$ be the homogeneous global-sum exponential family at scale L , and let $\Phi_{\ell \rightarrow L}$ be an admissible coarse-graining channel. For fine-scale multipliers λ set $\sigma := \Phi_{\ell \rightarrow L}(\omega_\ell(\lambda))$ and define the **closure defect**

$$\varepsilon_{\ell \rightarrow L}(\lambda) := \inf_{\lambda'} D(\sigma \parallel \omega_L(\lambda')),$$

with D the quantum relative entropy. When the infimum is attained at a unique λ^* , set $R_{\ell \rightarrow L}(\lambda) := \lambda^*$.

Lemma 2.6b (I-projection residual bound). If the constrained operators at scale L together with the identity are linearly independent, then $\lambda' \mapsto D(\sigma \| \omega_L(\lambda'))$ is smooth and strictly convex; for faithful σ the minimizer $\lambda^* = R_{\ell \rightarrow L}(\lambda)$ exists, is unique, and is characterized by moment matching $\langle S_c \rangle_{\omega_L(\lambda^*)} = \langle S_c \rangle_\sigma$ for every constrained operator S_c ; and

$$\|\sigma - \omega_L(R_{\ell \rightarrow L}(\lambda))\|_1 \leq \sqrt{2\varepsilon_{\ell \rightarrow L}(\lambda)},$$

so the family reproduces every bounded expectation up to $\|B\|\sqrt{2\varepsilon_{\ell \rightarrow L}(\lambda)}$. The separately declared exact optimizer-pushforward interface is the statement $\varepsilon_{\ell \rightarrow L}(\lambda) = 0$ along the realized branch. In that case $\sigma = \omega_L(R_{\ell \rightarrow L}(\lambda))$, and the induced refinement map is this moment-matching I-projection.

Proof. Writing $\log \omega_L(\lambda') = -\sum_c \lambda'_c S_c - \log Z_L(\lambda')$, the objective is $-S(\sigma) + \sum_c \lambda'_c \langle S_c \rangle_\sigma + \log Z_L(\lambda')$, and the Hessian of $\log Z_L$ is the Duhamel (Kubo–Mori) covariance matrix of the constrained operators, positive definite under the stated independence; this gives strict convexity, uniqueness, and the moment-matching first-order condition, while attainment for faithful σ is standard finite-dimensional exponential-family duality [55]. The displayed residual bound is the quantum Pinsker inequality $D(\rho \| \tau) \geq \frac{1}{2} \|\rho - \tau\|_1^2$ [56] evaluated at λ^* , and $\varepsilon_{\ell \rightarrow L}(\lambda) = 0$ forces $\sigma = \omega_L(\lambda^*)$ by strict positivity of relative entropy off the diagonal. QED.

Acceptance check. A finite two-lattice test implementing this package (the constraint/multiplier count against the displayed N_{con} -dimensional family on successive lattices, the moment-matching projection, the residual bound, a generic decimation channel with strictly positive closure defect, and the exactly closed transverse-field subfamily where $R_{\ell \rightarrow L}$ acts as the identity) is included with the repository sources.

Strong conditional mixing premise. Let G_ℓ be the UV cell graph. Assume local dimension at most q , interaction degree at most Δ_0 , interaction diameter at most r_0 graph layers, and

$$\sup_x \sum_{X \ni x} \|\beta \Phi_\ell(X)\| \leq J_0$$

with constants independent of the cut and refinement stage. Let m_ℓ be the resolved collar width in graph layers and set $\delta = m_\ell \ell$. For the faithful Gibbs state, define

$$\mathbf{J}_\rho(A : D \mid B) := \log \rho_{ABD} + \log \rho_B - \log \rho_{AB} - \log \rho_{BD}.$$

The strong mixing premise is a boundary-anchor expansion

$$\mathbf{J}_\rho(A_\delta : D_\delta \mid B_\delta) = \sum_{z \in \partial_{r_0}^{\text{UV}} C} E_{z, \ell, \delta}, \quad \|E_{z, \ell, \delta}\|_\infty \leq \kappa e^{-(m_\ell - r_0)/\zeta},$$

with κ, ζ uniform in the cut, boundary condition, and stage. This conditional or matrix mixing condition is stronger than ordinary two-point exponential clustering. Local Gibbs form, a Lieb–Robinson bound, or a Hamiltonian spectral gap does not imply it for a general noncommuting Gibbs state.

Theorem 2.5 (finite-range conditional Gibbs mixing gives collar recovery).

Under Lemma 2.6 and the strong conditional mixing premise,

$$I_\omega(A_\delta : D_\delta \mid B_\delta) \leq \kappa |\partial C|_{\text{UV}} e^{-(m_\ell - r_0)/\zeta} \leq c |\partial C|_{\text{UV}} e^{-\delta/\xi_\ell}$$

with

$$\xi_\ell = \zeta \ell, \quad c = \kappa e^{r_0/\zeta}.$$

Proof. Embedding the marginal logarithms in the tripartite algebra gives

$$I(A : D \mid B)_\rho = \text{Tr}[\rho_{ABD} \mathbf{J}_\rho(A : D \mid B)].$$

The boundary expansion and trace-norm duality bound this expectation by the sum of the boundary term norms. The second form follows from $\delta = m_\ell \ell$. QED.

For one uniform family, the sharp sufficient limit condition is

$$\frac{\delta}{\xi_\ell} - \log |\partial C|_{\text{UV}} \longrightarrow +\infty.$$

If

$$|\partial C|_{\text{UV}} \leq a(\ell_0/\ell)^p, \quad \delta(\ell) \geq \zeta(p + \eta)\ell \log(\ell_0/\ell), \quad \eta > 0,$$

then $\delta \rightarrow 0$, $\delta/\ell \rightarrow \infty$, and

$$I(A_\delta : D_\delta \mid B_\delta) \leq ca(\ell/\ell_0)^\eta \longrightarrow 0.$$

The ratio $\delta/\ell \rightarrow \infty$ alone is insufficient. For $\ell_n = \ell_0 e^{-n^2}$, $\delta_n = \zeta n \ell_n$, and $|\partial C_n|_{\text{UV}} = e^{n^2}$, that ratio diverges while the upper envelope grows as $e^{n^2 - n}$.

A finite receipt records the interaction range and norm bound, treatment of global sector terms, boundary-cell count, graph separation, regional CMI in nats, matrix-defect norm, predeclared κ, ζ , log-envelope slack, Fawzi–Renner error, held-out cuts, and the rate margin $\delta/(\zeta \ell) - \log |\partial C|_{\text{UV}}$. Fitted constants and local random-triplet CMI remain finite proxies. They do not certify a cofinal limit.

This bound is the quantitative hinge for constructive gluing.

2.0.2 Concrete UV realization: quantum link models

Section 2.3 internalizes the regulator package: the fixed-cutoff type-I algebra and boundary fixed-point structure are the realized form of the screen net plus overlap gluing. Quantum link models are included here only as an explicit microscopic example of that structure, not as a separate axiom layer.

UV regulator. Triangulate S^2 at scale ℓ_{UV} , giving vertices v , oriented links ℓ , and plaquettes p . Refinement corresponds to $\ell_{\text{UV}} \rightarrow 0$ with increasing lattice size.

Degrees of freedom. Attach to every oriented link ℓ a **finite-dimensional** Hilbert space $\mathcal{H}_\ell$. In ordinary Wilson lattice gauge theory, the continuum/refinement-limit edge description is modeled by $L^2(G)$ (infinite-dimensional for continuous G), but the microscopic OPH regulator is instead a **quantum link model** with finite-dimensional link Hilbert spaces that preserve gauge symmetry in operator form [75]. Optionally attach matter Hilbert spaces $\mathcal{H}_v$ at vertices. Then:

$$\tilde{\mathcal{H}}_{\text{total}} = \bigotimes_{\ell} \mathcal{H}_\ell \otimes \bigotimes_v \mathcal{H}_v,$$

finite-dimensional on any finite lattice. This is a concrete realization of the extended type-I presentation used in Section 2.3.

Boundary gluing as Gauss-law invariants. Define a local gauge transformation group G_v at each vertex v acting on incident links (and matter at v). Physical states satisfy:

$$|\psi\rangle \in \mathcal{H}_{\text{phys}} \iff U(g_v)|\psi\rangle = |\psi\rangle \quad \forall v, g_v \in G_v.$$

Equivalently: $\mathcal{H}_{\text{phys}} = \tilde{\mathcal{H}}_{\text{total}}^{\prod_v G_v}$.

Region algebras. For any region $R \subset S^2$, define an extended Hilbert space $\tilde{\mathcal{H}}_R$ from the links/vertices in R . The **boundary gauge group** $G_{\partial R}$ acts on the cut degrees of freedom (the "half-links" ending on ∂R). Define:

$$\mathcal{A}_{\text{inv}}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R)^{G_{\partial R}}.$$

This is the concrete quantum-link instance of the lifted fixed-point presentation used in Section 2.3. The same gauge constraints then induce the sector-preserving EC algebra on collars.

Why EC follows immediately. Take a cap C and a collar B_δ around ∂C . Because the *only* coupling between inside and outside is through the boundary gauge constraint, the collar Hilbert space decomposes into superselection blocks labeled by boundary irreps:

$$\mathcal{H}_{B_\delta} \cong \bigoplus_{\alpha} (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

with center generated by the projectors P_α . This is precisely the Schur-lemma mechanism of Theorem 2.3. The labels α are the familiar edge-mode / electric-flux labels appearing whenever one factorizes gauge theories across an entangling cut [76]. Exact Markovity depends on the state and is not forced by the decomposition alone; nor does exact Markovity align the HJPW factors with these edge-mode labels, which is the separate Markov-split alignment hypothesis of Section 2.3.

Dynamics and MaxEnt. The natural Hamiltonian is a 2+1D lattice gauge Hamiltonian on the screen worldvolume: plaquette ("magnetic") terms, electric terms on links, vertex Gauss terms as constraints, plus local matter couplings. In quantum link form this is finite-dimensional per link while behaving like gauge theory in the continuum limit. Then the MaxEnt assumption becomes concrete: the MaxEnt state is a Gibbs state $\rho \propto e^{-\sum_i \lambda_i O_i}$ with quasi-local O_i , precisely the local-Gibbs regime.

Geometry and G . This microphysics naturally supplies the emergent geometric objects:

- **Edge entropy / area operator:** $L_C = \sum_{\alpha} (\log d_{\alpha}) P_{\alpha}$ becomes "log of boundary irrep dimension" in the gauge link model.
- **Newton constant G :** the conversion factor between edge entropy density per boundary UV cell and macroscopic geometric area.

Thus area is an operator living in the center of the boundary algebra, because in gauge systems the center is where the cut labels live.

Scope of this example. The quantum-link realization makes the fixed-cutoff matrix/fixed-point bookkeeping concrete. Modular flow on caps becomes geometric conformal dilation with the 2π KMS normalization through the support-visible BW scaling theorem on the geometric subnet. Viable architectures for this include holographic quantum error-correcting codes [77] and quantum

double / string-net Hamiltonians [78], but any use of QECC distance, min-cut resilience, or Knill–Laflamme correction requires the corresponding code subspace, logical-operator, error-family, and recovery certificate.

2.0.3 Conformal-modular fixed-point microphysics

On the faithful/interior local finite-constraint branch of Axiom 3 with identity-proportional reference, the logarithm of the selected state is a quasi-local UV generator. A separately declared optimizer-pushforward and refinement-tail interface places the branch inside one common finite-dimensional multiplier family. At each regulator stage the patch and cap algebras are type-I matrix algebras. The finite regulator class is not refinement-closed, so the scaling-limit observer algebra may leave that class. The fixed-cutoff analysis proves the local-Gibbs result and propagation. Endpoint-Lipschitz control of *reduced* modular Hamiltonians requires the additional exact-Markov/central-interface and endpoint-control branch stated below; it does not follow from finite-range locality alone. The cross-scale interface supplies refinement stability. The geometric cap pair with geometric modular flow is obtained in the support-visible sense: the target algebra is the extracted geometric subnet, the support-visible modular matrix elements converge under regularization, and weak-*/GNS extraction plus support-readable modular covariance and ordered cut-pair rigidity gives the geometric cap automorphism.

Local finite-constraint information-projection branch. The constraint family $\mathcal{C}$ is generated by finitely many gauge-invariant local densities $\{O_a(x)\}$ of UV range $O(\ell_{UV})$, constrained through their homogeneous global sums $\langle \sum_x O_a(x) \rangle = C_a$. Axiom 3 selects the fixed-stage information projection. Retention of the same finite label set under refinement belongs to the separate optimizer-pushforward and refinement-tail interface.

Theorem 2.6 (Interior local constraints imply a local-Gibbs form). If the MaxEnt constraints are expectations of finitely many quasi-local operators $\{O_a\}$ with bounded support size at scale ℓ_{UV} , and the prescribed moment vector lies in the relative interior of the attainable moment set so that the optimizer is faithful on the retained sector, then the entropy maximizer is

$$\omega \propto \exp\left(-\sum_a \lambda_a O_a\right),$$

so the MaxEnt generator $H_{\text{MaxEnt}} = -\log \omega$ is a UV-range quasi-local sum. This is exactly the local-Gibbs form used later.

Proof. The finite-dimensional interior exponential-family theorem gives finite Lagrange multipliers λ_a . Boundary optima require restriction to their support or extended infinite multipliers and are not included in this faithful branch. QED.

Derived propagation control on the same branch. Because H_{MaxEnt} is a finite-range or quasi-local sum on a finite type-I regulator net, standard Lieb–Robinson estimates [37] apply to the automorphism group it generates, or to any branch generator lying in the same bounded-support algebraic closure. Thus there is a finite propagation velocity v_{LR} and constants C, ξ such that for local observables A_X, B_Y ,

$$\|[\tau_t(A_X), B_Y]\| \leq C \|A_X\| \|B_Y\| \min(|X|, |Y|) e^{-(d(X,Y) - v_{LR}|t|)/\xi}.$$

Additional endpoint-control branch. Locality of the global Gibbs generator does not by itself make reduced interval modular Hamiltonians local; the four-qubit counterexample in Theorem 4.3d rules out that inference. On the exact-Markov/central-interface branch, assume in addition a specified endpoint-to-regulator-cell map and a uniform local norm/site-density bound such that,

after subtracting the central endpoint block, the operator-norm sum of terms changed between allowed intervals I, I' is at most $L|I'\Delta I|$. On this explicitly named branch, the local split and that geometric norm bound give bounded-interval endpoint-Lipschitz matrix elements,

$$|\langle \psi, (K[I'] - K[I])\phi \rangle| \leq C_{\psi, \phi, I_{\max}} |I'\Delta I|,$$

used later in the null-modular bridge. Thus quasi-local propagation follows on the faithful local-Gibbs branch, while reduced-modular endpoint control is a separate exact-Markov/central-interface receipt with the stated endpoint geometry and uniformity.

Refinement-stable multiplier branch. The separately declared optimizer-pushforward and refinement-tail interface preserves the same finite constraint family under coarse-graining and places the regulator states in one common finite-dimensional multiplier family. This is a substantive renormalization condition because coarse-graining a finite-range Gibbs family generically generates interactions outside a fixed density list. The “refinement-stable” language used later means persistence along the stable or fixed branch of this multiplier family. This state-side notion is enough to compare realized states across cutoffs. `FixedCutoffBosonicSectorCategory` constructs the fixed-stage tensor-generated category and its forgetful fiber. `RefinementFunctorAndFiberDescent` constructs the refinement pullbacks and compatible fibers only on a cofinal tail carrying the compact-gauge refinement receipt. The cofinal compact-gauge witness uses the same receipt. The separate finite Standard Model route uses complete reversible response and endogenous overlap transport to force the local Standard Model gauge Lie algebra. Incidence and target-blind readback derive the carrier module and $R = -J$. The finite conditional fixture supplies the charge-conjugate rank-15 projector pair, hypercharge lattice, common $\mathbb{Z}_6$ kernel, and maximal faithful matter image. Source derivation of that current, matter action, and global quotient is not supplied. Under the complete-band and operational-cost premises, the exact screen theorem selects the rank-three response band. Physical family attachment, extra-sector completeness, and the charged-lepton and electroweak quantitative branches carry their own premises. Axiom 3 supplies neither the finite current realization nor the sector-colimit data or laboratory attachment.

Scaling limit and algebraic type. The regulator presentation gives a family of finite type-I algebras

$$\mathcal{A}_\ell(C) \cong \mathcal{B}(\mathcal{H}_{C, \ell}).$$

A scaling limit of this family need not be type I. In the continuum-QFT case of interest one expects the local limit algebra $\mathcal{A}_\infty(C)$ to be non-type-I, typically type III. Accordingly the fundamental modular datum in the limit is the automorphism group of the pair $(\mathcal{A}_\infty(C), \omega_\infty^C)$, not a density matrix inside $\mathcal{A}_\infty(C)$.

Corollary 2.6 (Geometric modular action on paired-certified caps). For any support-visible extracted scaling-limit geometric cap pair $(\mathcal{A}_\infty^{\text{geo}, \text{sv}}(C), \omega_\infty^{\text{geo}, C})$ emitted on a branch carrying both the finite cap-normal certificate and the independently complete same-branch multiresolution algebra-state comparison package, let $\alpha_{\lambda_{\widehat{C}}(s)}$ be the automorphism induced by the independently normalized BW-framed cap flow. Then

$$\sigma_t^{\omega_\infty^{\text{geo}, C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

If the limit cap algebra happens to be type I, this may be written as

$$K_C = 2\pi B_C + Z_C, \quad Z_C \in Z(\mathcal{A}(C))_{\text{sa}}.$$

In the generic continuum case, the automorphism identity is the complete statement.

Proof. This is The spacetime and Einstein paper’s finite cap-net BW theorem applied to the extracted geometric cap pair. The finite cap-normal input supplies BW framing, support-order faithfulness, held-out oriented cross-ratio rigidity, the geometric support flow, independently normalized geometric 2π -KMS comparison, and wrong-normalization controls. The separate multi-resolution algebra-state comparison input supplies the complete algebra-state reference tower, cap-family-uniform mixed-GNS convergence, comparison maps, and modular support covariance. QED.

Support envelope from net regularity (not an alternative derivation). The outer-regularity / minimal-support condition used later lets one read a support envelope for the modular image from the scaling-limit net. By itself it does not establish modular covariance; that equality comes from the paired certificate in Corollary 2.6, or must be assumed as an additional two-sided net-automorphism property.

(NR) Outer regularity / minimal support. For any operator O , the intersection of all connected regions P with $O \in \mathcal{A}(P)$ is again a connected region, denoted $\text{supp}(O)$.

Proposition 2.7 (Modular support-envelope inclusion). Under (NR), define for any region $R \subset C$

$$f_t^C(R) := \bigcup_{O \in \mathcal{A}(R)} \text{supp}(\sigma_t^{\omega, C}(O)).$$

Then

$$\sigma_t^{\omega, C}(\mathcal{A}(R)) \subseteq \mathcal{A}(f_t^C(R)).$$

Equality does not follow from (NR). It requires, for example, a two-sided order-isomorphism of the local net compatible with σ_t and σ_{-t} , together with the relevant additivity/local-generation property. Those are substantive covariance data, not consequences of knowing minimal operator supports.

Proof. For every $O \in \mathcal{A}(R)$, (NR) assigns $\sigma_t^{\omega, C}(O)$ a minimal support contained in $f_t^C(R)$; isotony therefore places that operator in $\mathcal{A}(f_t^C(R))$. Taking all such operators gives the stated inclusion. Surjectivity onto the entire envelope algebra is not supplied by the fact that σ_t is an automorphism of the global algebra. QED.

Null-surface modular structure. On the same support-visible scaling branch and extracted geometric-subnet branch, the null-surface modular machinery narrows as follows:

- **Fixed-cutoff null-strip bridge.** The null-strip package proves transferred cut-center data, the theorem-local inherited left/right strip-split condition needed for the spatial-collar-type tensor decomposition, exact-or-controlled four-term strip additivity on one inherited strip model, and renormalized endpoint-Lipschitz control up to the weak tail generator. On the scaling branch the geometric action supplies the half-sided relation; the separately registered common-vector and relative-commutant standardness receipts make it a standard half-sided modular inclusion whose Borchers–Wiesbrock consequence is an explicit positive null-translation generator on its Stone domain with the affine half-line modular relation. The same half-line family fixes the generator’s endpoint-derivative identity internally; its local null-stress reading consumes the separately supplied effective-locality and same-source bridge.
- **Derived half-sided modular relation.** Nested null half-line algebras satisfy the half-sided covariance and containment relation on the declared geometric scaling branch by Corollary 5.2e. To obtain a *standard* half-sided modular inclusion, the common reference vector must additionally be cyclic and separating for both half-line algebras and cyclic for their relative

commutant. On that separately registered standardness receipt, Borchers–Wiesbrock identifies the positive null-translation generator on its Stone domain together with the affine half-line modular relation [41].

- **Weak continuity and finite variation.** The bounded-interval and half-line endpoint-Lipschitz control follows on the exact-Markov/central-interface branch only after the specified endpoint-to-cell map and uniform norm/site-density bound of the endpoint-control clause. It then defines the weak tail generator at fixed cutoff. The support-visible scaling theorem together with the extracted geometric cap pair supplies the continuum null-generator setting in which that weak-tail data can be matched to the geometric null modular action of the relativistic phase.

Constraint set specification. On the local finite-constraint branch, the “correct fixed-cap constraint set” becomes explicit: constraints are the local conserved charges of the symmetries used in the derivation:

1. **Edge/cap label constraints:** fix the distribution of collar-sector labels, equivalently $\langle L_C \rangle$ for each cap size, giving the area term.
2. **Gauge charges:** fix boundary flux or charge operators.
3. **Geometric (conformal) charges:** fix the expectation of the conformal Killing charges that preserve the cap, i.e. the generator B_C or its microscopic lattice approximation.

MaxEnt therefore selects the unique finite-stage invariant state compatible with those conserved charges. The support-visible BW scaling theorem is the continuum statement: if the limit algebra is non-type-I, the geometric modular action on that branch is outer rather than an inner density-matrix identity.

Focusing boundary. QNEC has rigorous QFT proofs in broad settings [51, 53], with hypotheses (null-surface locality and analyticity, modular-flow/causality/defect-OPE structure) that have not been verified on the reconstructed net, and QFC is strictly stronger than QNEC [52]. Quantum focusing is an optional gravity interface and is not consumed by the Einstein theorem:

- EC plus the declared state, edge-reference, and entropy-split premises identify $S_{\text{gen}} = S_{\text{bulk}} + \langle L_C \rangle$ on the finite branch (Section 5.4).
- The null-stress and Einstein theorem branch supplies the kinematic inputs required by an internal QNEC/QFC proof; that derivation is not supplied.

Summary. The CMFP package therefore separates the dependency structure into two layers:

- **Fixed-stage and cross-scale consequences:** Axiom 3 with the declared local constraint and reference package gives the local-Gibbs form and quasi-local propagation on the faithful/interior branch. Bounded-interval reduced-modular endpoint-Lipschitz control additionally consumes the exact-Markov/central-interface split and the specified endpoint-to-cell uniform bound. The separate optimizer-pushforward interface supplies the realized state-side refinement branch along which later transport questions are asked.
- **Support-visible scaling statement:** the scaling limit emits the support-visible geometric cap pair and ordered cut-pair rigidity holds on it. On that branch the limit algebra may be non-type-I, the modular action may be outer, and the 2π normalization follows. The later null half-sided-inclusion bridge additionally consumes its registered common-vector and relative-commutant standardness receipts.

The theorem route works on the support-visible quotient. The realized transported cap-local system packages the geometric cap-local test family, the projectively compatible transported marginal family, and the asymptotic transport-equivalence certificate. The regularized transport theorem below replaces the unavailable full-algebra lower spectral floor. Local weak-* extraction and GNS gluing then emit the support-visible scaling-limit cap pair, support-readable modular covariance reads the modular group as a cap-local support map, and ordered cut-pair rigidity collapses the residual cap-preserving conformal freedom to the unique hyperbolic subgroup.

Proposition 2.6a (Recoverability is not modular geometry: common-floor collapse countermodel). Exact or asymptotically exact Markov recovery at finite cutoff does not by itself imply a full-algebra lower spectral floor along refinement. In a two-dimensional matrix algebra, let

$$\rho_n = \begin{pmatrix} e^{-n} & 0 \\ 0 & 1 - e^{-n} \end{pmatrix}.$$

Each ρ_n is faithful at finite n , and tensoring it with any fixed finite exact-Markov collar factor gives a full-rank exact-Markov collar family. Nevertheless $\lambda_{\min}(\rho_n) = e^{-n} \rightarrow 0$, so there is no positive lower bound along the refinement family. On the off-diagonal matrix unit E_{12} , the modular generator carries the logarithmic gap

$$[-\log \rho_n, E_{12}] = n E_{12} + O(1)E_{12},$$

and the unregularized modular transport has no finite common-floor limit on that direction. Thus collar Markovity and finite-stage faithfulness are not enough for the false stronger full-algebra BW lift. The observer-facing theorem uses the support-visible regularized replacement, with the exact-Markov comparison family checked only on fixed collars and made replacement-independent after weak-*/GNS extraction.

Theorem 2.6b (regularized support-visible modular transport). Fix a local collar model of finite dimension $d_{m,\delta}$. Let ρ_n be the transported physical collar marginal, $\hat{\rho}_n$ the exact-Markov comparison marginal on the same collar model, and $\Delta_n = \|\rho_n - \hat{\rho}_n\|_1$. For $a > 0$, set $K_a(\rho) = -\log(\rho + a\mathbf{1})$. For every bounded collar observable O in the support-visible algebra,

$$|\mathrm{Tr} \rho_n O(K_a(\rho_n) - K_a(\hat{\rho}_n))| \leq \|O\| \left(\frac{4\Delta_n}{a} + d_{m,\delta}a + 4\Delta_n |\log a| \right).$$

Consequently, if $a_n \downarrow 0$ is chosen with

$$\Delta_n/a_n \rightarrow 0, \quad d_{m,\delta}a_n \rightarrow 0, \quad \Delta_n |\log a_n| \rightarrow 0,$$

then the regularized support-visible modular matrix elements converge on that fixed collar model.

Proof sketch. On the spectral interval $[a, \infty)$, the logarithm is operator-Lipschitz at the scale used above by its integral representation. Splitting the comparison into the support above a , the a -tail, and the trace-distance error gives the displayed bound. The three displayed conditions make the three error terms vanish. QED.

Definition 2.6c (explicit multiresolution reference branch). The finite-to-continuum bridge is evaluated on a declared reference branch, not on an unstructured sequence of finite plots. A regulator is

$$r = (m, L, b),$$

where m is the refinement depth, L is the finite support or volume scale, and b is a boundary/phase label. The screen branch uses nested icosahedral subdivisions of S^2 ; Euclidean branches use dyadic

mesh $a_m = a_0 2^{-m}$, with an additional temporal spacing a_t when a transfer matrix is present. Each added cell, collar, edge-center, or support-visible channel j carries

$$D_j = \bigoplus_{\alpha \in S_j} M_{d_{j,\alpha}}(\mathbb{C})$$

and a faithful detail state τ_j . At regulator r ,

$$\widetilde{M}_r = \bigotimes_{j \in \mathcal{J}_r} D_j, \quad \widetilde{\omega}_r = \bigotimes_{j \in \mathcal{J}_r} \tau_j.$$

A finite-depth local presentation circuit $\Gamma_r : \widetilde{M}_r \rightarrow M_r$ changes from multiresolution coordinates to the OPH patch, port, collar, and gauge coordinates. For $r \preceq s$, write

$$\widetilde{M}_s = \widetilde{M}_r \otimes D_{s \setminus r}, \quad \tau_{s \setminus r} = \bigotimes_{j \in \mathcal{J}_s \setminus \mathcal{J}_r} \tau_j.$$

The canonical refinement and coarse-graining maps are

$$\iota_{rs}(A) = \Gamma_s \left[\Gamma_r^{-1}(A) \otimes \mathbf{1} \right],$$

and

$$Q_{sr}(X) = \Gamma_r \left[(\text{id} \otimes \tau_{s \setminus r})(\Gamma_s^{-1}(X)) \right].$$

The embedded conditional expectation is $E_{sr} = \iota_{rs} Q_{sr}$. The reference state is $\widehat{\omega}_r = \widetilde{\omega}_r \circ \Gamma_r^{-1}$.

Theorem 2.6d (exact modular-compatible reference tower). The data

$$(M_r, \widehat{\omega}_r, \iota_{rs}, E_{sr})_{r \preceq s}$$

form a directed tower. The maps ι_{rs} are unital injective $*$ -homomorphisms with $\iota_{st} \iota_{rs} = \iota_{rt}$. The Q_{sr} are faithful state-preserving completely positive coarse-grainings with $Q_{tr} = Q_{sr} Q_{ts}$. The maps E_{sr} are faithful conditional expectations onto $\iota_{rs}(M_r)$, and

$$\widehat{\omega}_s \circ \iota_{rs} = \widehat{\omega}_r, \quad \widehat{\omega}_r \circ Q_{sr} = \widehat{\omega}_s, \quad \widehat{\omega}_s \circ E_{sr} = \widehat{\omega}_s.$$

The finite reference modular groups are also exactly compatible:

$$\widehat{\sigma}_t^{\widehat{\omega}_s}(\iota_{rs} A) = \iota_{rs}(\widehat{\sigma}_t^{\widehat{\omega}_r}(A)).$$

This theorem uses no full-algebra spectral floor uniform in r ; in bare coordinates it is only the identity $A \mapsto A \otimes \mathbf{1}$, partial expectation against faithful detail states, and conjugation by Γ_r, Γ_s .

Theorem 2.6e (canonical renormalization and correlation Cauchy bound). Let $\mathfrak{A}$ be the C^* -inductive limit with embeddings $j_r : M_r \rightarrow \mathfrak{A}$. There is a unique state $\widehat{\omega}$ with $\widehat{\omega} \circ j_r = \widehat{\omega}_r$, and the maps Q_{sr} induce state-preserving conditional expectations

$$\mathbb{E}_r : \mathfrak{A} \rightarrow j_r(M_r)$$

such that $\|\mathbb{E}_r(A) - A\| \rightarrow 0$ for every $A \in \mathfrak{A}$. The finite-regulator realization of a continuum test observable is

$$\mathcal{R}_r(A) = j_r^{-1}(\mathbb{E}_r A).$$

For $\|A_k\| \leq M$,

$$\left| \widehat{\omega}_r \left(\prod_{k=1}^n \mathcal{R}_r(A_k) \right) - \widehat{\omega} \left(\prod_{k=1}^n A_k \right) \right| \leq M^{n-1} \sum_{k=1}^n \|A_k - \mathbb{E}_r A_k\|.$$

Thus the operator mixing matrix and additive subtraction at finite cutoff are outputs of $\mathcal{R}_r$, not free fits to the target continuum answer. Independent lattice-spacing, volume, and boundary sweeps must publish the corresponding martingale-tail or physical-error envelopes before finite regulator data are read as continuum evidence.

Theorem 2.6f (compact-time modular convergence on fixed local algebras). Let $\rho_{s\downarrow r}^{\text{phys}}$ be the density matrix of the physical state at fine stage s , restricted to the embedded fixed local algebra M_r , and let

$$\epsilon_{s,r} := \|\rho_{s\downarrow r}^{\text{phys}} - \widehat{\rho}_r\|_1.$$

For a regularizer $\eta_s > 0$, define

$$K_{s\downarrow r}^{(\eta_s)} = -\log(\rho_{s\downarrow r}^{\text{phys}} + \eta_s \mathbf{1}).$$

If $\lambda_r = \lambda_{\min}(\widehat{\rho}_r) > 0$, then for every $A \in M_r$ and every $T < \infty$,

$$\sup_{|t| \leq T} \left\| e^{-itK_{s\downarrow r}^{(\eta_s)}} A e^{itK_{s\downarrow r}^{(\eta_s)}} - \sigma_t^{\widehat{\omega}_r}(A) \right\| \leq 2T\|A\| \left[\frac{\epsilon_{s,r}}{\eta_s} + \log \left(1 + \frac{\eta_s}{\lambda_r} \right) \right].$$

Hence $\eta_s \rightarrow 0$ and $\epsilon_{s,r}/\eta_s \rightarrow 0$ give uniform-on-compact-time convergence on each fixed support-visible local algebra. For a countable cofinal local test tower, pointwise $\epsilon_{s,r} \rightarrow 0$ admits a diagonal subsequence and one cutoff schedule. The phrase “finite-stage modular action” below means this transported, regularized local action; it does not mean the unregularized full finite-cap modular group.

Cap-family uniformity clause. For application to the finite cap-net BW theorem, the compact-time estimate is required uniformly over the declared finite nondegenerate cap family and the fixed separating support-visible test tower. The operator-norm estimate then implies the quadratic mixed-GNS Cauchy residual because $\|XD\|_{2,\omega} \leq \|X\|\|D\|_{2,\omega}$. The reference tower supplies the analytic engine; the finite cap-normal certificate adds the support-order, BW-frame, oriented cross-ratio, geometric 2π -KMS, and wrong-scale controls.

Theorem 2.6g (finite positive transfer and Lorentzian handoff condition). On Euclidean repair branches, let $P_{C,r}$ be weighted resampling inside the fibers of the complete repaired visible datum for active collar C . Ref. [14] proves that this finite operator is the orthogonal conditional expectation and gives the exact support, equal-fiber-row, and weighted detailed-balance matrix receipt that recognizes it. A concrete active collar repair kernel is identified with $P_{C,r}$ only when its independently derived transition matrix passes that receipt; constructing the matrix from the target formula is not a verification. Choose $c_{C,r} \geq 0$, and set

$$L_r^{\text{rep}} = \sum_C c_{C,r} (I - P_{C,r}), \quad T_r(a_t) = e^{-a_t L_r^{\text{rep}}}.$$

Then $L_r^{\text{rep}} = L_r^{\text{rep}*} \geq 0$, $0 < T_r(a_t) \leq I$, $T_r(a_t)\mathbf{1} = \mathbf{1}$, and the stationary Euclidean path measure is reflection positive:

$$\mathbb{E}_r[(\Theta F)F] \geq 0$$

for positive-time cylinder functions F . A Lorentzian unitary statement requires this finite positivity plus either exact shell-additive direct-limit semigroup identities or a Mosco / strong-resolvent

substitute, including vacuum-projection and intertwiner convergence. Weak-* state convergence alone is not a transfer theorem.

BW-side closure boundary. Proposition 2.6a blocks the stronger unregularized full-algebra conclusion from the proved finite-cutoff package. Theorems 2.6c–2.6g supply the explicit reference-tower certificate needed for the finite-to-continuum reading: continuum correlations come from the conditional-expectation renormalization map, compact-time modular group convergence is only for transported regularized physical marginals on fixed support-visible local algebras, and Lorentzian transfer needs reflection positivity or an equivalent positive-transfer certificate. Theorem 4.2 then uses exactly the observer-facing support-visible content to close the BW/geometric cap-pair automorphism statement. Without the multiresolution regulator certificate, finite regulator data remain regulator evidence. A black-box AQFT/BW reconstruction route would require separately verifying conformal-net properties such as locality, additivity, duality, standardness, positive energy, and suitable modular inclusions.

2.1 Overlap Consistency and Gluing

The constructive part of overlap consistency is the tree-gluing theorem below. The structural part is the origin of the gluing redundancy itself. On the ordinary or central-defect branch, gauge-as-gluing is the finite-regulator overlap redundancy of local chart presentations: once the finite-dimensional overlap algebras and their allowed chart-change groupoid are put in the block-complete spatial presentation specified below, overlap-consistent rechartings of a cut form a compact unitary transition system. The collar theorem of Section 2.3 should therefore be read with its boundary group G_Σ understood as shorthand for this spatialized compact boundary action, not as an automatic consequence of an arbitrary algebra representation. On the genuinely noncentral branch, the corresponding declared weak gluing data are encoded by a compact crossed-module change system. Block permutations and other groupoid data are retained rather than silently treated as inner automorphisms. A second structural point is UV underdetermination. If each patch Hilbert space is tensored with an inert finite ancillary factor and the observable patch algebras are embedded as $\mathcal{A}(P) \otimes \mathbf{1}$, then the physical observables, collar conditional mutual information, carried Markov errors, and quotient normal forms are unchanged. So the physical UV branch is fixed only modulo gauge or implementation hiding together with such ancillary stabilization, not a unique microscopic presentation; the unique theorem-grade object is the induced family of terminal expectation functionals on the declared physical observable algebras, not a preferred microscopic representative.

2.1.1 Constructive gluing on tree covers

Theorem 3.1 (tree gluing). Let a rooted tree of patches satisfy a tree-ordered overlap structure and a tripartite factorization (A_k, B_k, C_k) at step k . If a target state ρ^* obeys $I(A_k : C_k \mid B_k) \leq \varepsilon_k$, then there exist recovery maps $\mathcal{R}_k$ such that

$$\|\rho_{A_k B_k C_k}^* - (\text{id}_{A_k} \otimes \mathcal{R}_k)(\rho_{A_k B_k}^*)\|_1 \leq \delta_k,$$

with

$$\delta_k = 2\sqrt{1 - e^{-\varepsilon_k}} \leq 2\sqrt{\varepsilon_k},$$

The iteratively glued state $\hat{\rho}$ satisfies

$$\|\hat{\rho} - \rho^*\|_1 \leq \min\left(2, \sum_{k=2}^n \delta_k\right).$$

Proof. Induct on k . The recovery error contracts under CPTP maps, so the errors add. QED.

2.1.2 Gauge-as-gluing and loops

At finite regulator scale, the fixed-cutoff gauge-as-gluing package identifies the overlap-consistency redundancy of local chart presentations. Choose block-complete faithful finite-dimensional spatial presentations of the overlap algebras on a connected cut Σ , and require the allowed overlap-chart isomorphisms to be represented by their unitary normalizers. A finite-dimensional C^* -algebra is a direct sum of full matrix blocks: its star automorphisms permute isomorphic central blocks and act inner within each block. They are spatial in the canonical block-complete presentation, but need not be inner in the algebra itself (an abelian two-block swap is the elementary counterexample). Fixing a reference chart and retaining any allowed block permutations, the compact closure of the subgroup generated by the spatial recharting unitaries is the boundary gluing group K_Σ ; before fixing the reference chart, the same data form a compact unitary groupoid.

Proposition 3.2a (Spatialized gauge-as-gluing at finite regulator). Let a finite regulator chart be chosen for the patches meeting along a connected interface Σ , together with the block-complete spatial presentations and unitary-normalizer realization of the allowed chart-change groupoid just stated. Then overlap consistency determines a compact unitary transition system on the cut data. On the ordinary or central-defect branch, after a reference chart is fixed, this transition system reduces to a compact boundary group K_Σ , and when the triple-overlap defect is central its projective composition law lifts to a genuine action of a compact central extension $\widehat{K}_\Sigma$. On this realized presentation, gauge names the overlap redundancy; the spatialization data are explicit hypotheses rather than consequences of finite dimensionality alone.

Proof. The structure theorem for finite-dimensional C^* -algebras decomposes each chart algebra into full matrix blocks. An allowed star isomorphism permutes isomorphic central blocks and is inner on each matched block; the chosen block-complete presentation implements the combined operation by a unitary normalizer. The subgroup generated by those unitaries has compact closure inside a finite-dimensional unitary group. If triple-overlap defects are central, the resulting projective composition law lifts to a central extension. QED.

Lemma 3.2b (trees vs loops). If the patch adjacency graph is a tree, one can choose local charts h_i so that the overlap labels satisfy $g_{ij} = h_i^{-1}h_j$ on all edges. If loops exist, the loop holonomy

$$H(\gamma) = g_{i_1 i_2} g_{i_2 i_3} \cdots g_{i_n i_1}$$

transforms by conjugation at the chosen basepoint under local frame changes. Its conjugacy class, and therefore every class-function value (including every one-dimensional character value), is invariant. In particular, nontriviality is invariant and obstructs global trivialization. QED.

Proposition 3.2ba (Finite path ratio, character phase, and cyclic quantization). For reversal-compatible group-valued edge labels on a typed finite path system, the ordered transport ratio of two paths with common endpoints is exactly the holonomy of the loop obtained by following the first and reversing the second. Every one-dimensional unitary character sends this identity to an exact relative phase. A vertex rechart conjugates based holonomy, so holonomy itself is invariant for an abelian structure group and the character phase is invariant for an arbitrary group. There is an explicit four-vertex path/face control whose two declared triangular faces have trivial holonomy while a separate undeclared, unfilled loop and its two-arm relative character phase are nontrivial. Finally, if the loop holonomy factors through a supplied $\mathbb{Z}_n$ sector, every character phase is an n th root of unity.

Scope. The result is machine checked for supplied finite paths, edge labels, character, faces, and cyclic-sector factorization. The path/face control declares exactly two triangular faces; it neither

claims flatness on undeclared triangles nor proves a puncture or noncontractibility statement. No theorem here constructs the connection from observer records or identifies the phase with a physical gauge flux, spacetime loop, charge, laboratory arm, detector, or interference fringe.

Corollary 3.2c (Collar consequence on the ordinary or central-defect branch). Let $B_\delta = B_L \cup B_R$ be a collar around a cap boundary Σ , and set $\widehat{K}_\Sigma = K_\Sigma$ on the ordinary branch. Then the EC theorem of Section 2.3 has the derived interpretation

$$H_{B_\delta} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\widehat{K}_\Sigma} \cong \bigoplus_{\alpha} (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

with

$$Z(A_{\text{EC}}(B_\delta)) = \bigoplus_{\alpha} \mathbb{C} \mathbf{1}_\alpha.$$

The right half-collar carries the contragredient action because it sees the inverse transport across the same cut. Exact Markov normal forms used later require the additional state hypothesis $I_\omega(A_\delta : D_\delta \mid B_\delta) = 0$ *together with* the Markov-split alignment hypothesis of Section 2.3, or the explicitly stated idealized recoverability limit supplying both; the block decomposition itself is kinematic and follows from the derived boundary action.

Proof sketch. Decompose the left boundary data into irreps $(V_\alpha \otimes H_{b_L^\alpha})$ of $\widehat{K}_\Sigma$ and the right boundary data into the dual modules $(V_\beta^* \otimes H_{b_R^\beta})$. Then Schur's lemma leaves a singlet only when $\alpha = \beta$, producing the displayed direct sum. QED.

2.1.3 Loop obstruction class (central defect)

On the central-defect subbranch, identify the overlap centers with one fixed abelian unitary coefficient group Z_Σ and assume overlap transport acts trivially on it. Choose unitary overlap implementers normalized by $U_{ji} = U_{ij}^{-1}$ and define central defects z_{ijk} at the implementer level by

$$\Omega_{ijk} := U_{ij}U_{jk}U_{ki} = z_{ijk} \in Z_\Sigma,$$

or equivalently $U_{ij}U_{jk} = z_{ijk}U_{ik}$. This implementer-level relation records the projective multiplier; the automorphism-level expression $\text{Ad}(z_{ijk})$ alone would be the identity and would not determine z_{ijk} .

Then $\{z_{ijk}\}$ is an untwisted Čech 2-cocycle with coefficients in Z_Σ , and its cohomology class $[z]$ is gauge invariant. Central defects do not obstruct the collar block decomposition: they only replace K_Σ by its central extension $\widehat{K}_\Sigma$. The condition $[z] = 0$ is exactly the strictifiability of the projective triangle defect. It is not by itself global path independence: at least one resulting strict edge 1-cocycle must also have trivial represented holonomy around every closed overlap loop. With a nontrivial action on varying centers, the corresponding statement uses the twisted/local-system Čech differential instead. (A full proof for the fixed-coefficient subbranch appears in Section 6.4 below, in the algebra-net language.)

2.1.4 Non-central obstruction (2-group cocycle)

When defects are not central, the natural coefficient data is a crossed module $(H \rightarrow G)$ with an action of G on H by conjugation. Here G is the reconstructed gauge group, and H is the unitary group acting on edge multiplicity spaces, with boundary map $\partial : H \rightarrow G$.

A crossed module is a homomorphism $\partial : H \rightarrow G$ together with an action of G on H such that

$$\partial(g \triangleright h) = g \partial(h) g^{-1}, \quad \partial(h) \triangleright h' = h h' h^{-1}.$$

On a good cover $\{P_i\}$, a weakly coherent gluing is encoded by:

$$g_{ij} : P_{ij} \rightarrow G, \quad h_{ijk} : P_{ijk} \rightarrow H,$$

obeying the 2-cocycle conditions

$$g_{ij} g_{jk} = \partial(h_{ijk}) g_{ik},$$

and on quadruple overlaps,

$$h_{jkl} h_{ikl} = (g_{ij} \triangleright h_{ikl}) h_{ijk}.$$

Gauge changes act by 1- and 2-cochains in the standard way for crossed-module cohomology. Write

$$q_\Sigma = [(g, h)] \in \check{H}^2(N_\Sigma, H_\Sigma \rightarrow G_\Sigma)$$

for the full crossed-module gluing orbit. It classifies the equivalence orbit of the pair (g, h) but does not in general determine a unique ordinary G_Σ -valued 1-cocycle class after strictification, because the map below need not be injective. Let

$$\begin{aligned} \iota_\Sigma : \check{H}^1(N_\Sigma, G_\Sigma) &\longrightarrow \check{H}^2(N_\Sigma, H_\Sigma \rightarrow G_\Sigma), \\ [g] &\longmapsto [(g, 1)] \end{aligned}$$

be the natural strict-locus map. Define the separate $\{0, 1\}$ -valued strictification obstruction by

$$\begin{aligned} o_\Sigma^{(2)}(g, h) = 0 &\iff q_\Sigma \in \text{im}(\iota_\Sigma) \\ &\iff [(g, h)] \text{ has a representative } (g^{\text{str}}, 1) \\ &\quad \text{with } g_{ij}^{\text{str}} g_{jk}^{\text{str}} = g_{ik}^{\text{str}}. \end{aligned}$$

and set $o_\Sigma^{(2)}(g, h) = 1$ otherwise.

Theorem 3.4 (non-central higher-defect strictification). The higher associator defect of (g_{ij}, h_{ijk}) is removable iff $o_\Sigma^{(2)}(g, h) = 0$. In that case a higher-gauge change gives $h_{ijk} = 1$ and a genuine G -valued edge 1-cocycle. Strict endpoint-only transport additionally requires that at least one allowed strict representative have trivial represented holonomy. The full orbit q_Σ need not determine a unique ordinary H^1 class.

Proof sketch. Removing the higher associator corresponds to $h_{ijk} = 1$ and $g_{ij} g_{jk} = g_{ik}$. Crossed-module coboundaries preserve the full orbit, so this strictification exists exactly when that orbit meets the image of the ordinary G -valued 1-cocycle locus, which is the definition of $o_\Sigma^{(2)} = 0$. The map ι_Σ need not be injective: H -valued edge changes can relate strict representatives and alter their ordinary holonomy through ∂ . Thus the invariant residual test is whether at least one allowed strict representative has trivial represented holonomy, as in Theorem 3.4b. QED.

The central-defect case is the abelian truncation with H central and trivial action, which reduces to Section 3.3. A genuinely noncentral class is the point where the fixed-cutoff collar theorem upgrades from the ordinary-group package to the higher-gauge one below.

Corollary 3.4a (Fixed-cutoff higher-gauge EC and transportability). For a finite regulator chart on a connected cut Σ , the genuinely noncentral branch admits a compact crossed-module change system

$$\mathcal{T}_\Sigma = C^1(N_\Sigma, H_\Sigma) \rtimes C^0(N_\Sigma, G_\Sigma).$$

The physical higher-gauge collar is

$$\mathcal{H}_B^{2g} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\mathcal{T}_\Sigma} \cong \bigoplus_{\lambda} (\mathcal{H}_{b_L^\lambda} \otimes \mathcal{H}_{b_R^\lambda}),$$

with

$$Z(\mathcal{A}_{2g}(B)) = \bigoplus_{\lambda} \mathbb{C} \mathbf{1}_\lambda,$$

and the full orbit q_Σ is invariant under local rechartings and classifies the fixed-cutoff genuinely noncentral gluing datum. Its separate invariant property $o_\Sigma^{(2)} = 0$ holds iff the higher associator is removable. The resulting strict representatives can have different ordinary G_Σ -valued 1-holonomies, so transportability is the separate existential test of Theorem 3.4b rather than a uniquely defined holonomy attached to q_Σ .

Proof sketch. Pair-overlap rechartings are inner, while triple-overlap associators strictify to compact crossed-module data. Finite-dimensional unitary $\mathcal{T}_\Sigma$ -modules decompose semisimply, and Schur matching leaves only diagonal left/right blocks. Crossed-module cohomology classifies the higher defect; Theorem 3.4b separately tests the ordinary loop holonomy that remains after strictification. QED.

Theorem 3.4b (TransportabilityFromOverlapGluing). Fix a connected cut Σ and a finite regulator charting nerve N_Σ . Transport of a collar charge is defined by the path composite in the overlap recharting groupoid: along $p = (i_0 \rightarrow \cdots \rightarrow i_m)$, the charge block is moved by $U_p = U_{i_{m-1}i_m} \cdots U_{i_0i_1}$. This construction uses the overlap unitary transition system itself, not an added DHR transportability assumption.

After a central or crossed-module strictification, let U^{str} denote a resulting genuine edge 1-cocycle. Write $\mathcal{C}_\alpha$ for the full transported collar-charge block, including its boundary carrier and multiplicity/intertwiner data, and define

$$\text{Hol}_{\alpha, U^{\text{str}}} : \pi_1(|N_\Sigma|, i_*) \longrightarrow U(\mathcal{C}_\alpha), \quad [\gamma] \longmapsto U_\gamma^{\text{str}}|_{\mathcal{C}_\alpha}.$$

On the ordinary branch, transport is strictly path-independent iff every closed overlap loop has trivial holonomy on the collar-sector block. On the central branch, elementary triangle moves accumulate the central cocycle z_{ijk} . Strict path-independent transport exists iff $[z]_\Sigma = 0$ and there is a central 1-cochain strictification for which $\text{Hol}_{\alpha, U^{\text{str}}} = 1$. The first condition removes the projective triangle defect; the second removes the residual ordinary loop action. If $[z]_\Sigma = 0$ but the second condition fails, the lifts form a genuine 1-cocycle and transport is path dependent.

On the genuinely noncentral branch, path comparisons are H_Σ -valued 2-morphisms in the crossed module $H_\Sigma \rightarrow G_\Sigma$. Strict ordinary transport exists iff $o_\Sigma^{(2)} = 0$ and at least one crossed-module strictification gives a genuine G_Σ -valued 1-cocycle with trivial represented holonomy on $\mathcal{C}_\alpha$. If $o_\Sigma^{(2)} \neq 0$, the sector has higher transport rather than ordinary compact-group DR transport. If $o_\Sigma^{(2)} = 0$ but every allowed strict representative has nontrivial represented G_Σ -holonomy, the higher associator has strictified but ordinary path dependence remains. The full orbit $q_\Sigma = [(g, h)]$ need not determine a unique ordinary H^1 class and is not itself the strictification test. Thus “zero obstruction” below means the combined associator-strictification and trivial-holonomy criterion, not vanishing of triangle or higher-associator data alone. QED.

Triangle-free cycle check. Let $N_\Sigma = C_n$, $n \geq 4$, have no filled 2-simplices. On $\mathcal{C}_\alpha = \mathbb{C}^2$, put identity transport on every oriented cycle edge except one, where the sector action is the non-scalar unitary $V = \text{diag}(-1, 1)$, and use inverse transports on reversed edges. All central 2-data and higher associator data are vacuously trivial, so $[z]_\Sigma = 0$ and $o_\Sigma^{(2)} = 0$. Its ordered cycle holonomy is

$V \neq 1$, which no central rephasing removes. For a genuinely noncentral instance, take the crossed module

$$\mathrm{SU}(2) \hookrightarrow \mathrm{U}(2)$$

with conjugation action and the standard action on $\mathcal{C}_\alpha$. The band $\mathrm{U}(2)/\mathrm{SU}(2) \cong \mathrm{U}(1)$ is detected by the determinant, and the cycle has band holonomy $\det V = -1$. An $\mathrm{SU}(2)$ -valued edge change has determinant one and a vertex-frame change only conjugates the based loop, so no allowed strict representative has trivial holonomy. The two routes between the endpoints of the distinguished edge differ, and the combined criterion rejects both the central and noncentral assignments.

2.2 Modular Flow and Lorentz Kinematics

2.2.1 Modular additivity in the Markov collar limit

Consider a collar tripartition $A : B : D$ around a cap boundary, with the EC decomposition of Section 2.3 understood. Define, for a faithful reference state ω ,

$$\Delta K(\omega) := K_{ABD}(\omega) - K_{AB}(\omega) - K_{BD}(\omega) + K_B(\omega).$$

Exact modular additivity is not a consequence of EC alone. It is available only on the EC-aligned exact Markov set, or along a controlled fixed-cutoff family that approaches that set on one fixed collar model. Four quantities must therefore be kept separate:

1. the raw collar conditional mutual information $I(A : D \mid B)$;
2. the constructive Fawzi–Renner comparison error

$$r_{\mathrm{FR}}(\varepsilon) := 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon};$$

3. the fixed-collar exact-Markov replacement modulus

$$\delta_{A:B:D}^{\mathrm{M}}(\varepsilon) := \sup \left\{ \inf_{\sigma \in \mathfrak{M}_{A:B:D}} \|\rho - \sigma\|_1 : I(A : D \mid B)_\rho \leq \varepsilon \right\},$$

where

$$\mathfrak{M}_{A:B:D} := \{\sigma_{ABD} : I(A : D \mid B)_\sigma = 0\}.$$

4. the EC-aligned subset and a family-specific aligned replacement modulus,

$$\mathfrak{M}_{A:B:D}^{\mathrm{EC}} := \{\sigma \in \mathfrak{M}_{A:B:D} : \sigma \text{ satisfies MSA on the preselected EC split}\},$$

$$\delta_{\mathcal{F};A:B:D}^{\mathrm{M,EC}}(\varepsilon) := \sup \left\{ \inf_{\sigma \in \mathfrak{M}_{A:B:D}^{\mathrm{EC}}} \|\rho - \sigma\|_1 : \rho \in \mathcal{F}, I(A : D \mid B)_\rho \leq \varepsilon \right\}$$

for the declared controlled collar family $\mathcal{F}$. Convergence of the generic modulus in item 3 does not imply convergence of this aligned modulus; $\delta_{\mathcal{F}}^{\mathrm{M,EC}} \rightarrow 0$ is a separate family hypothesis.

Lemma 4.1a (EC-aligned exact Markov implies exact additivity up to a central term). If $I(A : D | B)_\omega = 0$ and ω satisfies the Markov-split alignment hypothesis of Section 2.3, then the state takes the EC-aligned HJPW block form

$$\omega_{ABD} = \bigoplus_{\alpha} p_{\alpha} \omega_{Ab_L^{\alpha}}^{(\alpha)} \otimes \omega_{b_R^{\alpha}D}^{(\alpha)},$$

and $\Delta K(\omega)$ is central. On the canonical HJPW block model one may take

$$\Delta K(\omega) = 0.$$

Equivalently, there exists a central operator $K_{\partial, ABD}(\omega) \in Z(\mathcal{A}(B))$ such that

$$K_{ABD}(\omega) = K_{AB}(\omega) + K_{BD}(\omega) - K_B(\omega) + K_{\partial, ABD}(\omega).$$

Proof. By alignment, the HJPW blocks are the EC blocks. On each block the modular Hamiltonians of ABD , AB , BD , and B are the logarithms of tensor-product states with the same classical block weight p_{α} . The blockwise logarithms therefore cancel exactly in the combination $K_{ABD} - K_{AB} - K_{BD} + K_B$. If one keeps the blockwise endpoint-label bookkeeping explicit rather than absorbing it into the canonical block identification, the remainder is a direct sum of block constants, hence central in $Z(\mathcal{A}(B))$. Without alignment, exact Markovity cancels the four logarithms on the state's own HJPW blocks, but the resulting block constants need not lie in $Z(\mathcal{A}(B))$, so the central form used here is unavailable. QED.

Proposition 4.1b (Controlled exact-Markov replacement on a fixed collar). Because the state space is compact and conditional mutual information is continuous in finite dimension,

$$\delta_{A:B:D}^M(\varepsilon) \longrightarrow 0 \quad (\varepsilon \downarrow 0).$$

Thus small collar CMI converges to the exact HJPW normal form only in this controlled fixed-cutoff sense; the manuscript does not claim a dimension-free one-shot trace-norm bound from small $I(A : D | B)$ directly to an exact Markov state. The limit points are exact Markov states in the CMI-zero sense; identifying their factorization with the preselected EC factors is the Markov-split alignment hypothesis of Section 2.3, which is carried separately and is not produced by small CMI (the Bell-pair counterexample of Section 2.3 has zero CMI at fixed distance from every EC-aligned state).

Theorem 4.1b' (Finite-collar Markov replacement stability). On one fixed faithful finite-dimensional collar model, the qualitative modulus above can be made collar-local and quantitative. If all relevant marginals stay above a chosen floor $\lambda_* > 0$, then there are constants $C_{A:B:D, \lambda_*} > 0$ and $\theta_{A:B:D, \lambda_*} > 0$, depending only on that collar model and floor, such that

$$d_M(\rho) \leq C_{A:B:D, \lambda_*} I(A : D | B)_{\rho}^{\theta_{A:B:D, \lambda_*}}.$$

This is a fixed-collar Łojasiewicz-type rate for the analytic function $I(A : D | B)$ on the faithful state manifold. It is not a dimension-free stability theorem for arbitrary tripartite quantum systems.

Corollary 4.1c (Carried EC-aligned collar defect operator). Let $\omega_{\varepsilon} \in \mathcal{F}$ satisfy $I(A : D | B)_{\omega_{\varepsilon}} \leq \varepsilon$, assume the declared family has an EC-aligned replacement modulus tending to zero, and choose $\sigma_{\varepsilon} \in \mathfrak{M}_{A:B:D}^{\text{EC}}$ with

$$\|\omega_{\varepsilon} - \sigma_{\varepsilon}\|_1 \leq \delta_{\mathcal{F}; A:B:D}^{\text{M, EC}}(\varepsilon).$$

Define the carried defect operator

$$\mathfrak{D}_{A:B:D}(\omega_{\varepsilon}, \sigma_{\varepsilon}) := \Delta K(\omega_{\varepsilon}) - \Delta K(\sigma_{\varepsilon}).$$

Then

$$K_{ABD}(\omega_\varepsilon) = K_{AB}(\omega_\varepsilon) + K_{BD}(\omega_\varepsilon) - K_B(\omega_\varepsilon) + K_{\partial,ABD}(\sigma_\varepsilon) + \mathfrak{D}_{A:B:D}(\omega_\varepsilon, \sigma_\varepsilon),$$

where $K_{\partial,ABD}(\sigma_\varepsilon) = \Delta K(\sigma_\varepsilon)$ is central. For every bounded observable X supported on $A \cup B$,

$$|\text{Tr}[X(\omega_\varepsilon - \sigma_\varepsilon)]| \leq \|X\|_\infty \delta_{\mathcal{F};A:B:D}^{\text{M,EC}}(\varepsilon).$$

If the relevant marginals of ω_ε and σ_ε are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\mathfrak{D}_{A:B:D}(\omega_\varepsilon, \sigma_\varepsilon)\|_\infty \leq 4\lambda_*^{-1} \delta_{\mathcal{F};A:B:D}^{\text{M,EC}}(\varepsilon).$$

Independently, Fawzi–Renner recovery supplies a constructive comparison state $\omega_\varepsilon^{\text{rec}}$ with

$$\|\omega_\varepsilon - \omega_\varepsilon^{\text{rec}}\|_1 \leq r_{\text{FR}}(\varepsilon), \quad r_{\text{FR}}(\varepsilon) := 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}.$$

This $O(\varepsilon^{1/2})$ term is the finite-stage observable error. The generic modulus $\delta_{A:B:D}^{\text{M}}(\varepsilon)$ justifies replacement only by some exact Markov state on its state-dependent HJPW split. The separately assumed $\delta_{\mathcal{F};A:B:D}^{\text{M,EC}}(\varepsilon)$ is the quantity that justifies the EC-aligned normal form used in later geometric modular arguments.

Accordingly, the manuscript’s later exact collar formulas take one of two forms:

1. literal exactness when the reference state is exact Markov and EC-aligned on the relevant collar; or
2. a controlled collar family for which $\delta_{\mathcal{F};A:B:D}^{\text{M,EC}}(\varepsilon_\delta) \rightarrow 0$, with the constructive Fawzi–Renner remainder and the carried operator defect $\mathfrak{D}_{A:B:D}$ kept explicit at finite stage.

Theorem 4.1d (Finite-stage modular-defect propagation and dimension quarantine).

For any fixed downstream branch calculation that uses finitely many collar or strip modular-additivity identities, replacing each exact identity by its controlled finite-stage form changes every bounded downstream modular observable O by at most

$$\mathcal{P}_O\left(\{r_{\text{FR}}(\varepsilon_j)\}, \{\delta_{\mathcal{F};j}^{\text{M,EC}}(\varepsilon_j)\}, \{\eta_j^{\text{reg}}\}\right),$$

where $\mathcal{P}_O \rightarrow 0$ as all listed errors vanish. The polynomial-continuity modulus depends only on the declared fixed collar models, support-visible dimensions, bounded observable class, faithful floors or regularization schedules, and bounded modular-parameter intervals. No BW/Lorentz, null-modular, or Einstein scaling step uses a dimension-free trace-norm stability theorem from small CMI directly to exact Markov normal form.

2.2.2 Theorem: BW_{S^2} on the extracted geometric subnet

The collar analysis proves a fixed-cutoff statement on the finite type-I regulator net: the reduced cap state has a literal density matrix, its modular Hamiltonian exists, and its nonadditive part is confined to a shrinking collar up to carried errors. The Lorentz claim is therefore not a fixed-cutoff matrix-algebra identity and not the slogan “finite cells imply Lorentz invariance.” Its target is the support-visible refinement-limit geometric cap pair $(\mathcal{A}_\infty^{\text{geo,sv}}(C), \omega_\infty^{\text{geo},C})$. The branch theorem is the conditional implication

$$\text{FiniteCapBWCertificate} + \text{ModularAlgebraStateData} \implies \sigma_t^{\omega_\infty^{\text{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

The finite cap-normal certificate consists of cap-normal density and nondegeneracy, BW framing, support-order faithfulness, geometric support-flow group and continuity control, held-out oriented cross-ratio convergence, independently normalized geometric 2π -KMS comparison, and wrong-normalization separation. The independent multiresolution algebra-state comparison package supplies the algebra-state reference tower, common comparison maps, cap-family-uniform regularized modular transport, mixed-GNS convergence, and modular support covariance. The support-visible theorem below uses both inputs; it does not require a type-I continuum algebra or a full-algebra unregularized common spectral floor. Bare finite consensus implies neither input.

Fix a cap $C \subset S^2$ and a shrinking collar family $(A_\delta, B_\delta, D_\delta)$ around ∂C . Write

$$\varepsilon_\delta := I(A_\delta : D_\delta \mid B_\delta)_\omega, \quad r_{\text{FR}}(\varepsilon_\delta) := 2\sqrt{1 - e^{-\varepsilon_\delta}} \leq 2\sqrt{\varepsilon_\delta},$$

and, on each fixed faithful collar model,

$$\eta_\delta^{\text{M}} := 4\lambda_*^{-1} \delta_{\mathcal{F}_\delta; A_\delta: B_\delta: D_\delta}^{\text{M,EC}}(\varepsilon_\delta),$$

where $\lambda_* > 0$ is the lower spectral bound for comparing modular Hamiltonians. All exact collar formulas in this subsection are therefore to be read either literally at EC-aligned exact Markovity or along a controlled collar family satisfying

$$\delta_{\mathcal{F}_\delta; A_\delta: B_\delta: D_\delta}^{\text{M,EC}}(\varepsilon_\delta) \rightarrow 0 \quad (\delta \downarrow 0),$$

with $r_{\text{FR}}(\varepsilon_\delta)$ and η_δ^{M} carried explicitly.

For each cap C , let $\lambda_C(s) \subset \text{Conf}^+(S^2)$ denote the standard cap-preserving conformal one-parameter subgroup, normalized so that the null blow-up near a smooth cut acts by $v \mapsto e^{-s}v$, and let $\alpha_{\lambda_C(s)}$ denote the induced automorphism of the scaling-limit cap net.

Theorem 4.2 (Finite BW_{S^2} cap-net scaling). For a nondegenerate BW-framed cap $\hat{C} = (C, n_C, p_C^-, p_C^+)$, consider an OPH realization on an observer-supporting refinement branch that satisfies Axioms 1–3 and carries the named fixed-cutoff regulator, collar, consensus, and optimizer-pushforward interfaces. Assume the finite cap-normal BW certificate above and, independently, the complete multiresolution reference tower and algebra-state comparison package of Theorems 2.6c–2.6f on the same branch. The support-visible geometric cap net then admits a weak-*/GNS scaling-limit cap pair $(\mathcal{A}_\infty^{\text{geo,sv}}(C), \omega_\infty^{\text{geo},C})$, whose modular automorphism group is

$$\sigma_t^{\omega_\infty^{\text{geo},C}} = \alpha_{\lambda_{\hat{C}}(2\pi t)}.$$

At finite regulator stage the nonadditive part of

$$K_C^{(\delta)} := -\log \rho_C^{(\delta)}$$

is confined to the shrinking collar up to the carried errors $r_{\text{FR}}(\varepsilon_\delta)$, the EC-aligned fixed-collar replacement bound η_δ^{M} , and the regularized support-visible modular transport bound. The support-visible modular limit is independent of the EC-aligned exact-Markov replacement sequence after projective compatibility and GNS quotienting, and support-readable modular covariance turns the limiting modular group into a cap-local support map before the certificate's framed-cap and oriented cross-ratio rigidity clauses are applied. No separate cap-isotropy/ $\text{SO}(2)$ -equivariance selector, finite-cell Lorentz-invariance premise, or full-algebra unregularized common floor is used. Record/pointer and interface-inert auxiliary registers are outside the extracted geometric subnet. If the scaling-limit cap algebra is type I, the same automorphism identity may be represented by

$$K_C = 2\pi B_C + Z_C, \quad Z_C \in Z(\mathcal{A}(C))_{\text{sa}};$$

otherwise the automorphism identity is the full statement and the action is outer. Consequently, under the paired same-branch finite cap-normal and complete multiresolution algebra-state comparison inputs, the transported regularized modular action on every fixed support-visible geometric local algebra converges uniformly on compact modular-parameter intervals to the geometric cap-dilation action. Without both inputs, this theorem makes no BW or finite-to-continuum modular claim. Even with them, it does not assert compact-time convergence of finite-regulator modular groups beyond the declared support-visible certificate.

Proof. Markov locality localizes the modular defect to the shrinking collar, with carried finite-stage errors $r_{\text{FR}}(\varepsilon_\delta)$ and η_δ^{M} . Proposition 2.6a shows why a full-algebra unregularized floor is unavailable in general. Theorem 2.6b supplies the matrix-element replacement on fixed collar models, and Theorem 2.6f supplies the stronger compact-time group estimate only for transported regularized local actions with the stated reference-tower certificate. Projective replacement compatibility and the support-visible convergence theorem ensure that the emitted modular limit is independent of the chosen EC-aligned exact-Markov comparison family. Support-visible cap-pair extraction on the local GNS support quotient uses weak-* compactness of finite-stage state spaces and the consensus refinement mechanism to supply local limit states, then applies GNS construction to obtain the scaling-limit cap pair. Support-readable modular covariance reads the modular group as a cap-local support map. The finite cap-normal certificate supplies support-order faithfulness, BW framing, held-out oriented cross-ratio rigidity, and compact-time equicontinuity, so the support flow is the framed cap-preserving conformal flow up to normalization:

$$\sigma_t^{\omega_\infty^{\text{geo},C}} = \alpha_{\lambda_C(\kappa_C t)}$$

for some $\kappa_C > 0$, with the certificate-selected frame understood in λ_C . The certificate then passes the independently normalized geometric 2π -KMS condition to $s \mapsto \alpha_{\lambda_C(s)}$. KMS uniqueness, together with wrong- β separation and nontriviality, fixes $\kappa_C = 2\pi$. Thus

$$\sigma_t^{\omega_\infty^{\text{geo},C}} = \alpha_{\lambda_C(2\pi t)}.$$

If the limit algebra is type I, this automorphism identity is represented by $K_C = 2\pi B_C + Z_C$ with Z_C central; otherwise the automorphism identity is the full statement and the action is outer. QED.

Definition 4.2a (BW-branch observer-relative modular ordering). On the branch satisfying the hypotheses of Theorem 4.2, the modular automorphism parameter t of the extracted cap pair $(\mathcal{A}_\infty^{\text{geo},\text{sv}}(C), \omega_\infty^{\text{geo},C})$ supplies a dimensionless ordering for that observer's accessible algebra-state pair. Physical time requires an observer-readable transition, event correspondence, and calibrated clock instrument. No claim about arbitrary operational clocks, global time, or the full problem of time follows from the geometric modular-flow theorem alone.

Scope note. The theorem fixes the 2π normalization and the carried refinement limit on the support-visible extracted geometric cap pair under the paired finite cap-normal and independently complete same-branch multiresolution algebra-state comparison inputs. The UV-side scaffold is the realized transported geometric cap-local system, regularized support-visible modular transport on fixed local collars, support-visible cap-pair extraction on the local GNS support quotient, support-readable modular covariance, BW framing, oriented cross-ratio rigidity, and geometric KMS normalization. No separate cap-isotropy input is used, and the conformal support-map clause is the explicit scaling-limit branch condition rather than a finite-regulator Lorentz premise. Compact-time convergence of finite-regulator modular groups is available only through the transported-local certificate of Theorem 2.6f; a black-box conformal-net BW reconstruction would be an additional validation route, not a hidden premise.

2.2.3 Theorem: cap-normal H^3 reconstruction

Theorem 4.3 (Cap-normal H^3 reconstruction, imported from the spacetime and Einstein derivation). On the support-visible round-cap BW branch, write the celestial null section as

$$q(\Omega) = (1, \Omega), \quad \Omega \in S^2.$$

For a nondegenerate oriented round cap

$$C(\mathbf{c}, \alpha) = \{\Omega : \mathbf{c} \cdot \Omega \geq \cos \alpha\}, \quad 0 < \alpha < \pi,$$

define

$$n_C = (\cot \alpha, \csc \alpha \mathbf{c}).$$

Then, for the time-first Lorentz form $\eta = -dt^2 + d\mathbf{x}^2$,

$$\eta(n_C, n_C) = 1, \quad \eta(n_C, q(\Omega)) = \frac{\mathbf{c} \cdot \Omega - \cos \alpha}{\sin \alpha}.$$

Thus $\eta(n_C, q(\Omega)) = 0$ exactly on ∂C , with positive sign in the cap interior and negative sign outside. For $g \in \text{Conf}^+(S^2)$ with Lorentz representative Λ_g ,

$$\Lambda_g q(\Omega) = \omega_g(\Omega) q(g\Omega), \quad n_{gC} = \Lambda_g n_C.$$

Consequently

$$\text{Conf}^+(S^2) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1),$$

oriented round caps form the de Sitter normal space

$$\text{Cap}_{\text{round}}^{\text{or}}(S^2) \cong dS_3^{\text{cap}} \cong \text{SO}^+(3, 1)/\text{SO}^+(2, 1),$$

and future unit timelike observer frames form the distinct hyperbolic space

$$H^3 \simeq \text{SO}^+(3, 1)/\text{SO}(3), \quad \dim H^3 = 3.$$

Each cap normal determines the geodesic half-space

$$\mathcal{H}_C^+ = \{u \in H^3 : \eta(n_C, u) \geq 0\},$$

not a preferred observer point.

Proof. This is the cap-normal reconstruction theorem in The spacetime and Einstein paper, which supplies the projective-null-cone model, the cap-normal bijection, Lorentz equivariance, the H^3 observer-frame hyperboloid, the ideal-boundary identification, and the cap/half-space duality. The BW theorem supplies the cap-preserving conformal subgroup and its 2π modular normalization; the cap-normal theorem supplies the explicit projective Lorentz representation and does not rederive BW from bare finite consensus. QED.

Corollary 4.3a (Frame-space boundary). The theorem proves the exact canonical Lorentz/ H^3 observer-frame hyperboloid on the stated branch. Its points are future unit timelike frames, not event positions. It does not populate a physical spacetime, construct a chart-blind neutral bulk, derive a physical curvature radius R_H , produce stress-energy, or close the Einstein branch-entry umbrella. Record-conditioned cap responses can estimate frame data under the certificate imported next; event location, object families, neutral bulk, scale, stress, and Einstein branch entry are separate receipts. Theorems 4.3c–4.3h below import The spacetime and Einstein

paper's geometry-producer, null-net-standardness, source-causal precursor and continuum, stress, entropy-bridge, and composed branch-entry packets. Their composition is conditional on one source-derived common-domain tower with certified cofinal tails, universal coupling, a vacuum reference, and independent physical scale readouts. No source-derived tower carrying that complete packet is constructed here.

Theorem 4.3b (Conditional record-conditioned H^3 frame estimate). Fix a quotient-visible record token i and clock slice t . On the frame space of Corollary 4.3, suppose its record-conditioned modular cap responses

$$R_i(C, t, O) = \omega_{i,O} \left(\sigma_t^{C,O}(M_{C,0,O}) \right), \quad \omega_{i,O}(A) = \frac{\omega_O(P_{i,O} A P_{i,O})}{\omega_O(P_{i,O})},$$

admit the calibrated frame-local factorization

$$y_{i,j} = F_j(X_i(t)) + e_{i,j}, \quad F_j(X) = \psi_j \left(\frac{\eta(X, n_j)}{R_H} \right),$$

on a compact frame region $\Omega \subset H_{R_H}^3$. Assume the finite response map has quantitative observability

$$\alpha \bar{d}(X, Y) \leq |F(X) - F(Y)|_W \leq L \bar{d}(X, Y), \quad \alpha > 0,$$

and the total error obeys $|e_i|_W \leq \sigma_i$. If $\mathcal{N}_\varepsilon$ is an ε -net of Ω and $\hat{X}_i(t)$ is a residual minimizer up to tolerance τ_i , then the conditioned feasible frame support obeys the conservative enclosure

$$S_i(t) \subseteq B_H \left(\hat{X}_i(t), R_H \left[\frac{L}{\alpha} \varepsilon + \frac{2}{\alpha} \sigma_i + \frac{1}{\alpha} \tau_i \right] \right).$$

The theorem does not assert that every point of this output ball fits the data. Exact continuous data have a unique minimizer. Finite noisy data produce the displayed frame ball; a unique finite frame value is licensed only when the certified residual gap $\Delta_{\text{loc},i}(t) > 0$. If that gap fails, the fail-closed protocol returns AMBIGUOUS, meaning that no unique frame value is certified; it is not a proof that the mathematical data have multiple minimizers. This theorem does not locate an event, populate the event base, resolve unlabeled mixtures, identify particle species, produce stress-energy, or establish chart-blind neutral bulk.

Theorem 4.3c (Quotient-intrinsic geometry producer and dimension-selection boundary). On a refinement tower of repaired quotient normal forms, form at each level the support-visible incidence complex: vertices are the quotient-visible patch classes and simplices are the jointly supported patch sets. Each level is computable from its normal form and is invariant under repair schedule and gauge. Refinement generally changes the literal complex, so the tower must additionally carry certified simplicial refinement maps; whenever preservation of topology is used, their realizations must be certified subdivisions or homeomorphisms. If this compatible tower carries the computable receipts (i) spherical incidence (connected closed orientable combinatorial surface with Euler characteristic 2), (ii) disk/mesh nondegeneracy, (iii) modular cross-ratio Cauchy convergence, and (iv) independently normalized geometric 2π -KMS comparison with wrong-normalization separation, then the refinement limit produces the round S^2 , its conformal structure, the oriented round-cap family with cap normals, and every clause of the finite cap-normal BW certificate, with quantitative error envelopes. Conversely, four finite repair systems with one rewrite signature (three patches in the seed record, one repair step) have incidence complexes realizing S^2 , T^2 , the 2-skeleton of $\partial\Delta^4$, and a wedge of two spheres, and the scalar KMS receipt admits any modular temperature β : repair confluence alone does not select topology, dimension, or modular normalization. Orientation framing and the MGNS state enter

as separately typed receipts; no countermodel artifact for that pair is exhibited. The screen S^2 is therefore *produced on the receipt branch*; the receipt set is one sufficient certificate, but no semantic-minimality theorem or full-tower countermodel matrix is supplied. Producers that reach an oriented S^2 support limit by another computable function of the normal forms, such as the source-derived incidence nerve of the carrier-support bridge, are untouched by these countermodels. (Spacetime and Einstein paper, geometry-producer packet; machine receipts accompany the public geometry code.)

Theorem 4.3d (Null-net standardness, derived translations, and the four-translation assembly). On the producer branch, the null half-line/interval blow-up net at a produced cap boundary point lives on one common GNS space, and: (i) the reference vector is cyclic and separating for every half-line used, given faithful finite-stage states on the interior MaxEnt branch, the mixed-GNS Cauchy clause, and a cofinal cyclicity receipt uniform over every half-line in the construction; boundary MaxEnt solutions require a support-restricted treatment and do not supply faithfulness by themselves; (ii) the produced geometric cap flow derives the half-sided covariance relation without assuming translation unitaries, but the Borchers–Wiesbrock step additionally consumes standardness for every nested pair—cyclicity and separation for both half-line algebras and cyclicity for their relative commutant; interval cyclicity is therefore a registered standard-relative-commutant receipt, not a consequence of one half-line receipt; (iii) weak additivity of the null net is a separate cofinal receipt used for the generated net, rather than a consequence of finite-stage faithfulness; (iv) on the exact-Markov/central-interface branch, with a specified endpoint-to-cell map and a uniform local norm/site-density bound, the reduced interval modular Hamiltonians split into local terms plus uniformly bounded interface blocks (endpoint-Lipschitz), while an explicit four-qubit nearest-neighbour Gibbs witness shows finite-range locality alone does *not* imply this (machine-verified); (v) on the standard half-sided-inclusion and modular-intersection receipt branches, the derived positive null translations and modular dilations generate Möbius covariance and the projective bounded-interval action; and (vi) the per-direction positive generators commute, transform with the null-vector weight under the produced Lorentz action, satisfy the dependent-family linearity relation, and assemble into one four-parameter translation representation with joint spectrum in the closed future cone: a positive-energy Poincaré representation. (Spacetime and Einstein paper, null-net standardness packet.)

Theorem 4.3e (Source-derived finite 1 + 3 precursor and effective-continuum criterion). An authenticated semantic-event log supplies its finite event carrier and strict generated read-from order. Its canonical source height is computed from that parent relation: roots have height zero and every nonroot has one plus the maximum height of its direct parents. Lean proves strict increase on parent edges and generated ancestry, bounds every authenticated direct-parent chain ending at an event by that event’s height, and constructs an attaining chain. Thus source height is exactly the longest authenticated-parent-chain length. Let V_{src} be the exact rank-three positive source Gram quotient. Then $W_{\text{src}} = \mathbb{R} \oplus V_{\text{src}}$ has real dimension four, and $Q(t, x) = t^2 - g_{\text{src}}(x, x)$ has inertia $(1, 3)$. Every source-unit direction gives the future-null vector $(1, u)$, realizing the exact equivalences $\text{SourceUnitDirection} \simeq S^2 \simeq \text{FutureNullRay}$. For a supplied spatial readback $x(e)$ and positive height scale τ , the placement $F(e) = (\tau h_L(e), x(e))$ carries authenticated parenthood, and hence generated ancestry, into the future cone whenever every parent displacement obeys the source-metric speed bound. If every future-cone comparison between placed events is supported by reflexive generated ancestry, order and cone agree exactly. Lean derives this support from two concrete finite checks: spatial readback separates each equal-height layer, and every increasing-height pair unsupported by ancestry obeys the strict spacelike inequality. Antisymmetry then derives injectivity, every source interval agrees with the corresponding cone interval on the placed image, and composition with the source-carrier equivalence gives a sep-

arated one-chart finite causal-chart interface. This global finite chart is not an open manifold atlas. Thus the ambient target-carrier dimension and Lorentz signature are constructed without a freely declared rank-four chart, fitted cone, or supplied placement rank; the intrinsic poset dimension, spatial placement, speed/converse conditions, physical interpretation, and clock remain separate data. The exact-embedding route to a smooth effective manifold uses one cofinal source-selected family carrying physical event/link identification, exact two-way cone/order agreement, refinement compatibility, dense and isotropic source- S^2 link directions, a calibrated $\#I/\rho \rightarrow \text{Vol}(I)$ law, independent dimension/manifoldlikeness and height/profile tests, stable thickened-antichain topology or homology, and a specified distinguishing Lorentzian limit with uniqueness control. On that certificate, causal order fixes conformal geometry and the calibrated measure fixes scale. Smooth Einstein promotion additionally requires either same-family tensor-curvature reconstruction converging to the Einstein tensor or the independently premised continuum small-ball/null-balance identification, together with physical stress, Ward/Bianchi, coupling, and remainder data on the same family. Scalar-curvature convergence alone is diagnostic. No physical instance of this certificate is supplied. (Spacetime and Einstein paper; Lean theorems and independently verified simulation evidence.)

A source-record order and count limit. In the continuous rank-three carrier, conservative Fibonacci records give q^3 distinct spatial sites. Repeat this population at every layer and let each event read all preceding-layer sites inside a radius a_q , including itself. With spatial fill distance $h_q = o(a_q)$, $a_q \rightarrow 0$, and model duration a_q/c , actual read paths approach the speed- c causal cone. Equal-volume cell assignments with vanishing displacement error make normalized raw counts converge to $dt d^3x$; the ordering fraction in interior timelike diamonds tends to $1/10$. This is a flat effective spacetime theorem for a specified population, read law and model clock (Spacetime and Einstein paper). It supplies neither their native dynamical selection nor their identification with a matter action or laboratory clock.

The same histories admit a retrospective clock. For intervals approaching fixed interior timelike diamonds, the fourth root of their inclusive count ratio converges to their proper-time ratio. A reference interval fixes the unit, and complete ancestry access supplies the count. This readout needs no oscillator or numerical timestamps. Operational covariance gives a separate cone criterion: a nonzero closed convex pointed cone invariant under the source icosahedral rotations and all boosts along one axis must be one of the two Lorentz cones. A time orientation selects the future cone. The covariance must act on possible influences and readouts. Algebraic direction maps alone do not establish that operational premise[15].

For Einstein promotion there are two distinct sufficient routes at this step: an explicit same-family tensor-curvature reconstruction converging to the smooth Einstein tensor, or a separately premised continuum small-ball/null-balance identification. Convergence of a scalar-curvature estimator is an independent diagnostic and is insufficient by itself.

Every finite semantic log also admits an explicit auxiliary separated one-way placement in the source $(1+3)$ carrier. Enumerating events along one source axis and taking height scale one larger than the event count makes the event map injective, sends generated precedence into the future cone, and makes distinct same-height events spacelike. The enumeration is arbitrary and generally introduces cone comparisons unsupported by ancestry; it is not an order-reflecting, source-selected, or physical placement.

The provenance grammar has the complementary exact expressivity theorem. Every finite decidable irreflexive transitive relation R compiles into an abstract authenticated semantic log whose direct parent relation and generated strict order are both exactly R ; predecessor cardinality supplies the rank. The generic parent relation is all of R rather than its Hasse reduction alone. The relation is supplied to the compiler, and the per-event snapshots are not proved to arise from one threaded executor. This is a universal finite-poset representation result, not a dynamics,

manifoldlikeness, or continuum theorem.

An exact four-event Boolean diamond inhabits the order-faithful finite interface: its parent edges are null and its two independent branches are spacelike. This is a machine-checked non-chain causal-set-like witness, not a physical continuum or an empirical dimension measurement.

The finite provenance order is causal-set-like rather than a physical causal set. Lean packages reflexive authenticated ancestry as a partial order and proves every interval finite at each finite cutoff, exactly realizing the abstract order-theoretic causal-set axioms. Standard causal set theory starts from a locally finite order interpreted as proto-causality [1, 2]. The exact-embedding continuum route requires an order embedding whose image has the approximately Poisson count density needed for a causal-set faithful embedding, together with manifoldlikeness and dimension [3, 4, 5]. Under the relevant distinguishing hypotheses, continuum causal order fixes the conformal geometry [12, 13]; the volume datum fixes the conformal factor. The Lorentz-invariant causal-set approximation uses Poisson sprinkling [11]; the OPH commit log is not a Poisson sprinkling, and its event count is not a spacetime volume. The finite source-causal theorem supplies the (1+3) precursor, not a continuum geometry. It derives neither a physical order embedding nor the count-to-volume density law that together constitute a causal-set faithful embedding. It also establishes neither continuum existence nor the causal-set Hauptvermutung from the finite OPH order. The finite causal/Lorentz precursor instead constructs the ambient carrier from an independent real axis and the exact rank-three source carrier. A spatial event readback, canonical source height, and positive `timeScale` then give a concrete placement; a local speed bound proves one-way cone compatibility, while equal-height spatial separation plus strict spacelikeness of unsupported increasing-height pairs supply converse cone support and exact two-way agreement. Antisymmetry then derives injectivity, while the two-way equivalence preserves every source interval exactly on the placed image. The event-count measure is distinct from public capacity N . The finite join does not constitute spacetime. Physical event/link identification, refinement, manifoldlikeness, topology, independent dimension tests, count-volume calibration, and uniqueness are required before the continuum can be read as emergent rather than fundamental. The exact source-direction bridge gives $\text{SourceUnitDirection} \simeq S^2 \simeq \text{FutureNullRay}$ for the rank-three source Gram quotient and fixes the finite (1+3) carrier and cone after adjoining an independent real axis. Positive `timeScale` times canonical source height enters only through event placement; supplied event-local Lorentz frames preserve the rest metric. This is an algebraic direction and causal-carrier result, not a physical sky or signal theorem. The effective reading additionally requires the physically faithful images of authenticated local read-from links to cover this S^2 densely and isotropically; that coverage is not supplied. The measure calibration separately requires $\#I/\rho \rightarrow \text{Vol}(I)$. Refinement must preserve order, directions, and count density while soldering rank-three neighborhoods with $|B_{\text{car}}(r)| \asymp r^3$ over a scaling window. This is compatibility of the four inputs, not a fifth one. Neither the local rank-three carrier nor the seeded global finite wiring determines event-base dimension.

The theorem-status composition is exact at the finite interface and conditional thereafter. The source-native causal-chart interface uses semantic commits as the event carrier and the generated reflexive order definitionally; it uses no separate population map or duplicate precedence field. The finite compatibility theorem constructs the ambient (1+3) carrier, proves its rank and source-unit/null identities, and proves the one-way or two-way cone result under the explicit placement conditions. Supplied chart transports retain their bounded algebraic covariance role. No physical placement, count-density field, or refinement family is constructed. Physical faithfulness, link-direction coverage, manifoldlikeness, topology, volume, and refinement conditions remain independently testable.

The exact same-event-type source-order Einstein theorem, in a separately supplied 3 + 1 tensor interface, replaces an all-null balance hypothesis by nine supplied balances on fixed algebraic source-

direction representatives. Given supplied symmetric geometry/stress fields indexed by the finite event type, four discrete step maps, Ward/Bianchi identities, step connectedness, a base event, and an algebraic coupling, exact nine-direction tomography derives all-null balance, the pointwise metric ambiguity, its constancy, and the Einstein-form relation. The minimal premise bundle contains no normalization, vacuum-calibration, Newton-constant, or universal-source field. The generated order and the independently derived $1 + 3$ carrier use the same event type, but no provenance link selects a tensor field, step, or balance. The theorem does not identify tensor-coordinate differences with those of the constructed carrier or identify the supplied fields with smooth curvature or physical stress-energy. That reading requires the source-causal continuum, count-volume, convergent curvature/stress, small-ball, coupling, and scale receipts on one family.

Proposition 4.3e0 (Bounded algebraic event-frame soldering). Let raw record representatives carry an actual coincidence setoid and a readback into the intrinsic Hermitian Lorentz module that is constant on coincidence classes. The readback descends uniquely. Given one time-oriented affine Lorentz overlap cocycle and, separately, a supplied chart-coordinate family satisfying its affine overlap law, coordinate displacements transform through the cocycle's linear part, their Lorentz intervals are chart independent, and the reverse transition and triple-overlap laws follow from the same cocycle. Future-null displacements descend to covariant celestial directions. One base-chart future-unit frame selection induces compatible frames in every chart, and Lorentz-orthogonal rest projections transport by exact rest-metric isometries. The rank-three source Gram quotient is linearly and isometrically equivalent to the standard internal rest fiber, as a candidate local readout. The standalone quotient-descent theorem does not construct this atlas coordinate family or identify the descended readback with it. A finite two-event/two-chart control has nonzero affine translation and a nonzero future-null displacement. A separate three-record control proves that a reflexive symmetric pairwise-overlap relation need not be transitive, so it cannot replace the coincidence setoid. This bounded algebraic adapter assumes the supplied chart-coordinate overlap law, coincidence setoid, separation, physical cone attachment, refinement naturality, and operational clock. It can be used after a source-causal continuum certificate has selected an effective atlas, but it does not construct the event carrier and does not supply physical causality, manifoldlikeness, topology, count-volume calibration, or smooth curvature.

Proposition 4.3ea (Finite signed-seam operator realization). A declared sign-twisted derivative on a finite seam complex with unit-counting measure gives a positive semidefinite kinetic operator. Its kernel consists of parallel sections; on a connected frustrated component that kernel is zero. The finite realization with 8,662 sites and 11,816 seams has 38 frustrated triangles. Exact boundary-rank elimination gives lift-ambiguity rank 3,117, and deterministic integer sections check adjoint, kinetic, covariance, gluing and boundary identities. The separate rank-three response and declared fifteen-state charge assumptions define a conditional 45-dimensional internal factor. The declared operator $D_\sigma \otimes I_{45}$ retains the gap of D_σ with multiplicity 45. These are properties of the specified finite operator and matter table. They select neither a spacetime chart nor a physical matter action, and do not identify the domain with the twelve-port Spin construction. The microphysics paper gives the finite algebra and independent replay [24].

Theorem 4.3f (Finite null tomography and conditional local-stress upgrade). A symmetric bilinear form on $(\mathbb{R}^4, \eta)$ is determined by its null-null values up to a multiple of η , and a directional scalar family arises from a tensor iff it satisfies the finite dependent-family linearity relations; nine generic null directions reconstruct the trace-free part with Lipschitz stability. The released code verifies this finite linear algebra on declared and synthetic charge families. Per-direction values violating one linearity relation admit no rank-two reconstruction, and scalar collar CMI alone cannot define a rank-two source.

The physical upgrade is conditional. If one common source-derived endpoint-charge family on

the event region satisfies the linearity relations, admits the declared same-source identification with local null-stress charges, carries one common normalization and Poincaré-covariant local transformation law, and satisfies the weak Ward identity, then tomography defines a symmetric semiclassical $\langle T_{ab} \rangle$, with null projections equal to those charges and ambiguity ϕg_{ab} plus the declared improvement terms. Locality, covariance, the charge/stress identification, and weak conservation are separate weak-Ward and null-stress matching supplies; they are not consequences of finite tomography or of the synthetic reconstruction receipt. Universal coupling to the geometric branch is a further named premise. (Spacetime and Einstein paper, stress packet; finite tomography receipts accompany the public geometry code.)

Theorem 4.3g (Repaired entropy bridge and absolute Einstein equation). On the central-interface branch, write the state as a direct sum of bulk states tensored with maximally mixed edge factors. Its entropy splits exactly into $S_{\text{bulk}} = H(p) + \sum_{\alpha} p_{\alpha} S(\rho_{\alpha}^{\text{bulk}})$ and $S_{\text{edge}} = \sum_{\alpha} p_{\alpha} \log d_{\alpha}$. The finite first law is $\delta S = 2\pi\delta\langle B_C \rangle + \delta\langle Z_C \rangle$, with $Z_C = \sum_{\alpha} (\log d_{\alpha}) P_{\alpha}$. Hence $\delta S_{\text{bulk}} = 2\pi\delta\langle B_C \rangle$ and $\delta S_{\text{edge}} = \delta\langle Z_C \rangle$ throughout the declared central-interface class, including variations of the sector weights. The changing stress channel is supplied by the exact MaxEnt envelope identity $dS/dt = \lambda = 2\pi$, giving the Clausius balance $\delta S_{\text{edge}} = \lambda\delta t - 2\pi\delta\langle B_C \rangle$ on the coupled variation class. Reaching the geometric balance additionally requires the declared fixed-cap generalized-entropy stationarity interface, the conditional same-source local-stress and weak-Ward bridge, the continuum diamond kernel, and the fixed-volume small-ball area identity. One common scaling family must carry the conditional-mixing, boundary-count, fixed-collar, sharp-rate, and $o(\ell^4)$ remainder bounds. Coverage of all local timelike directions follows from population plus the produced Lorentz orbit. The vacuum-reference premise fixes the first-variation integration constant. Independent scale readouts with trivial positive-rescaling stabilizer identify the structural coupling. Under all of these premises, $G_{ab} + \Lambda g_{ab} = 8\pi G\langle T_{ab} \rangle$ per connected component. Without the vacuum reference, the result fixes the residue tensor only up to one covariantly constant metric term. (Spacetime and Einstein paper, entropy-bridge and small-diamond packets.)

Theorem 4.3h (Conditional Einstein branch composition). Bare finite consensus is not Einstein-complete: the same finite-consensus reduct admits extensions in which the Einstein equation holds and extensions in which it fails. The valid gravity result is conditional. Let one source-derived repaired quotient tower carry deterministic geometry, modular, causal-order, stress, entropy, and scale readouts on a common domain, with commuting refinement diagrams. Assume the finite source-causal precursor, geometry and null-net receipts; cap-interior modular data; one physical source-causal continuum certificate on a cofinal family with either same-family tensor-curvature reconstruction or the independently premised continuum small-ball/null-balance identification, for which every relevant remainder is $o(\ell^4)$; the universal-coupling and vacuum-reference receipts; and two independent scale readouts with trivial positive-rescaling stabilizer. The stress-tomography, entropy, curvature-identification, timelike-polarization, Ward, and Bianchi results then compose to

$$G_{ab} + \Lambda g_{ab} = 8\pi G\langle T_{ab} \rangle$$

on each connected event component. At the curvature-identification step, the causal-set route requires a same-family tensor-curvature reconstruction converging to G_{ab} ; the alternative continuum route instead premises the small-ball/null-balance identification independently. Convergence of a scalar-curvature estimator alone establishes neither route. The exact finite layer includes the nine-direction tomography frame with determinant 8192/27 and the entropy and MaxEnt identities. Conditional on the continuum diamond-kernel formula and the smooth fixed-volume small-ball area identity, exact arithmetic gives the coefficients $8\pi^2\ell^4/15$ and $-4\pi\ell^4/15$; the continuum formulas themselves remain explicit analytic premises. The finite operator of Proposition 4.3ea does not supply the geometric modular, stress, entropy, scale, or asymptotic data of this theorem. No source

tower satisfying the joint continuum antecedent is constructed here, and no semantic-minimality theorem for that antecedent is claimed. (Spacetime and Einstein paper, composed branch-entry theorem.)

The register surface of this composition is inhabited. For every declared repair law and every horizon record over it, an explicit value of the typed premise bundle is machine-checked: its modular flow supply is the finite thermodynamic first-law datum on the simplex tangent space at the declared reference, with the boost and central charges of the record, so the first-law clause of the composed conclusion at this value is the first-order entanglement first law of the horizon-thermality surface on the same variation space. The other supplies are synthetic witnesses at the demo tier: the two-patch consensus tower with six typed readouts extending the committed event and geometry fragment, a sixteen-point four-direction torus chart with a nonconstant divergence-free stress field and the geometry defined as $\kappa T_{ab} + \Lambda \eta_{ab}$, a strict audit manifest evaluated on that chart, a shrinking family with ℓ^5 error channels, and small-ball reals solved from the three small-ball identities. The Einstein clause, the small-ball arithmetic, the universal-source clause, and the vacuum reference hold at this value by definition of those witnesses and select nothing. A nonvacuity receipt exercises the clauses the data can fail on nonzero data, and a counterexample shows that the split of the horizon record is the load-bearing input: with a uniform reference and zero central charge in place of the split, the same boost charge fails the modular-generator clause on the same variation. The value certifies joint satisfiability of the premises and places the composed branch on the thermodynamic variation space; it attaches no physical metric, stress tensor, horizon, temperature, or Newton constant, and realized nonemptiness on a source-generated tower is not established.

2.3 Gravity from Fixed-Cap Generalized-Entropy Stationarity

2.3.1 Cap first law

Section 4.2 fixes the cap modular statement first at the automorphism level. On the extracted geometric subnet, the scaling-limit cap modular flow is geometric with the standard 2π normalization,

$$\sigma_t^{\omega_\infty^{\text{geo},C}} = \alpha_{\lambda_C(2\pi t)}.$$

If the realized scaling-limit cap algebra happens to be type I, one may choose a density matrix $\rho_C^{\omega_\infty^{\text{geo},C}}$ and modular Hamiltonian

$$K_C := -\log \rho_C^{\omega_\infty^{\text{geo},C}},$$

and then for a perturbation $\rho(\varepsilon)$ with $\rho(0) = \omega_\infty^{\text{geo},C}$ the first-law identity reads

$$\delta S_C = \delta \langle K_C \rangle = 2\pi \delta \langle B_C \rangle.$$

This displayed variation is the special type-I/fixed-central-sector form of $K_C = 2\pi B_C + Z_C$, where the central term has no first-order contribution on the chosen sector. In the generic continuum/non-type-I case, Section 4.2 does not supply an inner operator identity; the theorem there is the geometric modular automorphism statement, so the cap first-law formula is restricted to this special realization.

2.3.2 Null-surface modular bridge

This subsection extends the fixed-cutoff weak-tail-generator boundary to the derived positive null-translation stage, the exact endpoint-derivative identity for its half-line generator, and a conditional local-charge reading. At fixed cutoff one first transfers cut-center data to regulated null strips, then

imposes the extra inherited split condition needed for the spatial-collar-type tensor decomposition, obtains exact or controlled strip additivity on one inherited strip model, and derives a weak tail generator for the renormalized half-line family. On the scaling-limit geometric-cap branch of Theorem 4.2, the null half-line blow-up net then inherits geometric dilation and therefore the half-sided covariance relation. Together with the separately supplied common-vector and relative-commutant standardness receipts, this is a standard half-sided modular inclusion, so Borchers–Wiesbrock supplies an explicit positive null-translation generator on its Stone domain. Bounded-interval formulas are discharged by the E0.5 affine/projective kernel below, and the tensor upgrade carries the null-invisible metric ambiguity. Accordingly the null modular bridge consists of:

- null-cut center transfer yielding the central cut-label decomposition, together with the extra inherited split condition needed for the spatial-collar-type tensor decomposition;
- exact or controlled four-term strip additivity on one fixed inherited strip model;
- endpoint-Lipschitz control for the renormalized half-line family and the resulting weak tail generator;
- the geometrically derived half-sided relation plus the registered standardness receipts on the null half-line blow-up net, and the resulting Borchers positive translation generator.

The later density-upgrade template is kept below as an explicit downstream template rather than as a fixed-cutoff theorem proved in this subsection. Lemma 5.2f makes the positive null-translation generator itself explicit, and Theorem 5.2g records the Borchers unitary group, positivity and self-adjointness on the Stone domain, the corrected affine commutator relation, and the exact half-line modular-Hamiltonian identity $K_a = K_0 - 2\pi a P_\Omega$. Identifying that endpoint derivative with a local null-stress charge additionally consumes an effective-locality and same-source matching premise. Bounded-interval formulas are downstream.

Proposition 5.2a (Null-cut center transfer and inherited split). Fix a regulated null tripartition

$$I_- = (v_1, v_2), \quad J = (v_2, v_3), \quad I_+ = (v_3, v_4),$$

with cuts $\Gamma_- := \{v = v_2\}$ and $\Gamma_+ := \{v = v_3\}$. Under the fixed-cutoff realized presentation, assume that the two null cuts inherit the ordinary or central-defect boundary-redundancy data used in the spatial collar branch, so that in a compatible type-I regulator presentation one has

$$\begin{aligned} \tilde{\mathcal{H}}_{I_-} &\cong \bigoplus_{\alpha_-} W_{\alpha_-} \otimes \mathcal{H}_{i_-^{\alpha_-}}, \\ \tilde{\mathcal{H}}_J &\cong \bigoplus_{\alpha_-, \alpha_+} W_{\alpha_-}^* \otimes \mathcal{H}_{j^{\alpha_-, \alpha_+}} \otimes W_{\alpha_+}, \\ \tilde{\mathcal{H}}_{I_+} &\cong \bigoplus_{\alpha_+} W_{\alpha_+}^* \otimes \mathcal{H}_{i_+^{\alpha_+}}, \end{aligned}$$

where opposite sides of each cut carry inverse transport. If the strip algebra is the commutant of the transported cut actions, then

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-, \alpha_+} \mathcal{B}(\mathcal{H}_{j^{\alpha_-, \alpha_+}}), \quad Z(\mathcal{A}(J)) = \bigoplus_{\alpha_-, \alpha_+} \mathbb{C} P_{\alpha_-, \alpha_+}.$$

If, in addition, each multiplicity space factors as

$$\mathcal{H}_{j^{\alpha_-, \alpha_+}} \cong \mathcal{H}_{j_L^{\alpha_-, \alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-, \alpha_+}},$$

with $\mathcal{A}(I_- \cup J)$ acting blockwise only on $\mathcal{H}_{i_-}^{\alpha_-} \otimes \mathcal{H}_{j_L}^{\alpha_-, \alpha_+}$ and $\mathcal{A}(J \cup I_+)$ acting blockwise only on $\mathcal{H}_{j_R}^{\alpha_-, \alpha_+} \otimes \mathcal{H}_{i_+}^{\alpha_+}$, then

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-, \alpha_+} \mathcal{B}(\mathcal{H}_{j_L}^{\alpha_-, \alpha_+}) \otimes \mathcal{B}(\mathcal{H}_{j_R}^{\alpha_-, \alpha_+}),$$

and

$$\mathcal{H}_{I_- \cup J \cup I_+} \cong \bigoplus_{\alpha_-, \alpha_+} \mathcal{H}_{i_-}^{\alpha_-} \otimes \mathcal{H}_{j_L}^{\alpha_-, \alpha_+} \otimes \mathcal{H}_{j_R}^{\alpha_-, \alpha_+} \otimes \mathcal{H}_{i_+}^{\alpha_+}.$$

Proof. Complete reducibility gives the displayed decompositions. Commuting with the two transported cut actions forces strip operators to preserve the sector pair (α_-, α_+) and act only on the multiplicity space; Schur's lemma kills the off-diagonal intertwiners and leaves the direct-sum algebra above. Taking invariants across both cuts in the glued tripartition again uses Schur's lemma and leaves exactly the matching sector pairs. The left/right split of the multiplicity spaces is additional structure; it is not forced by the center transfer alone. QED.

Corollary 5.2b (Exact or controlled four-term null modular relation on an inherited strip model). On one fixed finite-dimensional strip model satisfying Proposition 5.2a, define

$$\mathfrak{M}_{I_-:J:I_+} := \{\sigma : I(I_- : I_+ \mid J)_\sigma = 0\},$$

and

$$\delta_{I_-:J:I_+}^{\text{M}}(\varepsilon) := \sup \left\{ \inf_{\sigma \in \mathfrak{M}_{I_-:J:I_+}} \|\rho - \sigma\|_1 : I(I_- : I_+ \mid J)_\rho \leq \varepsilon \right\}.$$

Also define the EC-aligned subset $\mathfrak{M}_{I_-:J:I_+}^{\text{EC}}$ by requiring the strip MSA on the inherited split. For a declared controlled strip family $\mathcal{F}_J$, let

$$\delta_{\mathcal{F}_J; I_-:J:I_+}^{\text{M,EC}}(\varepsilon) := \sup \left\{ \inf_{\sigma \in \mathfrak{M}_{I_-:J:I_+}^{\text{EC}}} \|\rho - \sigma\|_1 : \rho \in \mathcal{F}_J, I(I_- : I_+ \mid J)_\rho \leq \varepsilon \right\}.$$

The generic exact-Markov modulus need not control this aligned one; convergence of the latter is a separate controlled-family hypothesis. For any strip state η , let

$$\Delta K_J(\eta) := K_{I_- \cup J \cup I_+}(\eta) - K_{I_- \cup J}(\eta) - K_{J \cup I_+}(\eta) + K_J(\eta).$$

If the strip reference state ω is exact Markov and EC-aligned on the inherited split (the strip form of the Markov-split alignment hypothesis of Section 2.3), then $\Delta K_J(\omega)$ is central, and on the canonical inherited HJPW model one may take

$$\Delta K_J(\omega) = 0.$$

Equivalently, there exists a central operator $K_{\partial, J}(\omega) \in Z(\mathcal{A}(J))$ such that

$$K_{I_- \cup J \cup I_+}(\omega) = K_{I_- \cup J}(\omega) + K_{J \cup I_+}(\omega) - K_J(\omega) + K_{\partial, J}(\omega).$$

If instead $\omega \in \mathcal{F}_J$ and $I(I_- : I_+ \mid J)_\omega \leq \varepsilon$, assume the aligned modulus tends to zero and choose $\tilde{\omega}_J \in \mathfrak{M}_{I_-:J:I_+}^{\text{EC}}$ with

$$\|\omega - \tilde{\omega}_J\|_1 \leq \delta_{\mathcal{F}_J; I_-:J:I_+}^{\text{M,EC}}(\varepsilon).$$

Define

$$\mathfrak{D}_J(\omega, \tilde{\omega}_J) := \Delta K_J(\omega) - \Delta K_J(\tilde{\omega}_J).$$

Then

$$K_{I_- \cup J \cup I_+}(\omega) = K_{I_- \cup J}(\omega) + K_{J \cup I_+}(\omega) - K_J(\omega) + K_{\partial, J}(\tilde{\omega}_J) + \mathfrak{D}_J(\omega, \tilde{\omega}_J),$$

with $K_{\partial, J}(\tilde{\omega}_J) = \Delta K_J(\tilde{\omega}_J) \in Z(\mathcal{A}(J))$. Every bounded observable on $I_- \cup J$ or $J \cup I_+$ then differs from the exact-Markov reference by at most

$$\|O\|_\infty \delta_{\mathcal{F}_J; I_-: J: I_+}^{\text{M,EC}}(\varepsilon).$$

If the relevant strip marginals are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\mathfrak{D}_J(\omega, \tilde{\omega}_J)\|_\infty \leq 4\lambda_*^{-1} \delta_{\mathcal{F}_J; I_-: J: I_+}^{\text{M,EC}}(\varepsilon).$$

Independently, Fawzi–Renner gives a recovered comparison state with trace-norm error

$$r_{\text{FR}}(\varepsilon) = 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}.$$

Thus the exact four-term strip relation is available only at EC-aligned exact Markovity or in a controlled strip family on one fixed inherited strip model with $\delta_{\mathcal{F}_J; I_-: J: I_+}^{\text{M,EC}}(\varepsilon_J) \rightarrow 0$, while $\mathfrak{D}_J$ is carried explicitly at finite stage.

Proof. On the inherited split of Proposition 5.2a, the EC-aligned exact-Markov case is the same HJPW block calculation as in the spatial collar theorem: alignment identifies the HJPW blocks with the inherited blocks, the four logarithms cancel blockwise on the canonical model, and any residual bookkeeping term is central. Without alignment, HJPW factorizes the state only over its own state-dependent split, and the centrality of ΔK_J in $Z(\mathcal{A}(J))$ is not available. The controlled replacement is the same fixed-model compactness argument used for ordinary collars, after relabeling $A, B, D \mapsto I_-, J, I_+$; the operator-norm bound on $\mathfrak{D}_J$ is the corresponding relabeling of the modular-transport estimate. QED.

Definition (Renormalized null modular functional). For a null interval I on generator Ω , let $K_{\partial}(I, \Omega)$ denote the central endpoint-label term singled out by Corollary 5.2b on the inherited strip model, or by its controlled exact-Markov replacement when that model is used as reference. Define

$$\widetilde{K}[I, \Omega] := K[I, \Omega] - K_{\partial}(I, \Omega), \quad \widetilde{K}_a(\Omega) := \widetilde{K}[(a, \infty), \Omega].$$

Proposition 5.2c (Endpoint-Lipschitz null modular families and weak tail generator). Work on the exact-Markov/central-interface inherited-strip branch, and assume the endpoint-to-regulator-cell map and uniform local norm/site-density bound of the endpoint-control clause in Section 2.7, uniformly on the allowed compact endpoint window. For renormalized modular Hamiltonians on one null generator,

$$\widetilde{K}[I, \Omega] := K[I, \Omega] - K_{\partial}(I, \Omega), \quad \widetilde{K}_a(\Omega) := \widetilde{K}[(a, \infty), \Omega],$$

the additional branch-internal endpoint-control gives:

$$|\langle \psi, (\widetilde{K}[(a', b'), \Omega] - \widetilde{K}[(a, b), \Omega])\phi \rangle| \leq C_{\psi, \phi, \Omega}(|a' - a| + |b' - b|)$$

for bounded intervals in a compact endpoint window, and

$$|\langle \psi, (\widetilde{K}_{a'}(\Omega) - \widetilde{K}_a(\Omega))\phi \rangle| \leq C_{\psi, \phi, \Omega}|a' - a|$$

for half-lines. Hence

$$f_{\psi, \phi}(a) := \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle$$

is locally Lipschitz and therefore absolutely continuous, and its distributional derivative defines the weak tail generator

$$\langle \psi, q(a, \Omega) \phi \rangle := -\frac{1}{2\pi} \partial_a f_{\psi, \phi}(a).$$

For $a < b$,

$$\langle \psi, (\widetilde{K}_b(\Omega) - \widetilde{K}_a(\Omega)) \phi \rangle = -2\pi \int_a^b \langle \psi, q(v, \Omega) \phi \rangle dv.$$

$q(a, \Omega)$ is a weak tail generator rather than a local operator-valued density; its identification with the positive self-adjoint Borchers generator occurs only after Corollary 5.2e and Lemma 5.2f.

Proof. The assumed endpoint-control estimate applies to the renormalized family after removal of the central endpoint term. For bounded intervals, the symmetric-difference length is bounded by $|a' - a| + |b' - b|$; for half-lines it is exactly $|a' - a|$. Therefore the matrix-element functions are Lipschitz in the endpoints. A Lipschitz function on a compact interval is absolutely continuous, so $f_{\psi, \phi}(a)$ has an L_{loc}^∞ derivative and obeys the fundamental theorem of calculus. Defining q as the rescaled negative derivative gives the displayed integral relation. QED.

Downstream c.12–c.13 boundary. The fixed-cutoff bridge established above is the end of this subsection's proved content.

Lemma 5.2d (Downstream density-upgrade template). If, in addition, the weak tail generator $q(a, \Omega)$ is weakly differentiable in a and both $f_{\psi, \phi}(a)$ and $q_{\psi, \phi}(a)$ vanish at $+\infty$, then one may define a density

$$\langle \psi, p(a, \Omega) \phi \rangle := -\partial_a \langle \psi, q(a, \Omega) \phi \rangle = \frac{1}{2\pi} \partial_a^2 \langle \psi, \widetilde{K}_a(\Omega) \phi \rangle$$

and obtain the half-line formula

$$\langle \psi, \widetilde{K}_a(\Omega) \phi \rangle = 2\pi \int_a^\infty (v - a) \langle \psi, p(v, \Omega) \phi \rangle dv.$$

This is a downstream density-upgrade step and is not proved from the fixed-cutoff strip arguments above.

Null half-line blow-up net. Fix a smooth cut point and choose affine coordinate v on generator Ω so that the cut sits at $v = 0$. For $a \geq 0$, write

$$H_a := (a, \infty), \quad \mathcal{M}_a(\Omega) := \overline{\bigvee_{a < c < d < \infty} \mathcal{A}((c, d), \Omega)}.$$

By isotony of the null interval net,

$$a \leq b \implies \mathcal{M}_b(\Omega) \subseteq \mathcal{M}_a(\Omega).$$

Corollary 5.2e (Derived half-sided modular relation on null half-lines). On the scaling-limit geometric-cap branch of Theorem 4.2, assume that the common reference vector is cyclic and separating for every half-line algebra used. The blow-up modular action near a smooth entangling cut acts on the null coordinate by

$$v \mapsto e^{-2\pi t} v.$$

Therefore, for every $a \geq 0$,

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) = \mathcal{M}_{e^{-2\pi t}a}(\Omega).$$

Hence, for every $a > 0$,

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) \subseteq \mathcal{M}_a(\Omega) \quad (t \leq 0),$$

so the inclusion $\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega)$ obeys the half-sided modular relation. After the harmless convention change $t \mapsto -t$, this has the standard positive-time orientation. The relation is derived from the null-interval structure, isotony, and the scaling-limit geometric action. It becomes a *standard* half-sided modular inclusion only when the reference vector is also cyclic for $\mathcal{M}_a(\Omega)' \cap \mathcal{M}_0(\Omega)$; that relative-commutant standardness is a separate receipt, not a consequence of the geometric action.

Proof. Blow up the cap modular flow of Theorem 4.2 near a smooth cut and restrict to the chosen null generator. In the tangent limit the cap-preserving flow becomes the null dilation $v \mapsto e^{-2\pi t}v$. For every bounded interval $(c, d) \subset H_a$, this sends $\mathcal{A}((c, d), \Omega)$ to $\mathcal{A}((e^{-2\pi t}c, e^{-2\pi t}d), \Omega)$, and taking the von Neumann closure of the interval net yields the displayed identity for $\mathcal{M}_a(\Omega)$. If $t \leq 0$, then $e^{-2\pi t}a \geq a$, so $H_{e^{-2\pi t}a} \subseteq H_a$, and isotony gives $\mathcal{M}_{e^{-2\pi t}a}(\Omega) \subseteq \mathcal{M}_a(\Omega)$. QED.

Lemma 5.2f (Positive null-translation generator). Fix $a > 0$ and assume that the common reference vector is cyclic and separating for both algebras below and cyclic for $\mathcal{M}_a(\Omega)' \cap \mathcal{M}_0(\Omega)$. For the resulting standard half-sided modular inclusion

$$\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega) \quad (a > 0),$$

let $\Delta_0(\Omega)$ be the modular operator of the standard pair $(\mathcal{M}_0(\Omega), \omega)$ and define $K_0(\Omega) := -\log \Delta_0(\Omega)$. Borchers–Wiesbrock then yields a unique strongly continuous one-parameter unitary group

$$U_\Omega(a) = e^{iaP_\Omega}, \quad a \in \mathbb{R},$$

such that

$$U_\Omega(a)\omega = \omega, \quad U_\Omega(a)\mathcal{M}_b(\Omega)U_\Omega(a)^* = \mathcal{M}_{a+b}(\Omega) \quad (a, b \geq 0),$$

and whose generator P_Ω is positive and self-adjoint on the Stone domain

$$D(P_\Omega) := \left\{ \psi \in \mathcal{H} : \lim_{a \rightarrow 0} \frac{U_\Omega(a)\psi - \psi}{ia} \text{ exists} \right\}.$$

In addition,

$$\Delta_0(\Omega)^{it}U_\Omega(a)\Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t}a), \quad \Delta_0(\Omega)^{it}P_\Omega\Delta_0(\Omega)^{-it} = e^{-2\pi t}P_\Omega,$$

and on the common invariant analytic core of $K_0(\Omega)$ and P_Ω ,

$$[K_0(\Omega), P_\Omega] = -i2\pi P_\Omega.$$

If $\Delta_a(\Omega)$ denotes the modular operator of $(\mathcal{M}_a(\Omega), \omega)$ and $K_a(\Omega) := -\log \Delta_a(\Omega)$, then

$$\Delta_a(\Omega) = U_\Omega(a)\Delta_0(\Omega)U_\Omega(a)^*, \quad K_a(\Omega) = U_\Omega(a)K_0(\Omega)U_\Omega(a)^*,$$

so

$$K_a(\Omega) = K_0(\Omega) - 2\pi a P_\Omega$$

as a quadratic-form identity on $D(K_0(\Omega)) \cap D(P_\Omega)$. Equivalently,

$$\langle \psi, (K_b(\Omega) - K_a(\Omega))\phi \rangle = -2\pi(b - a)\langle \psi, P_\Omega\phi \rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$. In the canonical normalization of Corollary 5.2b, the weak endpoint derivative from Proposition 5.2c is therefore exactly the Borchers generator. Bounded-interval modular-Hamiltonian formulas are downstream and require the separate interval-preserving branch recorded in Theorem 5.2g.

Proof. Corollary 5.2e gives the half-sided modular relation, and the hypotheses of this lemma supply standardness of the inclusion. Borchers–Wiesbrock then yields the unique strongly continuous unitary group $U_\Omega(a)$, its positive self-adjoint generator P_Ω , and the affine covariance relation with the modular group; Stone’s theorem gives the displayed domain formula. Differentiating

$$\Delta_0(\Omega)^{it}U_\Omega(a)\Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t}a)$$

first at $a = 0$ and then at $t = 0$ on the common analytic core gives

$$[K_0(\Omega), P_\Omega] = -i 2\pi P_\Omega.$$

Because $U_\Omega(a)\omega = \omega$ and $U_\Omega(a)\mathcal{M}_0(\Omega)U_\Omega(a)^* = \mathcal{M}_a(\Omega)$, uniqueness of modular data for the translated standard pair gives the displayed formulas for $\Delta_a(\Omega)$ and $K_a(\Omega)$. Differentiating $K_a(\Omega) = U_\Omega(a)K_0(\Omega)U_\Omega(a)^*$ in a yields $dK_a/da = -2\pi P_\Omega$ as a quadratic-form identity, and integrating from 0 to a gives the stated half-line modular-Hamiltonian relation. QED.

Theorem 5.2g (Half-line generator and conditional EFT null-stress reading). In the canonical normalization of the preceding half-line construction,

$$\langle \psi, P_\Omega \phi \rangle = -\frac{1}{2\pi} \frac{d}{da} \Big|_{a=0^+} \langle \psi, \widetilde{K}_a(\Omega) \phi \rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$. If that same renormalized half-line family admits an effective local-stress description and a same-source identification, its null-stress charge is represented by the identical endpoint derivative,

$$Q_{kk}(\Omega) := -\frac{1}{2\pi} \frac{d}{da} \Big|_{a=0^+} \widetilde{K}_a^{\text{eff}}(\Omega),$$

so

$$P_\Omega = Q_{kk}(\Omega)$$

as a quadratic-form identity on the common domain. The OPH half-line theorem fixes P_Ω ; reading it as a local null-stress charge consumes the effective-locality and same-source identification premise rather than deriving that premise. Bounded-interval transport through the affine-covariant kernel $g_I(v)$ is discharged by E0.5 below, and reconstruction of a full tensor from the directional charges is subject to the null-invisible metric ambiguity.

Boundary of the null bridge. The null bridge is an explicit theorem branch rather than an automatic fixed-cutoff identity. Its required data are:

- transferred cut-center data alone give the central sector decomposition of the null strip rather than the left/right HJPW split;
- the extra decomposition-inheritance condition of Proposition 5.2a is exactly what upgrades those sectors to the same collar-type block structure used in the spatial branch;
- the fixed-cutoff null-strip bridge is exact for exact Markov strip states on that inherited decomposition, and otherwise uses a controlled limit with the carried defect operator $\mathfrak{D}_J$;
- the exact-Markov/central-interface branch, together with its specified endpoint-to-cell map and uniform norm/site-density bound, gives endpoint-Lipschitz control strong enough to define a weak tail generator for renormalized half-lines; the scaling-limit geometric-cap branch of Theorem 4.2 then derives the half-sided relation on the half-line blow-up net, while common-vector and relative-commutant standardness remain separate receipts;
- with those standardness receipts, the Borchers–Wiesbrock theorem supplies $P_\Omega \geq 0$, its Stone domain, and $K_a = K_0 - 2\pi a P_\Omega$;

- the half-line generator and its endpoint-derivative identity are fixed inside the bridge itself; the local null-stress reading requires the separately supplied effective-locality and same-source bridge, while bounded-interval transport and the later tensor upgrade are downstream.

This is the route from null-strip factorization to the Einstein branch, with the positive half-line generator proved inside the null modular bridge and its physical stress reading conditional on the named EFT bridge.

2.3.3 Conditional modular-energy stress-tensor reading on null half-lines

The effective-theory input here is the standard stress-tensor representation that names the operator fixed by the null bridge. In the canonical normalization of the preceding null-half-line construction, the positive null generator is the endpoint derivative of the renormalized half-line modular family:

$$\langle \psi, P_\Omega \phi \rangle = -\frac{1}{2\pi} \frac{d}{da} \Big|_{a=0^+} \langle \psi, \widetilde{K}_a(\Omega) \phi \rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$.

If that same half-line family admits the supplied effective local-stress description and same-source identification, its local null-stress charge is represented by the identical endpoint derivative,

$$Q_{kk}(\Omega) := -\frac{1}{2\pi} \frac{d}{da} \Big|_{a=0^+} \widetilde{K}_a^{\text{eff}}(\Omega),$$

so on the common quadratic-form domain

$$P_\Omega = Q_{kk}(\Omega).$$

Thus the null generator has the displayed local null-stress reading conditional on an effective description that admits a local stress tensor and identifies the endpoint derivative with the same source family. The algebraic half-line construction alone does not supply that effective-locality or source-identification bridge. In a UV CFT regime, or another local Lorentzian regime with local modular Hamiltonians, the conditional reading is the standard null-stress charge associated with the chosen null generator.

For comparison, in a CFT vacuum on a ball one has the familiar local formula

$$H_\zeta = \int_\Sigma T_{ab} \zeta^b d\Sigma^a,$$

so, under that EFT premise, the conditional identification is the null-half-line version of the same modular-energy locality.

The downstream boundary is narrower. The E0.5 bounded-interval kernel transports this half-line statement to bounded null intervals with affine-covariant kernel $g_I(v)$, and the null-to-tensor upgrade determines T_{ab} only up to the familiar ambiguity

$$T_{ab} \mapsto T_{ab} + \phi g_{ab}.$$

The generator is fixed inside the null bridge; its physical local-stress interpretation remains conditional on the stated EFT bridge.

2.3.4 Localized generalized entropy from Markov + MaxEnt

Using the collar decomposition, exact Markovity, and MSA, the declared central-interface branch has the following edge-aligned Markov normal form. On the nonexact branch, Theorem 2.5 supplies a recovered comparison state and a vanishing CMI envelope only under strong conditional mixing; convergence to an exact Markov state is a fixed-collar statement, and alignment with the displayed edge factors remains a separate MSA condition. Under those conditions MaxEnt selection within each edge sector gives

$$\rho_C = \bigoplus_{\alpha} p_{\alpha} \left(\rho_{\text{bulk},C}^{\alpha} \otimes \frac{\mathbf{1}_{\text{edge}}^{\alpha}}{d_{\alpha}} \right).$$

The entropy splits as

$$S(\rho_C) = H(p_{\alpha}) + \sum_{\alpha} p_{\alpha} S(\rho_{\text{bulk},C}^{\alpha}) + \sum_{\alpha} p_{\alpha} \log d_{\alpha}.$$

Convention: Throughout this paper, "log" denotes the natural logarithm ($\ln$), so entropies are measured in **nats** (1 nat = $1/\ln 2 \approx 1.443$ bits). This is standard in thermodynamics and QFT; the Bekenstein-Hawking formula $S = A/4G$ uses nats. When clarity requires it, we write $\log_2$ explicitly for bits.

Define

$$S_{\text{bulk}}(C) := H(p_{\alpha}) + \sum_{\alpha} p_{\alpha} S(\rho_{\text{bulk},C}^{\alpha}),$$

and the central area operator

$$L_C := \sum_{\alpha} (\log d_{\alpha}) P_{\alpha}.$$

Then

$$S_{\text{gen}}(C) := \text{Tr}(\rho L_C) + S_{\text{bulk}}(C).$$

Newton coupling as an edge-entropy area-law readout.

In the collar double-scaling limit, the edge contribution becomes extensive along the entangling surface $\Sigma = \partial C$:

$$\text{Tr}(\rho L_C) \approx N_{\Sigma} \cdot \bar{\ell}(t), \quad \bar{\ell}(t) := \sum_{\alpha} p_{\alpha} \log d_{\alpha},$$

where N_{Σ} is the number of UV cut elements covering Σ and $\bar{\ell}(t)$ is the **single-cell edge entropy** from the heat-kernel distribution (Theorem 6.20). Similarly, the geometric area is extensive:

$$A(C) \approx N_{\Sigma} \cdot a_{\text{cell}},$$

where a_{cell} is the area per UV cut element in the emergent metric.

Matching these expressions gives the geometric area-law coupling:

$$G_{\text{geom}} = \frac{a_{\text{cell}}}{4 \bar{\ell}(t)}$$

where:

- a_{cell} is fixed operationally from the UV correlation/mixing length ξ via $a_{\text{cell}} \sim \xi^2$ (from the exponential mixing hypothesis),
- $\bar{\ell}(t) = \sum_R p_R(t) \log d_R$ is computed from the heat-kernel edge distribution with $p_R \propto d_R e^{-t\lambda_R}$.

Explicitly:

$$\bar{\ell}(t) = \frac{\sum_R d_R e^{-t\lambda_R} \log d_R}{\sum_R d_R e^{-t\lambda_R}}.$$

On the OPH gravity branch this formula does not back-solve the scale. The selected no- G scale certificate supplies $\gamma_\star = \ell_\star \nu_{\text{Cs}}/c$, equivalently $B_\star = 3\pi/\ell_\star^2$. The local cell readings are

$$a_{\text{cell}} = P\ell_\star^2, \quad \bar{\ell}_{\text{shared}} = P/4,$$

so the Newton area-law readout gives

$$G_{\text{geom}} = \frac{P\ell_\star^2}{4(P/4)} = \ell_\star^2.$$

The SI display is $G_{\text{SI}} = c^3 \ell_\star^2 / \hbar$.

2.3.5 Finite MaxEnt identities and the fixed-cap generalized-entropy stationarity interface

The proved finite MaxEnt envelope identities hold within the declared state family. The calculation does not, by itself, identify a variation of the state constraints with a geometric area variation or prove stationarity of their sum. The Einstein branch therefore assumes the following joint state-geometry interface: for its admissible variations preserving cap labels (fixed size and charges),

$$\delta S_{\text{gen}}(C) = 0.$$

Once this balance is supplied, the split above and the proved finite first law for the bulk term give

$$\delta S_{\text{gen}}(C) = \delta \langle L_C \rangle + \delta \langle K_{\text{bulk}} \rangle.$$

2.3.6 No-smuggling dependency discharge for the Einstein bridge

Theorem E0 (Einstein bridge dependency discharge). Write **E** for the Einstein-branch premise package: the three OPH axioms together with the collar-recovery, edge-normalization, and generalized-entropy interfaces consumed by this branch, each an explicit hypothesis rather than an axiom. Let one source-derived cofinal OPH regulator tower carry the typed common-domain geometry, modular, event, stress, entropy, and scale readouts of Theorem 4.3h, together with its finite, asymptotic, and physical-identification premises. On that branch, **E** and the additional receipts supply the inputs used by the Einstein bridge:

- geometry readout through quotient normal forms via the screen fold $\chi_{S,r} \circ n_r \circ \pi_r$;
- Lorentz/H3 readout from round cap pairs, since $\text{Conf}^+(S^2) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1)$ and $H^3 \simeq \text{SO}^+(3, 1)/\text{SO}(3)$;
- 2π -normalized support-visible BW flow on the extracted geometric subnet;
- the positive null generator on the same branch and, under the separately supplied effective-locality and same-source bridge, its local null-stress charge reading;

- the bounded-interval kernel of Lemma E0.5;
- fixed-cap generalized-entropy stationarity from the separately declared stationarity interface;
- the standard fixed-volume small-ball area identity after geometry readout;
- $o(\ell^4)$ remainder control by Lemma E0.6; and
- all local timelike directions by the cap-pair coverage of Lemma E0.7.

Thus the finite consensus reduct supplies quotient normal forms. The typed branch receipts supply the geometric, modular, stress, entropy, coverage, tail, coupling, vacuum, and scale inputs needed by the Einstein bridge. No source-derived tower carrying all of those inputs is constructed here. Bare finite consensus does not imply the Einstein equation.

Lemma E0.5 (Conditional bounded-interval kernel). On the null blow-up of the support-visible geometric branch, assume the effective-locality and same-source bridge that identifies the proved half-line endpoint derivative with a local null-stress charge. Together with endpoint-Lipschitz renormalized interval control and affine/projective covariance, this gives

$$K_{(a,b)}^{\text{null}} = 2\pi \int_a^b \frac{(v-a)(b-v)}{b-a} T_{kk}(v) dv + R_{(a,b)}.$$

The remainder $R_{(a,b)}$ is the carried endpoint/collar term of the same refinement family. In the small-ball limit this is the ball modular kernel used below.

Lemma E0.6 (Uniform remainder control). Let (r, ℓ_r, δ_r) be one declared cofinal family on which Theorem 2.5 holds uniformly over the event region and cap family, and set

$$\varepsilon_r = c|\partial C_r|_{\text{UV}} e^{-\delta_r/\xi_r}.$$

Assume that the fixed-collar Markov replacements have one modulus $\delta_r^{\text{M}} \leq C_{\text{M}} \varepsilon_r^\theta$, with $C_{\text{M}} < \infty$ and $\theta > 0$, and that for some

$$s > \max\{8, 4/\theta\}, \quad \omega_r \longrightarrow +\infty,$$

the same family satisfies

$$\frac{\delta_r}{\xi_r} - \log(c|\partial C_r|_{\text{UV}}) \geq s \log(\ell_0/\ell_r) + \omega_r.$$

If the interval-kernel residual is also $o(\ell_r^4)$ on this family, then

$$2\sqrt{\varepsilon_r} = o(\ell_r^4), \quad \delta_r^{\text{M}} = o(\ell_r^4),$$

and the endpoint, collar, Markov, and long-wavelength derivative remainders do not alter the Einstein coefficient. Indeed, $\varepsilon_r \leq (\ell_r/\ell_0)^s e^{-\omega_r}$; the two conclusions follow from $s/2 > 4$ and $s\theta > 4$. Mere pointwise convergence of a collar error does not supply this common-family rate, and no per-radius diagonal choice is used.

Lemma E0.7 (Cap-pair timelike coverage). On the cap-normal H^3 branch, every future timelike vector v has $r = \sqrt{-\eta(v, v)}$ and $u = v/r \in H^3$. For any unit $s \in u^\perp$, the vectors

$$q_\pm = u \pm s$$

are future null and

$$v = \frac{r}{2}(q_+ + q_-).$$

Thus normalized future null directions in observer skies cover every local timelike direction once the observer-frame chart is supplied. This does not say that two unnormalized projective rays alone

select a unique observer frame; the observer normalization or equivalent causal-diamond scale data is also required.

Proposition E2 (Conditional recovered-core Ward closure). Suppose the finite tomography hypotheses reconstruct a tensor from one compatible family of null and timelike charges on the recovered core. Assume separately that its same-source local charge family obeys the weak Ward balance: signed fluxes on opposite faces of infinitesimal recovered-core diamonds cancel with a controlled remainder, and this balance converges in the recovered charts to the distributional divergence law. Then $\nabla^a \langle T_{ab} \rangle = 0$ weakly, and pointwise on a smooth branch. Charge compatibility and tensor tomography alone imply neither the face cancellation nor its continuum limit.

2.3.7 Einstein equation from fixed-cap generalized-entropy stationarity

This subsection is a branch theorem. The finite consensus reduct supplies quotient normal forms; Theorem E0 discharges the recovered-core premise package **E** needed by the Einstein bridge: geometry readout, support-visible BW modular covariance, the conditional local null-stress reading under effective-locality and same-source matching, bounded-interval transport, the declared fixed-cap generalized-entropy stationarity interface, the fixed-volume small-ball area variation, carried-remainder control, and the timelike scalar-to-tensor upgrade.

In the $d=4$ scaling regime, the small-ball bridge follows conditionally once the effective-locality, same-source, endpoint-control, and affine/projective hypotheses of E0.5 are supplied. The additional standard geometric input is the fixed-volume area-variation identity for a small geodesic ball. The kernel shape then follows on the extracted geometric subnet: a sufficiently small cap is assumed to admit the tangent causal-diamond description D_ℓ , whose preserving conformal Killing field is

$$\xi_{D_\ell} = \frac{1}{2\ell} \left((\ell^2 - r^2 - t^2) \partial_t - 2t x^i \partial_i \right).$$

On the $t=0$ slice its lapse is

$$\xi_{D_\ell} \cdot n = \frac{\ell^2 - r^2}{2\ell}.$$

The bounded-interval kernel combines the geometric cap generator with the affine/projective transport of the half-line stress bridge, so the cap-modular theorem and the stress bridge together yield

$$\delta S_{\text{bulk}}(C) = 2\pi \int_{B_\ell} \frac{\ell^2 - r^2}{2\ell} \delta \langle T_{00} \rangle d^3x + \delta \langle E_{C,\ell}^{(\eta)} \rangle.$$

If $\delta \langle T_{00} \rangle$ is approximately constant across the ball and the carried remainder is $o(\ell^4)$, then

$$\delta S_{\text{bulk}}(C) = \frac{8\pi^2 \ell^4}{15} \delta \langle T_{00} \rangle + O(\ell^5 \partial T) + o(\ell^4).$$

Stationarity of generalized entropy therefore gives the observer-covariant scalar relation

$$\delta \left[(G_{ab} + \Lambda g_{ab}) u^a u^b \right] = 8\pi G u^a u^b \delta \langle T_{ab} \rangle$$

for the local diamond rest-frame four-velocity u^a . In the adapted rest frame $u^a = (1, 0, 0, 0)$, this is

$$\delta \langle G_{00} + \Lambda g_{00} \rangle = 8\pi G \delta \langle T_{00} \rangle.$$

2.3.8 Overlaps supply all timelike directions

This scalar-to-tensor upgrade is internal once the Lorentz branch is in place. Overlapping observers through the same bulk point realize every local timelike four-velocity u . Define

$$Y_{ab} := G_{ab} + \Lambda g_{ab} - 8\pi G \langle T_{ab} \rangle.$$

The local-diamond rest-frame relation says

$$u^a u^b \delta Y_{ab} = 0$$

for every such u . In local inertial coordinates,

$$u^a = \gamma(1, v^i), \quad |\vec{v}| < 1,$$

so

$$0 = u^a u^b \delta Y_{ab} = \gamma^2 \left(\delta Y_{00} + 2v^i \delta Y_{0i} + v^i v^j \delta Y_{ij} \right)$$

for all $|\vec{v}| < 1$. The polynomial therefore vanishes coefficientwise, hence

$$\delta Y_{00} = 0, \quad \delta Y_{0i} = 0, \quad \delta Y_{ij} = 0.$$

So the full tensor variation vanishes:

$$\delta Y_{ab} = 0.$$

Along any connected scaling branch of reference states, Y_{ab} is therefore constant. The only purely local freedom left by the null reconstruction is the metric term, and fixing that term on one maximally symmetric reference state gives

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle.$$

Thus the tensor upgrade does not require a separate EFT handoff once the geometric-cap theorem, the null bridge, and the overlap-complete timelike family are in place.

2.3.9 Non-tunable numerical constants

The gravity chain yields specific numerical constants as rigid outputs of the axiom chain.

The 2π KMS normalization. From Theorem 4.2, cap modular flow on the support-visible scaling-limit geometric cap pair is fixed by the KMS condition to the standard 2π normalization. This is the same rigidity that fixes Unruh/Hawking temperature normalization.

The geometric coefficient $\Omega_{d-2}/(d^2 - 1)$. This coefficient appears in both (a) the CFT-ball modular Hamiltonian weight integral and (b) the geometric area-variation identity. It is an exact integral identity:

$$\int_{B_\ell^{d-1}} \frac{\ell^2 - r^2}{2\ell} d^{d-1}x = \frac{\Omega_{d-2} \ell^d}{d^2 - 1}.$$

In $d = 4$:

$$\frac{\Omega_2}{4^2 - 1} = \frac{4\pi}{15} \approx 0.8377580409572781.$$

This is the reason prefactors cancel cleanly when going from $\delta S_{\text{gen}} = 0$ to the Einstein equation (leaving $8\pi G$ with the 2π fixed by Theorem 4.2).

What is predicted. The framework cleanly separates:

- **Non-tunable constants:** 2π (KMS period), $\Omega_{d-2}/(d^2 - 1)$ (geometric coefficient), the existence of the Einstein form.
- **Global closure constants:** G is the conversion between edge entropy and geometric area on the emitted local gravity branch. Λ is not fixed by the local reference-state variation data alone; stable correctable-record closure plus identification of the record carrier with de Sitter horizon entropy fixes $\Lambda_* \ell_*^2 = 3\pi/N_*$, while the SI curvature scale also uses the selected OPH scale certificate.

2.3.10 Quantitative Markov error and controlled corrections

The role of this subsection is to keep three distinct quantities separate:

1. the raw collar conditional mutual information

$$\varepsilon_\delta := I(A_\delta : D_\delta \mid B_\delta)_\omega;$$

2. the constructive recovery error

$$r_{\text{FR}}(\varepsilon_\delta) := 2\sqrt{1 - e^{-\varepsilon_\delta}} \leq 2\sqrt{\varepsilon_\delta};$$

3. the exact-Markov replacement error on one fixed faithful collar model,

$$\eta_\delta^{\text{M}} := 4\lambda_*^{-1} \delta_{A_\delta : B_\delta : D_\delta}^{\text{M}}(\varepsilon_\delta),$$

where $\lambda_* > 0$ is the lower spectral bound needed to compare modular Hamiltonians.

The exact identity for the modular defect expectation is unchanged:

$$\langle \Delta K_\delta \rangle_\omega = -I(A_\delta : D_\delta \mid B_\delta)_\omega = -\varepsilon_\delta,$$

with

$$\Delta K_\delta := K_{A_\delta B_\delta D_\delta} - K_{A_\delta B_\delta} - K_{B_\delta D_\delta} + K_{B_\delta}.$$

What changes is the interpretation of later “exact” collar formulas. For each fixed finite-dimensional collar one may choose an exact Markov reference state σ_δ with

$$\|\omega_\delta - \sigma_\delta\|_1 \leq \delta_{A_\delta : B_\delta : D_\delta}^{\text{M}}(\varepsilon_\delta),$$

and a constructive recovered comparison state $\omega_\delta^{\text{rec}}$ with

$$\|\omega_\delta - \omega_\delta^{\text{rec}}\|_1 \leq r_{\text{FR}}(\varepsilon_\delta).$$

These two comparisons do different jobs:

Bounded-observable control. For every bounded observable X supported on $A_\delta \cup B_\delta$,

$$|\text{Tr}[X(\omega_\delta - \omega_\delta^{\text{rec}})]| \leq \|X\|_\infty r_{\text{FR}}(\varepsilon_\delta),$$

and

$$|\text{Tr}[X(\omega_\delta - \sigma_\delta)]| \leq \|X\|_\infty \delta_{A_\delta : B_\delta : D_\delta}^{\text{M}}(\varepsilon_\delta).$$

The first bound is constructive and dimension-free; the second is the fixed-collar route to the exact Markov set.

Modular-additivity control. If the relevant marginals are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\Delta K_\delta(\omega) - \Delta K_\delta(\sigma_\delta)\|_\infty \leq \eta_\delta^M.$$

Since $\Delta K_\delta(\sigma_\delta)$ is central by exact Markovity, the finite-stage modular defect is within η_δ^M of the exact splice or additivity value.

Therefore the manuscript's exact collar identities arise as limits with a carried remainder

$$\eta_\delta := r_{\text{FR}}(\varepsilon_\delta) + \eta_\delta^M,$$

together with the separate long-wavelength derivative remainders present in the small-ball expansion. Exact identities at finite collar width require exact Markovity; otherwise one works with η_δ -controlled approximations and then lets $\eta_\delta \rightarrow 0$ in the controlled collar limit.

Finite-cutoff Einstein remainder bound. In the small-ball rest-frame calculation, write

$$\delta S_C = \frac{8\pi^2 \ell^4}{15} \delta \langle T_{00} \rangle + \delta \langle E_C^{(\delta)} \rangle + O(\ell^5 \partial T),$$

where $E_C^{(\delta)}$ packages the carried collar remainder. Then the declared fixed-cap generalized-entropy stationarity interface, together with the proved finite MaxEnt identities, gives

$$\delta \langle G_{00} + \Lambda g_{00} \rangle = 8\pi G \delta \langle T_{00} \rangle + \mathcal{E}_\delta,$$

with

$$\mathcal{E}_\delta := \frac{15G}{\pi \ell^4} \delta \langle E_C^{(\delta)} \rangle + O(\ell \partial T),$$

and, by the bounds above,

$$|\delta \langle E_C^{(\delta)} \rangle| \leq C_C \eta_\delta$$

for some collar-dependent constant C_C on the fixed faithful model. Equivalently, before the controlled collar and small-ball limits, one may write

$$|\mathcal{E}_{\ell,\delta}| \leq C_1 r_{\text{FR}}(\varepsilon_\delta) + C_2 \delta_{A_\delta:B_\delta:D_\delta}^M(\varepsilon_\delta) + C_3 \eta_\delta^{\text{reg}} + C_4 \ell \|\partial T\| + o_\delta(1) + o_\ell(1),$$

with constants depending only on the fixed collar model, the bounded observable class, and the small-ball chart. Thus the exact Einstein relation is the controlled limit, not a finite-cutoff identity inherited from approximate Markovity.

If one wishes to package this carried remainder as an effective anomalous energy density, the natural definition is

$$\delta \langle T_{00}^{\text{anom}} \rangle := \frac{15}{8\pi^2 \ell^4} \delta \langle E_C^{(\delta)} \rangle.$$

Then

$$\delta \langle G_{00} + \Lambda g_{00} \rangle = 8\pi G \delta \langle T_{00} \rangle + \delta \langle T_{00}^{\text{anom}} \rangle + O(\ell \partial T).$$

This is a bookkeeping definition of the finite-stage remainder, not an upgrade of the recovered-core theorem. Its significance is exactly that it is controlled:

$$|\delta \langle T_{00}^{\text{anom}} \rangle| \leq \frac{15 C_C}{8\pi^2 \ell^4} \eta_\delta.$$

On a separate conditional dark/anomaly branch, this carried scalar remainder may be identified with a repair stress only after a quotient-invariant source-localization receipt identifies the same collar operator with the small-ball Einstein remainder, and after a positive causal packet lift supplies the full tensor. In SI units that continuation has rest-energy normalization

$$T_A^{ab}u_a u_b = \frac{15\hbar c}{8\pi^2 \ell^4} R_{r,\ell} + O(\epsilon_{\text{loc}}),$$

where $R_{r,\ell}$ is measured in nats and ℓ is the proper geodesic small-ball radius. The scalar quantity $I(A : D \mid B)$ alone is not a stress tensor: packet factorization, causal directions, an allocation rule, a rest frame, source-localization map, and transport law are additional continuation data.

The framework carries the Markov error through three distinct controls:

1. $r_{\text{FR}}(\varepsilon_\delta)$ controls bounded observables at one stage;
2. $\delta_{A_\delta:B_\delta:D_\delta}^{\text{M}}(\varepsilon_\delta)$ controls convergence to the exact Markov normal form on a fixed collar;
3. η_δ^{M} controls modular-Hamiltonian replacements once faithfulness is assumed;
4. the Einstein-branch remainder is a carried $O(\eta_\delta)$ term, not a silent exact identity at finite cutoff.

This is the concrete bridge from “axioms about screens” to “precision GR predictions plus a disciplined carried remainder”.

2.3.11 Focusing/QNEC boundary on the null-modular branch

The focusing content does not follow from Axioms 1–3. The two focusing statements on the null-modular branch require the additional hypotheses stated below. Quantum focusing is an optional gravity interface and the Einstein theorem does not consume it.

What the branch supplies. The fixed-cutoff null bridge of §5.2 supplies transferred cut-center data and exact-or-controlled strip additivity on the inherited strip model. The geometric scaling limit supplies the half-sided relation; on the separately registered common-vector and relative-commutant standardness receipts, Borchers–Wiesbrock supplies its translation generator and the affine relation $[K, P] = -i2\pi P$. These are kinematic inputs a QNEC statement consumes; they are not a proof of it or an identification of (P) with a local stress charge.

Why relative-entropy monotonicity does not yield QNEC. Monotonicity under restriction orders the relative entropies $S(\rho_{R(\lambda)} \parallel \omega_{R(\lambda)})$ along nested null cuts; it fixes the sign of the first λ -derivative and says nothing about the second. A monotone function of the cut need not be convex, so the step from monotonicity to $d^2 S_{\text{rel}}/d\lambda^2 \geq 0$ is unavailable. The entanglement first law $\delta S = \delta \langle K \rangle$ is a first-order state variation at fixed algebra and reference state; the QNEC second null shape variation is a different object, and expanding the first law to second order in the cut does not produce it. The published QNEC proofs supply exactly the missing machinery: null-surface locality and analyticity hypotheses in the free and super-renormalizable settings [51], and modular-flow, causality, and defect-OPE inputs in the general QFT setting [53]. None of those hypotheses is verified on the reconstructed patch net.

Why QFC is not derived from QNEC. The Quantum Focusing Conjecture is strictly stronger than QNEC: the foundational QFC statement obtains QNEC as its locally parallel limit [52], and no implication runs the other way. QNEC together with the Einstein branch (Theorem 5.1) and the Raychaudhuri identity controls the diagonal, stationary-cut part of the generalized-expansion variation and leaves the off-diagonal, shear, counterterm, and general nonstationary-cut content of QFC untouched. The optional quantum-focusing interface therefore carries no independent internal support from this branch.

Claim boundary. The null-modular branch supplies the half-sided modular structure but not an internal QNEC proof. Such a proof additionally requires the QFT-side hypotheses named above on the reconstructed net and the off-diagonal QFC content. Any theorem that consumes focusing must name the optional quantum-focusing interface explicitly.

Theorem 5.1 (OPH Einstein-branch closure in the scaling regime). Under the premise package E on a recovered-core, support-visible scaling branch satisfying the E0 dependency discharge above, the fixed-cap generalized-entropy stationarity condition implies the rest-frame first-variation relation

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G \delta\langle T_{00} \rangle.$$

If this relation holds for all local directions and reference states in the locally Lorentzian scaling regime, the timelike scalar-to-tensor theorem yields the semiclassical Einstein equation modulo the expected Λg_{ab} ambiguity. The capacity branch closes that metric term globally under stable whole-fiber correctable-record closure, the unique finite-size slack zero, and the horizon-record identification.

Proof sketch. The recovered-core branch supplies the exact hypothesis package required by the Jacobson-type bridge: quotient-safe geometry readout, 2π -normalized BW cap flow, the conditional same-source local null-stress charge, bounded-interval ball kernel with $o(\ell^4)$ remainder, the declared fixed-cap generalized-entropy stationarity interface, the fixed-volume small-ball area identity, and all timelike directions. The cap first law identifies δS_C with $\delta\langle K_C \rangle$, while the null bridge together with its effective-locality and same-source premise and the bounded-interval kernel relates the bulk modular variation to the required null stress-tensor charge on the diamond. Since the reference state is maximally symmetric, the fixed-volume small-ball area identity enters at first order in the perturbation, and the declared joint state-geometry balance for admissible fixed-cap variations, combined with the proved MaxEnt envelope identities on the cap-label-preserving state family, yields the displayed rest-frame relation. The timelike-coverage lemma supplies all local timelike directions, the scalar-to-tensor theorem upgrades the relation to the tensor equation modulo Λg_{ab} , and the capacity branch closes the remaining metric term under its separately typed stable-capacity and horizon-record premises. QED.

2.3.12 Black-hole structural statements and continuation-level spectroscopy

This subsection separates two result classes. The structural black-hole package is a four-step chain: fixed-cutoff edge-center collar decomposition on the exact-Markov or idealized exact-recoverability horizon carrier, the small-CMI recoverability reading of interior encoding on that same carrier, Hawking/KMS normalization on the geometric modular branch, and the discrete area spectrum together with the leading semiclassical Schwarzschild first-law relation for an actual sector change. Finite same-boundary records and finite reconstruction thresholds do not by themselves supply physical radiation entropy, exterior time, evaporation flux, a geometric island, a radiative quotient, or an asymptotic waveform readout. Any clean comb structure, QNM selector, Page-type linewidth estimate, PBH burst template, Kerr/LIGO horizon spectroscopy template, or Page-curve/island closure is continuation-level and uses continuation inputs stated below.

Derived structural statements.

Area eigenvalues from edge sectors. The central area operator (Section 5.4) is

$$L_C = \sum_{\alpha} (\log d_{\alpha}) P_{\alpha},$$

where $d_\alpha \in \mathbb{N}$ is the dimension of the edge Hilbert space in sector α . With the normalization $\text{Tr}(\rho L_C) = \langle A \rangle / 4G_{\text{geom}}$, the area eigenvalues are

$$A_\alpha = 4G_{\text{geom}} \log d_\alpha = 4\ell_\star^2 \ln d_\alpha,$$

where $\ell_\star^2 = \hbar G_{\text{SI}} / c^3$ is the scale-readout area displayed as the Planck area. Since d_α is a positive integer, **areas are discretely spaced** with logarithmic gaps.

Hawking emission energy quantization. For a Schwarzschild black hole with $A(M) = 16\pi G^2 M^2 / c^4$, define the signed change of the hole for a transition from d_{before} to d_{after} by

$$\Delta_{\text{BH}} A = 4\ell_\star^2 \ln(d_{\text{after}} / d_{\text{before}}).$$

The differential first law, evaluated on the initial background, gives the leading semiclassical relation

$$\Delta_{\text{BH}}(Mc^2) = k_B T_H \ln(d_{\text{after}} / d_{\text{before}})$$

with $T_H = \hbar c^3 / (8\pi G k_B M)$, whose 2π normalization is fixed by the BW_{S^2} tangent-limit normalization of Theorem 4.2. This is not an exact finite-difference identity: using the exact Schwarzschild mass–area relation, an emission with entropy loss $\ln k$ has $E_{\text{emit}} = Mc^2 [1 - \sqrt{1 - \ln k / S_{\text{BH}}}]$, whose expansion is $k_B T_H \ln k [1 + O(\ln k / S_{\text{BH}})]$.

Continuation-level spectroscopy templates. The remainder of this subsection is not part of the recovered core. It records what follows only if one adds extra discrete-horizon selection rules and, where stated, standard semiclassical inputs such as evaporation-power models, QNM/transition identifications, or greybody matching.

Integer transitions. If one imports the additional rule that dominant emission steps reduce the edge dimension by an integer factor k , then $d_{\text{before}} = k d_{\text{after}}$, so k must divide the initial dimension. The signed black-hole entropy change is $-\ln k$, while the positive entropy loss is $\ln k$. To leading semiclassical order the emitted spectrum becomes a discrete comb:

$$\Delta E_k = k_B T_H \ln k, \quad \Delta f_k = \frac{c^3}{16\pi^2 G M} \ln k.$$

Structural condition: comb vs. generic discreteness. The log-integer *comb* structure requires the imported dynamical assumption that such integer-ratio emission steps dominate. A fixed d_{before} admits only its divisors; the displayed all- k ladder is a union of candidate continuations, not a source-produced transition set. If generic transitions between arbitrary integers dominate instead, the set of $|\ln(d_{\text{after}} / d_{\text{before}})|$ values becomes a dense log-rational set that may appear quasi-continuous after folding in linewidths and astrophysical effects. What the recovered finite edge-sector structure supplies, under the stated generalized-entropy and horizon-area dictionaries, is the *discrete area spectrum*; the clean comb pattern follows only with that selection rule.

Continuation-level template (Discrete Hawking spectrum). If the additional integer-transition selection rule is realized, the Hawking emission spectrum consists of discrete lines with spacing $\Delta E_k = k_B T_H \ln k$, where k is an integer characterizing the dominant sector transitions, instead of a continuous thermal profile.

Mass-independent linewidth-to-spacing ratio (continuation-level estimate). Using Page's semiclassical calculation for emission power $P(M) = p_0 \hbar c^6 / (G^2 M^2)$ with $p_0 \approx 2 \times 10^{-4}$, the emission rate is $\dot{N} \approx P / \langle E \rangle$ where $\langle E \rangle = a k_B T_H$ with $a \sim \mathcal{O}(1-10)$. The natural linewidth $\Gamma \sim \hbar \dot{N}$ divided by the level spacing gives:

$$\frac{\Gamma}{\Delta E_k} \approx \frac{64\pi^2 p_0}{a \ln k},$$

a function of (a, k) . At $k = 2$ the declared range $a \in [1, 10]$ spans 1.8% to 18%; at $k = 3$, 1.1% to 11%. A few-percent value therefore requires the particle content and greybody model that pin a inside a narrower interval (for instance $a \approx 4\text{--}6$ at $k = 2$); no such model is selected here. The fraction is mass-independent at every (a, k) .

Connection to quasinormal modes (interpretive continuation). The highly-damped Schwarzschild quasinormal modes have asymptotic real part (Motl, 2002):

$$\text{Re } \omega \rightarrow \frac{c^3}{8\pi GM} \ln 3.$$

This matches exactly the $k = 3$ transition frequency $\Delta E_3/\hbar$. If one adopts a Bohr-type identification between quantum transition frequencies and asymptotic QNM frequencies, this selects

$$\Delta A = 4\ell_\star^2 \ln 3 \approx 4.39 \ell_\star^2$$

as the fundamental area quantum.

Scope statement. The area quantization follows from the edge-sector structure (derived). The $k = 3$ selection requires the additional interpretive identification with QNM frequencies (not derived from axioms). The linewidth prediction uses standard semiclassical inputs.

Numerical examples. For $\Delta f_k = (c^3/16\pi^2 GM) \ln k$:

- **M = 30 M_⊙:** $k=2$ at 29.7 Hz, $k=3$ at 47.1 Hz
- **M = 1 M_⊙:** $k=2$ at 891 Hz, $k=3$ at 1412 Hz
- **M = 10¹² kg (primordial):** $k=2$ at 7.3 MeV, $k=3$ at 11.6 MeV

Within the integer-transition continuation, these frequencies track $k_B T_H \ln k$ exactly and are in principle distinguishable from a continuous thermal spectrum.

Continuation-level PBH burst search template.

Within the discrete-horizon continuation, the Hawking comb would provide a distinctive gamma-ray template. A template-level discriminant is **log-integer energy ratios**: if two emission lines are observed at energies E_2 and E_3 , their ratio must satisfy

$$\frac{E_3}{E_2} = \frac{\ln 3}{\ln 2} \approx 1.585$$

exactly, independent of black hole mass. Within that continuation template, the ratio is fixed once the log-integer rule is assumed.

Available instruments and energy coverage. The $k = 2$ line energy $E_2 = k_B T_H \ln 2$ determines which instruments can see a given BH mass:

Instrument	Energy band	BH mass range ($k=2$ in band)
Fermi GBM (BGO)	0.15–40 MeV	2×10^{11} – 5×10^{13} kg
Fermi LAT	0.1–300 GeV	2×10^7 – 7×10^{10} kg
H.E.S.S.	0.1–100 TeV	7×10^4 – 7×10^7 kg
LHAASO-WCDA	1–15 TeV	5×10^5 – 7×10^6 kg

Detector resolution vs. intrinsic linewidth. The continuation-level Schwarzschild linewidth-to-spacing estimate is $\Gamma/\Delta E_k = 64\pi^2 p_0/(a \ln k)$, spanning 1.8%–18% at $k = 2$ over the declared $a \in [1, 10]$ (mass-independent at every (a, k)). Published detector energy resolutions:

- Fermi GBM: $< 10\%$ (0.1–1 MeV), $\sim 4\%$ at 10 MeV (BGO)
- Fermi LAT: $< 10\%$ (1–100 GeV)
- H.E.S.S.: $\sim 15\%$ (TeV)
- LHAASO-WCDA: $\sim 33\%$ (TeV)

Within this template, the comb could be resolvable with GBM/LAT; at TeV energies it would appear as moderately broad bumps rather than sharp lines.

Search protocol. A dedicated OPH-comb search would:

1. Select burst-like candidates (10–120 s time windows, matching existing PBH burst search protocols).
2. Fit each candidate with null model (smooth continuum) vs. OPH comb model (peaks at $E_k = E_0 \ln k$ convolved with detector response).
3. Scan over the single scale parameter $E_0 = k_B T_H$ (equivalently, BH mass).
4. Require at least two lines satisfying log-integer ratio to claim detection.
5. Correct significance for trials (time windows $\times$ sky positions $\times$ E_0 scan).

Observational context. Dedicated PBH burst searches (H.E.S.S., LHAASO) report **no significant bursts**. An OPH-specific comb-template analysis of archival data would:

- Set upper limits on OPH-comb PBH burst rates
- Demonstrate direct testability of the discrete spectrum prediction
- Provide constraints comparable to or stronger than generic PBH burst limits

Data availability. Fermi GBM provides public Time-Tagged Event (TTE) burst data; Fermi LAT provides public photon event lists with documented analysis workflows. H.E.S.S. has a small public test data release.

Continuation-level GW horizon spectroscopy template for Kerr remnants.

The same continuation-level discrete-horizon branch extends to gravitational wave observables. For Kerr black holes, the thermodynamic first law fixes the leading small-transition relation. For emission the hole changes by $\delta E_{\text{BH}} = -\hbar\omega$ and $\delta J_{\text{BH}} = -\hbar m$, while the imported integer-division rule has $\delta S_{\text{BH}} = -\ln k$. Hence

$$\hbar(\omega - m\Omega_H) = k_B T_H \ln k, \quad k \in \{2, 3, 4, \dots\}$$

up to finite-step corrections from the changing Kerr background. Under those additional discrete-horizon assumptions, this gives the **GW horizon spectroscopy comb**: a continuation-level set of candidate resonant frequencies. The integer-division selection and its all- k union are imported, not selected by the edge-sector spectrum alone. A physical QNM or ringdown claim needs a separate exterior-background bridge, a radiative quotient, a finite operator converging to the continuum perturbation generator, future-horizon and future-null-infinity boundary conditions, and an asymptotic readout map. A finite repair spectrum or fitted clock scale is not that bridge.

Kerr line frequencies. For a remnant with mass M and dimensionless spin $\chi = a_*/M$, define the spin correction factor:

$$g(\chi) = \frac{2\sqrt{1-\chi^2}}{1+\sqrt{1-\chi^2}}, \quad \Omega_H(M, \chi) = \frac{c^3}{2GM} \cdot \frac{\chi}{1+\sqrt{1-\chi^2}}.$$

The line frequencies are:

$$f_{k,m}(M, \chi) = \frac{m\Omega_H(M, \chi)}{2\pi} + \frac{c^3}{16\pi^2 GM} g(\chi) \ln k$$

Within this continuation template, once LIGO/Virgo infers (M, χ) for a remnant, the line pattern is fixed by the inferred remnant parameters and the assumed discrete-horizon rule. The line pattern belongs to a continuation template outside recovered core.

Within-line KMS response, not a cross-line prior. In this continuation template, the GR greybody absorption response may be combined with a declared bin width $\Delta\omega_k \approx \omega_T \ln(1 + 1/k)$, where $\omega_T = k_B T_H / \hbar$. KMS detailed balance then supplies the net absorption-minus-stimulated-emission factor at each candidate frequency:

$$W_{k,\ell m}^{\text{net}} = \Gamma_{\ell m}^{\text{GR}}(\omega_{k,m}) \cdot \Delta\omega_k \cdot \frac{k-1}{k}$$

where $\Gamma_{\ell m}^{\text{GR}}$ is the standard GR greybody factor and the $(k-1)/k$ factor follows from KMS detailed balance with $e^{(\omega - m\Omega_H)/T_H} = k$. This factor does not determine transition rates across different k , and it cannot be normalized into a cross- k prior without an additional rate model.

Universal stacking coordinate. Define the dimensionless rescaled frequency:

$$x := \frac{GM}{c^3 g(\chi)} (\omega - m\Omega_H).$$

Then the predicted line locations collapse to universal constants:

$$x_k = \frac{\ln k}{8\pi} \quad (k = 2, 3, 4, \dots)$$

Numerically: $x_2 = 0.02758$, $x_3 = 0.04371$, $x_4 = 0.05516$, $x_5 = 0.06404$.

Stacking test. Multiple BBH events can be mapped to this universal x coordinate and stacked only after fixing a common frame. Observed hertz must use $M_{\text{det}} = (1+z)M_{\text{source}}$ in both Ω_H and the thermal offset. Under a prospectively normalized comb submodel, its peaks then align across events with different (M, χ) ; detector noise need not stack coherently.

Comparison to existing work. Prior area-quantization searches [79] used parameterized models with one free spacing constant. The imported continuation considered here has multiple candidate lines with exact $\ln k$ ratios. The $(k-1)/k$ detailed-balance factor is a within-line net-response factor, not a derived hierarchy of transition probabilities.

Numerical example (GW170608). Remnant parameters: $M_f \approx 18.0M_\odot$, $\chi_f \approx 0.69$. For $m = 2$, the horizon rotation frequency is $m\Omega_H/(2\pi) \approx 719$ Hz. The **thermal comb spacing** (the part that encodes the area quantization) is:

k	$\Delta f_k := \frac{c^3 g(\chi)}{16\pi^2 GM} \ln k$ (Hz)	KMS net-response factor ($k-1$)/ k
2	41.6	0.500
3	65.9	0.667
4	83.2	0.750
5	96.5	0.800

k	$\Delta f_k := \frac{c^3 g(\chi)}{16\pi^2 GM} \ln k$ (Hz)	KMS net-response factor ($k - 1$)/ k
6	107.5	0.833

The displayed values are source-frame frequencies. Detector-frame frequencies are lower by $1+z$, equivalently obtained by replacing M_f with $M_{\text{det}} = (1+z)M_f$ throughout. A stacking analysis uses the rescaled coordinate $x = GM(\omega - m\Omega_H)/(c^3 g(\chi))$ with all quantities in one frame, which maps the thermal spacing to universal constants $x_k = \ln k/(8\pi)$ regardless of the rotation offset.

Template-matching criterion. After rescaling by (M, χ) , spectral features in this continuation template must satisfy the offset-subtracted ratio

$$\frac{f_{k,m} - m\Omega_H/(2\pi)}{f_{2,m} - m\Omega_H/(2\pi)} = \frac{\ln k}{\ln 2}$$

exactly, equivalently $x_k/x_2 = \ln k/\ln 2$ in the universal coordinate defined above, independent of a common scale and offset. The full frequencies $f_{k,m}$ do not satisfy $f_k/f_2 = \ln k/\ln 2$ at nonzero Ω_H : the additive rotation offset breaks the ratio, and only the thermal comb pieces carry it. A null result can challenge only the finite support or normalized prior fixed before comparison. It cannot exclude the unrestricted integer family merely by ruling out $2 \leq k \leq 12$, and it does not by itself test the derived area-spectrum statement.

2.3.13 Classical mechanics on the conditional GR branch

If the physical source-causal continuum, curvature/stress, scale, and convergence premises promote the finite Einstein-form identity to the standard smooth Einstein equation, its ordinary GR consequences apply. This section records the textbook dust and weak-field reductions on that conditional branch; it is not an additional derivation from the informational order.

Stress-energy conservation is automatic. The contracted Bianchi identity is geometric:

$$\nabla^a G_{ab} = 0.$$

Combined with the Einstein equation, this implies:

$$\nabla^a \langle T_{ab} \rangle = 0.$$

Geodesic motion from dust limit. For pressureless classical matter ("dust"), $T^{ab} = \rho u^a u^b$. Conservation yields:

$$\nabla_a(\rho u^a u^b) = 0 \quad \Rightarrow \quad u^b \nabla_a(\rho u^a) + \rho u^a \nabla_a u^b = 0.$$

Projecting orthogonally to u^b using $h^b_c = \delta^b_c + u^b u_c$ kills the first term, giving:

$$\rho u^a \nabla_a u^b = 0 \quad \Rightarrow \quad u^a \nabla_a u^b = 0.$$

This is the geodesic equation: free classical bodies follow spacetime geodesics.

Newtonian limit from weak-field GR. Take the weak-field, slow-motion limit with metric:

$$g_{00} \approx -(1 + 2\Phi/c^2), \quad g_{0i} \approx 0, \quad g_{ij} \approx \delta_{ij}(1 - 2\Phi/c^2),$$

and velocities $|\mathbf{v}| \ll c$. Then $G_{00} \approx 2\nabla^2 \Phi/c^2$ (leading order), and $T_{00} \approx \rho c^2$. The Einstein equation reduces to:

$$\nabla^2\Phi = 4\pi G\rho.$$

Geodesic motion reduces to:

$$\ddot{\mathbf{x}} = -\nabla\Phi.$$

These are Newton's gravitational law and Newton's second law. They are controlled limits of the supplied smooth GR branch, rather than consequences of the finite provenance poset alone.

One finite piece of this limit is machine-checked on the committed carrier. On the three-dimensional completion of the exact record frame, the ball volume scales as radius to the carrier dimension by the generic finite-dimensional volume theorem, the shell content at positive radius is $4\pi r^2$ with the exponent equal to the carrier dimension minus one, and any declared spherically symmetric strength profile whose product with the shell content is a constant charge equals $Q/(4\pi r^2)$ exactly, with $f(r_1)r_1^2 = f(r_2)r_2^2$ shell-independent (`Lean/Geometry/InverseSquareShellLaw.lean`). The inverse-square exponent is therefore derived from the committed carrier dimension rather than assumed, while the radial readout and the shell-flux normalization are declared premises on the audited register. No join of this shell law to the composed Einstein branch is supplied because the screen-carrier-to-physical-position attachment is absent. The weak-field reduction above otherwise remains a standard prose derivation without a finite receipt.

This shell law lands on the rest space of every declared observer frame. For each future unit-timelike frame of the declared Hermitian Lorentz module, an explicit boost-shear map is a machine-checked linear equivalence from the completion carrier onto the frame's Lorentz-orthogonal rest space, and the pullback of the positive-definite rest metric along it is exactly the Euclidean coordinate form; at the standard frame the map reduces to the candidate bridge that places the source Gram quotient in the standard rest fiber (`Lean/Geometry/RestFiberShellTransport.lean`). Under the same two declared premises the radial readout pulled back through the equivalence is the carrier norm, the strength at a nonzero rest vector is $Q/(4\pi r^2)$ with r the rest radius, the exponent reads as the rest-space dimension minus one with the dimension proved through the transport equivalence, and every oriented Lorentz chart transport carries the strength field exactly. One composed identity carries the screen chain into the fiber: the finite source Gram of two integer record controls equals the rest metric of their transported difference at every declared frame. The frame is a declared choice and no uniqueness of the identification is claimed. No attachment of the rest fiber to physical position or corresponding Einstein-branch join is supplied.

Precision classical predictions. On a branch that identifies the finite tensor fields with smooth physical curvature and stress and supplies the required source-causal continuum, scale, and curvature-convergence receipts, the standard GR precision toolbox applies (post-Newtonian expansion, lensing, time delay, etc.). These are inherited consequences of that physical identification, not independent predictions of the informational order.

Selected precision predictions (in the regime where the GR derivation applies):

Light bending by mass M : For impact parameter b ,

$$\Delta\theta = \frac{4GM}{c^2b}.$$

For the Sun with $b \approx R_\odot$: $\Delta\theta \approx 1.751$ arcsec.

Mercury perihelion advance: Per orbit,

$$\Delta\varpi = \frac{6\pi GM}{a(1-e^2)c^2}.$$

Using Mercury's orbital parameters: $\Delta\varpi \approx 42.98$ arcsec/century.

Gravitational redshift: Between two radii in a static potential,

$$\frac{\Delta\nu}{\nu} \approx \frac{\Delta\Phi}{c^2}.$$

For the Sun (surface to infinity): $z \approx 2.12 \times 10^{-6}$.

These predictions are fixed functions of G and known source parameters, and are confirmed observationally to high precision. The framework contains them automatically on the derived Einstein branch.

2.3.14 Precision gravity predictions and experimental bounds

The pure Einstein linearization makes classical dispersion and polarization statements that can be confronted with the tightest available experimental bounds. This section translates those branch-scoped statements into the observables that experiments constrain; it does not infer a quantum graviton pole from the classical equation.

Speed of gravitational waves. On a suitable background, the two transverse-traceless classical modes of the pure Einstein linearization propagate on the same invariant null cone as the Maxwell modes:

$$\frac{c_{\text{GW}} - c_\star}{c_\star} = 0 \text{ exactly on this action/background branch.}$$

Published bound (GW170817 + GRB 170817A multi-messenger):

$$-3 \times 10^{-15} < \frac{c_{\text{GW}} - c}{c} < +7 \times 10^{-16} \quad (90\% \text{ credibility}).$$

For a source at ~ 40 Mpc, this fractional difference corresponds to only a few seconds of propagation-time mismatch across $\sim 10^8$ years of travel.

Einstein quadratic mass parameter. On the additional pure two-derivative Einstein–Hilbert action branch, the transverse-traceless quadratic operator has no Fierz–Pauli hard-mass parameter:

$$\mu_{\text{FP,hard}}^2 = 0.$$

This is an action-content statement, not a graviton-particle mass prediction from diffeomorphism redundancy alone. A quantum mass or pole claim requires Definition 6.18, while diffeomorphism-invariant Stueckelberg/bimetric completions and extra-field theories lie outside the pure-Einstein field-content branch.

Published bound (GW dispersion analysis, PDG 2025):

$$m_{\text{grav,disp}} \leq 1.76 \times 10^{-23} \text{ eV}/c^2 \quad (90\% \text{ credibility; model-dependent dispersion fit}).$$

This corresponds to a reduced Compton wavelength $\bar{\lambda}_C \gtrsim 1.6 \times 10^{16}$ m, i.e., order ~ 1.6 light-years.

No dipole radiation. Many modified gravity theories predict extra channels (scalar/vector) producing dipolar radiation at $(-1)\text{PN}$ order. The derived GR limit predicts no such channel.

Published bound (GW170817 inspiral phasing, PDG 2025):

$$-4 \times 10^{-6} < \delta\hat{p}_{-2} < 2 \times 10^{-5} \quad (90\% \text{ credibility}).$$

Tensor polarizations on the selected field-content branch. The pure Einstein linearization contains two tensor polarizations. It does not prove that every broader diffeomorphism-invariant EFT lacks additional scalar, vector, or massive modes. Pure non-tensor hypotheses are disfavored by observational constraints, and mixed tensor-scalar/vector models are tightly constrained.

Equivalence principle tests. Additional null checks from the derived GR structure:

- Universality of free fall (space tests): precision $\sim 10^{-15}$
- Nordtvedt parameter ($\eta = 4\beta - \gamma$): $(0.47 \pm 0.55) \times 10^{-4}$
- Binary pulsar radiative damping (PSR J0737-3039): 0.999963 ± 0.000063

2.3.15 State–observable error propagation and the dynamical bridge

On the additional pure-Einstein action/background branch, the framework states the classical equalities recorded in Section 2.3.14. Finite Markov/recovery control supplies a separate and more limited quantitative statement. Two error classes must remain separate. The *state–observable error class* concerns expectation values of specified bounded local observables in a recovered state; the Markov/recovery machinery controls this class. The *dynamical-parameter error class* concerns spectral and dispersion properties of an evolution generator: propagation speeds, particle-pole masses, and correlator poles. The trace-distance chain alone does *not* control those quantities.

The key quantitative hook. From Theorem 3.1, if the target state satisfies

$$I(A_k : C_k \mid B_k) \leq \varepsilon_k,$$

then, writing ρ for the target state and σ for its recovered comparison state, the recovery map gives the trace-norm bound

$$\|\rho - \sigma\|_1 \leq \delta_k := 2\sqrt{1 - e^{-\varepsilon_k}} \leq 2\sqrt{\varepsilon_k}.$$

State–observable bound. Trace distance gives immediate bounds on expectation-value errors for the declared operator class: bounded operators supported in the recovered region, compared at the time of recovery. Using the standard dual norm inequality:

$$|\langle O \rangle_\rho - \langle O \rangle_\sigma| \leq \|O\|_\infty \|\rho - \sigma\|_1 = 2\|O\|_\infty D(\rho, \sigma),$$

where $D(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$ is the trace distance. This inequality is exactly as strong as it looks and no stronger: the norm $\|O\|_\infty$ is the Lipschitz constant only for bounded O , and the comparison is an equal-time statement about states.

Exponential decay from the mixing hypothesis. Writing w for collar width, so it is not confused with the recovery error δ_k , the mixing assumption (Section 2.3) provides

$$I_\omega(A_w : D_w \mid B_w) \leq C_{\text{mix}} |\partial C|_{\text{UV}} e^{-w/\xi}.$$

Combining these gives an explicit precision dial for bounded local observables:

$$\delta_{\text{step}} \lesssim 2\sqrt{C_{\text{mix}} |\partial C|_{\text{UV}}} \cdot e^{-w/(2\xi)}.$$

For an operator normalized by $\|O\|_\infty \leq 1$, an independently chosen absolute expectation-error target τ_O is guaranteed at one step if $\delta_{\text{step}} \leq \tau_O$; the theorem therefore gives the sufficient CMI condition $\varepsilon \leq (\tau_O/2)^2$. The target τ_O applies only to the state–observable bound and cannot be set by importing a gravitational-wave speed tolerance.

What the dial does not certify: dynamical parameters. The gravitational-wave propagation speed and any graviton pole or dispersion-mass parameter are not bounded local observables. They are spectral/dispersion properties of the dynamical generator (poles of retarded correlators, or long-time wave-packet phase and group velocities), so the trace-distance inequality supplies neither their norm nor their Lipschitz constant. Two states can be close in trace distance at one time while being vacua of generators with different dispersion relations, and a small equal-time expectation error does not control phase accumulation over the $\sim 10^8$ -year propagation baseline of GW170817. Consequently, $\delta \lesssim 10^{-15}$ is *not* a theory-side certificate for the multimessenger bound $|c_{\text{GW}}/c - 1| \lesssim 10^{-15}$ of Section 2.3.14, and the recovery error must not be presented as the precision dial for that published bound. The ideal classical relation $c_{\text{GW}} = c_\star$ and the vanishing pure-Einstein hard parameter $\mu_{\text{FP,hard}}^2 = 0$ belong to the additional action-level branch recorded there; neither stability away from that branch nor a quantum graviton pole follows from the trace-distance chain.

Required dynamical-parameter bridge. A sufficient conditional theorem can be stated precisely, but its hypotheses have not been derived from the Markov remainder in this paper. Let $\mathcal{B} = [k_{\min}, k_{\max}]$, with $0 < k_{\min} < k_{\max}$, be the declared experimental wave-number band. For each tensor helicity, assume a C^1 self-adjoint finite-regulator generator $H_\nu^{\text{TT}}(k)$ and the GR reference generator $H_0^{\text{TT}}(k)$, expressed in angular-frequency units ($\hbar = 1$) on one specified common finite-dimensional helicity fiber or after a declared C^1 unitary identification of the two fibers. Assume that the reference TT tensor-mode eigenvalue $\omega_0(k) = c_\star k$ is simple in the helicity block and is separated from the rest of the spectrum by a uniform gap $g_{\mathcal{B}} > 0$. Define

$$M_{\mathcal{B}} := \frac{1}{c_\star} \sup_{k \in \mathcal{B}} \left\| \partial_k H_0^{\text{TT}}(k) \right\|_\infty$$

and the dimensionless dynamical error

$$\eta_{\text{dyn},\nu} := \max \left\{ \frac{1}{g_{\mathcal{B}}} \sup_{k \in \mathcal{B}} \left\| H_\nu^{\text{TT}}(k) - H_0^{\text{TT}}(k) \right\|_\infty, \frac{1}{c_\star} \sup_{k \in \mathcal{B}} \left\| \partial_k H_\nu^{\text{TT}}(k) - \partial_k H_0^{\text{TT}}(k) \right\|_\infty \right\}.$$

For $\eta_{\text{dyn},\nu} < 1/4$, let $\omega_\nu(k)$ be the continued isolated eigenvalue and define its group velocity by $c_{\text{GW},\nu}(k) := \partial_k \omega_\nu(k)$. The standard isolated-eigenvalue projector estimate and the Hellmann–Feynman formula then give, uniformly over $\mathcal{B}$,

$$\sup_{k \in \mathcal{B}} \left| \frac{c_{\text{GW},\nu}(k)}{c_\star} - 1 \right| \leq L_{\mathcal{B}} \eta_{\text{dyn},\nu}, \quad L_{\mathcal{B}} := 1 + 4M_{\mathcal{B}}.$$

Indeed, if $P_\nu(k)$ and $P_0(k)$ are the rank-one spectral projectors, then $\|P_\nu - P_0\|_\infty \leq 2\eta_{\text{dyn},\nu}$, hence $\|P_\nu - P_0\|_1 \leq 4\eta_{\text{dyn},\nu}$. Differentiating the isolated eigenvalue therefore bounds the group-velocity change by the derivative perturbation plus $4M_{\mathcal{B}} c \eta_{\text{dyn},\nu}$. This displays a dimensionless constant derived from a specified generator and uniform over the declared band. If $q := L_{\mathcal{B}} \eta_{\text{dyn},\nu} < 1$, then stationary propagation at fixed group velocity over path length D obeys the group-delay bound

$$|\Delta t| \leq \frac{D}{c_\star} \frac{q}{1 - q}.$$

The same bound applies to varying propagation only if it holds pointwise along the entire path; on a cosmological path, the redshifted wave number must remain inside $\mathcal{B}$. A wave-packet phase bound likewise requires uniform control of the inverse dispersion $k_\nu(\omega)$ on the corresponding frequency band. A model-dependent particle dispersion-mass bound is a separate spectral estimate at $k = 0$; it is not a corollary of the band-limited speed estimate.

The OPH-specific bridge requires three additional conditions:

1. H_ν^{TT} , or an equivalent retarded two-point kernel, is derived from the finite and scaling OPH dynamics;
2. a norm-resolvent, quadratic-form, or correlator estimate has the form $\eta_{\text{dyn},\nu} \leq C_{\mathcal{B}} r_N$, where $r_N := \min(2, \sum_{k=2}^N \delta_k)$ is the accumulated trace-norm recovery bound of Theorem 3.1 for the associated N -step construction and $C_{\mathcal{B}}$ is derived rather than assumed; and
3. the band-uniform dispersion estimate applies to the detector band and propagation baseline, including phase accumulation where used by the likelihood.

Only after the second item is proved may one combine the estimates to obtain

$$\sup_{k \in \mathcal{B}} \left| \frac{c_{\text{GW},\nu}(k)}{c} - 1 \right| \leq L_{\mathcal{B}} C_{\mathcal{B}} r_N.$$

No estimate $\eta_{\text{dyn},\nu} \leq C_{\mathcal{B}} r_N$ is proved here. Substituting δ_k , δ_{step} , or any equal-time trace-distance error for $\eta_{\text{dyn},\nu}$ without that bridge is invalid.

Precision summary. The framework provides:

1. The ideal pure-Einstein linearization statements $\mu_{\text{FP,hard}}^2 = 0$ and $c_{\text{GW}} = c_\star$, without a quantum graviton-pole conclusion.
2. Translation of those ideal equalities into the specific observables experiments constrain (Section 2.3.14).
3. Explicit bounds on how far expectation values of bounded local observables in the recovered state can drift, using the conditional mutual information $\rightarrow$ trace distance $\rightarrow$ observable error chain.

It therefore provides quantitative error control on recovered states and bounded observables. No theory-side certificate is supplied for the published GW-speed or model-dependent particle dispersion-mass bounds, and no derived dynamical-parameter certificate is supplied.

2.3.16 Dark-sector response from the modular anomaly

The conditional dark-sector continuation maps scalar repair occupation to a canonical pair (n, θ) . Its lattice action is a proposed dynamical completion rather than a recovered-core theorem.

The candidate anomaly-tensor slot T_{ab}^{anom} is available only after the finite covariant source packet of Section 5.9 supplies source localization, CMI-to-modular-source matching when that route is used, rank-two reconstruction, conservation, normalization, and universal coupling. On that packet it supplies one structural ingredient for a possible dark-sector continuation without introducing additional particle species. Raw scalar collar CMI supplies only a recovery diagnostic.

Conditional source identification. Once the source packet has identified an anomalous local modular energy, its rest-frame normalization is

$$\langle T_{00}^{\text{anom}} \rangle = \frac{15}{8\pi^2} \cdot \frac{\delta \langle K_C^{(\text{anom})} \rangle}{\ell^4}$$

The quantity can be called dark only on a branch whose packet also establishes gravitational coupling and absence of an electromagnetic coupling. These properties motivate the dark-sector interpretation; neither they nor the observational identification follow from the collar-CMI theorem.

Conditional repair-charge action. The phase equation supplies a conserved repair current with explicit boundary flux. Its dilute homogeneous phase has $\rho_R \propto a^{-3}$, while the cubic condensed link energy gives $a_R = \sqrt{a_b a_0}$ and $v^4 = G M_b a_0$ on the spherical deep branch. The same action

couples coherent matter through $q_\star \chi_\nu^{\text{can}} S_{\text{coh}}^{\text{can}}$ and carries the field stress required for momentum closure.

Scope. The canonical pair, compact-phase completion, baryonic source map, dimensional couplings, full constitutive law, relativistic refinement limit, abundance, and physical likelihoods are not supplied. A compact neutral coherent source has no monopole and therefore cannot close the declared single-monopole source ansatz. Bound neutral composites, dipole or higher-multipole sources, and topological or solitonic architectures are not excluded.

Cosmology/Boltzmann contract. The conditional background scaling above is not an FLRW perturbation kernel. A cosmological dark/anomaly claim must instead expose the variables needed by an Einstein–Boltzmann implementation:

$$\bar{\rho}_A(a), \quad \bar{\rho}_{A,\text{eq}}(a), \quad w_A(a), \quad c_{s,A}^2(k,a), \quad \sigma_A(k,a), \quad Q_A^\mu, \quad B_A(k,a), \quad \Gamma_{\text{rec}}(k,a).$$

The first, source-side no-go is that $\bar{\rho}_A(a)$, $\bar{\rho}_{A,\text{eq}}(a)$, and $B_A(k,a)$ alone do not define a physical Boltzmann source. A physical source must be emitted by a finite covariant collar-packet parent

$$\mathcal{P}_r[X, g] = (C_r, Z_r, A_r, R_r, G_r, \pi_r[X], L_r[X], Q_r, D_r),$$

whose finite packet states split the anomaly and any repair recipient, whose repair generator and re-action channels close the total stress tensor, and whose causal response data set the pressure, sound speed, anisotropic stress, exchange current, and repair rate. The current registered Boltzmann contract accepts either of two non-conflated local source routes: fixed-reference anomalous modular energy, or nonlinear CMI stress with a CMI-to-modular-source matching theorem or hypothesis and explicit residual bounds. A hybrid or different mechanism lies outside those two routes and must provide its own typed source contract and closure receipts. Raw collar CMI is otherwise a recoverability diagnostic, not the $15/(8\pi^2)$ BW source. B_A is gauge-invariant only when built from the anomaly-frame baryon density $n_b^{(A)} = -u_{A\mu} J_b^\mu$. If the exchange branch is nonzero, explicit recipient stress and an equal-and-opposite exchange current are mandatory; otherwise the repair term is not a closed stress-energy source. A transition spectral number is only $\gamma_{\text{repair step}}$ without active-fiber, physical-clock, and common-parent response-pole receipts for a physical Γ_{rec} . A physical likelihood comparison also requires declared source, solver, dataset, covariance, and nuisance provenance.

Canonical finite covariant collar-packet parent. For a regulator r and background (X, g) , the source object is

$$\mathcal{P}_r[X, g] = (\mathcal{C}_r, g_r, e_r, U_r, Z_r, \omega_r, p_r, f_r, \Phi_r, \mathcal{R}_r, \mathcal{G}_r, \pi_r, \text{Read}_r).$$

Here $\mathcal{C}_r$ is a declared physical-causal regulator complex, not the source-derived informational read-from poset. It carries oriented faces, cell four-volumes, causal adjacency, and finite parallel transports $U_{F \rightarrow c}$; e_r is the tetrad/local-frame data; $Z_r = \bigsqcup_s Z_{s,r}$ is the packet state space for anomaly, recipient, radiation, standard matter, and any interaction sector; ω_r, p_r, f_r are invariant weights, local momenta, and occupations; Φ_r is the face flux; $\mathcal{R}_r$ is the reaction channel list; $\mathcal{G}_r$ is the finite gauge/quotient data; π_r is the restriction/refinement map; and Read_r is the local observable readout. With signature $(-+++)$, every packet used in a kinetic stress branch must satisfy $g_{ab} p_z^a p_z^b = -m_z^2$, $p_z^0 > 0$, and $\omega_z f_{c,z} \geq 0$, with the invariant measure convention declared explicitly.

Scalar-to-stress nonuniqueness. A scalar collar row, even a finite row for $\rho_A(a)$, $\rho_{A,\text{eq}}(a)$, or $B_A(k,a)$, does not determine a stress tensor. Many inequivalent packet parents can have the

same scalar density while differing in pressure, sound speed, anisotropic stress, exchange current, gauge-invariant perturbation variables, and causal response. Scalar rows are therefore insufficient as physical CMB, lensing, growth, or S_8 inputs unless the parent supplies the missing stress and response variables.

Finite MaxEnt lift. When only finitely many one-cell source readings $e_x(u_i) = T_{ab}^A(x)u_i^a u_i^b$ are available, a stress lift is not unique until the branch supplies a finite selection rule. The admissible MaxEnt lift minimizes no hidden observational residual: it maximizes the declared packet entropy subject to the source readings, mass shells, conserved charges, local frame covariance, and refinement restrictions. The lift is physical only when held-out timelike probes pass a quadraticity/tomography test and the variational stress agrees with the packet-moment stress in the predeclared norm.

Packet stress and exchange closure. For a kinetic parent,

$$T_{I,r}^{ab}(c) = \frac{1}{V_c} \sum_{z \in Z_{I,r}(c)} \omega_z f_{c,z} p_z^a p_z^b, \quad J_{I,r}^a(c) = \frac{1}{V_c} \sum_{z \in Z_{I,r}(c)} \omega_z f_{c,z} p_z^a.$$

More general packet parents may add an internal stress Σ_z^{ab} , but then Σ_z^{ab} is part of the primitive parent data and must obey the same local-frame and refinement checks. The finite transport/reaction equation must imply

$$\text{Div}_r T_{I,r}^{ab} = Q_{I,r}^b, \quad \sum_I Q_{I,r}^b = 0.$$

If the repair exchange is nonzero, a row labelled “recipient” is not enough: the recipient stress T_R^{ab} , exchange current Q_R^b , and closure residual $Q_A^b + Q_R^b$ must be emitted and certified. If exchange is off, the parent instead emits a repair-exchange-off receipt.

Response, rate, and gauge receipts. The finite causal response must provide a finite domain of dependence, subluminal characteristic bound, retarded support residual, and response stability residual. The transition-matrix diagnostic may be called $\gamma_{\text{repair step}}$ until the active fiber, conserved-sector decomposition, physical clock $\Delta\tau_{\text{physical}}$, and common parent response pole are certified. Only then may one write

$$\Gamma_{\text{rec}} = -\text{Re } \lambda_{\text{active}} / \Delta\tau_{\text{physical}}.$$

Gauge checks must be stated in gauge-invariant or rest-frame variables; raw Newtonian/synchronous agreement is a diagnostic, not a gauge-independence receipt.

Certificate hierarchy. The finite parent certificate is the conjunction of primitive evidence, not a producer-declared Boolean. The source side must certify the route and entropy unit, modular nonadditivity, localization and stress tomography, finite-packet kinematics and mass shell, covariant transport and four-momentum, stress readout and variational agreement, local-frame and quotient invariance, total stress and exchange-current closure, cosmological gauge invariance, finite domain of dependence, subluminal and retarded response, stability under refinement, and the cold-dark-matter limit. Frozen source, solver, likelihood, and data hashes belong to the separate comparison contract. They do not belong to the parent theorem and cannot rescue a failed parent.

The same source-provenance requirement applies to the CMB-source inputs

$$\eta_R, \quad \gamma_{\text{repair step}} \text{ or certified } \Gamma_{\text{rec}}, \quad A_\zeta, \quad q_{\text{IR}}, \quad \ell_{\text{IR}}, \quad B_A(k, a), \quad \rho_A(a), \quad N_{\text{CRC}}.$$

Each one must have a source-provenance dependency node recording the source report, parents, no-CMB-data flag, and measurement-use flags. Contradictory provenance, such as declaring no CMB data use while also fitting to Planck, fails closed. Reducers must pool additive sufficient statistics globally before any nonlinear estimate of tilt, amplitude, inverse precision, rank, condition number, isocurvature leakage, B_A , ρ_A , or Γ_{rec} ; raw transition diagnostics are $\gamma_{\text{repair step}}$. Shard-local nonlinear averages and shard-local `any()` likelihood rollups are diagnostic only. N_{CRC} is the stable capacity-closure invariant unless a separate additive capacity schema proves disjoint coverage. The cold transported limit is the check case: when exchange, pressure, sound-speed, and anisotropic stress corrections are turned off, the anomaly slot must reduce to a CDM-like component before recombination. Any nonzero-field response, late-time growth suppression, or S_8 -relief branch has to be emitted, under this registered contract, by the finite-collar parent evaluator through $B_A(k, a)$ and $\Gamma_{\text{rec}}(k, a)$ after the physical-clock and response receipts pass, not fitted as a free environmental kernel to CMB, weak-lensing, SPARC, or cluster data. The compressed H_0 , Ω_m , σ_8 , and S_8 rows are plumbing diagnostics for a low- H_0 , Planck-like branch, not theorem-grade cosmology.

Finite cosmological source and prediction eligibility. The physical cosmology condition is the conjunction

$$\text{CMB ready} = \text{SOURCE} \wedge \text{SCALE} \wedge \text{PARENT} \wedge \text{KERNEL} \wedge \text{INIT} \wedge \text{TRANSFER} \wedge \text{FREEZE} \wedge \text{LIKE}.$$

SOURCE is the source-only DAG with transitive measurement-taint rejection and globally pooled sufficient statistics; **SCALE** is the physical mode/clock/lift bridge; **PARENT** is the finite covariant dark/anomaly parent with packet kinematics, mass shell, packet stress readout, reaction-channel four-momentum, recipient stress and exchange-current closure for nonzero exchange, total stress closure, local-frame/quotient invariance, cosmological gauge invariance, finite domain of dependence, subluminal retarded response, response stability, refinement, and CDM-limit recovery receipts; **KERNEL** emits physical ρ_A , $\rho_{A,\text{eq}}$, B_A , and certified Γ_{rec} ; the source-only ρ_A row additionally requires the anomaly abundance selector

$$\mathcal{E}_{A,r} = \mathbb{E}_{\mu_{A,r}^{\text{rel}}} [\mathbb{L}_{A,r}]$$

and its no-data-use and refinement receipts; **INIT** supplies regular initial modes and Einstein constraints; **TRANSFER** supplies Boltzmann well-posedness and numerical convergence; **FREEZE** fixes the model generation; and **LIKE** is the official likelihood execution. While this conjunction is false, physical CMB, BAO, lensing, growth, RSD, and S_8 outputs are diagnostic or conditional, not recovered-core predictions.

This branch is a conditional completion contract, not a physical dark-matter theory.

Falsifiability. The repair-charge condensate requires a source-derived action, relativistic limit, and joint galaxy, Solar-System, lensing, cluster, and cosmological likelihood. These constructions are not supplied, so the falsification program contains no cosmological target or verdict.

2.3.17 De Sitter holography: static patch vs boundary-at-infinity

A natural question arises: how does this framework relate to the “unsolved problem” of de Sitter holography?

What the usual dS holography problem is. When people say “dS holography is unsolved,” they typically mean that we do not have anything as sharp as AdS/CFT, where the bulk has a timelike asymptotic boundary supporting a well-defined dual CFT with a precise dictionary. For de Sitter, there is no asymptotic timelike boundary in the static patch where one can simply place the dual theory. The classic dS/CFT proposal at future infinity has familiar difficulties, including non-unitarity worries and complex conformal weights.

Static-patch/horizon-screen setup. The framework begins with an observer’s static patch and its horizon screen S^2 , building a net of subregion algebras on that screen. At finite cutoff those algebras are type-I regulators. The Lorentz statement, when invoked, is a support-visible scaling-limit statement about the refinement-limit observer net, and that limit may leave the regulator class. By Theorem 2.2.2, the realized scaling-limit cap modular action on the extracted geometric cap pair is geometric and, in the non-type-I case of interest, generally outer.

This is therefore a fundamental fork away from AdS/CFT-style holography:

AdS/CFT	This framework
Codimension-1 boundary at infinity	Codimension-2 horizon screen (S^2)
Specified boundary field theory	Observer patches with typed overlaps
Conformal dual in specified constructions	Patch algebras plus geometric and physical premises
Negative Λ in AdS	Positive Λ on the declared de Sitter branch

This aligns with the static-patch/complementarity intuition in the dS literature, where the fundamental description is patch-based and different static patches are related by consistency rules, not by a single global boundary theory.

The mechanism: Λ as global capacity, not local physics. Once the separately supplied effective-locality and same-source bridge identifies the directional null charges with one symmetric local stress tensor, finite null tomography determines that tensor only up to an additive metric term. Vacuum-energy or cosmological-constant shifts are therefore invisible to this conditional local null-data route. The conditional Einstein equation obtained under the declared fixed-cap generalized-entropy stationarity interface is correspondingly fixed only up to Λg_{ab} . Accordingly, this null-data route cannot fix Λ without independent trace-sensitive or global input. A source-selected vacuum potential or non-null local equations could supply such input.

Conditional theorem boundary: correctable public-record closure. Freeze a capacity carrier $\mathcal{H}_{\text{cap},r,D}$ with $D = \dim \mathcal{H}_{\text{cap},r,D}$ and $N = \log D$. For every terminal world q , compatible record atoms, endogenous reachability, a frozen publicness policy, and the complete source-supplied global checkpoint family define a compound confusability graph G_q . The exact public capacity and whole-fiber readback are

$$M_0(q) = \alpha(G_q), \quad \mathfrak{F}_{r,0}(D) = \{M_0(q) : q \in \tilde{\Omega}_{r,D}\}.$$

When the entire nonempty terminal fiber has one common readback, define $M_0(\mathfrak{U}_N) := \hat{F}_{r,0}(e^N)$. The universe-level closure is then

$$\boxed{N = \log M_0(\mathfrak{U}_N)}.$$

The notation is typed: M_0 is always a multiplicative record count, while N is its logarithm. Stable direct closure and its exact finite-size selector are

$$\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \quad s(D) := \log D - \log M_0(D), \quad s(D_\star) = 0 < s(D) \quad (D \neq D_\star).$$

On the exact reversible branch every checkpoint generator is injective, so

$$M_0(q) = |X_{\text{reach}}(q)|,$$

reducing the finite computation to exact CSP or model counting. This supplies an exact implementation route for the source-only finite public checkpoint packet $\mathcal{C}_{r,D}^{\text{pub}}$.

The fixed- $D = 24$ source packet realizes the complete declared terminal fiber inside its declared finite source category. A separate all-rung counterfamily shares base agreement, positivity, the carrier bound, and finite incidence, reversible-action, semantic-projection, composition, extension, and oriented-record fiber-product controls. Reversible, copy-collapse, two-class, and hidden-spectator completions give different exact fixed sets. Lean proves the all-rung arithmetic consequence of the base, positivity, and carrier-bound structure. The executable certificate checks the remaining finite controls, and an independent simulator reconstructs held-out finite rungs. This does not lift the complete A1–A3 terminal fibers, atom maps, joint kernels, meaning maps, feasible sets, and regulator controls across capacity, so it is not a verdict on the complete source class.

The local null-data route is blind to metric-term shifts, so the Einstein branch is fixed locally only modulo Λ_{gab} . Conditional on stable direct closure and the independent horizon–record identification,

$$N_\star = \log D_\star = \frac{A_{\text{dS}}}{4\ell_\star^2}, \quad \Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}.$$

With the selected scale certificate the same branch has the display

$$G_{ab} + \frac{3\pi}{GN_\star} g_{ab} = 8\pi G \langle T_{ab} \rangle.$$

The independent common screen/electroweak load-carrier identification tests the same closed N against the weak/Higgs load through

$$R_{\text{EW}}(P, N) = \alpha_U(P) \log\left(\frac{N}{\pi}\right) - \frac{6\pi}{P} = 0, \quad N_{\text{bridge}} = \pi \exp\left[\frac{6\pi}{P\alpha_U(P)}\right].$$

Neither physical bridge constructs the direct capacity map.

Capacity readout. The branch uses N_{scr} as the entropy capacity. The bare radius-squared ratio is $N_{\text{patch}} = (r_{\text{dS}}/\ell_P)^2$, and

$$N_{\text{scr}} = \pi N_{\text{patch}} = \frac{3\pi}{\Lambda \ell_P^2}.$$

The observed late-time scale gives $N_{\text{patch}} \simeq 1.05 \times 10^{122}$, with a Planck- Λ central entropy capacity $N_\Lambda = 3.3129270981 \dots \times 10^{122}$. The conditional source-forward electroweak bridge value $N_0 = 3.5321315434 \dots \times 10^{122}$ is about 6.6 percent higher. The separately premised finite-presence and Poisson candidates are $3.2920978773 \dots \times 10^{122}$ and $3.3000722254 \dots \times 10^{122}$. Exact positive composition countermodels select neither action from the finite source. Their comparisons are retrospective.

Fixed-capacity shock sign. The pure de Sitter shock parameter obeys

$$\mu^2 = (d-2)\kappa r_c = d-2 = \lambda_{\ell=1}(S^{d-2}),$$

so the cosmological constant cancels and the smooth isometry modes lie at zero shock eigenvalue. For finite sector dimensions, maximizing

$$S_{\text{gen}}(p, d) = -\sum_i p_i \log p_i + \sum_i p_i \log d_i$$

over p gives $p_i = d_i/M$ and $S_{\text{gen}}^{\text{max}} = \log M$. The associated logarithmic area observable $\mathcal{A} = \sum_i (d_i/M) \log d_i$ has symmetric-point Hessian

$$\text{Hess } \mathcal{A}|_{d_i=d} = \frac{1}{nd^2} \left(I - \frac{2}{n} J \right)$$

on the positive-real relaxation of the integer dimensions. Its curvature is positive on fixed- M tangent directions and negative in the homogeneous direction; the gradient at the symmetric point is nonzero. The physical sign comes from a one-sided transfer. Assigning a fraction f of a fixed total horizon–observer capacity to the observer uniformly depletes the horizon and gives

$$\Delta S_{\text{gen}}^{\text{max}} = \Delta \mathcal{A} = \log(1 - f) < 0.$$

The finite identity applies to admissible integer depletions, and its positive-real interpolation is strictly decreasing. Under the independent horizon-area and fixed-budget dictionaries, this is the de Sitter time-advance sign studied in Ref. [49]. It is a boundary maximum along the transfer direction, rather than an interior maximum at fixed horizon capacity. Identifying the finite capacity transfer with observer mass is an additional physical dictionary.

On the icosahedral carrier, the spectrum

$$\{-2, 0, 1 + 3/\sqrt{5}, 1 + \sqrt{5}\}$$

is conditional on two further statements: the A_5 rotation triplet is an exact gauge sector, and the shock kinetic term is the scaled nearest-neighbour port or edge-sector Laplacian. The regular line-graph identity makes the port and edge-sector low spectra equal. It adds the eigenvalue 10 with multiplicity 18 on the edge carrier. No shock coefficient, physical attachment, or generator for sector-dimension shocks is supplied. The finite heat-bath repair generator acts at fixed sector dimensions and therefore supplies none of these objects. The sign mechanism does not rescue the static-patch trace conjecture [23]. That obstruction tests the cited positive cyclic trace interpretation. Nontrace horizon-screen descriptions, separately constructed dimension-changing generators, and source-derived kinetic operators beyond the scaled nearest-neighbour graph remain outside these classes. None of these boundaries disproves OPH; the pure-de-Sitter normalization, finite entropy and transfer identities, and regular line-graph theorem survive.

What this solves vs. what it assumes. The model does **not** solve the classic “give me a unitary CFT at future infinity” problem. It does not aim there. It also does **not** prove that every refinement-stable MaxEnt branch lands in the static-patch geometric modular branch with emitted cap pair and standard modular action. What it does provide is a coherent route to **patch holography** in which de Sitter static patches are natural:

1. the fundamental object is a horizon screen in a static-patch description;
2. Λ is a capacity parameter tied to finite Hilbert-space dimension, not a locally reconstructible vacuum-energy term;
3. Einstein-like dynamics emerge up to Λg_{ab} ;
4. on the support-visible BW scaling branch, the realized scaling-limit cap modular action is geometric and may be outer on a non-type-I observer algebra.

BW-side boundary. Theorem 4.2 proves the support-visible Lorentz scaling statement. It avoids the false stronger route through a full-algebra unregularized common floor; the automorphism statement on the observer-facing geometric cap pair is the required static-patch content.

Many observers, one Λ . In this framework, each timelike observer is associated with a horizon patch rather than a single global description. On the conditional capacity branch, the dimensionless de Sitter capacity relation is the shared global constraint across overlap-consistent descriptions; its SI Λ display uses the selected scale certificate.

Summary. The model moves the holographic screen from “infinity” to an observer’s horizon. Direct N closure is a correctable public-record conditional construction, not a checkpoint-fixed or marginal-information count. The fixed- $D = 24$ packet is exact inside its declared finite source category. The bounded completion class has incompatible fixed sets. No universal all-rung membership in the complete A1–A3 capacity-source contract or executable-to-Lean bridge is supplied, so direct N is not evaluable on the incomplete source antecedent. Horizon saturation and the weak/Higgs common-load equation are independent downstream commuting squares. The weak/Higgs bridge and the late-time de Sitter coordinate differ by about 6.6 percent. A finite presence correction and a separate Poisson correction give the retrospective comparison coordinates $3.2920978773 \dots \times 10^{122}$ and $3.3000722254 \dots \times 10^{122}$. Exact compositional countermodels select neither correction from the finite source, so the named-law branch is not evaluable there. Neither is a physical capacity until a stronger source law and both bridges share a certified carrier.

2.4 Einstein-chain theorem and falsification boundary

The finite support and algebraic packets establish repair-invariant incidence, the spherical branch receipts, cap-normal identities, the Lorentz action on the celestial screen, exact finite entropy splitting on the central-interface branch, and the finite tomography identities used to reconstruct a trace-free symmetric tensor from null projections.

Authenticated semantic read-from provenance supplies the finite event carrier and its generated precedence order. Its canonical source height is the attained longest authenticated-parent-chain length and strictly increases on generated ancestry. Separately, every finite decidable strict partial order compiles into an abstract authenticated log whose direct parent relation and generated order are exactly the supplied relation. Thus the generic parent relation contains every supplied predecessor rather than Hasse covers alone; this proves grammar expressivity, not dynamical or physical selection. Independently, the exact rank-three positive source Gram quotient and a real axis define the four-dimensional ambient target carrier

$$\mathbb{R} \oplus V_{\text{src}}, \quad Q(t, x) = t^2 - g_{\text{src}}(x, x),$$

with Lorentz inertia $(1, 3)$; source-unit directions are exactly its future-null rays. Canonical source height enters only as the temporal coordinate of a positively scaled event placement. Every finite log has an explicit auxiliary injective forward-causal placement obtained from a finite enumeration along one source axis and a scale larger than its diameter. It is not order-reflecting or source-selected. A supplied physical-candidate spatial readback and edge speed bound give one-way cone compatibility. Equal-height spatial separation and strict spacelikeness of unsupported increasing-height pairs imply converse support and exact two-way order-cone equivalence. Antisymmetry derives injectivity; the two-way equivalence gives exact interval preservation. Physical event identity, the spatial readback, the order-faithful-placement conditions, and operational clock calibration remain separately testable. A controlled effective route uses vanishing causal discrepancy and compatible count-volume convergence. Conservative Fibonacci-record populations with a specified complete-neighbour read law have such a flat $1 + 3$ limit when their fill error is small relative to the shrinking read radius. Equal-mass assignments with vanishing displacement error give normalized event counts converging to volume. On fixed interior timelike diamonds, fourth roots of interval-count ratios recover proper-time ratios, with a reference interval fixing the unit and complete ancestry access supplying the readout. Within the class of nonzero closed convex pointed displacement cones, operational invariance under source rotations and every boost along one axis forces the Lorentz

cone up to time orientation. The covariance is an assumption about actual influences and readouts. Physical selection of the population, read law and common matter realization is separate from these conditional results.

The source-direction Einstein theorem works in a separately supplied $3 + 1$ tensor interface indexed by the same finite event type and takes balance on nine fixed algebraic inverse images of the coordinate null-tomography directions, symmetric geometry/stress fields in that tensor interface, discrete Ward and Bianchi identities, an algebraic coupling, four step maps, a base event, and connectedness. It derives all-null balance, the pointwise and constant metric ambiguities, and the Einstein-form tensor identity in that supplied tensor interface. The nine directions are not observed provenance links, physical rays, or sky samples, and the supplied fields are not thereby identified with curvature or physical stress-energy. The informational order shares the event type but is not used in the tensor calculation: no provenance link selects a direction, field, step, or balance. The distinguished nine-set is not selected by the source dynamics and is not invariant as a set under Lorentz or $SO(3)$ transformations. The matrix fields and discrete step maps remain supplied tensor data, and their coordinate differences are not identified with those of the constructed carrier. Ambient target-carrier dimension and signature are constructed independently by the source-causal carrier theorem; intrinsic poset dimension remains a continuum diagnostic. Neither those fields nor the physical premise bundle is constructed.

On the exact-embedding route, an effective Lorentzian manifold requires the source-causal continuum certificate of Theorem 4.3e: one physical refinement family with exact two-way order/cone agreement, calibrated count-volume, independent dimension and manifoldlikeness tests, stable topology, and a unique distinguishing Lorentzian limit. The smooth Einstein reading additionally requires a same-family tensor-curvature reconstruction converging to the smooth Einstein tensor, or the separately stated continuum small-ball/null-balance identification, plus modular and scaling control, physical stress, Ward/Bianchi, universal coupling, a vacuum reference, and physical $8\pi G$ scale identification on that same family. A convergent causal-set scalar-curvature estimator is an independent diagnostic and is insufficient by itself. Finite certificate programs verify their stated algebra and controls; no source-derived family carrying the full continuum antecedent is constructed here.

The implication requires its receipts on one cofinal branch, a rank-three source Gram quotient, tensor-linearity of the directional charges, the stated curvature or small-diamond remainder rate, and scale readouts with trivial common rescaling. The chosen causal route must meet its exact order-cone criterion or its controlled causal and volume error bounds.

A Technical Details for the Specialist Derivations

This appendix carries the gravity details that support the spacetime and Einstein derivation. It sharpens the Lorentz, Einstein, and cosmological-constant surfaces without widening the recovered core beyond its stated branch conditions.

A.1 Port-record metric completion

Theorem A.1 (Exact repair-response three-dimensional carrier). *Let A be the adjacency matrix of the declared twelve-port incidence, $L = 5I - A$, $T = I - L/60$, and let Q remove the constant port mode. Write P_{slow} , P_5 , and P_{fast} for the exact nonconstant spectral projectors of ranks $(3, 5, 3)$.*

For the complete equally weighted port-probe census,

$$C_n = QT^{2n}Q = r_{\text{slow}}^{2n}P_{\text{slow}} + (9/10)^{2n}P_5 + r_{\text{fast}}^{2n}P_{\text{fast}},$$

where

$$r_{\text{slow}} = \frac{55 + \sqrt{5}}{60}, \quad r_{\text{fast}} = \frac{55 - \sqrt{5}}{60}, \quad 0 < r_{\text{fast}} < \frac{9}{10} < r_{\text{slow}} < 1.$$

Trace-twelve normalization therefore gives

$$\frac{12C_n}{\text{tr } C_n} \longrightarrow 4P_{\text{slow}} =: G.$$

The intrinsic space $H = \text{ran } P_{\text{slow}}$ has dimension three, and the labeled vectors $v_p = 2P_{\text{slow}}e_p$ satisfy $\langle v_p, v_q \rangle = G_{pq}$.

Let

$$M_{\mathbb{Z}} = \mathbb{Z}[P]/\langle e_{-p} + e_p \rangle \cong \mathbb{Z}^6$$

be the cumulative signed port-record module. The limiting form on its real six-control extension has a three-dimensional radical. Its real quotient is isometric to H . The integer module has no nonzero metric kernel, its image is dense in H , and its metric completion is an abstract continuous three-dimensional Euclidean vector group. The construction is unique up to port-label-preserving isometry and selects no Cartesian axes.

Proof. The spectral polynomial of A gives the four exact projector ranks $(1, 3, 5, 3)$ and the displayed decomposition of C_n . The strict response order makes the two faster ratios vanish after normalization by the slow response. Vertex transitivity fixes the unit diagonal and hence $G = 4P_{\text{slow}}$. Projector symmetry and idempotence give the intrinsic Gram identity. The source Gram calculation identifies its real radical with the coordinate kernel. Irrationality of the golden ratio makes the integer map injective, and the raw coordinate image has finite index eight in $\mathbb{Z}[\varphi]^3$. Its density in $\mathbb{R}^3$ gives the completion. These finite identities, the limit, the radical calculation, injection, and density are machine checked. $\square$

Corollary A.2 (Conditional local position carrier). *Suppose cumulative signed port load is the carrier-position readout, the complete probe census and T belong to that same source record, and the normalized limiting Gram is operational distinguishability. Then the finite record architecture reconstructs one abstract continuous local three-dimensional Euclidean carrier instead of consuming a Cartesian three-space.*

Boundary. For finite n , C_n has rank eleven on the centered port space and rank six on the signed antipodal sector. The normalized limit must precede the radical quotient and metric completion. The theorem does not select the repair law, the cumulative record semantics, or the Gram topology from the adopted axioms. It supplies no faithful isometric completion action, overlap/refinement gluing, global or physical spacetime, preferred frame, physical scale, clock, field sector, or observable. Ordered record history and repair cost remain a separate fiber.

A.2 Lorentz and Einstein Bridge

Theorem A.3 (Conformal group isomorphism). *The orientation-preserving conformal group of S^2 is isomorphic to the connected Lorentz group:*

$$\text{Conf}^+(S^2) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1).$$

Theorem A.4 (Support-visible geometric modular flow on caps on the extracted geometric subnet). *Assume the OPH axioms, the derived fixed-cutoff collar package, and the support-visible BW scaling theorem on the extracted geometric subnet. That theorem requires the finite cap-normal support/flow certificate and an independently complete multiresolution algebra-state comparison package on the same tower. Then the scaling-limit modular automorphism group of the cap pair is geometric conformal dilation on that subnet, with the standard 2π normalization supplied by the finite cap-normal certificate. If the emitted scaling-limit cap algebra is type I, this may be written as $K_C = 2\pi B_C + Z_C$ with Z_C central; in the generic continuum case the theorem is the automorphism statement on a non-type-I cap algebra with outer geometric modular action. The collar replacements used in this branch are exact only at EC-aligned exact Markovity or in controlled fixed-collar families with $\delta_{\mathcal{F}}^{\text{M,EC}} \rightarrow 0$; small CMI supplies a Fawzi–Renner recovered comparison state, not a dimension-free one-shot exact Markov normal form.*

Definition A.5 (BW-branch observer-relative modular ordering). *On the branch satisfying the hypotheses of the preceding geometric modular-flow theorem, the modular automorphism parameter t of the extracted cap pair supplies a dimensionless ordering for that observer’s accessible algebra-state pair. Physical time requires an observer-readable transition, event correspondence, and calibrated clock instrument. No claim about arbitrary operational clocks, global time, or the full problem of time follows from the geometric modular-flow theorem alone.*

BW-side UV scaffold. The UV data on this branch are not a separate cap-isotropy or Euclidean-regularity selector. The fixed-cutoff cap algebras are type-I regulators, but the scaling-limit observer algebra may leave that class, and MaxEnt alone does not select the BW / canonical cap phase. The UV package is the realized transported geometric cap-local system together with the carried-collar schedule derived from the transported fixed-local-collar Markov/faithfulness datum on each fixed local collar model, followed by support-readable modular covariance and ordered cut-pair rigidity on the emitted scaling-limit geometric cap pair.

Null-strip completion boundary. The fixed-cutoff strip package is more specific than the generic inherited-strip summary. The null cuts first transfer the same cut-center data as the spatial collar branch, which fixes the central sector-pair decomposition of the strip algebra. The stronger left/right tensor decomposition used by the null modular bridge then requires the extra inherited strip-split condition on the multiplicity spaces, together with the exact-or-controlled Markov hypotheses on one fixed inherited strip model. On that same exact-Markov/central-interface model, a specified endpoint-to-cell map and uniform local norm/site-density bound make the renormalized half-line family endpoint-Lipschitz; this is additional endpoint-control data, not a consequence of finite-range locality alone. It hence defines the weak tail generator, and on the scaling-limit geometric-cap branch the half-line blow-up net carries the derived half-sided modular pair. Borchers–Wiesbrock then supplies the positive null-translation generator on its Stone domain together with the affine half-line modular relation $K_a(\Omega) = K_0(\Omega) - 2\pi a P_\Omega$, and the same half-line family fixes the generator and its endpoint-derivative identity internally. Its reading as a local null-stress charge additionally requires the effective-locality and same-source identification premise. Bounded-interval formulas are downstream of the E0.5 affine/projective kernel in the Einstein bridge.

Lemma A.6 (Null data ambiguity). *If a symmetric tensor X_{ab} satisfies*

$$X_{ab} k^a k^b = 0$$

for every null vector k , then $X_{ab} = \phi g_{ab}$ for some scalar ϕ .

Corollary A.7 (Conditional null-stress data leave the metric term invisible). *If the supplied effective-locality and same-source bridge identifies the directional generators with one symmetric local stress tensor, their null contractions determine that tensor only up to ϕg_{ab} . Under the remaining Einstein-branch premises, the local equation consequently retains the corresponding Λg_{ab} ambiguity.*

Theorem A.8 (Jacobson-type rest-frame relation). *In the local Lorentzian scaling regime, assume the declared fixed-cap generalized-entropy stationarity interface for admissible fixed-cap variations, together with the proved finite MaxEnt identities on the realized cap-label-preserving family. If the effective-locality and same-source bridge gives the half-line generator its local null-stress reading, the conditional bounded-interval transport and small-ball premises yield the rest-frame first-variation relation that drives the spacetime and Einstein paper's local Einstein branch; on the same scaling branch, if that rest-frame relation holds for all local observer four-velocities and all reference states, the spacetime and Einstein paper upgrades it to the full tensor equation by the explicit local quadratic-polarization argument in the main derivation.*

Theorem A.9 (Newton coupling from the scale certificate and edge entropy density). *The geometric coupling read by the Newton area law is*

$$G_{\text{geom}} = \frac{a_{\text{cell}}}{4\bar{\ell}(t)}.$$

On the OPH gravity row the scale is supplied first by $\gamma_\star = \ell_\star \nu_{\text{Cs}}/c$, equivalently $B_\star = 3\pi/\ell_\star^2$, as an independent scale certificate rather than a consequence of $P_\star$ and $N_\star$ alone. With $a_{\text{cell}} = P\ell_\star^2$ and $\bar{\ell}_{\text{shared}} = P/4$, this gives

$$G_{\text{geom}} = \ell_\star^2, \quad G_{\text{SI}} = \frac{c^3 \ell_\star^2}{\hbar}.$$

Here $\bar{\ell}_{\text{shared}}$ is a shared-cut density. It is not the logarithm of an independent cell Hilbert-space dimension and does not select a fixed primitive observer capacity.

A.3 Cosmological-Constant / Screen-Capacity Closure

Lemma A.10 (Vacuum energy blindness). *For any null vector k ,*

$$T_{kk}^{\text{vac}} = T_{ab}^{\text{vac}} k^a k^b = -\rho_{\text{vac}} g_{ab} k^a k^b = 0.$$

Vacuum-energy contributions therefore lie in the kernel of the null map $T_{ab} \mapsto T_{kk}$.

Proposition A.11 (Structural separation). *Within the structural split used here:*

1. *local modular/null data fix T_{ab} only up to ϕg_{ab} ; and*
2. *the dimensionless Λ -capacity relation is fixed by global screen capacity, not by local null data.*

So the large vacuum-energy bookkeeping of EFT is not itself the local quantity determining curvature on this branch.

Theorem A.12 (No Λ prediction from the declared local null data). *Within the null-modular reconstruction used here:*

1. *all T_{kk} data are unchanged under $T_{ab} \rightarrow T_{ab} + \phi g_{ab}$;*
2. *therefore Λ cannot be fixed by these local null-modular data alone; and*

3. the dimensionless Λ -capacity relation requires the global screen-capacity closure beyond the local null data. The stable closure target is $\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}$, with $N_{\text{CRC}} = \log D_\star$. The correctable public-record definition and conditional operational closure theorem close its operational public-section, correctable-code, capacity-bound, approximate-stability, and greatest-fixed-point implications under named premises. The full specification and receipt schema are carried by From Observer Consensus to Standard Physics. The finite record restrictions, reachability, publicness, global coupling, carrier, scalarization, extension, refinement, and sewing receipts are exact at $D = 24$ inside their declared finite source category. A bounded all-rung counterfamily has different exact fixed sets under shared base, positivity, carrier, and executable finite controls. This proves nonidentifiability for the bounded completion class. Universal all-rung membership in the complete A1–A3 capacity-source contract and an executable-to-Lean bridge are absent. Direct N is not evaluable on the incomplete source antecedent, and the bounded result does not determine the complete source class. A positive cosmic value requires a complete source antecedent, one physical zero, a universe-level carrier attachment, and the horizon-record identification. The electroweak relation is an independent comparison after F and does not construct it. SI curvature values additionally use the selected scale certificate.

A different typed local route carrying trace-sensitive or non-null overlap observables is not excluded and would require its own source and physical receipt.

Corollary A.13 (Benchmark cosmological readout on the capacity branch). *On the conditional capacity branch defined above, inserting the observed value $\Lambda \approx 1.09 \times 10^{-52} \text{ m}^{-2}$ gives*

$$r_{\text{dS}} = \sqrt{\frac{3}{\Lambda}} \approx 1.66 \times 10^{26} \text{ m}, \quad t_\Lambda = \frac{r_{\text{dS}}}{c} \approx 17.5 \text{ Gyr},$$

$$N_{\text{patch}} = \left(\frac{r_{\text{dS}}}{\ell_P}\right)^2 \approx 1.05 \times 10^{122}, \quad N_{\text{scr}} = S_{\text{dS}} = \pi N_{\text{patch}} \approx 3.31 \times 10^{122},$$

$$\Lambda \ell_P^2 \approx 2.85 \times 10^{-122}.$$

Proof. Immediate from the displayed static-patch and capacity relations. $\square$

Observed-age benchmark boundary. The branch quantity t_Λ is the de Sitter static-patch timescale. The usual cosmic age t_0 is a compare-only FLRW benchmark, with no additional capacity-theorem output. On the flat Λ CDM benchmark

$$t_0 = \frac{2}{3H_0\sqrt{\Omega_\Lambda}} \sinh^{-1}\left(\sqrt{\frac{\Omega_\Lambda}{\Omega_m}}\right)$$

one gets the standard comparison value $t_0 \approx 13.8 \text{ Gyr}$ at $H_0 \approx 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\Omega_\Lambda \approx 0.685$.

Lemma A.14 (FLRW curvature as visible scalar holonomy). *On a homogeneous-isotropic spatial slice with constant sectional curvature K , small spatial loop holonomy obeys*

$$\text{Hol}_{\square_{uv}} = \exp\left(K A_{\square} J_{uv} + O(A_{\square}^{3/2})\right).$$

For oriented nondegenerate loops small enough that the logarithm branch is fixed, define the normalized scalar readout

$$\kappa_{\square} := \frac{\langle J_{uv}, \log \text{Hol}_{\square_{uv}} \rangle}{A_{\square} \langle J_{uv}, J_{uv} \rangle}.$$

Then $\kappa_{\square} = K + O(A_{\square}^{1/2})$. On a visibly separated OPH refinement system, the limit of this normalized readout vanishes if and only if $K = 0$. The unnormalized holonomy itself tends to the identity for every finite K , so it is not a flatness test. Thus a flat FLRW branch is the zero-normalized-spatial-curvature-holonomy branch.

Flatness boundary. This holonomy statement names a conditional cosmology bridge only. It does not add a capacity-theorem output and does not solve the inflationary flatness or horizon problems. Selecting $K = 0$ requires an additional theorem or premise: a direct flatness theorem, a conditional cosmological-minimal-holonomy selector on a clocked FLRW boundary, or an explicit flat-branch assumption. No gauge- or matter-sector result supplied in this packet selects $K = 0$. Any gauge-, matter-, or dynamics-based flatness route needs its own theorem and physical receipt.

Theorem A.15 (Local/global cosmological-capacity closure). *Assume the spacetime and Einstein paper's local Einstein branch, stable direct public-record closure and its capacity coordinate*

$$\mathfrak{F}_{r,0}(D_{\star}) = \{D_{\star}\}, \quad N_{\text{CRC}} = \log D_{\star},$$

and the independent horizon-record identification readout

$$N_{\text{CRC}} = S_{\text{dS}},$$

the standard de Sitter entropy relation

$$S_{\text{dS}} = \frac{A_{\text{dS}}}{4G} = \frac{3\pi}{G\Lambda},$$

and the standard static-patch formulas

$$r_{\text{dS}} = \sqrt{\frac{3}{\Lambda}}, \quad t_{\Lambda} = \frac{r_{\text{dS}}}{c}.$$

Then the cosmological-constant package is one local/global theorem chain: local null data fix the Einstein branch only modulo Λg_{ab} , and with the selected scale certificate the same branch has the global display

$$G_{ab} + \frac{3\pi}{GN_{\text{CRC}}} g_{ab} = 8\pi G \langle T_{ab} \rangle,$$

while the capacity closure itself fixes the entropy and dimensionless capacity relations

$$S_{\text{dS}} = N_{\text{CRC}}, \quad A_{\text{dS}} = 4GN_{\text{CRC}}, \quad r_{\text{dS}} = \sqrt{\frac{3}{\Lambda}}, \quad t_{\Lambda} = \frac{r_{\text{dS}}}{c},$$

and the observed cosmic age is a downstream FLRW benchmark, with no additional theorem output.

Scope boundary. The hypotheses are exactly the local Einstein branch, stable whole-fiber public-record closure $\mathfrak{F}_{r,0}(D_{\star}) = \{D_{\star}\}$, horizon-record identification $N_{\text{CRC}} = S_{\text{dS}}$, the standard de Sitter entropy relation, and the standard static-patch formulas. The local null-data route does not by itself determine the global capacity; that value is fixed only on a discharged stable-public-record branch with horizon-record identification. Conditional on that branch, the capacity closure fixes $\Lambda_{\text{CRC}} \ell_{\star}^2 = 3\pi/N_{\text{CRC}}$; the SI static-patch scale additionally requires the selected scale certificate. CMB kernels, inflation-replacement claims, H_0/S_8 branches and dark/anomaly growth kernels require separate theorems or likelihood contracts. The baryogenesis anomaly/current theorem and gauge/deck no-go are separate from the global-capacity branch; the anomalous record attachment and physical CP-odd source generator are not supplied.

Theorem A.16 (Conditional OPH screen-spectrum theorem). *Let $(\mathcal{S}_r)$ be a cofinal finite spherical OPH screen system with a schedule-independent quotient-normal-form scalar q_r , removal of the background and dipole sector, and a target-free positive quadratic repair operator K_r . If $K_r \rightarrow K$, the source-selected finite scalar release energy satisfies $2E_{q,r}^{\text{src}}/d_r \rightarrow A_q$, and local MaxEnt is imposed at fixed expected quadratic release energy, then q_r converges in finite harmonic distributions to the centered Gaussian screen field with covariance $A_q K^{-1}$. Exact per-sample release energy would instead give a microcanonical ellipsoid. If the certified repair-scale measure is $t^{\theta/2} dt/\Gamma(1 + \theta/2)$, then*

$$K_{\text{asy}} = (-\Delta_{S^2})^{1+\theta/2}, \quad C_{\ell,\text{asy}}^q = A_q[\ell(\ell+1)]^{-1-\theta/2},$$

as the large- ℓ asymptotic model. The theorem-grade finite- ℓ conformal-shell family uses the normalized gamma-ratio precision

$$\kappa_\ell(\theta) = \frac{\Gamma(\ell+2+\theta/2)}{\Gamma(\ell-\theta/2)},$$

with

$$C_\ell^q = A_q \frac{\Gamma(\ell-\theta/2)}{\Gamma(\ell+2+\theta/2)}.$$

If q is additionally certified as the thin-shell pullback of a homogeneous curvature field with $\Delta_\zeta^2(k) = A_\zeta(k/k_\star)^{-\theta}$, the corresponding inverse amplitude relation is

$$A_\zeta = \frac{A_q^{\text{shell}} \Gamma(3/2 + \theta/2)}{\pi^{3/2} Z_\star^2 (k_\star D_\star)^\theta \Gamma(1 + \theta/2)}.$$

Suppose the same source construction emits a strongly continuous full-collar survival cocycle with generator density $P_\star/24$, together with the orientation-reversal half-collar identity. Then

$$\theta = \frac{P_\star}{48}, \quad n_s = 1 - \frac{P_\star}{48}, \quad \kappa_{\text{rep}}^{\text{edge}} = \frac{P_\star}{48(P_\star - \varphi)}.$$

A finite one-step survival value determines θ through $-\log u_q(\log b)/\log b$; it does not supply the infinitesimal generator receipt. If a scale-natural source embedding transports this cocycle to the physical covariance through $D_s^{-1}C_\zeta D_s = e^{-\theta s}C_\zeta$, the source family is $\Delta_\zeta^2(k) = A_\zeta(k/k_\star)^{-\theta}$. A single shell has an infinite-dimensional radial kernel, including positive ambiguities. Complete radial cross-covariances provide the independent tomography route. A Fawzi–Renner/Markov-collar observable estimate gives an upper bound on release observables, not an amplitude equality. No physical TT/TE/EE statement follows without independent Boltzmann transfer and likelihood closure.

Screen-spectrum boundary. The conditional theorem fixes the exact angular family. Its radial theorem gives physical source-dilation and cross-covariance tomography as separate uniqueness routes. A finite OPH source construction that emits the primitive collar ensemble, full-collar generator density, half-collar identity, conformal precision, and physical dilation-intertwiner or tomography receipt is not supplied. The thin-shell lift uses the gamma-ratio operator above; finite-width windows require a Bessel-kernel error certificate. Physical temperature and polarization transfer and a frozen likelihood contract are not supplied.

Separate pixel-side benchmarks. The companion pixel numbers

$$a_{\text{cell}} \approx 1.63 \ell_{\star}^2, \quad \ell_{\text{UV}} = \sqrt{a_{\text{cell}}} \approx 1.28 \ell_{\star}, \quad \bar{\ell} \approx 0.408$$

belong to the separate pixel-closure/local-readout package, outside the cosmological-parameter corollary itself. After the selected scale certificate is supplied, $\ell_{\star}$ is displayed as the usual Planck length.

A.4 Causal order on a source-metric population

The continuous rank-three repair carrier admits finite populations chosen by its own metric. A covering property of such a population also supplies a constructive causal approximation, with an explicit local read law and model clock. Let Ω be a convex subset of a real normed space, and let $S \subset \Omega$ satisfy

$$\forall x \in \Omega \quad \exists s \in S : \quad \|s - x\| \leq h, \quad h \geq 0.$$

For an edge radius $a > 2h$, put an event at every (j, s) , $j \in \{0, \dots, K\}$, $s \in S$. An event at $(j+1, t)$ reads every (j, s) with $\|t - s\| \leq a$. This includes the same-site read, so waiting is permitted. The event order is the reflexive transitive closure of these reads. Reading every permitted neighbour, keeping the full population at each layer, and assigning a duration $\Delta = a/c$ with $c > 0$ are hypotheses of this construction. They are substantive choices of dynamics and clock.

Proposition A.17 (Covering constructs causal paths). *For $s, t \in S$ and $k \geq 1$, with both layers present,*

$$\|t - s\| \leq k(a - 2h) \implies (j, s) \preceq (j + k, t) \implies \|t - s\| \leq ka. \quad (1)$$

If S is finite and nonempty, this layered poset has width exactly $|S|$. Every event at layer j has height j , counting edges from layer zero.

Proof. Divide the segment from s to t into k equal pieces, with division points $x_i = s + i(t - s)/k$. Convexity gives $x_i \in \Omega$. Keep the endpoints $p_0 = s$, $p_k = t$ and choose $p_i \in S$ with $\|p_i - x_i\| \leq h$ at every interior division point. Then

$$\|p_{i+1} - p_i\| \leq \|t - s\|/k + 2h \leq a.$$

These are actual permitted reads. For any permitted path the triangle inequality gives the opposite bound in (1). A layer is an antichain, and the same-site waiting chains partition all events into $|S|$ chains. This proves the width statement. Every read increases the layer by one, and a waiting chain reaches every layer, proving the height statement. $\square$

The path construction and outer displacement bound hold in arbitrary real normed spaces; `Lean/Geometry/SourceNetCausalCone.lean` formalizes them. The partial order, its equality with the reflexive transitive closure of the reads, the graph-distance characterization, and the width and height statements are formalized in `Lean/Geometry/SourceNetLayeredOrder.lean`. The volume estimates below are analytic results for the three-dimensional Euclidean repair carrier.

For model time $\tau = k\Delta$, the two radii in (1) are $c(1 - 2h/a)\tau$ and $c\tau$. Consider a family with

$$a_n \longrightarrow 0, \quad h_n/a_n \longrightarrow 0. \quad (2)$$

The read law agrees with the speed- c continuum causal order away from its null boundary. More explicitly, take two continuum events with positive time separation whose positions lie in Ω . Sample positions within h_n and times on the layer grid within Δ_n . A strict timelike margin eventually

exceeds the endpoint errors, the time rounding errors, and $2ch_n/a_n$ times the sampled duration; the inner implication then applies. A strict spacelike margin eventually exceeds the endpoint and time errors, and the outer implication forbids a path. This statement requires the sampled layers to exist in the chosen time window. It makes no assertion of exact order reflection for null-separated events at a finite resolution.

Volume from a controlled population measure. Suppose $\Omega \subset \mathbb{R}^3$ is bounded and convex, and S is finite. In addition to the h -covering population S , supply a measurable partition $(C_s)_{s \in S}$ of Ω , up to null sets, with positive masses $v_s = \text{vol}(C_s)$ and assignment error

$$y \in C_s \implies \|y - s\| \leq H \quad \text{for almost every } y. \quad (3)$$

Clipped Voronoi cells allow $H = h$. Equal-volume cells with a different assignment bound are also admitted; no Voronoi hypothesis is required. For $x \in S$ and $T = K\Delta > 0$, let I be the inclusive discrete interval between $(0, x)$ and (K, x) , and assign it spacetime mass

$$V_{S,\Delta}(I) = \Delta \sum_{(j,s) \in I} v_s.$$

The spatial assignment rule and the temporal factor are part of the measure. The limiting volume convention is $dt d^3x$. For the metric $c^2 dt^2 - |dx|^2$, its metric volume is c times this coordinate volume.

Proposition A.18 (Weighted Alexandrov-volume limit). *Assume the preceding hypotheses and $B(x, cT/2 + H) \subset \Omega$. Then*

$$\left| V_{S,\Delta}(I) - \frac{\pi c^3 T^4}{24} \right| \leq 4\pi(T + \Delta)(cT/2 + H)^2 \left(H + \frac{cTh}{a} \right) + \frac{\pi c^3 T^3 \Delta}{2}. \quad (4)$$

Thus $H_n \rightarrow 0$ and (2) imply the continuum Alexandrov-volume limit on aligned, bounded interior time windows. If all cells have a common mass v , this is the raw event-count statement $v\Delta |I| \rightarrow \pi c^3 T^4/24$.

Proof. Write $u_j = \min(j\Delta, T - j\Delta)$ and $r_j = cu_j$. The causal sandwich puts every population point within $c(1 - 2h/a)u_j$ of x in the interval, and puts every interval point within r_j . At the endpoint layers the interval contains precisely the single endpoint. The partition assignment therefore gives, at every layer,

$$\frac{4\pi}{3} (c(1 - 2h/a)u_j - H)_+^3 \leq \sum_{(j,s) \in I} v_s \leq \frac{4\pi}{3} (r_j + H)^3.$$

For the lower estimate, a point in the smaller ball is assigned to a population point inside the inner causal radius. For the upper estimate, the union of cells assigned to interval points lies in the larger ball. The stated interior buffer keeps these balls in the spatial window. The deviation of either radius from r_j is at most $H + cTh/a$. Since the derivative of $(4\pi/3)r^3$ is $4\pi r^2$, the error per layer is bounded by $4\pi(cT/2 + H)^2(H + cTh/a)$.

Finally, $F(t) = (4\pi/3)[c \min(t, T-t)]^3$ has Lipschitz constant $\pi c^3 T^2$, vanishes at both endpoints, and integrates to $\pi c^3 T^4/24$. Its left Riemann sum differs from its integral by at most $\pi c^3 T^3 \Delta/2$. The inclusive right endpoint contributes zero to this reference sum. Summing the spatial errors over $K + 1$ layers proves (4). $\square$

Proposition A.19 (Interior diamonds with moving tips). *Let $x, y \in S$, $\|y - x\| < cT$, and $T = K\Delta > 0$. Suppose the continuum diamond between $(0, x)$ and (T, y) has a positive spatial buffer inside Ω . Write*

$$D(t) = B(x, ct) \cap B(y, c(T - t)), \quad V(D) = \int_0^T \text{vol}(D(t)) dt.$$

For the inclusive discrete interval I between these tips,

$$|V_{S,\Delta}(I) - V(D)| \leq 8\pi(T + \Delta)(cT + H)^2 \left(H + \frac{2cTh}{a} \right) + 4\pi c^3 T^3 \Delta. \quad (5)$$

In particular, the same refinement conditions give weighted interval-volume convergence for arbitrary fixed interior timelike diamonds.

Proof. At layer j , the union of assigned cells contains the intersection of the two balls with radii $(c(1 - 2h/a)j\Delta - H)_+$ and $(c(1 - 2h/a)(T - j\Delta) - H)_+$, up to null sets. It is contained in the intersection with radii $cj\Delta + H$ and $c(T - j\Delta) + H$. Zero-radius balls contribute zero volume. Changing either reference radius changes the intersection only within that ball's spherical shell. Each radial error is at most $H + 2cTh/a$, and each outer radius is at most $cT + H$. The sum of the two shell bounds gives the spatial term in (5). The reference section volume has Lipschitz constant at most $8\pi c^3 T^2$: over a time increment, each of the two radii changes at speed c , and each ball has surface area at most $4\pi c^2 T^2$. Both endpoint sections have zero volume. The same Riemann-sum argument gives the temporal term. $\square$

A conservative-record family with a raw count law. The population and equal-mass hypotheses have an explicit realization in conservative source records. In the exact source-axis presentation, write $\varphi = (1 + \sqrt{5})/2$ and $L = 2/\sqrt{\varphi + 2}$. For consecutive Fibonacci numbers $q = F_n$, $p = F_{n+1}$, with $n \geq 4$, define

$$\xi_b = b\varphi - \lfloor b\varphi \rfloor, \quad 0 \leq b < q.$$

For a triple $b = (b_1, b_2, b_3)$, put $m_i = -\lfloor b_i\varphi \rfloor$ and take the integer record

$$z(b) = (b_2 - m_1, b_2 + m_1, b_3 - m_2, b_3 + m_2, b_1 - m_3, b_1 + m_3). \quad (6)$$

Its coordinate sum is $2(b_1 + b_2 + b_3)$. It therefore lies in the exact conservative seam-current image, whose integer section supplies a finite source word. Direct contraction with the source axes gives the readback

$$s(b) = L(\xi_{b_1}, \xi_{b_2}, \xi_{b_3}) \in \Omega = [0, L]^3.$$

These are actual source-record images. A regular mesh is used below only to prove a volume estimate; it does not replace the event positions.

Proposition A.20 (Source-record quadrature and four-dimensional counting). *The population $S_q = \{s(b) : 0 \leq b_i < q\}$ has exactly q^3 distinct sites. It admits a partition of Ω into cells of common volume $v_q = L^3/q^3$, assigned to the sites with error*

$$H_q = \frac{2\sqrt{3}L}{q}. \quad (7)$$

Consequently, for every Lipschitz function f on Ω ,

$$\left| v_q \sum_{s \in S_q} f(s) - \int_{\Omega} f(x) d^3x \right| \leq L^3 H_q \text{Lip}(f).$$

Apply the preceding complete-neighbour read law on S_q , with $a_q = L/\sqrt{q}$ and $\Delta_q = a_q/c$. Its density in the declared $dt d^3x$ convention is

$$\rho_q = \frac{q^3}{L^3 \Delta_q}. \quad (8)$$

Normalized raw event counts converge to volume on fixed bounded regions with volume-null boundary. On intervals whose endpoints converge to a strictly timelike pair and whose limiting diamond has an interior buffer in the spatial and temporal windows, the generated order intervals converge to that same diamond in count measure. In particular, for vertical tips separated by model time T ,

$$\frac{|I_q|}{\rho_q} \longrightarrow \frac{\pi c^3 T^4}{24}.$$

The local limiting order and measure are those of flat 1 + 3 spacetime. This conclusion concerns this declared source-record family and update law.

Proof. The Fibonacci recurrence gives $\gcd(p, q) = 1$ and $p - q\varphi = (-1)^n \varphi^{-n}$, hence $|\varphi - p/q| < q^{-2}$. Let k_b be the residue of bp modulo q . These residues permute $0, \dots, q-1$. For $b = 0$ the two floors below are zero. For $0 < b < q$, bp/q is at least $1/q$ from either adjacent integer, whereas $|b(\varphi - p/q)| < 1/q$. Therefore

$$\lfloor b\varphi \rfloor = \lfloor bp/q \rfloor, \quad \xi_b = \frac{k_b}{q} + b(\varphi - p/q), \quad \left| \xi_b - \frac{k_b}{q} \right| < \frac{1}{q}.$$

Irrationality of φ makes the orbit points distinct; the source readback then distinguishes all q^3 records.

Assign to $s(b)$ the rectangular cell $\prod_i [Lk_{b_i}/q, L(k_{b_i} + 1)/q]$, resolving cell-boundary ties arbitrarily. This is a partition up to null sets. A site need not belong to its assigned cell, but every coordinate displacement from a point of that cell is at most $2L/q$. This proves (7), the covering property and the Lipschitz estimate by integrating the displacement bound over each cell. Taking products with time cells $[j\Delta_q, (j+1)\Delta_q]$ proves convergence of the normalized event measure: for a Lipschitz test function on an aligned time window of length T , the error is at most

$$TL^3(H_q \text{Lip}_x(f) + \Delta_q \text{Lip}_t(f)).$$

An extra endpoint layer contributes at most $L^3 \Delta_q \|f\|_\infty$. Continuous approximation then gives the count statement for regions with volume-null boundary.

Use $h_q \leq H_q$ in the causal sandwich. Here $a_q \rightarrow 0$ and $H_q/a_q = 2\sqrt{3}/\sqrt{q} \rightarrow 0$; the sufficient positive-inner-cone condition from this bound holds when $q > 48$. For converging timelike tips, the order and sampling discrepancies are confined to shrinking neighbourhoods of the two limiting null boundaries. Their volume is zero. The uniform cell displacement and temporal error also tend to zero, so the normalized number of events in these shrinking neighbourhoods tends to zero. This proves interval-count convergence; for vertical tips it also follows directly from (4). The formula for the flat diamond is its elementary ball-section integral. $\square$

Proposition A.21 (The four-dimensional ordering fraction). *For the contained timelike intervals in Proposition A.20, let $N_q = |I_q|$ and let C_q count their strict causally ordered pairs. Then*

$$\frac{2C_q}{N_q(N_q - 1)} \longrightarrow \frac{1}{10},$$

the four-dimensional Myrheim–Meyer fraction [8].

Proof. The cell assignment gives convergence also for the product event measure. Interval-membership boundaries, the pairwise null-relation boundary and the coincidence diagonal have zero limiting measure. The causal sandwich confines disagreement to shrinking neighbourhoods of these boundaries. Hence the normalized strict-pair count tends to its continuum double integral; $N_q \rightarrow \infty$ removes the distinction between N_q^2 and $N_q(N_q - 1)$.

In units $c = 1$, place the limiting tips at $(0, 0)$ and $(T, 0)$ and write $V = \pi T^4/24$. The volume to the future of (t, x) within the interval is $\pi[(T - t)^2 - |x|^2]^2/24$. Thus the strict oriented-pair integral is

$$\frac{\pi^2}{6} \int_0^T \int_0^{\min(t, T-t)} r^2 ((T - t)^2 - r^2)^2 dr dt = \frac{\pi^2 T^8}{11520} = \frac{V^2}{20}.$$

The ordering fraction counts both orientations of distinct comparable pairs, giving $1/10$. Constant changes of time and volume units cancel in the ratio. This is an interval statistic; the same value is not asserted for a whole spatial box times a time window. $\square$

A sharper finite covering constant is available without changing this construction. Sort the one-dimensional orbit as $0 = \eta_0 < \dots < \eta_{q-1} < 1$ and put

$$r_q = \max \left\{ \frac{1}{2} \max_{i < q-1} (\eta_{i+1} - \eta_i), 1 - \eta_{q-1} \right\}, \quad h_q = \sqrt{3} L r_q.$$

The product population has exactly this covering radius in the closed cube. The path condition $2h_q < a_q$ is the exact quadratic-field inequality $12qr_q^2 < 1$. This sharper h_q concerns paths; the assignment estimate H_q in the count proof is separate.

The source-word section has coefficients growing at most linearly in q ; the finite implementation records their actual primitive lengths. Counting one distinguished event per site and layer is part of the stated law. Intermediate seam-word operations, read instructions and bookkeeping records are not silently included in the same density. The spatial populations are nested along increasing Fibonacci cutoffs. Their changing layer duration and edge radius do not make the spacetime histories induced prefixes; regulator convergence uses the common metric carrier and controlled cell assignments.

Source populations and physical interpretation. The finite-word source construction supplies arbitrarily fine metric nets in bounded convex windows with nonempty interior in the rank-three completion. A maximal separated subset retains a controlled fill distance and three-dimensional packing bounds. These statements supply the spatial hypotheses of Proposition A.17 under a specified resolution and population rule. Three-dimensional packing bounds alone do not supply equal-volume sampling. The count conclusion requires a partition satisfying (3) and, for unweighted counts, equal cell masses or a separate counting approximation.

The construction gives a causal approximation on source-metric points without inserting a Cartesian spatial table. The complete neighbour read law, repeated population, cell measure, and model duration are supplied. An implementation by accepted observer repairs must realize these inputs with its actual records and operation costs. A scalar action on the same population can have a different finite propagation relation; the causal cone proved here is not an identification of that action's stencil cone or its continuum response speed. Agreement with a field action and a physical clock is an additional common-model condition.

A.5 Protected spatial records and mutable repair states

A population attached to retained records and a population read from live repair amplitudes obey different evolution laws. The distinction can be characterized exactly for scalar seam averaging.

On a finite port set I , let T_{uv} replace the two endpoint loads by their arithmetic mean and leave the other loads unchanged. For coefficients $w_i \in \mathbb{R}$, define $R_w(N) = \sum_i w_i N_i$.

Proposition A.22 (Linear records preserved by repair). *For distinct endpoints u, v ,*

$$R_w(T_{uv}N) - R_w(N) = \frac{(w_u - w_v)(N_v - N_u)}{2}.$$

Thus R_w is invariant under T_{uv} for every input precisely when $w_u = w_v$. For a specified seam graph, invariance under every allowed repair is equivalent to constancy of w on each connected component. On a connected graph every such record has the form $R_w(N) = w_ \sum_i N_i$.*

Proof. Only the endpoint summands change. Substitution gives the displayed defect. Equality of their coefficients suffices for invariance. Conversely, use the input with value one at v and zero elsewhere. Along a graph path the endpoint equalities propagate, proving constancy on components. Grouping the resulting sum by components gives the claimed factorization. $\square$

The defect, both directions of the invariance criterion, propagation along arbitrary finite paths, and connected-component factorization are proved in `Lean/Geometry/SourceRecordProtection.lean`. Apply the scalar theorem coordinatewise to a vector readout. An active seam changes that readout whenever its two coefficient vectors differ. This describes the specified linear readout and autonomous pair-mean law. It excludes neither nonlinear records nor preserved memory, driven evolution, or a different repair law.

For the twelve-port carrier, the live rank-three readback has distinct coefficient vectors at distinct ports. It therefore changes at every active seam. This does not change the rank or metric of the response space: those belong to the repair operator. It changes the point represented by the live load inside that space. If internal repairs converge to a constant load, the centered rank-three readback tends to zero.

An exact finite execution distinguishes the two record choices on the golden population with $q = 13$. For each of its 2197 sites, the initialized loads use the signed antipodal split of the conservative record. The first lexicographically ordered unequal intra-carrier seam is averaged once; the zero record requires no update. All 2196 active updates preserve total load and strictly decrease its quadratic potential. Every active carrier's live spatial address changes, while the 2197 resulting live addresses are distinct. For the squared dimensionless read radius $1/13$, the number of directed metric-neighbour pairs including self-pairs changes from 294191 to 32249; 135123 unordered adjacency decisions change. The exact rational event history and independent quadratic-field distance census are provided in `code/source_population/`. This execution uses intra-carrier repairs only; it is not an implementation of routed inter-carrier reads.

There is also a simple invariant population extension. Give each carrier an immutable initialized address register and a mutable port-load register; let each repair write only the latter. The address population and its metric-neighbour graph are then unchanged under every repair word, even though the live amplitudes relax. The same execution retains all 2197 prepared addresses exactly. The golden population, its initialization and the protected write footprint are supplied inputs. The causal/count limit applies to these retained addresses when its read law and measure hypotheses hold. Availability of a conservative source word alone does not establish the preparation or preservation of its live readback. A maintained source population requires a stated protected-record or driven-population law.

A.6 Source-population quadrature and detector errors

Cell masses connect a finite source population to an integral on its metric carrier. A covering radius controls nearest-point geometry; an assignment radius controls quadrature for a specified measure. These radii can differ. For example, a golden-orbit point need not lie in its equal-volume permutation cell. The estimates below use the actual cell assignment and its measure.

Let Ω be a metric window with a finite positive Borel measure μ . Choose a finite measurable partition C_i , representatives $s_i \in \Omega$, and masses $v_i = \mu(C_i)$. A measurable convention assigns shared boundaries; null modifications do not affect the statements. Representatives need not belong to their cells. Let f take values in a real Banach space and be integrable on every cell. Set

$$Qf = \sum_i v_i f(s_i), \quad D_i = \int_{C_i} d(s_i, x) d\mu(x).$$

Proposition A.23 (Integrated cell-transport estimate). *Suppose $\|f(s_i) - f(x)\| \leq K_i d(s_i, x)$ for almost every $x \in C_i$, where $K_i \geq 0$ and the displacement is integrable. Then*

$$\left\| Qf - \int_{\Omega} f d\mu \right\| \leq \sum_i K_i D_i. \quad (9)$$

If $d(s_i, x) \leq H_i$ almost everywhere on C_i , the right-hand side is at most $\sum_i v_i K_i H_i$. In particular, for a globally K -Lipschitz function and a uniform assignment bound H ,

$$\left\| Qf - \int_{\Omega} f d\mu \right\| \leq \mu(\Omega)KH. \quad (10)$$

For a sequence of such partitions of the same measured window, $H_n \rightarrow 0$ therefore gives $Q_n f \rightarrow \int_{\Omega} f d\mu$ for each integrable Lipschitz test function. Neither nesting nor a fixed number of representatives is required.

Proof. The difference equals $\sum_i \int_{C_i} [f(s_i) - f(x)] d\mu(x)$. The norm of a Bochner integral is bounded by the integral of the norm. Apply the local Lipschitz bound and sum the resulting inequalities. The uniform estimate uses $D_i \leq v_i H_i$ and $\sum_i v_i = \mu(\Omega)$. Its right-hand side tends to zero under the stated refinement. The first-moment coefficient is sharp: on $[0, H]$, with Lebesgue measure, one representative at zero and $f(x) = x$, the error and displacement integral both equal $H^2/2$. $\square$

The Lean module `SourcePopulationQuadrature` proves the actual Bochner-integral estimates for finite cell measures, their specialization to measurable disjoint partitions, and convergence under a vanishing assignment bound. It also checks that a constant detector exposes any error in total assigned mass. The source coordinates, partition geometry and their displacement bounds must be established for each population. These hypotheses are not replaced by an assumed quadrature limit.

For the golden permutation cells of side L/q , the established assignment bound $H_q = 2\sqrt{3}L/q$ and equal masses L^3/q^3 instantiate (10). Product time cells of length Δ_q contribute the independent temporal error $TL^3\Delta_q \text{Lip}_t(f)$ on an aligned window of duration T . The spatial term is $TL^3H_q \text{Lip}_x(f)$. An unmatched endpoint layer contributes at most $L^3\Delta_q \|f\|_{\infty}$. This application retains the declared event population and time-cell convention; extra routing or bookkeeping events cannot be inserted into its density without another count estimate. The golden floor/permutation argument, spacetime null-boundary estimates and causal ordering-fraction integral are analytic arguments separate from the formal cell-transport theorem.

Proposition A.24 (Localized radial-kernel quadrature). *For $\epsilon > 0$, write*

$$w_x(y) = (1 - d(x, y)/\epsilon)_+, \quad g_x(y) = w_x(y)[f(x) - f(y)].$$

If f is a real K -Lipschitz function, then g_x is K -Lipschitz and vanishes outside the closed radius- ϵ ball about x . With an assignment map $P(y) = s_i$ on C_i satisfying $d(P(y), y) \leq H$, where $H \geq 0$, its quadrature error obeys

$$\left| \sum_i v_i g_x(s_i) - \int_{\Omega} g_x(y) d\mu(y) \right| \leq K \int_{\Omega \cap B(x, \epsilon + H)} d(P(y), y) d\mu(y) \leq KH \mu(\Omega \cap B(x, \epsilon + H)). \quad (11)$$

The same statement holds with closed balls; their boundaries are irrelevant for Euclidean volume.

Proof. The tent weight lies in $[0, 1]$ and satisfies $|w_x(y) - w_x(z)| \leq d(y, z)/\epsilon$. Order the two weights so that $w_x(y) \leq w_x(z)$. If both vanish the claim is immediate; otherwise $d(x, z) < \epsilon$. The identity

$$g_x(y) - g_x(z) = w_x(y)[f(z) - f(y)] + [w_x(y) - w_x(z)][f(x) - f(z)]$$

is bounded by $Kd(y, z)[w_x(y) + d(x, z)/\epsilon]$. The bracket is at most one because $w_x(y) \leq w_x(z) = 1 - d(x, z)/\epsilon$. The same proof with y, z exchanged covers the other order. If $d(x, y) \geq \epsilon + H$, both kernel terms $g_x(y)$ and $g_x(P(y))$ vanish. Integrate the pointwise bound over the remaining region. The tent Lipschitz estimates have Lean proofs; the displayed support-restricted integral applies the cell-transport argument. $\square$

For three-dimensional volume, multiplying (11) by $45/(\pi\epsilon^5)$ gives the quadrature contribution $60KH(\epsilon + H)^3/\epsilon^5$ in the scalar operator estimate, and hence $480KH/\epsilon^2$ when $H \leq \epsilon$. Actual local cell masses or integrated displacements can give smaller bounds than the enclosing-ball volume. They must be computed or bounded for the same assignment. A small fill radius does not justify replacing the equal-volume assignment radius. This estimate alone yields no vanishing operator error when $H = O(q^{-1})$ and $\epsilon = O(q^{-1/2})$.

Dual cells for a local tensor action. A different supplied action can use sorted source coordinates $0 = x_0 < \dots < x_N = L$, with interval lengths $\ell_j = x_{j+1} - x_j$. Half-interval endpoint weights and interior weights $(\ell_{j-1} + \ell_j)/2$ are the masses of the one-dimensional dual cells. Their tensor products satisfy the preceding transport theorem. For $f \in C^2([0, L])$, linear interpolation on each interval also gives the sharper analytic trapezoidal bound

$$\left| Qf - \int_0^L f(x) dx \right| \leq \frac{\|f''\|_{\infty}}{12} \sum_j \ell_j^3 \leq \frac{L\ell_{\max}^2}{12} \|f''\|_{\infty}.$$

Indeed the pointwise interpolation error is at most $\|f''\|_{\infty}(x - x_j)(x_{j+1} - x)/2$; integrate it on each interval. Successive application in three coordinates gives the tensor bound $L^3\ell_{\max}^2 \sum_{a=1}^3 \|\partial_a^2 f\|_{\infty}/12$. This applies to detector products and sampled mode Gram entries with their stated derivative bounds. Dual-cell weights and equal-volume source-count weights are different finite measures, each with its own comparison to volume. The trapezoidal argument does not improve the radial action by changing its weights implicitly, and it supplies no source-selection law, primitive timing rule or physical calibration.

A.7 Operational selection of the causal cone

The response-selected source Gram space E is three-dimensional. Its unit directions form a sphere, and adjoining a scalar line gives the quadratic form $Q(t, x) = t^2 - \|x\|^2$ on $\mathbb{R} \oplus E$. The finite proper carrier rotations act isometrically on E . Under natural feasibility and objective data with a unique normalized minimizer, the sixty directed seams have equal weight and select the corresponding translation average on the conservative record carrier. These are spatial and algebraic results. The following analytic theorem states an additional condition that selects an operational causal cone on that same carrier.

A displacement cone C contains zero and is closed under addition and nonnegative scaling. Suppose it represents a homogeneous operational order by $p \preceq q$ if and only if $q - p \in C$. Pointedness, $C \cap (-C) = \{0\}$, expresses antisymmetry. Operational covariance means that transformations of admissible histories, interventions and readouts preserve possible influence, with the corresponding linear action on event displacements. Merely changing a frame or relabeling celestial directions does not establish this condition. If C is instead the closed cone generated by finite operations, the conclusion concerns that closure; it does not assert that every displacement is a finite direct read.

For $u \in E$, define

$$X_u(t, x) = (\langle u, x \rangle, tu).$$

For a unit vector n , $B_n(s) = \exp(sX_n)$ is the boost of rapidity s : writing $x = x_{\parallel}n + x_{\perp}$,

$$t' = \cosh(s)t + \sinh(s)x_{\parallel}, \quad x'_{\parallel} = \sinh(s)t + \cosh(s)x_{\parallel}, \quad x'_{\perp} = x_{\perp}.$$

Theorem A.25 (Finite rotations and one boost direction). *Let $G \subset \text{SO}(E)$ be finite and irreducible, as for the proper icosahedral representation. Let $C \neq \{0\}$ be a closed convex pointed displacement cone invariant under $\text{diag}(1, R)$ for $R \in G$. Suppose, for one unit n , that $B_n(s)C = C$ for every real s . Then*

$$C = \{(t, x) : t \geq \|x\|\} \quad \text{or} \quad C = \{(t, x) : t \leq -\|x\|\}.$$

A positive time-axis vector in C selects the first alternative. It is enough to assume boost invariance for a sequence of nonzero rapidities tending to zero.

Proof. Irreducibility gives $\text{span}\{Rn : R \in G\} = E$. Conjugation by $\text{diag}(1, R)$ gives invariance under every $\exp(sX_{Rn})$. Write $u = \sum_j a_j R_j n$. Since $u \mapsto X_u$ is linear,

$$\exp(sX_u) = \lim_{m \rightarrow \infty} \left(\prod_j \exp\left(\frac{sa_j}{m} X_{R_j n}\right) \right)^m. \quad (12)$$

Indeed, the one-step product and $\exp(sX_u/m)$ have equal constant and linear Taylor terms, so their difference in matrix norm is $O(m^{-2})$. A telescoping sum over m factors bounds the total difference by $O(m^{-1})$; all intermediate products have a uniform exponential norm bound. Every finite product preserves C , and closedness passes this inclusion to the limit. Repeating with $-s$ gives equality. Thus every boost preserves C .

The finite average $|G|^{-1} \sum_{R \in G} R$ projects onto the fixed vectors of G , which are zero by irreducibility. Convexity therefore gives $(t, 0) \in C$ whenever $(t, x) \in C$. If $\|x\| > |t|$, boosts parallel to x can make the time coordinate positive or negative. Averaging these two images puts both signs of the time axis in C , contradicting pointedness. Hence every nonzero vector in C is causal.

A nonzero future causal vector has positive time coordinate. Its finite rotation average supplies a positive time-axis vector. Boosts and positive scaling generate every future timelike vector; closedness adds the null boundary. A past vector would similarly supply the opposite time axis,

contradicting pointedness. This gives the future cone. Starting with a past vector gives the past cone. Nontriviality ensures one case occurs.

Finally, if $s_k \neq 0$ tends to zero, for every $s \in \mathbb{R}$ choose integers m_k with $m_k s_k \rightarrow s$. Integer powers and inverses of $B_n(s_k)$ preserve C . Continuity and closedness give invariance under $B_n(s)$, reducing the last assertion to the preceding proof. $\square$

The theorem uses only two finite representation facts: the boost-axis orbit spans E , and the group average on E vanishes. It does not infer continuous spatial covariance from the finite carrier action. Its conclusion is a restriction: rotation covariance alone permits every cone $\{(t, x) : t \geq 0, \|x\| \leq vt\}$ with $v > 0$, and the stationary time-axis cone. Operational boost covariance excludes those alternatives except the unit-speed cone in the specified scalar coordinate.

The scalar coordinate requires an additive clock readout on actual histories. Once that attachment and the normalized boost action are fixed, the cone slope is fixed relative to them. Constructing the clock attachment is a separate dynamical requirement. The theorem does not select a layer tick, read weights, signal amplitudes or a field equation. The count construction in Section A.8 supplies a separate geometric duration readout on the declared order/count family.

At finite resolution, a bounded event set cannot carry a nontrivial continuous boost action by permutations. A physical use of the theorem therefore needs an operational continuum description or controlled comparisons across a varying family. Finite energy cutoffs and bounded windows must be transformed or restricted consistently. The existing algebraic Lorentz and celestial maps alone supply none of these intervention or clock attachments. The cone classification and product limit above are analytic proofs, rather than Lean formalizations of operational covariance.

A.8 A clock read from source interval counts

The equal-mass source populations of Proposition A.20 admit a clock defined by their records. Suppose an observer can read the complete ancestry of an upper event and identify a lower anchor. Its inclusive interval is $I(p, q) = \{e : p \preceq e \preceq q\}$. Choose a nondegenerate reference interval J in the same history and define

$$\mathcal{T}(I; J) = \left(\frac{|I|}{|J|} \right)^{1/4}. \quad (13)$$

This readout uses order and cardinality. It needs no numerical event labels, spatial coordinates, timestamps, absolute density or oscillator frequency. Complete ancestry access is a substantive readout hypothesis: a hash of an unavailable record does not provide its contents. The cost of collecting and counting the records is separate from the duration reconstructed by the readout.

Theorem A.26 (Count clocks and proper-time ratios). *Consider the declared source histories with their controlled order and equal-mass count limit. Let I_n, J_n approximate fixed interior timelike diamonds, with one common event weight $w_n > 0$, and let their limiting proper-time separations be $\tau_I, \tau_J > 0$. Then*

$$\mathcal{T}(I_n; J_n) \longrightarrow \frac{\tau_I}{\tau_J}.$$

The reference interval fixes the time unit; the clock origin is free.

Proof. For coordinate measure $dt d^3x$ and metric $c^2 dt^2 - |dx|^2$, a flat diamond of proper-time separation τ has volume $K\tau^4$, where $K = \pi c^3/24$. The vertical formula follows by integrating its ball sections. A Lorentz transformation has determinant one and sends a timelike tip displacement to its rest form, so the same formula holds for moving tips. The count-volume theorem gives $w_n |I_n| \rightarrow K\tau_I^4$ and $w_n |J_n| \rightarrow K\tau_J^4$. Taking their ratio and its positive fourth root proves the claim. Both the event weight and the volume constant cancel. $\square$

The supplied relation $\Delta = a/c$ enters the proof of the order and volume limit. It is absent from the clock's definition. This distinction allows a retrospective clock to be read from the same history without assuming an oscillator Hamiltonian. It does not select the population, read law or laboratory unit from accepted microscopic repairs.

Proposition A.27 (Finite clock enclosure). *Suppose $A = wN$, $B = wM$ and independently justified volume bounds obey $|A - V_I| \leq E_I$, $|B - V_J| \leq E_J$, with $B > E_J \geq 0$, $E_I \geq 0$ and $V_J > 0$. Then*

$$\left(\frac{(A - E_I)_+}{B + E_J} \right)^{1/4} \leq \frac{\tau_I}{\tau_J} \leq \left(\frac{A + E_I}{B - E_J} \right)^{1/4}. \quad (14)$$

The same formula uses counts N, M and errors $E_I/w, E_J/w$.

Proof. The true volumes belong to their stated intervals. Division by a positive denominator and the increasing fourth root preserve the two inequalities. The error bound in (5) supplies such volume errors when its geometric hypotheses hold. Counts alone do not supply those errors. $\square$

For relative volume errors $A = V_I(1 + d_I)$ and $B = V_J(1 + d_J)$, with $|d_I| \leq \eta_I < 1$ and $|d_J| \leq \eta_J < 1$, the measured clock divided by the exact proper-time ratio lies in

$$\left[\left(\frac{1 - \eta_I}{1 + \eta_J} \right)^{1/4}, \left(\frac{1 + \eta_I}{1 - \eta_J} \right)^{1/4} \right]. \quad (15)$$

Proposition A.28 (Unit freedom within the volume clock class). *A positive elapsed-time function $F(V)$ depending only on diamond volume and additive for successive collinear inertial intervals is $F(V) = \kappa V^{1/4}$ for some $\kappa > 0$.*

Proof. Set $g(t) = F(Kt^4)$ for $t > 0$. Inertial additivity gives $g(t + s) = g(t) + g(s)$. Positivity implies strict monotonicity. Additivity first fixes g on positive rationals; monotonicity and rational approximation give $g(t) = g(1)t$ for all positive real t . This is uniqueness within the stated class, not a premise about every physical pointer. $\square$

Accumulating duration along a curve. Let $\gamma(t) = (t, x(t))$ be C^2 on $[0, T]$, with $|x'(t)| \leq v < c$, $|x''(t)| \leq A_0$, and a common spatial buffer for its small diamonds. Segment durations are comparable to $\delta_n \rightarrow 0$. Suppose

$$\frac{\Delta_n}{\delta_n} \rightarrow 0, \quad \frac{H_n}{\delta_n} \rightarrow 0, \quad \frac{h_n}{a_n} \rightarrow 0,$$

and sampled tips have spatial and temporal errors $O(H_n + \Delta_n)$. Summing (13) over successive segment intervals, using the same fixed reference interval, converges to

$$\frac{1}{\tau_J} \int_0^T \sqrt{1 - |x'(t)|^2/c^2} dt. \quad (16)$$

Indeed, the two-shell estimate is uniformly $O(\delta_n^4 h_n/a_n + \delta_n^3(H_n + \Delta_n))$ per segment. The strict speed margin bounds each exact diamond volume below by a positive constant times δ_n^4 . Tip perturbations have the same relative error order. Equation (15) therefore controls all segment clocks by one vanishing relative error.

The continuum polygon error has an explicit bound. For $f(u) = \sqrt{1 - |u|^2/c^2}$ on $|u| \leq v$, its Hessian norm is at most $M_0 = [c^2(1 - v^2/c^2)^{3/2}]^{-1}$. Taylor expansion about each segment's average velocity has zero integrated linear term. Since the velocity differs from that average by at most $A_0\delta_i$, the chord and curve proper times differ by at most $M_0 A_0^2 \delta_i^3/2$. The total is at most $M_0 A_0^2 T(\max_i \delta_i)^2/2$. This proves (16). Fixed one-layer intervals do not satisfy these resolution hypotheses.

Finite records and calibration scope. The finite executable control parses writer identities, immutable register versions and values in the $q = 5$ history. Its 500 events give the three centered inclusive counts 2, 41, 80. The first two continuum diamonds lie inside the source window; the third does not. Exact dyadic fourth-root bounds, schedule permutation and opaque event relabeling check the readout. These coarse intervals provide no finite proper-time accuracy certificate. The source-pinned expected hash authenticates the replayed content relative to that commitment; it is not an external signature or a proof of physical provenance. Unrelated serial audit links are excluded from the semantic read order.

Lean formalizes the finite interval and root inequalities. The count-volume and curve limits above are analytic proofs. In this layered law every causal pair k layers apart has a path of exactly k reads, and every strict chain has at most k edges. Longest-chain height consequently reads coordinate duration independently of displacement. It is not a proper-time clock for moving tips in this family. The count construction concerns geometric duration reconstructed from accessible records; it neither sets the time cost of primitive seam words nor identifies physical clock readings without a common physical interpretation of the event law.

A.9 Observations of the causal manifold

An observer inside the source-record family of Section A.4 has access to the event order and to event cardinalities. This section records what such an observer reads from those two inputs at finite regulator, what the carriers produce of the family itself, and which of the finite statements are theorems.

The finite poset. For a Fibonacci level q the events are the pairs (j, s) of a round $j \leq K_q = \lceil \sqrt{q} \rceil$ and a site $s \in S_q$, and $(j, s) \preceq (j', t)$ exactly when $j \leq j'$ and the graph distance on the neighbour graph satisfies $d(s, t) \leq j' - j$. This relation is a partial order, it equals the reflexive transitive closure of the one-round reads, every round is an antichain, the width equals $|S_q| = q^3$, and every event at round j has height j (`Lean/Geometry/SourceNetLayeredOrder.lean`, theorems `layerPrec_isPartialOrder`, `layerPrec_iff_reflTransGen`, `walk_iff_walkDist_le`, `width_eq_card_sites`, `height_eq_layer`; the concrete read law of the family is the instance `precedes_iff_layerPrec`). The flat reference constants of the ordering fraction are exact: the oriented-pair integral of Proposition A.21 equals $T^8/1920$, so the fraction equals $1/10$, and the Myrheim–Meyer values in dimensions 2, 3 and 4 are $1/2$, $8/35$ and $1/10$ (`Lean/Geometry/OrderingFractionFourDimensional.lean`). The finite clock enclosure of Proposition A.27 is formalized in `Lean/Time/SourceCountClockEnclosure.lean`.

What the observer reads. On the interior vertical diamond of K_q rounds the inverted Myrheim–Meyer dimension reads 2.63, 2.93, 4.15, 3.61, 4.10 and 4.09 at $q = 5, 8, 13, 21, 34, 55$, with ordering fractions 0.306, 0.241, 0.088, 0.138, 0.092 and 0.093 against the flat value $1/10$; levels with even K_q approach the value from below and levels with odd K_q from above. The moving-tip diamonds of Proposition A.19 read 0.250, 0.135, 0.121, 0.115 and 0.106 at $q = 8, 13, 21, 34, 55$, dimension 2.89 to 3.93. The same read law on the two- and one-dimensional golden populations reads dimensions 3.02 and 2.01 at $q = 55$ against the constants $8/35$ and $1/2$, so the readout responds to the dimension of the population. The normalized count of the vertical diamond against $\pi c^3 T^4/24$ reads 0.193 against 0.198 at $q = 13$, 0.141 against 0.147 at $q = 34$ and 0.172 against 0.177 at $q = 55$. The count clock $(|I|/|J|)^{1/4}$ of two nested vertical diamonds reads 1.705, 2.088 and 1.907 at $q = 13, 34, 55$ against the model-time ratio 2. Pair counts are exact through $q = 34$ and stratified at $q = 55$; the structural fields at $q = 5, 8, 13$ agree byte for byte with the replay of Section A.4. The bounds

of Proposition A.18 exceed the diamond volume at these levels, and the finite values above are readings of the supplied law, not a demonstration of its limit.

What the carriers produce. The family is produced by the carriers themselves. One exact twelve-port carrier per site holds the record $z(b)$ as its port loads on the six antipodal axes and reads its position through the rank-three response $x = 2P_{\text{slow}}N$ of the rank-three source metric. The readback metric equals the source-axis contraction $s(b)$ with scale exactly one, an identity in $\mathbb{Q}(\sqrt{5})$ checked on every pair through $q = 21$ and on 10^6 seeded pairs at $q = 34$. Each carrier decides its neighbours from its own readback; the resulting relation equals the family's neighbour graph edge for edge at $q = 5, 8, 13, 21, 34$. The carriers then read one another for K_q rounds, every read an authenticated read-after-write event, and the precedence generated from those events with no declared parent lists equals the layered order on the centre diamond at every level, with strict pair counts 968, 4,244, 102,990, 2,904,889 and 47,881,819. A unit intervention at the centre in round zero changes exactly the future cone in every round. The operation costs are recorded: reads per round $2E + n$ for E edges and n carriers, at $q = 34$ a total of 157,922,880 reads and 275,128 writes.

Inference and boundary. Order and count determine conformal geometry and scale under the continuum causality hypotheses [12, 13]; that is the inference the observer makes from the readings above, and the finite readings are consistent with flat $1 + 3$ spacetime at every level examined. The population, the read law, the round duration and one counted event per carrier and round are supplied. Their selection by accepted native repairs, a physical clock, and the continuum limit are separate conditions rather than results of this section. Isotropy of link directions, interval abundance profiles and curvature estimators on the family are not evaluated here. The receipts, the carrier event logs and a standard-library generator that rebuilds the poset from the definitions are published as the causal-poset evidence package [24].

A.10 Spatial populations and a common scalar continuum action

The completed conservative source-record carrier is a Euclidean three-space under the stated response and metric hypotheses. Its whole dense record group is not a locally finite population. A finite-resolution selection and a transport law can be stated on that carrier without inserting a Cartesian lattice. This construction keeps both choices explicit.

Fix $r, \delta, \eta > 0$, a response-metric ball $\Omega = \overline{B}(0, r)$, and a finite directed seam-word endpoint menu $M \subset \Omega$ that is an η -net of Ω . Such menus exist: density approximates a slightly inward-shifted point by a source endpoint, compactness extracts finitely many approximants, and a finite maximum word budget contains all of them. This is algebraic endpoint availability, not a proof that a repair law accepts those words as moves. Choose a maximal δ -separated subset $S \subset M$, and set $h = \delta + \eta$. The resolution, window and tie rule are supplied; maximal means inclusion-maximal, not maximum cardinality or a selected energy minimum.

Proposition A.29 (Finite spatial population and metric transport). *The selected points form an h -net, and*

$$(r/h)^3 \leq |S| \leq (1 + 2r/\delta)^3.$$

For $R \geq h$ and $B(x, R) \subset \Omega$,

$$((R - h)/h)^3 \leq |S \cap B(x, R)| \leq (1 + 2R/\delta)^3.$$

Connect distinct points whose source distance is at most $a > 2h$, weighting edges by that distance. The graph is connected, degrees are bounded by $(1 + 2a/\delta)^3 - 1$, and its shortest-path metric satisfies

$$|s - t| \leq d_G(s, t) \leq \left(1 + \frac{2h}{a - 2h}\right) |s - t| + 2h.$$

Consequently $a \rightarrow 0$ and $h/a \rightarrow 0$ give uniform metric convergence on the fixed window. With $\eta \leq \delta$ and $a = 8\delta$, degrees instead stay at most 4912 and $d_G \leq 2|s - t| + 4\delta$; this latter estimate alone does not imply a Euclidean metric limit.

Proof. Maximality puts each menu point within δ of S , proving coverage. Disjoint radius- $\delta/2$ balls give the upper counts; covering the window, or $B(x, R - h)$, by radius- h balls gives the lower counts. For the graph, divide the segment from s to t into $k = \lceil |s - t|/(a - 2h) \rceil$ pieces. Convexity retains the segment in Ω ; approximate its interior division points within h , keeping its endpoints exact. Consecutive approximants are at most a apart and the path length is at most $|s - t| + 2hk$. The triangle inequality gives the lower bound. These estimates are independent of the choice among maximal sets. $\square$

The metric fixes relative three-volume. Let C_i be the Voronoi cell of $x_i \in S$, clipped to Ω , and $v_i = \text{vol}(C_i) > 0$. Thus $\sum_i v_i = V = \text{vol}(\Omega)$, and almost every point of C_i is at distance at most h from x_i . These are metric weights, not a physical event-count density. For $0 < h \leq \epsilon$ supply

$$a_{ij} = \frac{45v_i v_j}{\pi \epsilon^5} (1 - |x_i - x_j|/\epsilon)_+, \quad (Lq)_i = v_i^{-1} \sum_j a_{ij} (q_i - q_j).$$

The weighted inner product is $\langle p, q \rangle_v = \sum_i v_i p_i q_i$. Symmetry gives $\langle q, Lq \rangle_v = \frac{1}{2} \sum_{ij} a_{ij} (q_i - q_j)^2 \geq 0$. The common finite action is

$$S_h[q] = \frac{1}{2} \int \left[\|\dot{q}\|_v^2 - c^2 \langle q, Lq \rangle_v - m^2 \|q\|_v^2 \right] dt, \quad c > 0, \quad m \geq 0. \quad (17)$$

Its equation is $\ddot{q} + c^2 Lq + m^2 q = 0$. The coefficient m has inverse-time units in this convention; neither physical mass nor time units are inferred.

Proposition A.30 (Uniform scalar and detector limit). *Let $u_{tt} - c^2 \Delta u + m^2 u = 0$ be a smooth reference on the full carrier. On $[0, T]$, its spatial support has distance greater than ϵ from the complement of Ω . Write $M_1 = \sup |\nabla u|$ and $M_4 = \sup \|D_x^4 u\|_{\text{op}}$, with suprema over the full carrier and time window. Then*

$$|(Lu|_S)_i + \Delta u(x_i)| \leq \frac{15}{112} \epsilon^2 M_4 + \frac{120h(\epsilon + h)^3}{\epsilon^5} M_1 \leq \frac{15}{112} \epsilon^2 M_4 + \frac{960h}{\epsilon^2} M_1.$$

Put $B = c^2 \sqrt{V} [(15/112) \epsilon^2 M_4 + 960h \epsilon^{-2} M_1]$. Initialize the finite action with the sampled reference position and velocity. Then

$$\|\dot{q} - u_t|_S\|_v \leq tB, \quad \|q - u|_S\|_v \leq \frac{1}{2} t^2 B.$$

For a fixed C^1 detector b , the field readout $D_h = \sum_i v_i b(x_i) q_i$ and its continuum counterpart $D = \int_\Omega b u$ obey

$$|D_h(t) - D(t)| \leq \frac{1}{2} \sqrt{V} \|b\|_\infty t^2 B + Vh \|\nabla(bu(t))\|_\infty.$$

Thus $\epsilon \rightarrow 0$ and $h/\epsilon^2 \rightarrow 0$ yield a uniform finite-time observable limit. In normalized carrier units $\epsilon = h^{1/4}$ gives an $O(\sqrt{h})$ bound. Nested populations are unnecessary.

Proof. Compare L with the integral operator having kernel $45(1 - |z|/\epsilon)_+ / (\pi\epsilon^5)$. Odd Taylor terms cancel and

$$\int (1 - |z|/\epsilon)_+ z_1^2 dz = \frac{2\pi}{45}\epsilon^5, \quad \int (1 - |z|/\epsilon)_+ |z|^4 dz = \frac{\pi}{14}\epsilon^7.$$

The second moment gives $-\Delta$; the fourth-order Taylor remainder, bounded by $M_4|z|^4/24$, gives $15\epsilon^2 M_4/112$. For nearest-cell projection P , compare $g_x(y) = (1 - |x - y|/\epsilon)_+(u(x) - u(y))$ with $g_x(P(y))$. Splitting the difference at the larger kernel value uses a point within ϵ of x ; kernel and field Lipschitz bounds give $|g_x(P(y)) - g_x(y)| \leq 2hM_1$. This difference vanishes outside $B(x, \epsilon + h)$, giving the stated quadrature term. If $u(x) \neq 0$, the support buffer retains the whole kernel ball in Ω ; otherwise the integrand outside Ω is zero. No boundary term is omitted.

The sampled residual in the weighted norm is at most B . The positive operator $A = c^2 L + m^2 I$ has $\|\cos(t\sqrt{A})\| \leq 1$ and $\|\sin(t\sqrt{A})/\sqrt{A}\| \leq t$, with the zero eigenspace interpreted by continuity. Duhamel integration gives the two trajectory bounds. Cauchy–Schwarz and nearest-cell quadrature for bu give the detector bound. $\square$

The same cell argument gives $\|L\|_v \leq 960/\epsilon^2$, using twice the largest weighted degree. **MetricKernelEnergy** formalizes the finite positive-energy and mass-weighted degree inequalities; the geometric, integral and continuum arguments above are analytic proofs. A local discrete-action time integrator with a strict Courant bound may consume this operator and add its own time and initialization errors.

Smooth compact initial data admit the reference until their speed- c domain of influence reaches the window boundary. Local energy flux for the Klein–Gordon equation gives that speed; the nonnegative mass term changes energy density but not the characteristic cone. Applied to a smooth local preparation difference, the detector estimate bounds the finite response outside that cone. A nonzero continuum signal is resolvable when its lower bound exceeds the approximation and detector errors. Instantaneous small tails of the finite graph are not identified with detectable superluminal signals. This is an interior effective limit, not a theorem about arbitrary boundary conditions or exact microscopic order reflection.

The canonical variables $Q_i = \sqrt{v_i}q_i$ and $P_i = p_i/\sqrt{v_i}$, $p_i = v_i\dot{q}_i$, give the oscillator Hamiltonian $\frac{1}{2}(|P|^2 + Q^T \tilde{A}Q)$, where $\tilde{A} = M^{1/2}(c^2 L + m^2 I)M^{-1/2} \geq 0$ and $M = \text{diag}(v_i)$. Supplied canonical quantization on $L^2(\mathbb{R}^{|S|}, dQ)$ gives the same field readouts as multiplication operators. For $m > 0$ every mode has a Gaussian vacuum; at $m = 0$ zero modes remain free particles unless a separate restriction is declared. This finite Hilbert construction is not an interacting quantum continuum or a derivation of the Standard Model.

Coarsening, the volume prescription, radial force law, inertia, action coefficients and accessible observer variables remain added assumptions. The kernel graph may have growing degree. Algebraic source words alone do not implement its couplings as accepted primitive repairs or calibrate their duration. The construction supplies one conditional scalar effective world on the response carrier; its selection and physical interpretation remain separate obligations.

Proposition A.31 (Controlled boost comparison for scalar detectors). *Let Λ be a proper orthochronous Lorentz transformation of $c^2 dt^2 - |dx|^2$, and let u be a smooth Klein–Gordon solution on the full spacetime. Put $u^\Lambda = u \circ \Lambda^{-1}$. Suppose both references satisfy the support and derivative hypotheses of Proposition A.30 on the same $[0, T]$ and Ω , with a common positive interior buffer. Choose a real $f \in C_c^1(\mathbb{R}^{1+3})$ such that both f and $f^\Lambda = f \circ \Lambda^{-1}$ are supported in $(0, T) \times \Omega^\circ$. For $w = u, u^\Lambda$, let q_h^w solve the same action (17), with the same L, v_i, c, m , initialized by $w(0, x_i)$ and*

$\partial_t w(0, x_i)$. Define the spacetime-smeared readouts

$$\mathcal{D}_h(f; q) = \int_0^T \sum_i v_i f(t, x_i) q_i(t) dt, \quad \mathcal{D}(f; w) = \int_{\mathbb{R}^{1+3}} f w dt d^3x.$$

Using the preceding spatial derivative bounds $M_1(w), M_4(w)$, put

$$B_w = c^2 \sqrt{V} \left[\frac{15}{112} \epsilon^2 M_4(w) + 960 h \epsilon^{-2} M_1(w) \right],$$

and

$$\mathcal{E}_h(f, w) = \frac{\sqrt{V} B_w}{2} \int_0^T t^2 \|f(t, \cdot)\|_\infty dt + V h \int_0^T \|\nabla_x(f(t, \cdot) w(t, \cdot))\|_\infty dt.$$

Then

$$|\mathcal{D}_h(f^\Lambda; q_h^{u^\Lambda}) - \mathcal{D}_h(f; q_h^u)| \leq \mathcal{E}_h(f^\Lambda, u^\Lambda) + \mathcal{E}_h(f, u). \quad (18)$$

The right-hand side tends to zero for $\epsilon \rightarrow 0$ and $h/\epsilon^2 \rightarrow 0$. The conclusion is uniform over a bounded boost class when the common window, buffer and displayed derivative and detector bounds are uniform over that class.

Proof. Lorentz covariance of the Klein–Gordon equation makes u^Λ a solution with the same c, m . The preceding detector estimate applies at each time with $b = f(t, \cdot)$; its time integral is $|\mathcal{D}_h(f; q_h^w) - \mathcal{D}(f; w)| \leq \mathcal{E}_h(f, w)$. Since $\det \Lambda = 1$ and both detector supports lie in the comparison window, change of variables gives $\mathcal{D}(f^\Lambda; u^\Lambda) = \mathcal{D}(f; u)$. The two approximation bounds and the triangle inequality prove the claim. $\square$

The transformed preparations are supplied through the continuum reference on the transformed Cauchy surface; this comparison does not construct a boost map from one finite history or its initial data. The same estimate applies to smooth local preparation differences by linearity. Spacetime smearing is essential to the displayed comparison: a fixed-time detector instead transforms to a tilted hypersurface with its transformed measure. The result controls detector values for one common declared action, not exact finite-graph support, native source selection, or boosts of arbitrary finite records. Time integration and preparation errors, if present in an execution, require their own bounds.

A.11 A common source-population scalar and quantum detector packet

The prepared golden source addresses admit a local massive scalar action whose field edges fit inside the causal construction’s existing read radius. The action below uses adjacent sorted source coordinates, rather than the radial kernel. It therefore has its own coefficients and boundary condition. The source addresses are protected preparation data, distinct from the mutable field registers. No invariance of an address under arbitrary live port repair is assumed.

Put $L = 2/\sqrt{\phi + 2}$ and use coordinates $y = x/L$, model time $s = ct/L$, $\hbar = 1$ and dimensionless mass one. For $q = F_n$, sort $\xi_b = b\phi - \lfloor b\phi \rfloor$, $0 \leq b < q$, as $0 = x_0 < x_1 < \dots < x_{q-1} < 1$, and append $x_q = 1$ as a fixed Dirichlet boundary marker. This marker is also a source-coded point. The mutable interior population $\{x_1, \dots, x_{q-1}\}^3$ is contained in the original golden product population. Each coordinate is the normalized readback of the integer control $a = -\lfloor b\phi \rfloor, b$; the nodes are not replaced by their nearby uniform-grid labels.

Write $h_i = x_{i+1} - x_i$ and $w_i = (h_{i-1} + h_i)/2$ for interior nodes. Let $W = \text{diag}(w_i)$ and let K be the Dirichlet stiffness matrix with diagonal $h_{i-1}^{-1} + h_i^{-1}$ and off-diagonal $K_{i,i+1} = -h_i^{-1}$. On the tensor population use product masses and the action

$$\mathcal{S}_q = \frac{1}{2} \int \left[\|\dot{u}\|_{W^{\otimes 3}}^2 - \langle u, (I + \mathbf{L}_1 + \mathbf{L}_2 + \mathbf{L}_3)u \rangle_{W^{\otimes 3}} \right] ds, \quad \mathbf{L} = W^{-1}K. \quad (19)$$

Here $\mathbf{L}_d$ acts on coordinate d . Each varying field read is between adjacent source sites; fixed boundary values contribute on-site terms. The golden gap is at most $2/q$, so $h_i < q^{-1/2}$ for $q > 4$. The four finite certificates also check the stronger exact gap inequality $qh_i^2 < 1$ directly. Thus these field edges are permitted by the radius $a_q = L/\sqrt{q}$. This inclusion is not a construction of authenticated time-step events. Subcycling and the custody of a count-derived clock require an additional execution comparison.

Mass whitening gives the strictly positive real matrix

$$A_q = I + B \otimes I \otimes I + I \otimes B \otimes I + I \otimes I \otimes B, \quad B = W^{-1/2}KW^{-1/2}.$$

Canonical quantization is the oscillator Fock space with vacuum-subtracted Hamiltonian $d\Gamma(\omega_q)$, $\omega_q = A_q^{1/2} \geq I$. The comparison theory is the massive Dirichlet free scalar on the same unit cube, with $\omega = (I - \Delta_D)^{1/2}$. Its nine comparison modes are

$$e_k(y) = \sqrt{8} \sin(k\pi y_1) \sin(\pi y_2) \sin(\pi y_3), \quad \omega_k = \sqrt{1 + \pi^2(k^2 + 2)}, \quad 1 \leq k \leq 9.$$

The remaining continuum and lattice modes belong to their respective full Fock spaces; they are not discarded from an evolution residual.

A full frequency residual from small tridiagonal solves. Let S sample these modes with square roots of the tensor masses, $G = S^*S$, and suppose $\|G - I\| \leq d < 1$. Then $j = SG^{-1/2}$ is an isometry. Put

$$c_d = (1 - d)^{-1/2} - 1, \quad s_d = \sqrt{1 + d} c_d, \quad \Omega = \text{diag}(\omega_k).$$

For $f_k(x) = \sqrt{2} \sin(k\pi x)$ define $s_k = W^{1/2}f_k|_S$ and $r_k = W^{1/2}(\mathbf{L}f_k - (k\pi)^2 f_k)|_S$. For each k compute the two nonnegative quadratic forms

$$Q_{kj} = r_j^* [B + (1 + \omega_k^2)I]^{-1} r_j, \quad j \in \{k, 1\},$$

and set

$$b_k = \sqrt{Q_{kk}} \|s_1\|^2 + 2\sqrt{Q_{k1}} \|s_k\| \|s_1\|, \quad r = \left(\sum_k b_k^2 \right)^{1/2}.$$

The quadratic forms require only the positive tridiagonal solve $(K + (1 + \omega_k^2)W)y = W(\mathbf{L}f_j - (j\pi)^2 f_j)$.

Proposition A.32 (Full lattice leakage and coefficient-sensitive control). *With the definitions above,*

$$\begin{aligned} \|\omega_q j - j\Omega\| &\leq r(1 - d)^{-1/2} + 2\omega_9 s_d, \\ \|(\omega_q j - j\Omega)v\| &\leq \kappa(v) := \sum_k b_k |v_k| + (rc_d + 2\omega_9 s_d) \|v\|. \end{aligned} \quad (20)$$

The bound satisfies $\kappa(Dv) \leq \kappa(v)$ whenever D is diagonal and $|D_{kk}| \leq 1$. For $m = 1$,

$$\|\omega_q^{-1/2} jv - j\Omega^{-1/2} v\| \leq \frac{1}{2} \kappa(v). \quad (21)$$

Proof. For a positive scalar λ , $(\sqrt{\lambda} - \omega_k)^2 \leq (\lambda - \omega_k^2)^2 / (\lambda + \omega_k^2)$. Functional calculus applies this inequality to the full matrix A_q . The stiffness residual on $Se_k = s_k \otimes s_1 \otimes s_1$ is the sum of its three tensor residuals. Minkowski's inequality in the $(A_q + \omega_k^2)^{-1}$ norm, followed by $A_q + \omega_k^2 \geq B_d + (1 + \omega_k^2)I$, bounds the full frequency residual column by b_k . This step includes its component perpendicular to the sampled nine-mode space. Hence $\|\omega_q S - S\Omega\| \leq r$ and its action on v is bounded by $\sum_k b_k |v_k|$.

Write $G^{-1/2} = I + C$, with $\|C\| \leq c_d$ and $\|SC\| \leq s_d$. Expanding

$$\omega_q j - j\Omega = (\omega_q S - S\Omega)(I + C) + S[\Omega, C]$$

gives both bounds. In the commutator estimate use $\|S\|\|C\| \leq s_d$; no commutation of G and Ω is assumed. Diagonal contraction decreases every term in κ . Finally, the resolvent formula for the inverse square root gives

$$-\frac{1}{\pi} \int_0^\infty t^{-1/2} (\omega_q + t)^{-1} (\omega_q j - j\Omega) (\Omega + t)^{-1} v \, dt.$$

Its norm is at most $\kappa(v) \pi^{-1} \int_0^\infty t^{-1/2} (1+t)^{-2} dt = \kappa(v)/2$. $\square$

The isometry on the nine-mode comparison Fock space is $\Gamma(j)$. On its sector with at most N particles, the full generator residual is at most $N[r(1-d)^{-1/2} + 2\omega_9 s_d]$, by the sum of the N one-particle residuals on symmetric tensors. This is an actual generator estimate, not a bound on a compressed matrix. For the particular coherent states below, using their coefficients directly gives a substantially smaller comparison without an occupation cutoff.

Proposition A.33 (The same action has a free-field detector limit). *Along the golden refinements, the sampled isometries on every fixed finite Dirichlet sine band have vanishing full frequency residual. For any fixed finite time interval, the classical smeared responses to bounded, piecewise-continuous L^2 momentum preparations, and the coherent-state expectations of finitely many bounded smeared Weyl effects, converge to those of the massive Dirichlet scalar on the cube. Smearings and their products must be Riemann integrable; a finite collection of slab-boundary jumps satisfies this condition. Initial position is zero for the momentum preparations in this statement.*

Proof. Let $\ell = \max_i h_i \rightarrow 0$. Taylor's formula at an interior node gives

$$|\mathbb{L}f(x_i) + f''(x_i)| \leq \frac{h_{i-1}^2 + h_i^2}{3(h_{i-1} + h_i)} \|f'''\|_\infty \leq \frac{\ell}{3} \|f'''\|_\infty.$$

This includes the generally nonzero error from unequal adjacent gaps. The product trapezoidal rule has error at most $\ell^2 \sum_d \|\partial_d^2 F\|_\infty / 12$ on the unit cube. Boundary masses may be included: the Dirichlet sine products vanish there, so their contributions to the Gram matrix are zero. Consequently $G - I = O(\ell^2)$ on every fixed band and the full stiffness residual is $O(\ell)$. The square-root resolvent bound, or directly $(\sqrt{\lambda} + \omega_k)^{-1} \leq 1/2$, makes the frequency residual $O(\ell)$ as well. These are actual residual estimates for this action; no ratio $H/a_q^2 \rightarrow 0$ is required.

Finite-band Duhamel and the negative-power estimate give uniform finite-time convergence of $\omega_q^{-1} \sin(s\omega_q)$ between sampled finite sine combinations, as well as the corresponding classical velocity readouts. The graph kinetic energy norm is the same product-mass norm. For general stated smearings, first approximate them in L^2 by finite sine combinations. Their squared sampling errors converge to the squared L^2 errors by product Riemann quadrature. The uniform bounds $\|\omega_q^{-1/2}\|, \|\omega^{-1/2}\| \leq 1$ and unitarity control the tails independently of q and of time in the fixed

interval. Take $q \rightarrow \infty$ for a fixed sine approximation, then let its error tend to zero. This also applies to the compact slab functions below.

The same argument controls coherent amplitudes, Weyl parameters, their inner products and their squared norms. The Gaussian characteristic function therefore gives the asserted bounded quantum readout convergence; finite Weyl products use the Weyl relations and the same controlled inner products. The limiting Fock theory is built on the full continuum L^2 one-particle space. Each finite-band isometry has its own domain; there is no isometry of that entire infinite one-particle space into a single finite lattice space. This observable convergence does not assert an interacting continuum or an operator-norm limit of all local effects. $\square$

Compact preparation and a separated bounded detector. Normalize f to unit L^2 norm, where

$$f(y) = C \sin(\pi y_1)(1 + \cos(\pi y_1))^8 \sin(\pi y_2) \sin(\pi y_3), \quad g(y) = f(1 - y_1, y_2, y_3).$$

Both have exactly the nine modes above; denote their coefficient vectors by a and $\tilde{a}_k = (-1)^{k+1} a_k$. The actual smearings are

$$f_c = f \mathbf{1}_{y_1 \leq 9/20}, \quad g_c = g \mathbf{1}_{y_1 \geq 11/20}.$$

Their supports are separated by $1/10$ in the normalized source coordinate, or $L/10$ in the source metric. Prepare $\exp(4i\Phi(f_c))|0\rangle$ and measure $E_c = (I + \sin \Phi(g_c))/2$, with $0 \leq E_c \leq I$. Use the same sampled compact smearings and the same definitions in the finite graph oscillator theory. These are field instruments with spatial support, not a rank-one projector onto a named mode. The unperturbed vacuum probability and the prepared probability at time zero are both $1/2$.

For the smooth comparison, put $b = \|\Omega^{-1/2}a\|/\sqrt{2}$ and $\alpha = 4b$. Its exact induced probability is

$$P(s) - \frac{1}{2} = \frac{1}{2} e^{-b^2/2} \sin \left(4 \sum_{k=1}^9 (-1)^{k+1} a_k^2 \frac{\sin(\omega_k s)}{\omega_k} \right). \quad (22)$$

This follows from the coherent-state characteristic function. For two coherent amplitudes differing by at most d_α and two Weyl parameters differing by at most d_b , the corresponding effects obey the explicit probability bound

$$\mathcal{E}(b, \alpha, d_b, d_\alpha) = \frac{1}{4} d_b (2b + d_b) + d_b (\alpha + d_\alpha) + b d_\alpha. \quad (23)$$

Indeed, $|e^{-x} - e^{-y}| \leq |x - y|$ for $x, y \geq 0$ and $|\sin x - \sin y| \leq |x - y|$ control the vacuum factor and the coherent phase. All norms refer to the full one-particle Hilbert spaces after the isometric comparison.

For the graph versus continuum smooth comparison, take

$$d_b = \frac{\kappa(a)/2 + s_d}{\sqrt{2}}, \quad d_\alpha = 4d_b + \frac{4|s|}{\sqrt{2}} \kappa(\Omega^{-1/2}a).$$

Equation (21) controls the initial smearing maps. Duhamel's formula on the finite spectral comparison space controls evolution; $\kappa(e^{-is\Omega}v) = \kappa(v)$ makes the bound uniform in time. The graph unitary acts on every lattice mode.

Each compact-continuum smearing differs from its smooth version by squared L^2 norm $\theta = \int_{9/20}^1 |\sum_k a_k \sqrt{2} \sin(k\pi x)|^2 dx$. Thus its extra d_b is at most $\sqrt{\theta/2}$ and its extra d_α at most $4\sqrt{\theta/2}$, using $\omega \geq 1$. The graph tails are computed separately from their actual lumped masses. Their two values need not equal the continuum tail or each other. Apply (23) to both localization steps

and add their errors to the smooth comparison error. The continuum compact preparation has finite expected energy at most 8 above the vacuum. The graph energy is $8\|f_c|_S\|_{W^{\otimes 3}}^2$, which the certificate computes separately. A discontinuity at a slab boundary does not imply membership in the Hamiltonian's operator domain. Only its form domain and unitary norm preservation are used here; the Duhamel calculation uses the smooth finite-mode vector. For any $E > 0$, the continuum energy tail is at most $8/E$ by the spectral Markov inequality.

Theorem A.34 (Resolved compact scalar quantum response). *For the declared action, instruments and units above, the $q = 233$ golden population satisfies, uniformly for $19/20 \leq s \leq 1$,*

$$|P_{233,c}(s) - P_c(s)| < 0.050055, \quad |P_{233,c}(s) - \tfrac{1}{2}| > 0.137754.$$

The comparison error is smaller than the induced response throughout a nonzero time interval. The response is to a spatially nonuniform compact preparation at a spatially separated compact detector.

Proof. The executable certificate uses outward rational arithmetic, the exact golden coordinates, positive tridiagonal elimination and the bounds above. It encloses the continuum tail by 0.000222611 and the full nine-mode frequency residual by 0.053232. The coefficient-sensitive preparation residual is below 0.005360. One hundred closed time cells cover the entire interval; Taylor remainders enclose all trigonometric and exponential values. The final inequalities include both sampled localization tails and the continuum localization error. An independent verifier assembles edge fluxes, solves the positive systems in reverse order, and recomputes the probability bounds. The smaller populations $q = 13, 34, 89$ are reported without requiring them to satisfy the signal criterion. $\square$

The tensor calculation uses one-dimensional arrays of length 232, while the mathematical finite model has 232^3 oscillators and an infinite-dimensional Fock space. No full read/write trace at that population is claimed. The theorem is a free massive scalar comparison with prescribed axes, boundary, action, protected addresses and model time. It does not select these inputs from native repair, identify physical clock units, prove an interacting or charged continuum, or identify Hamiltonian substeps with the retained causal-count history.

References

- [1] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, *Space-time as a causal set*, Phys. Rev. Lett. **59**, 521–524 (1987). <https://doi.org/10.1103/PhysRevLett.59.521>.
- [2] S. Surya, *The causal set approach to quantum gravity*, Living Rev. Relativ. **22**, 5 (2019). <https://doi.org/10.1007/s41114-019-0023-1>.
- [3] L. Bombelli and D. A. Meyer, *The origin of Lorentzian geometry*, Phys. Lett. A **141**, 226–228 (1989). [https://doi.org/10.1016/0375-9601\(89\)90474-X](https://doi.org/10.1016/0375-9601(89)90474-X).
- [4] G. Brightwell and R. Gregory, *Structure of random discrete spacetime*, Phys. Rev. Lett. **66**, 260–263 (1991). <https://doi.org/10.1103/PhysRevLett.66.260>.
- [5] S. Major, D. Rideout, and S. Surya, *On recovering continuum topology from a causal set*, J. Math. Phys. **48**, 032501 (2007). <https://doi.org/10.1063/1.2435599>.
- [6] S. Major, D. Rideout, and S. Surya, *Stable homology as an indicator of manifoldlikeness in causal set theory*, Class. Quantum Grav. **26**, 175008 (2009). <https://doi.org/10.1088/0264-9381/26/17/175008>.

- [7] L. Glaser and S. Surya, *Towards a definition of locality in a manifoldlike causal set*, Phys. Rev. D **88**, 124026 (2013). <https://doi.org/10.1103/PhysRevD.88.124026>.
- [8] D. D. Reid, *Manifold dimension of a causal set: Tests in conformally flat spacetimes*, Phys. Rev. D **67**, 024034 (2003). <https://doi.org/10.1103/PhysRevD.67.024034>.
- [9] D. M. T. Benincasa and F. Dowker, *The scalar curvature of a causal set*, Phys. Rev. Lett. **104**, 181301 (2010). <https://doi.org/10.1103/PhysRevLett.104.181301>.
- [10] F. Dowker and L. Glaser, *Causal set d'Alembertians for various dimensions*, Class. Quantum Grav. **30**, 195016 (2013). <https://doi.org/10.1088/0264-9381/30/19/195016>.
- [11] L. Bombelli, J. Henson, and R. D. Sorkin, *Discreteness without symmetry breaking: A theorem*, Mod. Phys. Lett. A **24**, 2579–2587 (2009). <https://doi.org/10.1142/S0217732309031958>.
- [12] S. W. Hawking, A. R. King, and P. J. McCarthy, *A new topology for curved space-time which incorporates the causal, differential, and conformal structures*, J. Math. Phys. **17**, 174–181 (1976). <https://doi.org/10.1063/1.522874>.
- [13] D. B. Malament, *The class of continuous timelike curves determines the topology of spacetime*, J. Math. Phys. **18**, 1399–1404 (1977). <https://doi.org/10.1063/1.523436>.
- [14] B. Müller, J. Kim, D. Matscheko, and J. Hill, *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/observable_normal_forms.pdf.
- [15] B. Müller, A. Osika, M. Ponder, K. Xue, P. Nguyen, M. A. Visser, D. Matscheko, and B. de La Fournière, *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*, 2026. Available at [GitHub PDF](#).
- [16] B. Müller, A. Osika, M. Ponder, K. Xue, P. Nguyen, and D. Matscheko, *Deriving Standard Model Gauge Structure from Observer Overlap Consistency*, 2026. Available at [GitHub PDF](#).
- [17] B. Müller, K. Xue, J. Kim, K. A. Anirudha, D. Matscheko, and J. Hill, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/reality_as_consensus_protocol.pdf.
- [18] B. Müller, A. Osika, K. Xue, B. Cassie, and B. de La Fournière, *Federated Echoshedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/screen_microphysics_and_observer_synchronization.pdf.
- [19] B. Müller, A. Osika, M. Ponder, and K. Xue, *Deriving the Particle Zoo from Observer Consistency*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/deriving_the_particle_zoo_from_observer_consistency.pdf.
- [20] B. Müller and D. Matscheko, *Observer-Patch Holography and the Dark Sector: Modular Charge, the Anomalous Collar Source, and the Deep Galaxy Law*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/cosmology/oph_dark_matter_paper.pdf.

- [21] B. Müller and J. Hill, *Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/yang_mills_gap_clay_problem.pdf.
- [22] B. Müller, *Observer-Patch Holography as a String-Vacuum Selector: Observers, Clocks, Edge Strings, and the Bouchard-Donagi Witness*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/observer_patch_holography_as_string_vacuum_selector.pdf.
- [23] B. Müller, *The de Sitter Time-Advance Sign from a Finite Screen with Fixed Capacity*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/de_sitter_time_advance_sign_from_fixed_screen_capacity.pdf.
- [24] B. Müller, *OPH-FPE: finite simulator and receipt engine for Observer-Patch Holography physics experiments*, 2026. Source available at <https://github.com/muellerberndt/oph-physics-sim>. Large-run source revision: <https://github.com/muellerberndt/oph-physics-sim/tree/ce17921eb7504106fef1ba445e1349b5367aa676>. Exact-federation archive: https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence/exact_federation_L6_canonical_20260909. Closure-loop archive: https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence/closure_loop. Active refinement-readout revision: <https://github.com/muellerberndt/oph-physics-sim/tree/036b12b608a9ca4c19d69f5e951b3b83a7a51cd8>. Finite local-domain revision and data: <https://github.com/muellerberndt/oph-physics-sim/tree/550fe77dc67dc8bef2dd8927bcf72ec98cfa3506>, https://github.com/muellerberndt/oph-physics-sim/tree/550fe77dc67dc8bef2dd8927bcf72ec98cfa3506/data/local_domain. Exact repository-local archive and checker: https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence/local_domain. Causal-poset evidence package: https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence/source_net_causal_poset. Closure-loop evidence package: https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence/closure_loop. Evidence index: <https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence>.
- [25] Pragma Research, *Observer-Patch Holography: machine-checked proofs and executable certificates*, 2026. Available at <https://github.com/FloatingPragma/observer-patch-holography>.
- [26] Pragma Research, *Frozen Prediction Register and Primitive Twelve-Port Prediction Receipt*, 2026. Immutable source receipt: https://github.com/FloatingPragma/observer-patch-holography/blob/66176656dc1143f9ec50ba1a6e409c403545857f/code/a5_fingerprint/runtime/spin_six_primitive_port_prediction_receipt.json. Append-only custody packet: https://github.com/FloatingPragma/oph-meta/tree/8cc5261653e37cbca0e6017fcc95a9fe7f649963/falsification/frozen_targets/fz11_2026-07-31.
- [27] DESI Collaboration, “DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints,” 2025, arXiv:2503.14738. <https://arxiv.org/abs/2503.14738>. Official chains and checksum manifest: <https://data.desi.lbl.gov/public/papers/y3/bo-cosmo-params/README.html>.

- [28] M. Filzinger et al., “Improved Limits on the Coupling of Ultralight Bosonic Dark Matter to Photons from Optical Atomic Clock Comparisons,” *Physical Review Letters* **130**, 253001 (2023). <https://doi.org/10.1103/PhysRevLett.130.253001>.
- [29] F. Lelli, S. S. McGaugh, and J. M. Schombert, “SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves,” *Astronomical Journal* **152**, 157 (2016), arXiv:1606.09251. <https://arxiv.org/abs/1606.09251>. CDS/VizieR catalogue: <https://cdsarc.cds.unistra.fr/ftp/J/AJ/152/157/>.
- [30] S. S. McGaugh, F. Lelli, and J. M. Schombert, “Radial Acceleration Relation in Rotationally Supported Galaxies,” *Physical Review Letters* **117**, 201101 (2016), arXiv:1609.05917. <https://arxiv.org/abs/1609.05917>.
- [31] L. Onsager, “Reciprocal Relations in Irreversible Processes. I,” *Physical Review* **37**, 405–426 (1931), <https://doi.org/10.1103/PhysRev.37.405>.
- [32] M. S. Green, “Markoff Random Processes and the Statistical Mechanics of Time-Dependent Phenomena. II,” *Journal of Chemical Physics* **22**, 398–413 (1954), <https://doi.org/10.1063/1.1740082>; R. Kubo, “Statistical-Mechanical Theory of Irreversible Processes. I,” *Journal of the Physical Society of Japan* **12**, 570–586 (1957), <https://doi.org/10.1143/JPSJ.12.570>.
- [33] I. Csiszár, “I-Divergence Geometry of Probability Distributions and Minimization Problems,” *The Annals of Probability* **3**, 146–158 (1975), <https://doi.org/10.1214/aop/1176996454>.
- [34] J. E. Marsden and M. West, “Discrete Mechanics and Variational Integrators,” *Acta Numerica* **10**, 357–514 (2001), <https://doi.org/10.1017/S096249290100006X>.
- [35] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, and T. Schwetz, *NuFit-6.0: Updated global analysis of three-flavor neutrino oscillations*, JHEP **12** (2024) 216, arXiv:2410.05380, with the NuFIT 6.1 (2025) profile-table release at <https://www.nu-fit.org/?q=node/309>.
- [36] M. H. A. Newman, “On theories with a combinatorial definition of ‘equivalence’,” *Ann. of Math. (2)* **43** (1942), 223–243.
- [37] E. H. Lieb and D. W. Robinson, “The finite group velocity of quantum spin systems,” *Commun. Math. Phys.* **28** (1972), 251–257.
- [38] J. J. Bisognano and E. H. Wichmann, “On the duality condition for a Hermitian scalar field,” *J. Math. Phys.* **16** (1975), 985–1007.
- [39] J. J. Bisognano and E. H. Wichmann, “On the duality condition for quantum fields,” *J. Math. Phys.* **17** (1976), 303–321.
- [40] R. Brunetti, D. Guido, and R. Longo, “Modular Structure and Duality in Conformal Quantum Field Theory,” *Commun. Math. Phys.* **156** (1993), 201–219, arXiv:funct-an/9302008.
- [41] H.-W. Wiesbrock, “Half-Sided modular inclusions of von-Neumann-Algebras,” *Commun. Math. Phys.* **157** (1993), 83–92.
- [42] H. Araki and L. Zsidó, “Extension of the structure theorem of Borchers and its application to half-sided modular inclusions,” *Rev. Math. Phys.* **17** (2005), 491–543, arXiv:math/0412061.

- [43] D. Guido, R. Longo, and H.-W. Wiesbrock, “Extensions of Conformal Nets and Superselection Structures,” *Commun. Math. Phys.* **192** (1998), 217–244, arXiv:hep-th/9703129.
- [44] H.-W. Wiesbrock, “Modular Intersections of von-Neumann-Algebras in Quantum Field Theory,” *Commun. Math. Phys.* **193** (1998), 269–285.
- [45] W. G. Unruh, “Notes on black-hole evaporation,” *Phys. Rev. D* **14** (1976), 870–892.
- [46] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Phys. Rev. Lett.* **75** (1995), 1260–1263, doi:10.1103/PhysRevLett.75.1260, arXiv:gr-qc/9504004.
- [47] T. Jacobson, “Entanglement equilibrium and the Einstein equation,” *Phys. Rev. Lett.* **116** (2016), 201101, doi:10.1103/PhysRevLett.116.201101, arXiv:1505.04753.
- [48] G. W. Gibbons and S. W. Hawking, “Cosmological event horizons, thermodynamics, and particle creation,” *Phys. Rev. D* **15** (1977), 2738–2751, doi:10.1103/PhysRevD.15.2738.
- [49] Y. Chen, D. Stanford, H. Tang, and Z. Yang, “Negative shocks versus static patch holography,” arXiv:2607.14042 [hep-th] (2026). <https://arxiv.org/abs/2607.14042>
- [50] Planck Collaboration, “Planck 2018 results. VI. Cosmological parameters,” *Astron. Astrophys.* **641** (2020), A6, doi:10.1051/0004-6361/201833910, arXiv:1807.06209.
- [51] R. Bousso, Z. Fisher, J. Koeller, S. Leichenauer, and A. C. Wall, “Proof of the quantum null energy condition,” *Phys. Rev. D* **93** (2016), 024017, arXiv:1509.02542.
- [52] R. Bousso, Z. Fisher, S. Leichenauer, and A. C. Wall, “Quantum focusing conjecture,” *Phys. Rev. D* **93** (2016), 064044, arXiv:1506.02669.
- [53] S. Balakrishnan, T. Faulkner, Z. U. Khandker, and H. Wang, “A general proof of the quantum null energy condition,” *J. High Energy Phys.* **2019** (2019), 20, arXiv:1706.09432.
- [54] E. H. Lieb and M. B. Ruskai, “Proof of the strong subadditivity of quantum-mechanical entropy,” *J. Math. Phys.* **14** (1973), 1938–1941.
- [55] E. H. Wichmann, “Density matrices arising from incomplete measurements,” *J. Math. Phys.* **4** (1963), 884–896.
- [56] F. Hiai, M. Ohya, and M. Tsukada, “Sufficiency, KMS condition and relative entropy in von Neumann algebras,” *Pacific J. Math.* **96** (1981), 99–109.
- [57] M. Takesaki, “Conditional expectations in von Neumann algebras,” *J. Funct. Anal.* **9** (1972), 306–321.
- [58] D. Petz, “Sufficient subalgebras and the relative entropy of states of a von Neumann algebra,” *Commun. Math. Phys.* **105** (1986), 123–131.
- [59] D. Petz, “Sufficiency of channels over von Neumann algebras,” *Quart. J. Math.* **39** (1988), 97–108.
- [60] O. Fawzi and R. Renner, “Quantum conditional mutual information and approximate Markov chains,” *Commun. Math. Phys.* **340** (2015), 575–611, arXiv:1410.0664.
- [61] P. Hayden, R. Jozsa, D. Petz, and A. Winter, “Structure of states which satisfy strong subadditivity of quantum entropy with equality,” *Commun. Math. Phys.* **246** (2004), 359–374.

- [62] S. Doplicher and J. E. Roberts, “A new duality theory for compact groups,” *Invent. Math.* **98** (1989), 157–218.
- [63] S. Doplicher and J. E. Roberts, “Why there is a field algebra with a compact gauge group describing the superselection structure in particle physics,” *Commun. Math. Phys.* **131** (1990), 51–107.
- [64] T. Tannaka, “Über den Dualitätssatz der nichtkommutativen topologischen Gruppen,” *Tohoku Math. J.* **45** (1938), 1–12.
- [65] M. G. Krein, “A principle of duality for a bicomact group and a square block algebra,” *Dokl. Akad. Nauk SSSR* **69** (1949), 725–728.
- [66] H. Georgi and S. L. Glashow, “Unity of all elementary-particle forces,” *Phys. Rev. Lett.* **32** (1974), 438–441.
- [67] S. L. Glashow, J. Iliopoulos, and L. Maiani, “Weak interactions with lepton-hadron symmetry,” *Phys. Rev. D* **2** (1970), 1285–1292.
- [68] E. Witten, “An $SU(2)$ anomaly,” *Phys. Lett. B* **117** (1982), 324–328.
- [69] S. Dimopoulos, S. Raby, and F. Wilczek, “Supersymmetry and the scale of unification,” *Phys. Rev. D* **24** (1981), 1681–1683.
- [70] U. Amaldi, W. de Boer, and H. Fürstenau, “Comparison of grand unified theories with electroweak and strong coupling constants measured at LEP,” *Phys. Lett. B* **260** (1991), 447–455.
- [71] D. J. Gross and W. Taylor, “Two-dimensional QCD is a string theory,” *Nucl. Phys. B* **400** (1993), 181–208, arXiv:hep-th/9301068.
- [72] F. Peter and H. Weyl, “Die Vollständigkeit der primitiven Darstellungen einer geschlossenen kontinuierlichen Gruppe,” *Math. Ann.* **97** (1927), 737–755.
- [73] A. W. Knap, *Lie Groups Beyond an Introduction*, second edition, Progress in Mathematics **140**, Birkhäuser, 2002.
- [74] A. Bullivant, M. Calçada, Z. Kádár, P. Martin, and J. Faria Martins, “Topological phases from higher gauge symmetry in 3+1D,” *Phys. Rev. B* **95** (2017), 155118, arXiv:1606.06639.
- [75] S. Chandrasekharan and U.-J. Wiese, “Quantum link models: A discrete approach to gauge theories,” *Nucl. Phys. B* **492** (1997), 455–471, arXiv:hep-lat/9609042.
- [76] W. Donnelly and A. C. Wall, “Entanglement entropy of electromagnetic edge modes,” *Phys. Rev. Lett.* **114** (2015), 111603, arXiv:1412.1895.
- [77] F. Pastawski, B. Yoshida, D. Harlow, and J. Preskill, “Holographic quantum error-correcting codes: Toy models for the bulk/boundary correspondence,” *JHEP* **06** (2015), 149, arXiv:1503.06237.
- [78] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” *Phys. Rev. B* **71** (2005), 045110, arXiv:cond-mat/0404617.
- [79] D. Laghi, G. Carullo, J. Veitch, and W. Del Pozzo, “Quantum black hole spectroscopy: probing the quantum nature of the black hole area using LIGO-Virgo ringdown detections,” *Class. Quantum Grav.* **38** (2021), 095005, arXiv:2011.03816.

- [80] Clay Mathematics Institute, “Yang–Mills & the Mass Gap.” Available at <https://www.claymath.org/millennium/yang-mills-the-maths-gap/>.
- [81] A. Jaffe and E. Witten, “Quantum Yang–Mills Theory,” official Clay Mathematics Institute problem description. Available at <https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf>.
- [82] L. Wu, “Poincaré and transportation inequalities for Gibbs measures under the Dobrushin uniqueness condition,” *Annals of Probability* **34** (2006), 1960–1989, arXiv:math/0611635.
- [83] K. Osterwalder and R. Schrader, “Axioms for Euclidean Green’s functions,” *Communications in Mathematical Physics* **31** (1973), 83–112.
- [84] K. Osterwalder and R. Schrader, “Axioms for Euclidean Green’s functions II,” *Communications in Mathematical Physics* **42** (1975), 281–305.
- [85] Particle Data Group, “Review of Particle Physics,” *Phys. Rev. D* **110** (2024), 030001. Available at <https://pdg.lbl.gov/2024/download/db2024.pdf>.
- [86] Particle Data Group, “Electroweak Model and Constraints on New Physics,” Available at <https://pdg.lbl.gov/2024/reviews/rpp2024-rev-standard-model.pdf>.

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